

GE Healthcare

Signa HDx Series 1.5T TwinSpeed System Installation

Technical Publication



5166002
Rev 11

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation “**damage in shipment**” written on **all** copies of the freight or express bill **before** delivery is accepted or “signed for” by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period.

File a report with

- Call 1–800–548–3366 and use option 8.
- Contact your local service coordinator for more information on this process.

Rev. 08/15/2003

© 2006 – 20010 General Electric Company
GE Medical Systems, a General Electric Company, going to market as GE Healthcare
3000 N. Grandview Boulevard
Waukesha, Wisconsin 53188
USA

www.gehealthcare.com



WARNING

- THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.
- IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.
- DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.
- FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.

AVERTISSEMENT

- CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.
- SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.
- NE PAS TENTER D'INTERVENTION SUR LES ÉQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ÉTÉ CONSULTÉ ET COMPRIS.
- LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.

WARNUNG

- DIESES KUNDENDIENST-HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.
- FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.
- VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST-HANDBUCH ZU RATE GEZOGEN UND VERSTANDEN WURDE.
- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

AVISO

- ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS.
- SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.
- NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.
- LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.

ATENÇÃO

- ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.
- SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.
- NÃO TENHA TENTADO REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.
- O NÃO CUMPRIMENTO DESTA AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

AVVERTENZA

- IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.
- SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.
- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.
- NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

警告

- ・ このサービスマニュアルには英語版しかありません。
- ・ GEMS以外でサービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
- ・ このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないで下さい。
- ・ この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

注意：

- 本维修手册仅存有英文本。
- 非 GEMS 公司的维修员要求非英文本的维修手册时，客户需自行负责翻译。
- 未详细阅读和完全了解本手册之前，不得进行维修。
- 忽略本注意事项会对维修员，操作员或病人造成触电，机械伤害或其他伤害。

REVISION HISTORY

REV	DATE	PRIMARY REASON FOR CHANGE
1 . . .	Mar 7, 2006 . .	Initial Release.
2 . . .	Jul 11, 2006 . .	Section 1: Added LCD monitor adjustments to install flow chart (MRIhc15805). Section 6: Added Note 14 to cable maps for network availability for Magnet Monitor. Updated cable connections to rear pedestal, Sec 6–3–7–B. Added CAUTION note to not install O–ring and grease when connecting RF Body cables to penetration panel, Sec 6–4–2–D. Section 8: Added note informing the installer not to connect additional monitors to host computer prior to operating sustem software load (MRIhc15870).
3 . .	Feb 22, 2007 . .	Section 8: Added note that Humidity iButton can be installed in either direction. Appendix: Section 10–2–1, Corrected voltage ratings for ground cables 040 & 1254. Section 10–2–7, Added Run 2039 and added optional length for Run 2038.
4 . .	May 11, 2007 . .	Section 6: Updated cable maps to designate optional 16 Channel cable installation. Added Pang block and gradient cable connection information (CQA 13109555). Revised procedure for attaching and aligning rear pedestal and split bridge (CQA 13074448). Added note to Section 6–4–1 designating 16/32 Channel only cables. Section 8: Replaced existing OW table with new table, 5189137, and revised installation instructions. Updated iButton positioning and added block diagram in Environmental Sensor installation, Section 8–11. Section 10: Updated cable remarks for those cables used only in 16 Channel System.
5 . .	Aug 16, 2007 . .	Section 3: Added installation step for attaching the Canadian Registration Label. (PSR 13098263).
6 . . .	Oct 11, 2007 . .	Section 1: Added sub–section 1–3 for non–English label installation instructions. Section 6: Updated fiber optic connections in RFS cabinet for new CAM chassis, sub–section 6–2–6. Section 7: Revised Laser Alignment light warning label application for new covers with only a single laser light (ECR 2044782). Section 8: Added 1–wire Hub power cord connection options to J4, J5, or J6 (PQR 13097707).
7 . . .	Jun 5, 2008 . .	Changed title page and page headers to <i>Signa HDx Series 1.5T TwinSpeed System Installation</i> to reflect all product offerings in the HDx series. Removed Times Microwave stripping and cutting tools from Required Tools table in Sub–section 1–4 and moved information to Sub–section 6–4–2 as an orderable FRU. Section 6: Updated cable maps to show new GOC installation and the new Temperature and Humidity Sensor installation and cables. Corrected the J3 and J4 connection designations at the hybrid splitter (PQR 13172263). Section 8: Revised entire section for systems that are shipped after July 1, 2008 with new GOC. In Sub–section 8–3–10, revised temperature/humidity sensor power cord connection to J6 on power strip (PQR 13166741) and added power draw to Max Amp table in Sub–section 8–3–8 (PQR 13166207).
8 . .	Sep 18, 2008 . .	Section 1: Added note that the SPT Phantom Cart is not used with Mobile system (MRIhc34269). Added Sub–section 1–13 for Mobile installation flowchart. Section 2: Added notes about Mobile installation not using the pen panel wave guides or covers. Section 5: Added note for Mobile connecting coolant hoses to heat exchanger. Section 6: (ECO 2054786) Sub–section 6–1–1, updated cable maps by adding Run 1382. Sub–section 6–2–1, added Runs 1381 and 1382 to PDU connection table. Sub–section 6–3–2, added notes for Mobile gradient cable installation. Setion 7: Added note for Mobile installation of enclosure covers. Section 8: Sub–Section 8–2–7, corrected power and ground cable run numbers connected to OW. Sub–section 8–2–12, Added note about SCSI Tower being an option for HDxt and 15.x.
9 . .	Jun 17, 2009 . .	Section 7: Removed section on installation of rating plates (PQR 13246367). Added English label and information on attaching non–English label (PQR 13246299).
10 . .	Jun 4, 2010 . .	Section 6: Updated cable maps with new run numbers for Temp/Humidity cables (ECO 2055310) and added note that SCSI Tower is not supported for Z400 PC (ECO 2096124) Section 8: Added instructions for connecting cables to Z400 PC (ECO 2096124).
11 .	Oct 20, 2010 . .	Section 1: Revised Ferrous Material Handling warning. Inserted new Section 1–6 for Blower Box 200 gauss zone. Section 6: Revised Ferrous Material Handling warnings and instructions for installing blower box. Section 9: Added check box for inspecting Blower Box installation. (CAPA 1859293)

LIST OF EFFECTIVE PAGES

<u>PAGE</u>	<u>REV</u>	<u>PAGE</u>	<u>REV</u>	<u>PAGE</u>	<u>REV</u>	<u>PAGE</u>	<u>REV</u>	<u>PAGE</u>	<u>REV</u>
Title Page	11	TAB 3		TAB 6		7-2	8	8-16 to 8-28	7
Damage in Trans.	—	CABINET		CABLE & SYSTEM		7-3 to 7-7	7		
Direction 2128126	1*	POSITIONING		INSTALLATION		7-8 to 7-11	9	TAB 9	
A/B	11	2164904-3	—	2164904-6	—			SYSTEM	
i	7	3-1 to 3-4	7	6-1	11			MECHANICAL	
		TAB 4		6-2	7	TAB 8		CHECKLIST	
TAB 1		ACGD/PDU SET-UP		6-3 to 6-6	10	OPERATOR		2164904-9	—
GETTING STARTED		2164904-4	—	6-7 to 6-8	7	WORKSPACE		9-1	7
2164904-1	—	4-1 to 4-3	7	6-9 to 6-10	8	2164904-8	—	9-2	11
1-1 to 1-17	11			6-11 to 6-27	7	8-1 to 8-7	7	9-3 to 9-5	7
		TAB 5		6-28	10	8-8	8		
TAB 2		COOLING SYSTEM		6-29 to 6-45	7	8-9	7	TAB 10	
PENETRATION PANEL		INSTALLATION		6-46	8	8-10	10	APPENDIX	
2164904-2	—	2164904-5	—	6-47 to 6-51	7	8-11	7	2164904-10	—
2-1 to 2-4	7	5-1	7	6-52 to 6-72	11	8-12	10	10-1 to 10-15	7
2-4	8	5-2	8			8-13 to 8-14	7	10-16	10
2-5	7	5-3 to 5-5	7	TAB 7		8-15	10	10-17 to 10-18	7
				FINAL ENCLOSURE					
				INSTALLATION					
				2164904-7	—				
				7-1	9				

* This revision number/letter corresponds to the indicated document's revision control system.

TABLE OF CONTENTS

<u>SECTION</u>	<u>PAGE</u>
SECTION 1 – GETTING STARTED	1-1
SECTION 2 – PENETRATION PANEL	2-1
SECTION 3 – CABINET POSITIONING	3-1
SECTION 4 – HFD CABINET SET-UP	4-1
SECTION 5 – COOLING SYSTEM INSTALLATION	5-1
SECTION 6 – CABLE & SYSTEM INSTALLATION	6-1
SECTION 7 – ENCLOSURE INSTALLATION	7-1
SECTION 8 – OPERATOR WORKSPACE INSTALLATION	8-1
SECTION 9 – SYSTEM MECHANICAL CHECKLIST COMPLETION	9-1
SECTION 10 – APPENDIX	10-1

SECTION 1 – GETTING STARTED

TABLE OF CONTENTS

<u>SECTION</u>	<u>PAGE</u>
1-1 SIGNA EXCITE HD 1.5T TwinSpeed SYSTEM CONFIGURATION	1-2
1-2 SITE READY CHECK FOR MECHANICAL INSTALLATION	1-2
1-3 NON-ENGLISH LABEL INSTALLATION	1-2
1-4 REQUIRED TOOLS	1-2
1-5 LOCATION AND MOVEMENT OF BLOWER BOX	1-4
1-6 BASIC SYSTEM	1-4
1-7 SYSTEM OPTIONS	1-6
1-8 PRODUCT DELIVERY INSTRUCTIONS	1-6
1-9 DAMAGE IN TRANSPORTATION	1-7
1-10 PRODUCT LOCATOR	1-7
1-11 INSTALLATION PROCEDURE	1-8
1-11-1 INSTALLATION FLOW	1-8
1-11-2 INSTALLATION FLOWCHART EXPLANATION	1-8
1-12 INSTALLATION FLOWCHART	1-9
1-12-1 PREREQUISITES FOR SYSTEM INSTALLATION	1-9
1-12-2 SYSTEM MECHANICAL INSTALLATION	1-10
1-13 INSTALLATION FLOWCHART FOR MOBILE	1-14
1-13-1 PREREQUISITES FOR SYSTEM INSTALLATION	1-14
1-13-2 SYSTEM MECHANICAL INSTALLATION	1-15

1-1 SIGNA HDx 1.5T TwinSpeed SYSTEM CONFIGURATION

This manual provides instructions to complete the installation of Signa HDx (14.x) or HDxt (15.x) 1.5T TwinSpeed System.

The Fixed-Site installation flowchart is located in Section 1-12.

The Mobile installation flowchart is located in Section 1-14.

Additional options are installed according to the installation manual delivered with the option.

1-2 SITE READY CHECK FOR MECHANICAL INSTALLATION

Before equipment is delivered, the following must be complete to avoid delays and confusion.

- ☐ Pre-installation work must be complete. Refer to *Direction 5159902, Signa HDx, HDxt 1.5T TwinSpeed Pre-installation*, Preinstallation Checklist tab.
- ☐ Magnet Delivery and Installation complete. Refer to *Direction 5154839, GE HDe & HDx 1.5T LCC Magnet Delivery and Installation*.
- ☐ Magnet Subsystem should be installed. Refer to *Direction 2192624, GE 1.5T & 1.0T LCC Active Shield Magnet and Cryogen Subsystem*, Section 1.

1-3 NON-ENGLISH LABEL INSTALLATION

Components and accessories delivered to this site may have English labels attached to them. For non-English sites, optional multi-language labels have also been supplied.

If required, non-English labels should be attached over the English labels.

- For Surface Coils and Coil Cables – Apply the appropriate label over the English label. The label is included with the coil component.
- For all other labels, follow the label update procedures in the manual or on the applicable multi-language label sheet.

Verify at the end of the installation that the original English labels are covered by new labels representing the native language of the customer.

1-4 REQUIRED TOOLS



FERROUS MATERIAL HAZARD! IF MAGNET IS ENERGIZED, THE CRIMP TOOL, BLOWER BOX, AND OTHER TOOLS AND PARTS REQUIRED FOR THIS INSTALLATION THAT CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES. KEEP ALL FERROUS TOOLS AS FAR FROM THE MAGNET AS POSSIBLE.

Tools to install the Signa HDx or HDxt 1.5T TwinSpeed system are listed on the following page:

1-4 REQUIRED TOOLS (Continued)

TABLE 1-2
INSTALLATION EQUIPMENT

ITEM	GE PART NUMBER	DESCRIPTION
1	—	Ramp for removing cabinets from pallets for International shipments (See Note 1)
2	—	Wrecking bar
3	—	Claw hammer, 3/4 lb
4	46-271138G1	Restricted Access Control Kit. Contains two plastic warning signs for posting at site during installation and service activity.
5	—	4 foot or equivalent carpenter level
6	2319156	Aluminum platform ladder, 47.5 inches (1206.5mm) (See Note 1)
7	2134776	Gradient Cable Crimper/Stripper Kit (Note 2) consisting of: • 2134586 Cable stripping tool • 2134586-2 Stripping tool replacement blade • 2134587 Cable slicer • 46-282853P1 Ratcheting crimper • 2135839 1/2 inch terminals • 2135839-2 3/8 inch terminals
8	46-320273G3 or G4	Non-Magnetic Tool Kit – Universal (See Note 2) Both metric and inch Non-Magnetic Tool Kits needed. May substitute both of the following kits: • 46-320273G1 Non-Magnetic Tool Kit – Metric • 46-320273G2 Non-Magnetic Tool Kit – Inch
9	46-301450G1	Fiber optic connector repair kit (See Note 2)
10	46-198094P1	Wrist grounding strap
11	—	Volt Meter
12	—	Extension cords, with ground conductor
13	—	Power strip, grounded type, with minimum of five outlets
14	—	Plastic or aluminum flashlight
15	—	Assorted crimp tools.
16	—	Non-magnetic level
17	—	Non-magnetic tape rule, 12 ft
18	—	Assorted drill bits
19	—	Inspection mirror
20	—	Hobby and utility knives
Note		
1	Supplied as part of Signa.	
2	Supplied by GE until turnover of system to customer, then available as part of a GE Cryogen and/or Service Contract.	

1-5 LOCATION AND MOVEMENT OF BLOWER BOX



FERROUS MATERIAL HAZARD!

KEEP BLOWER BOX AS FAR FROM THE MAGNET AS POSSIBLE. THE BLOWER BOX REQUIRES TWO PEOPLE TO MOVE AND MUST BE INSTALLED OUTSIDE THE 200 GAUSS ZONE.

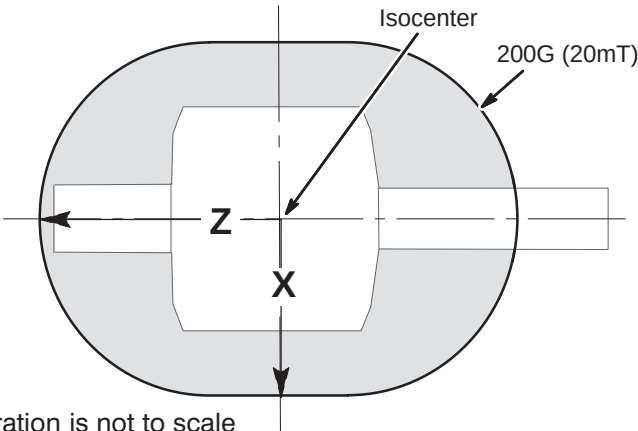


Illustration is not to scale

Approximate Distance from isocenter to 200G Line		
	X	Z
1.5T	150 cm (59 in) ¹	205 cm (81 in) ¹
Unshielded Magnet	Measure ²	
Note 1: These distances represent maximum distances for the magnet strength listed. Use gauss meter if needed. Note 2: For unshielded magnets, the magnetic field must be measured using a gauss meter.		

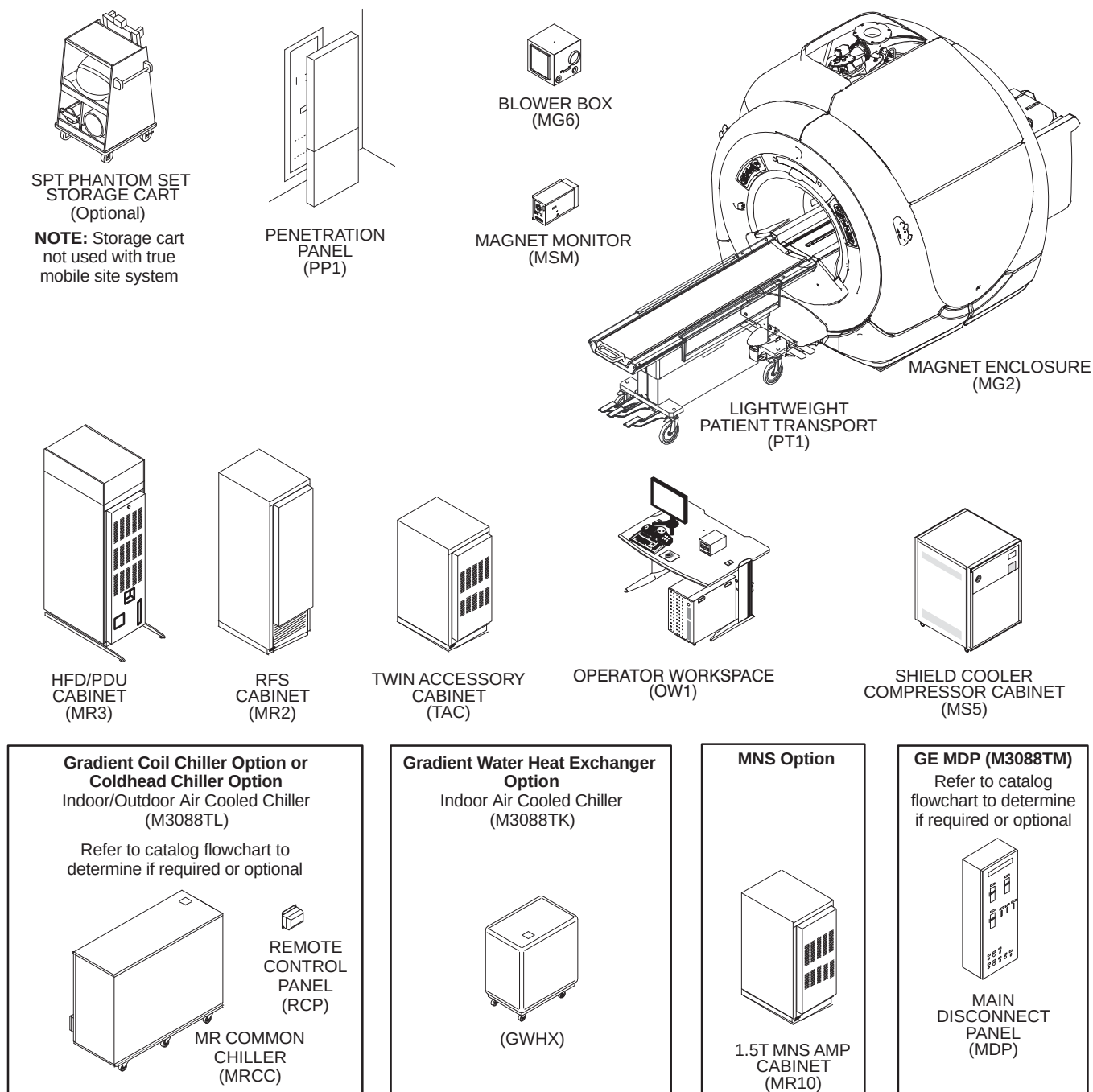
1-6 BASIC SYSTEM

The basic Signal HDx 1.5T TwinSpeed system, Illustration 1-1, consists of the following major equipment:

- System cooling equipment (See Illustration 1-2)
- 1.5T (15 kilogauss) LCC Active Shield Magnet with Magnet Enclosure
- Magnet Accessories
- Shield/Cryo Cooler Compressor Cabinet
- TwinSpeed Gradient (TRM) and RF body coils and 3.0T General Purpose Head Coil
- Main Disconnect Panel (GE supplied or customer supplied)
- System electronics cabinets:
 - HFD Cabinet containing HFD Gradient drivers and Power Distribution Unit module with unregulated transformer 200/380/400/415/480 Volt, 50/60 Hz with power filter
 - EXCITE RFS Cabinet containing the Multi Generational Data Acquisition (MGD) chassis with boards for either 8 or 16 Channels, components for the Volume Reconstruction Engine (VRE), Remote Radio Frequency (RRF) chassis, Driver module, RF Amplifier, and power distribution.
 - Twin Accessory Cabinet containing the Gradient Switch and the vacuum pump for Quiet Technology Magnet Enclosure
- Operator Workspace equipment: GOC Computer Cabinet with Linux PC, Mouse and Mouse Pad, LCD panel, and chair
- Pneumatic Patient Alert System

1-6 BASIC SYSTEM (Continued)

- Patient Transport Table and cradle
- Patient accessories such as: a phantom kit, patient log book, head cushion and sponges, chin and forehead straps, body wedges, knee cushions, and security/restraint straps
- Gating accessories which include: patient cardiac leads, peripheral gating probe, and respiratory bellows



SIGNA EXCITE 1.5T SYSTEM
ILLUSTRATION 1-1

1-7 SYSTEM OPTIONS

This section lists options for the Signa HDx or 1.5T TwinSpeed system which have site preparation impact.

- 700 Volt Amp UPS for MR Magnet Monitor Equipment (E4504AG)

Note

The GE MDP (M3088TH) must be ordered and installed when the UPS for MR Magnet Monitor (E4504AG) is installed. The GE MDP provides Emergency Off control of the output of the UPS for system safety.

- Advantage Workstation (AW)

Note

LCD monitor is recommended for a AW to be located near the Signa system Operator Workspace due to gauss field proximity limits.

- Bar Code Reader (M1090PM) for use in conjunction with HIS/RIS software option.
- Additional Patient Transport table with cradle for WideOpen Enclosure (M1090TL).
- Multi-Nuclear Spectroscopy (MNS) options.
- Other hard copy devices and patient accessories.

1-8 PRODUCT DELIVERY INSTRUCTIONS

The "Shipping Document" lists catalog numbers delivered. Review and confirm that order is delivered complete. Check impact on installation schedule if Catalogs and/or packing boxes are missing and/or noted as shipped short.

Labels that are attached to the outside of packing boxes summarize box contents. The labels are color coded for different installation areas as follows:

- Green – Equipment Room
- Purple – Operator's Room
- Orange – Magnet Room
- Yellow – Post-installation (Phantoms, Coils).

"Product Delivery Instructions" (PDI) specify box contents, part numbers, and shipping procedure. The PDI is numbered according to the catalog number (for example, PDI – M3333TA is for the Fixed Site Installation Kit). Lists of items included with each box are detailed by separate checklists, or a separate sheet that provides a summary of that box contents. Refer to PDI and packing lists for information specific to your shipment.

A set of service and operator manuals is delivered with the Fixed Site Kit. Refer to checklists packed with "Technical Publication" boxes for a list of delivered documents.

1-9 DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation “**damage in shipment**” written on **all** copies of the freight or express bill **before** delivery is accepted or “signed for” by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period.

File a report with

- Call 1-800-548-3366 and use option 8.
- Contact your local service coordinator for more information on this process.

Rev. 08/15/2003

1-10 PRODUCT LOCATOR

The Global Install Base Database (also known as Product Locator System) tracks shipment, trans-shipment, and field location of the serialized models. There are now two methods for submitting Product Locator information.

1. At this time, for **U.S. ONLY**, the preferred method of submitting information is the *FE Site Verification Web Site* at:
<http://gein2.med.ge.com/gib>

The FE Site Verification consists of three components that are available on the web from the main menu. They are:

- Install/deinstall product locator model and serial numbers
- Add/modify ship to address information
- Update CARES FE data for primary/secondary FEs

To obtain a copy of the FE training tutorial for using this website, a downloadable copy is available at the following:
Product Locator Support Central Page – <http://supportcentral.ge.com/15563>

2. One “Shipping Card” is filled out and submitted when shipped (extra cards are supplied for trans-shipments between storage and distribution points), and the “Installation Card” and extra shipment cards are attached.

Verify that serial and model number on each rating plate matches installation card numbers before removing installation card. Note that there may be one or more shipment cards and bar code labels with the installation card. These shipment cards are used to trace the transfer of serialized units between various inventory storage and distribution points until the product reaches its final installation destination. Process just the installation card and discard any extra shipment cards and labels.

1-11 INSTALLATION PROCEDURE

1-11-1 Installation Flow

Note

All on-site construction must be completed before equipment is delivered and installation starts. Attempting to install the system while construction is being completed will impact installation efficiency and further delay site completion. Making sure that all preinstallation and construction work is completed before equipment is delivered will usually result in an earlier turnover date.

The flow chart on the next page should be followed for an orderly and efficient installation of the Signa HDx 1.5T TwinSpeed system. Note that many procedures may be performed in parallel and may be performed in any order according to the specific situation of each site.

Note

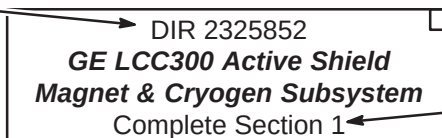
The flow chart references the Tab number and/or Section number for the respective installation procedures. These procedures are similar to cable maps in that they are to be removed from this binder and used at the location where hardware installation is being performed. The Procedures must be returned to the binder for future reference upon completion of the installation.

This installation flowchart has been developed assuming that all Signa HDx 1.5T TwinSpeed equipment has been delivered together. Make sure that every part required is available before starting.

1-11-2 Installation Flowchart Explanation

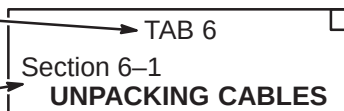
Shown below are three examples of the flow chart Installation Procedures and an explanation of the areas within the procedures.

Use the correct Magnet Manual that applies to the System being installed to locate the installation procedure listed below



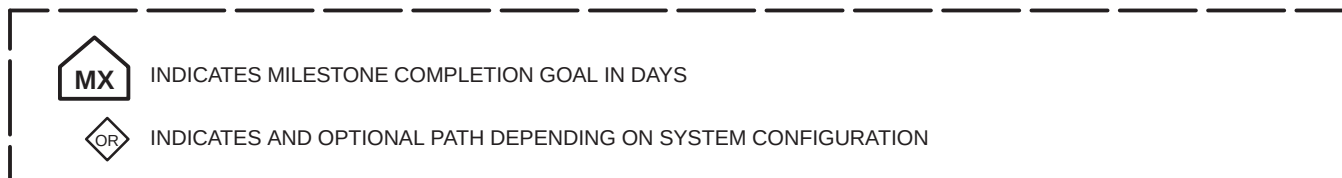
Section within manual to be completed.

Tab within this manual where the Section listed below is located



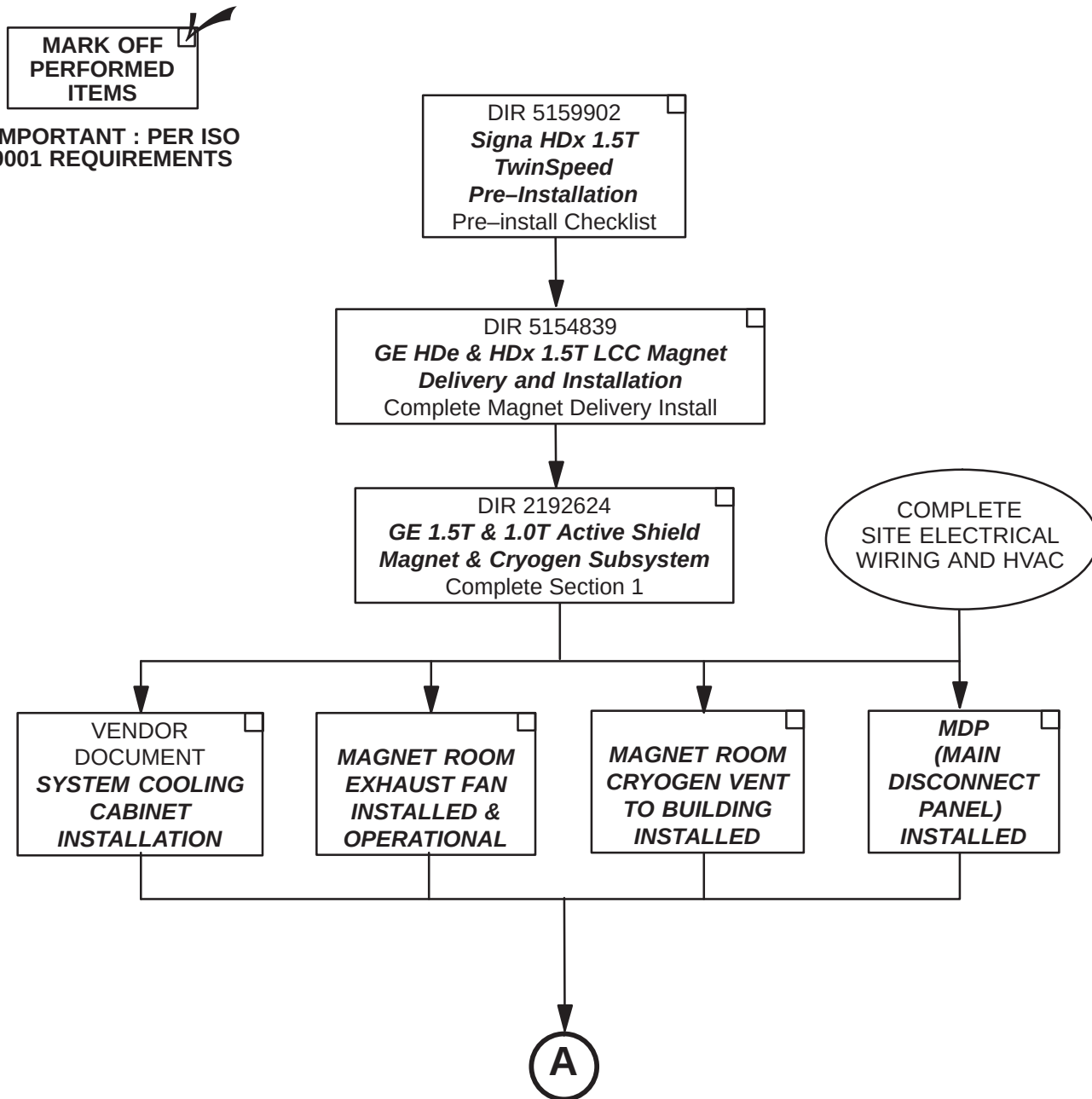
Section within the Tab where the Installation Procedure is located

LEGEND FOR FOLLOWING INSTALLATION FLOWCHART

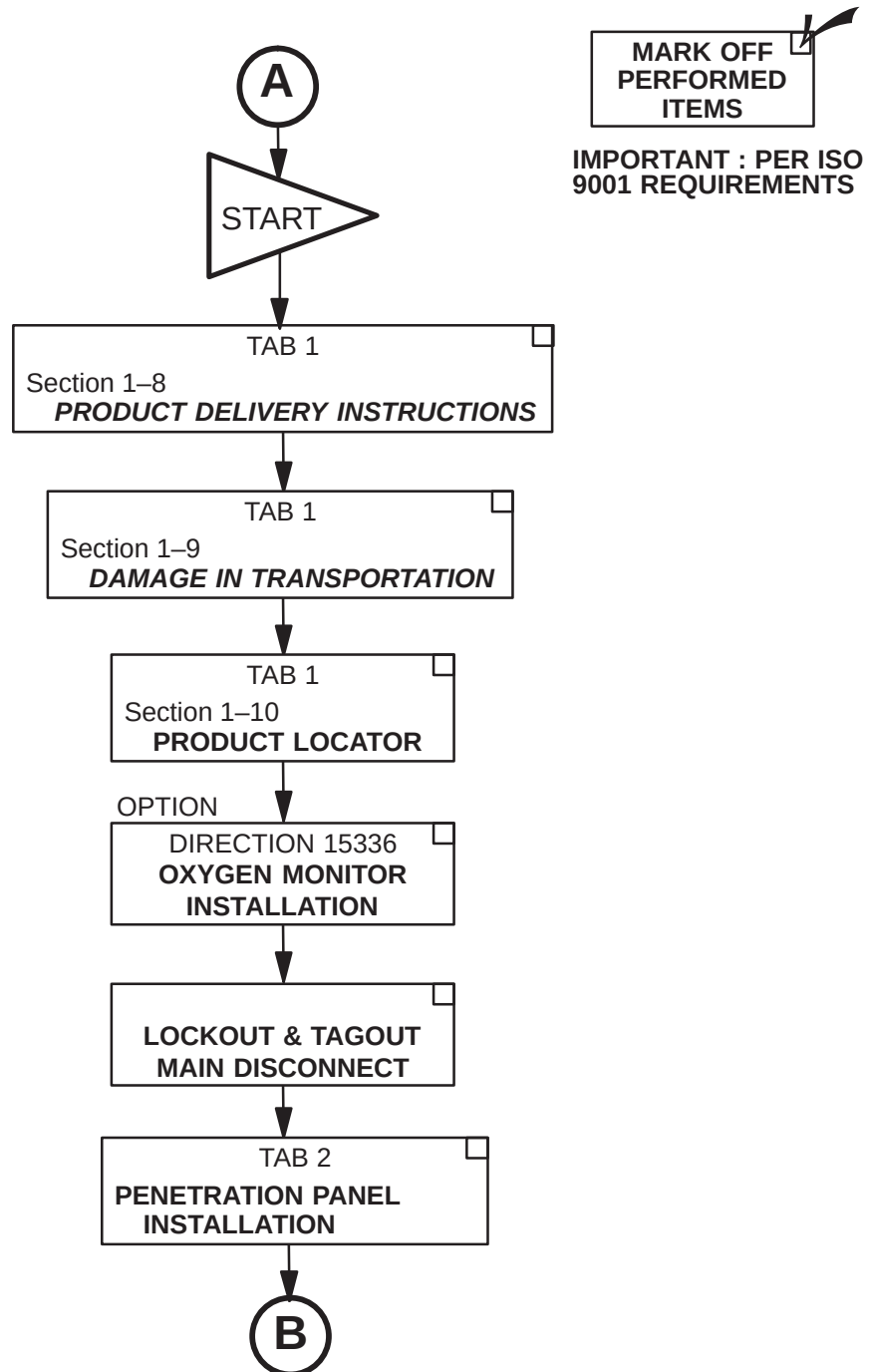


1-12 INSTALLATION FLOWCHART FOR FIXED SITE

1-12-1 PREREQUISITES FOR SYSTEM INSTALLATION

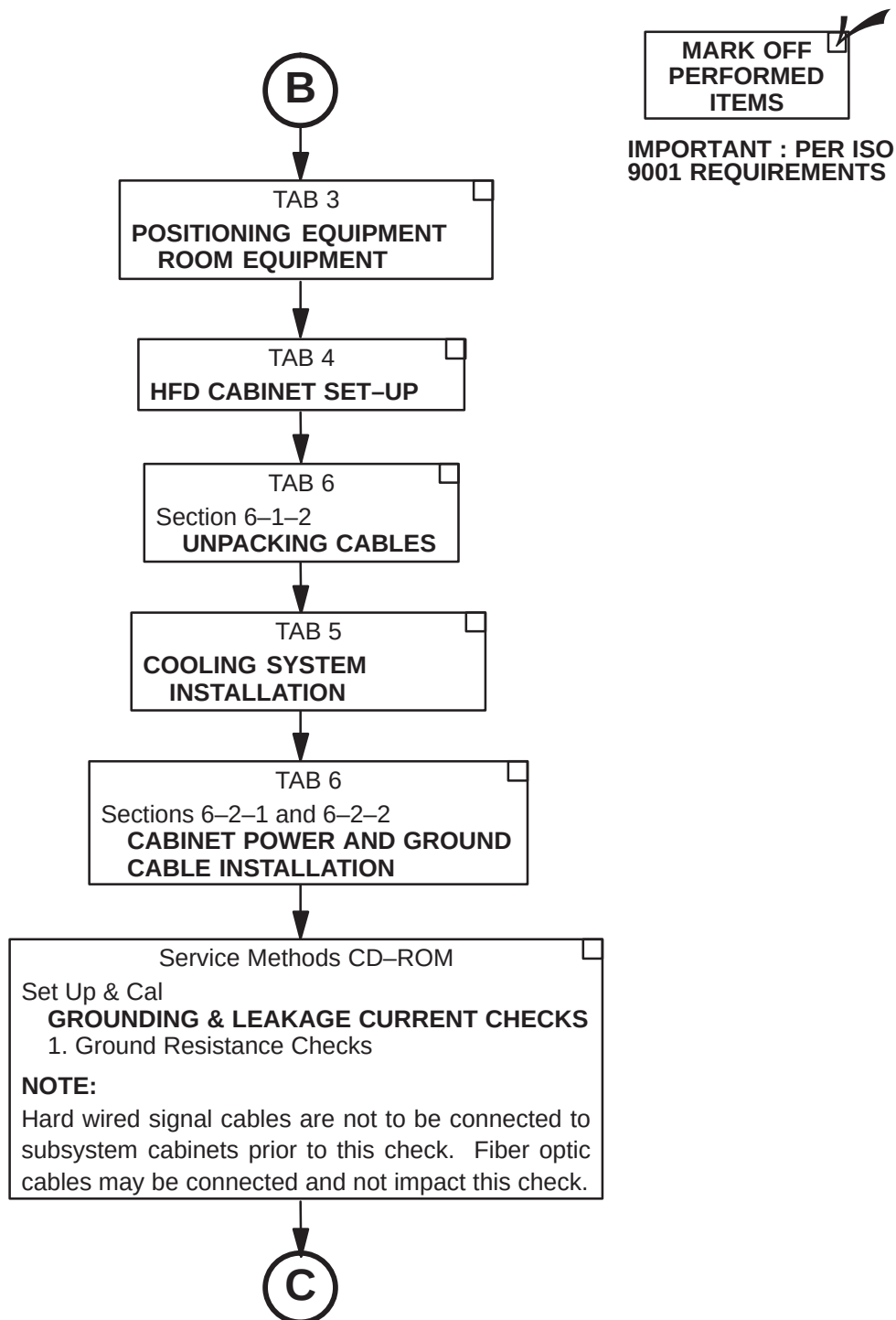
Site Installations Required Prior to System Mechanical Installation

1-12-2 SYSTEM MECHANICAL INSTALLATION

Procedures to be Completed by Mechanical Installers

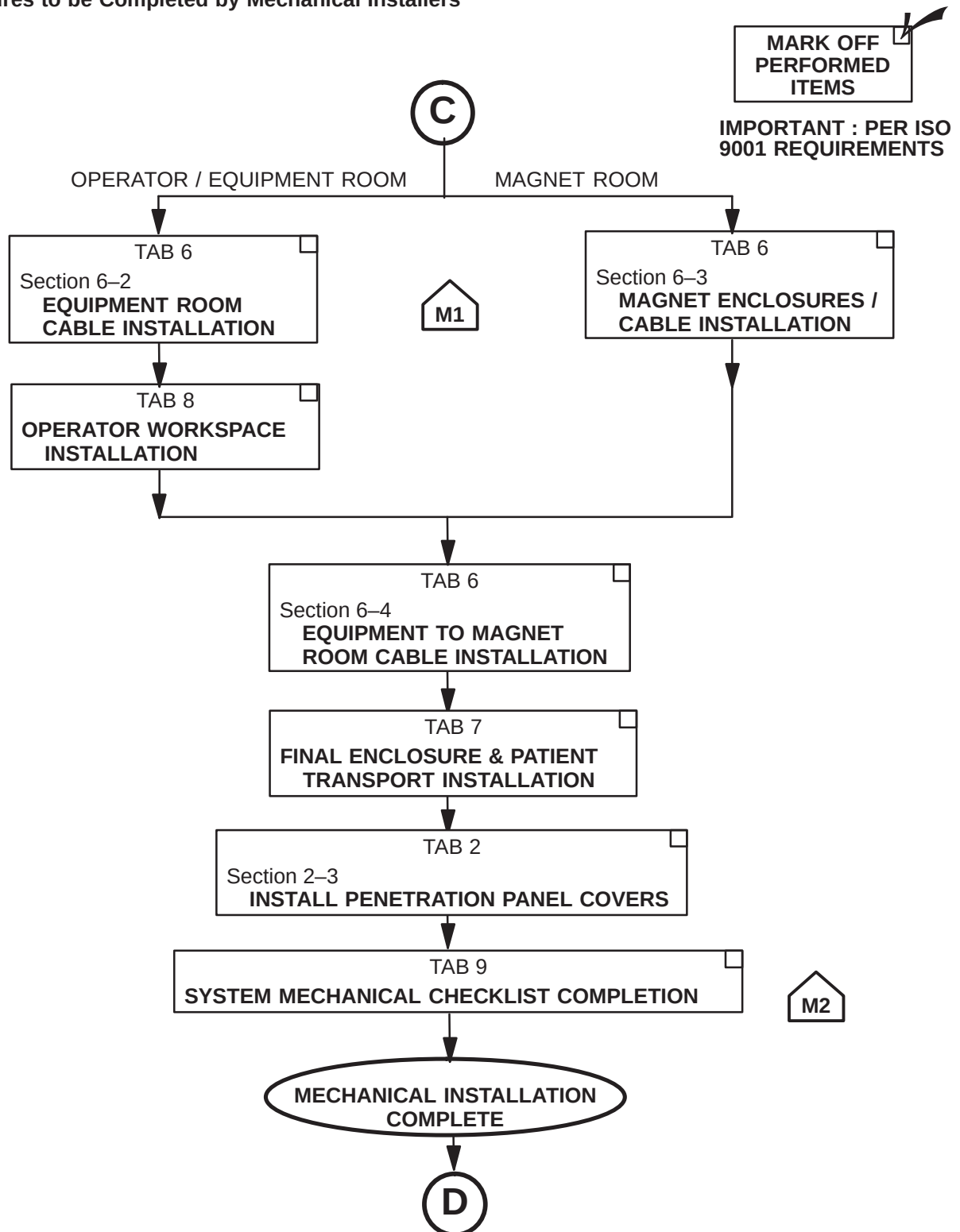
1-12-2 SYSTEM MECHANICAL INSTALLATION (Continued)

Procedures to be Completed by Mechanical Installers



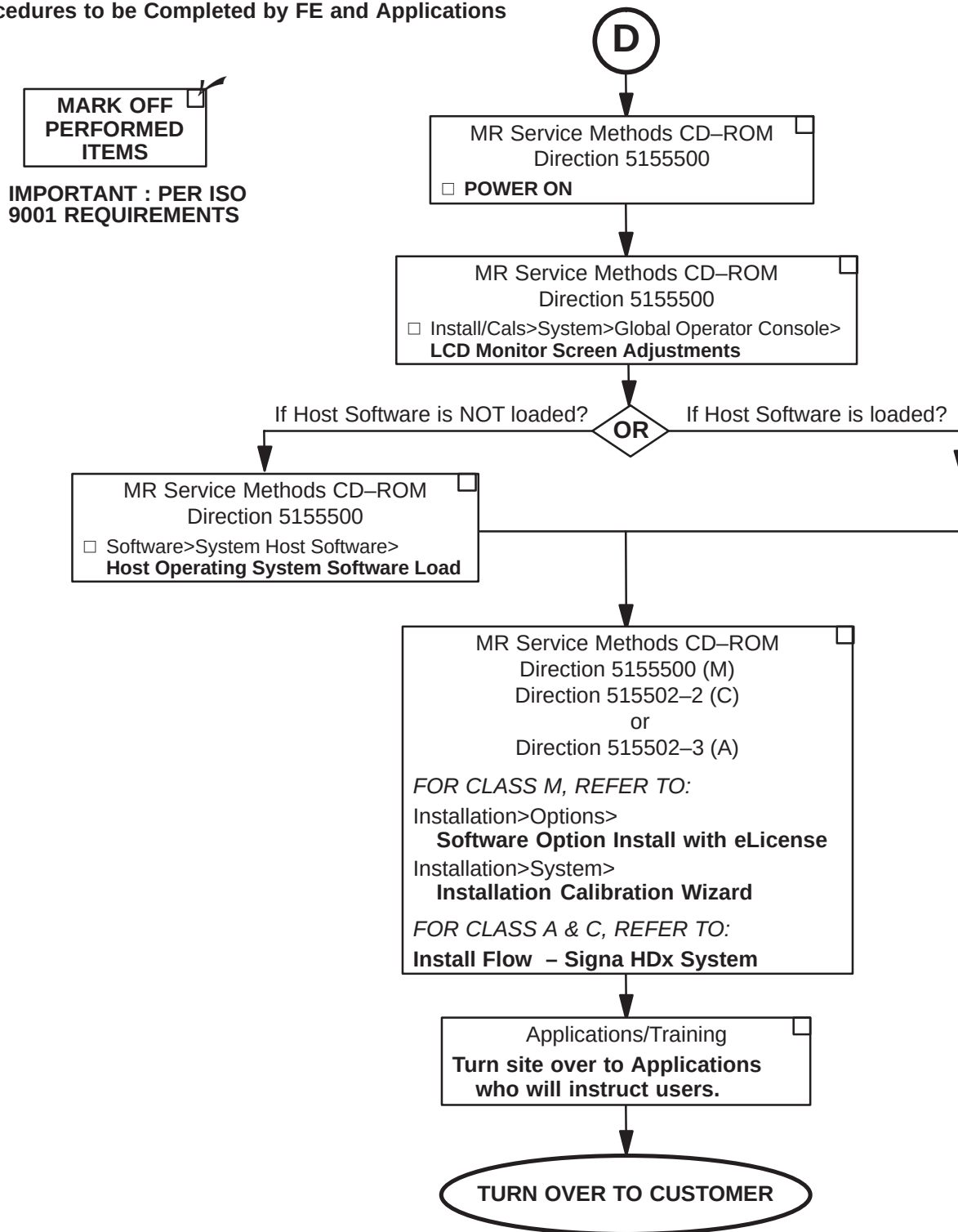
1-12-2 SYSTEM MECHANICAL INSTALLATION (Continued)

Procedures to be Completed by Mechanical Installers



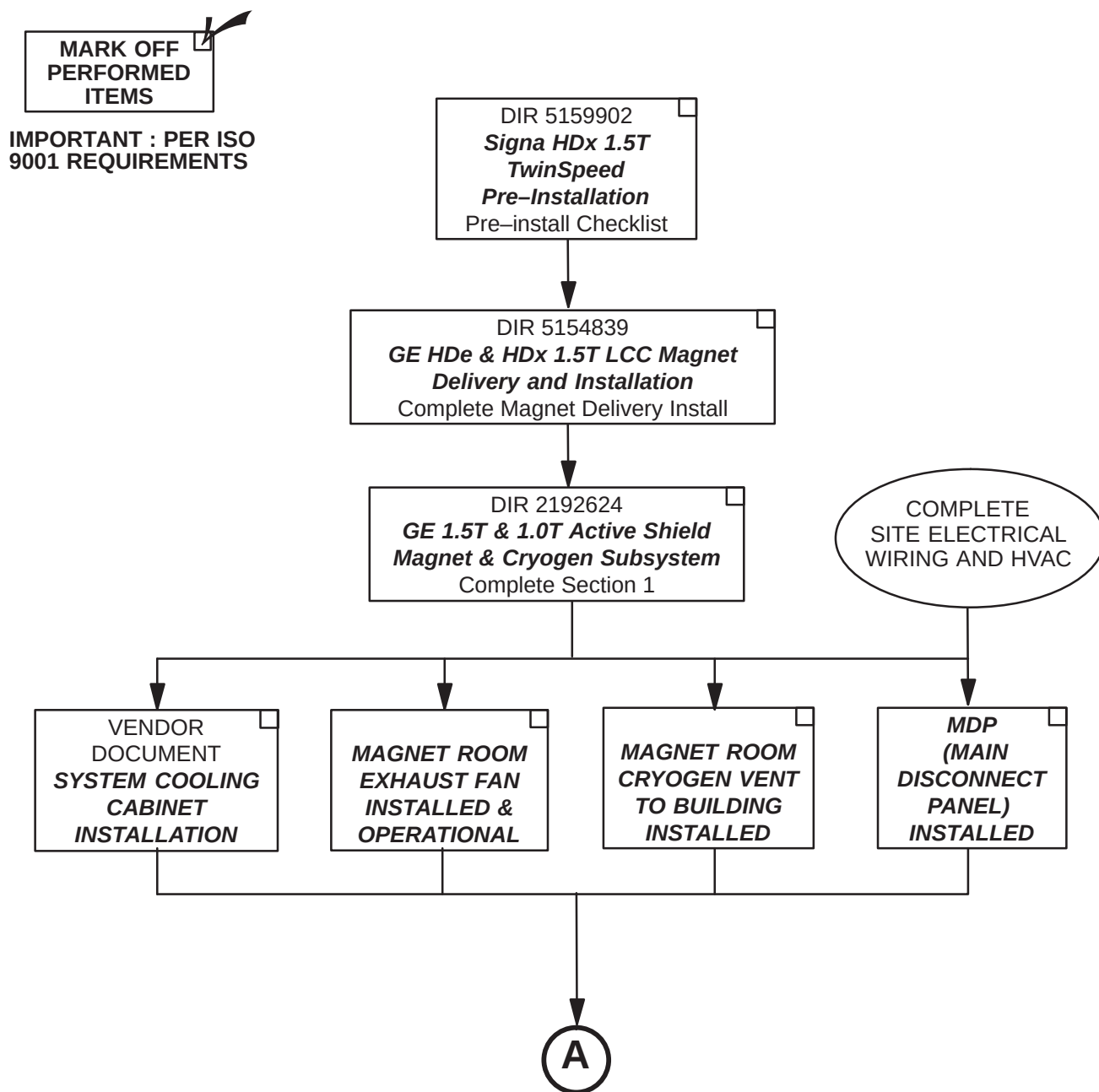
1-12-2 SYSTEM MECHANICAL INSTALLATION (Continued)

Procedures to be Completed by FE and Applications



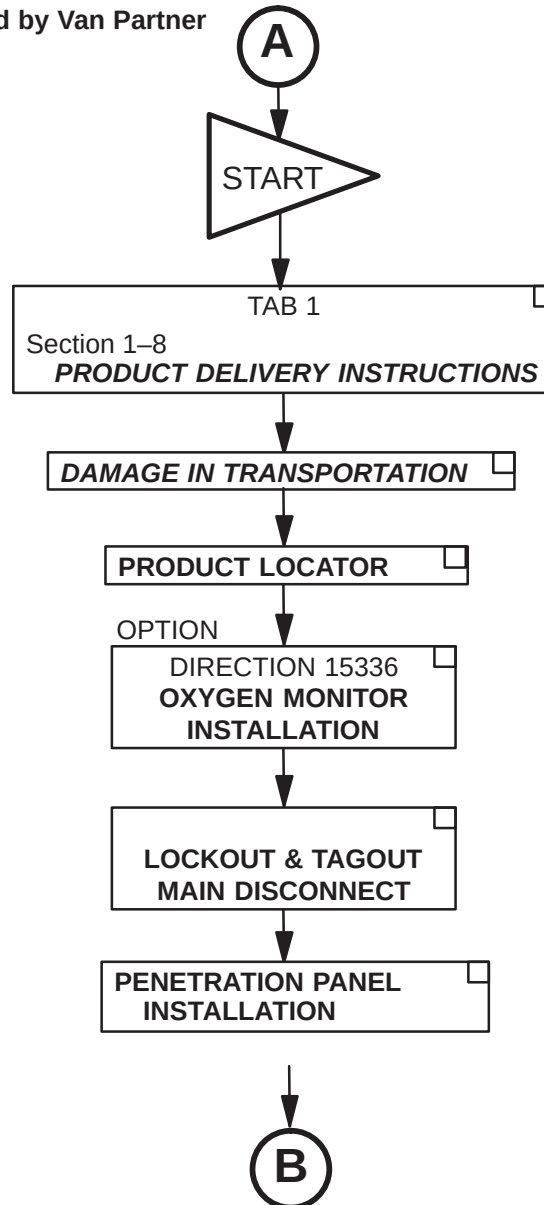
1-13 INSTALLATION FLOWCHART FOR MOBILE

1-13-1 SYSTEM MECHANICAL INSTALLATION

Site Installations Required Prior to System Mechanical Installation

1-13-2 SYSTEM MECHANICAL INSTALLATION

Procedures to be Completed by Van Partner

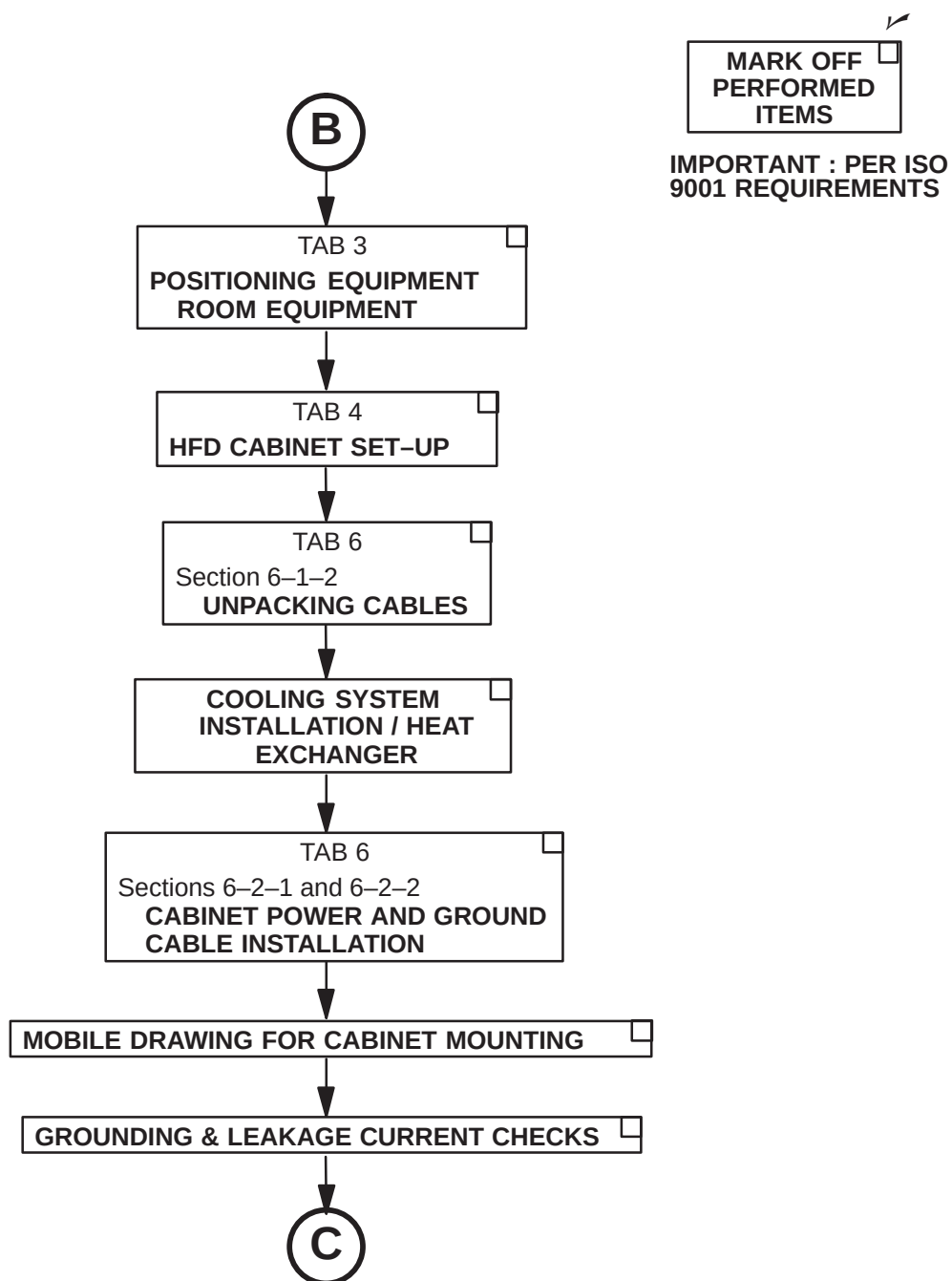


MARK OFF
PERFORMED
ITEMS

IMPORTANT : PER ISO
9001 REQUIREMENTS

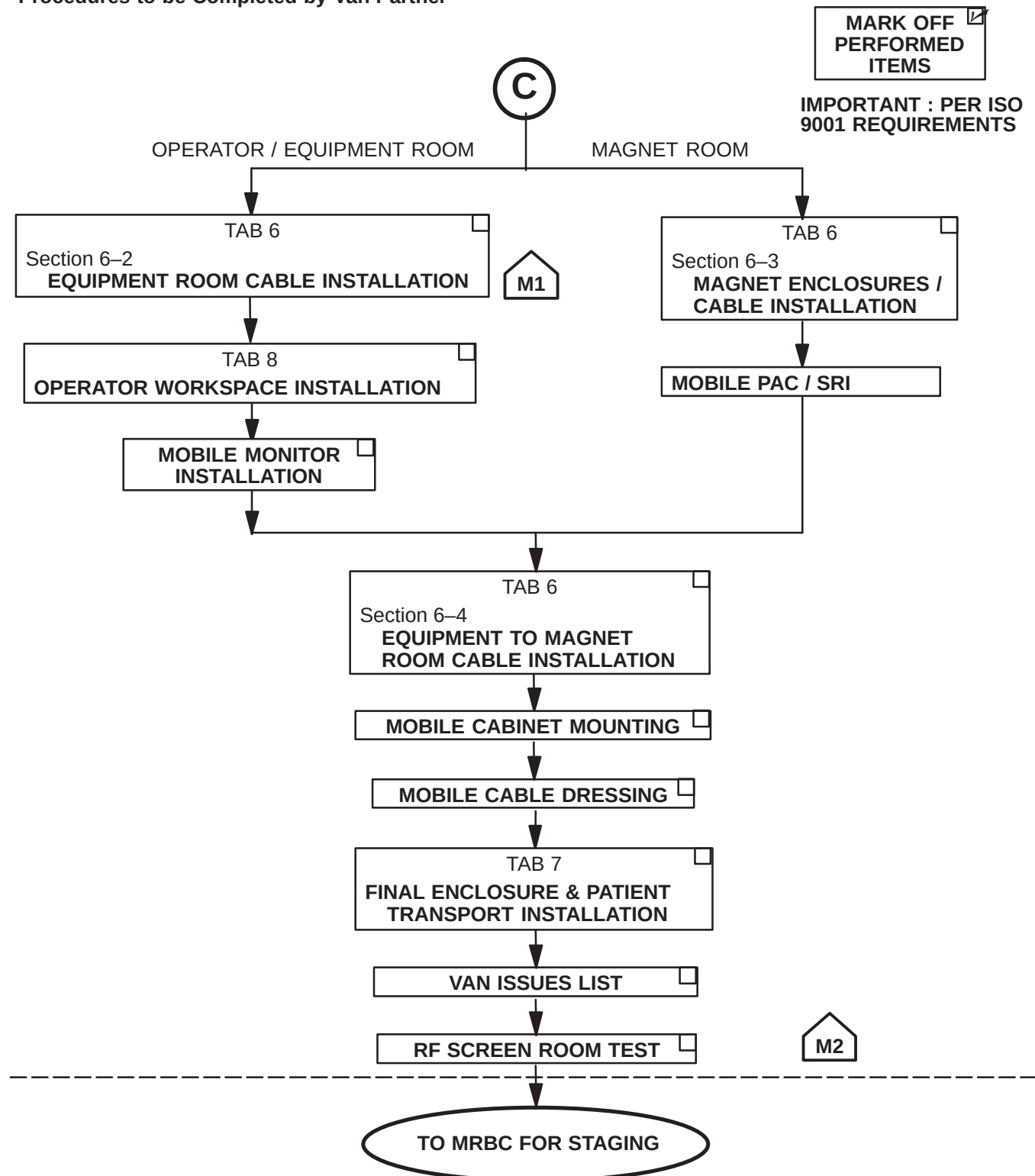
1-13-2 SYSTEM MECHANICAL INSTALLATION (Continued)

Procedures to be Completed by Van Partner



1-13-2 SYSTEM MECHANICAL INSTALLATION (Continued)

Procedures to be Completed by Van Partner



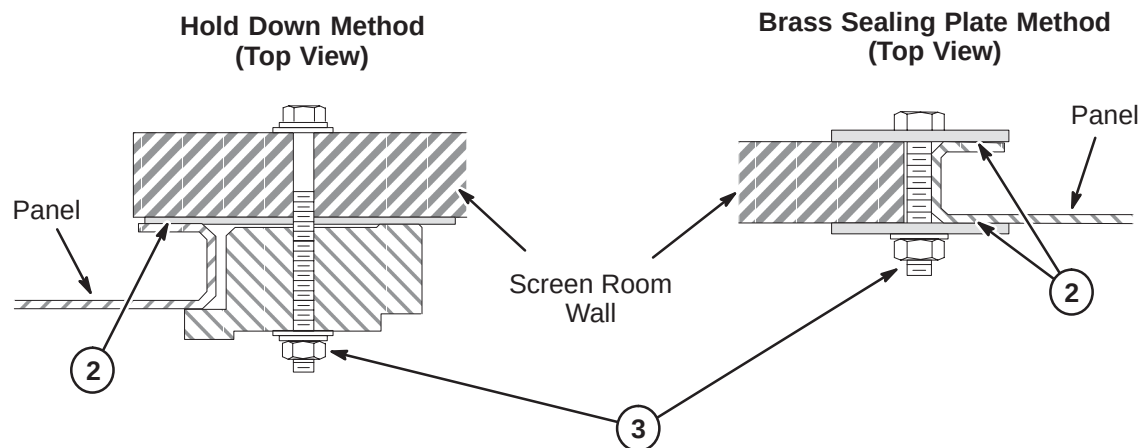
SECTION 2 – PENETRATION PANEL INSTALLATION

TABLE OF CONTENTS

<u>SECTION</u>	<u>PAGE</u>
2-1 PREPARE SCREEN ROOM OPENING FOR PENETRATION PANEL	2-2
2-2 INSTALL PENETRATION PANEL	2-3
2-3 INSTALL PENETRATION PANEL COVERS	2-4
2-3-1 INSTALL PENETRATION PANEL STAND-OFF HARDWARE	2-4
2-3-2 INSTALL PENETRATION PANEL COVERS (After Cable Installation Complete) ..	2-5

2-1 PREPARE SCREEN ROOM OPENING FOR PENETRATION PANEL

1. If still in place, remove vendor supplied blank panel. Save mounting hardware of either method shown below for installation of Penetration Panel.
2. To make sure of good conductivity, use a fine sandpaper or scouring pad to clean surface of screen room wall which will be in contact with Penetration Panel. **Do not sand surface of Penetration Panel as it is coated.**
3. Grade 8 bolts with black oxide coating should be used as mounting hardware. Tighten to torque of 20 lb–ft (1.5 to 2 turns after lock washer flattens).



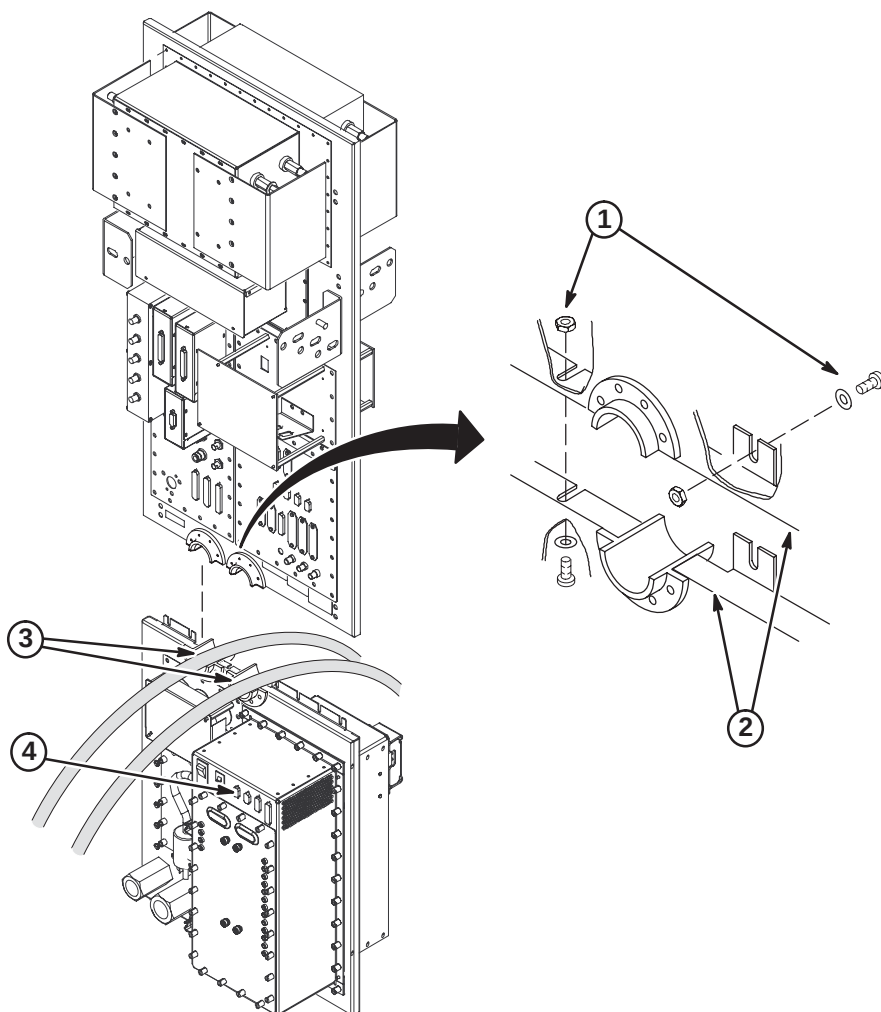
2-2 INSTALL PENETRATION PANEL

Install Penetration Panel using same method and hardware for mounting of blank panel removed in previous procedure.

Note

Install the Penetration Panel Floor plate that is supplied with the cover kit prior to routing and connecting cables to Penetration Panel. Refer to Installation Drawing furnished with the Cover Kit.

1. Assemble Penetration Panel around helium lines (Runs 621 and 622) using:
 - (7) #10-32 Nylock Lock Nuts (46-208937P6)
 - (7) 0.2 ID, 0.5 OD Flat Washers (1000904P425)
 - (7) #10-32 x 3/8 Screws (46-208560P82)
2. Seal all joints with copper foil tape with a conductive adhesive (3M #1181 or 46-258218P4, 5, or 6) for RF-tight integrity.
3. Fill inside of J60 and J61 guides with Bronze Wool (46-318068P1) that was supplied with magnet.
4. CAN Terminator (2303317) is supplied with Penetration Panel. It must be connected to J119 on TAC Cabinet interface panel



2-3 INSTALL PENETRATION PANEL COVERS

Note

The Penetration Panel Cover can be installed only after all connections and System installation has been completed.

Note

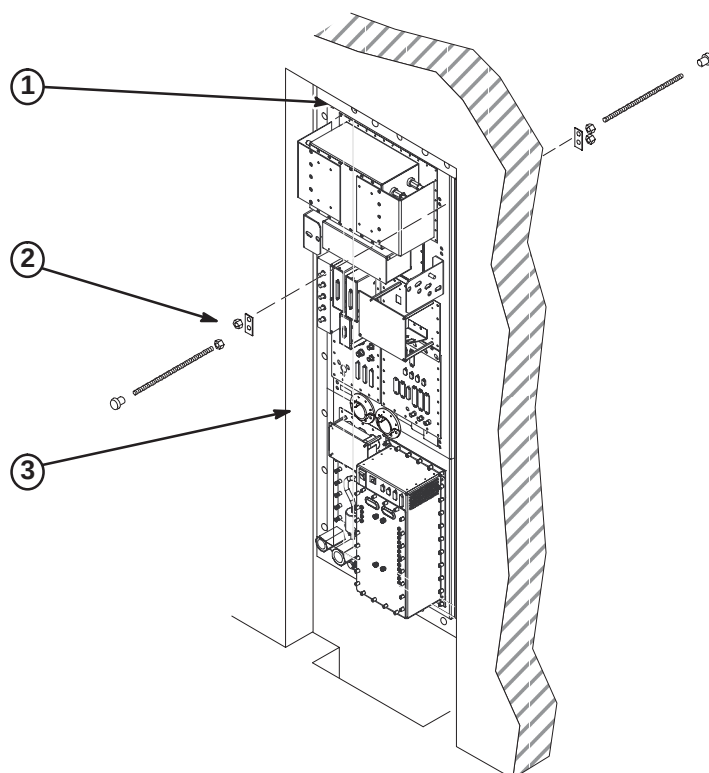
Mobile vans may not use Penetration panel wave guides for routing gas lines.

Note

Mobile vans may not use Production PEN panel covers.

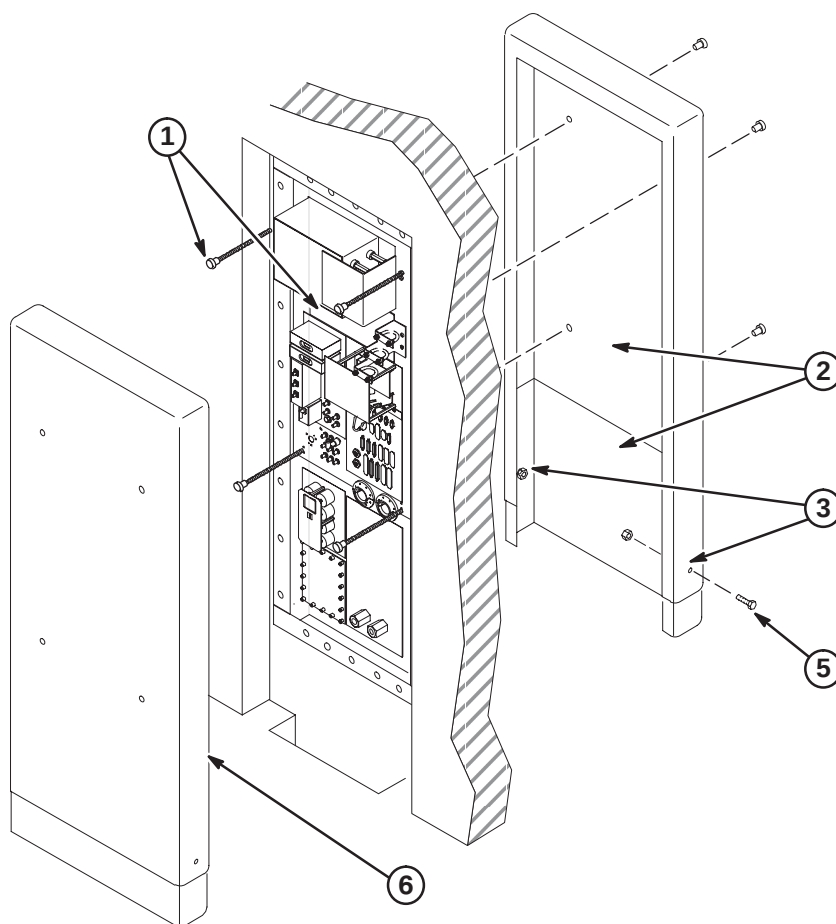
2-3-1 INSTALL PENETRATION PANEL STAND-OFF HARDWARE

1. Verify that cover will enclose finished wall opening. The distance from the top of Penetration Panel to wall opening must be less than 4-1/2 inches (114mm). If not, new mounting stud holes in Penetration Panel may be required.
2. Place (8) Reinforcing Plates (46-301211P1) over each hole pair on both sides of Penetration Panel and attach using:
 - (16) 5/16-18 x 9/16 Brass Nut (46-208935P9)
 - (8) 5/16-18 x 18 inch Threaded Rod (46-301213P2)
 - (8) 5/16-18 x 1 inch Barrel Nut (46-301212P1)
3. Measure distance from face of Pen Panel to equipment room wall surface. Add six inches (152mm) to obtain length threaded rod should extend from panel.
4. Repeat step 3 procedure for exam room cover.
5. Adjust threaded rods with attached barrel nuts to the previously calculated dimensions from steps 3 and 4. Secure to panel by tightening brass nuts on both sides of Penetration Panel.



2-3-2 INSTALL PENETRATION PANEL COVERS (After Cable Installation Complete)

1. Remove all barrel nuts from both sides of Penetration Panel.
2. In the Magnet room, temporarily install Top (46-306936P1) and Bottom (46-306964P1) Cover assembly on threaded rods and secure with four barrel nuts.
3. Adjust position of bottom cover section and mark two lower fastener holes.
4. Remove covers and drill hole in lower cover at each mark.
5. Attach top and bottom covers with furnished 10-32 nuts and screws.
6. Repeat procedure for Equipment room cover.
7. After all installation procedures are completed in Magnet and Equipment rooms, install assembled covers on threaded rods and secure with four barrel nuts.



SECTION 3 – CABINET POSITIONING

TABLE OF CONTENTS

<u>SECTION</u>	<u>PAGE</u>
3-1 CABINET INSTALLATION REQUIREMENTS	3-2
3-2 POSITION HFD CABINET AND ATTACH LABEL	3-2
3-3 POSITION RFS CABINET AND ATTACH LABEL(S)	3-3
3-4 POSITION TAC CABINET AND ATTACH LABEL	3-4

3-1 CABINET INSTALLATION REQUIREMENTS

Positioning of cabinets is critical for installing cables and post-installation system operation. Refer to architectural site layouts for:

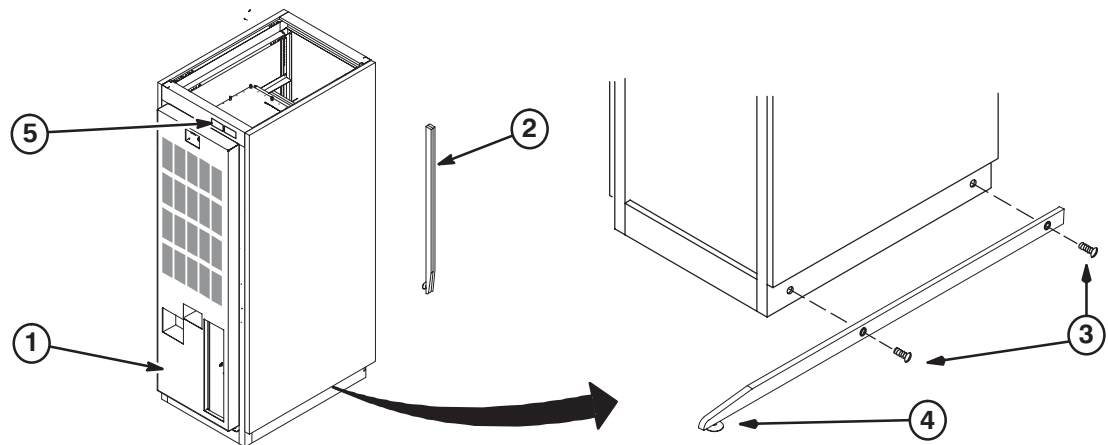
- Location of all cabinets and related equipment.
- For seismic anchoring (where required)

3-2 POSITION HFD CABINET AND ATTACH LABEL

WARNING!

DUE TO WEIGHT OF CABINET, AT LEAST TWO PEOPLE ARE REQUIRED TO MOVE CABINET

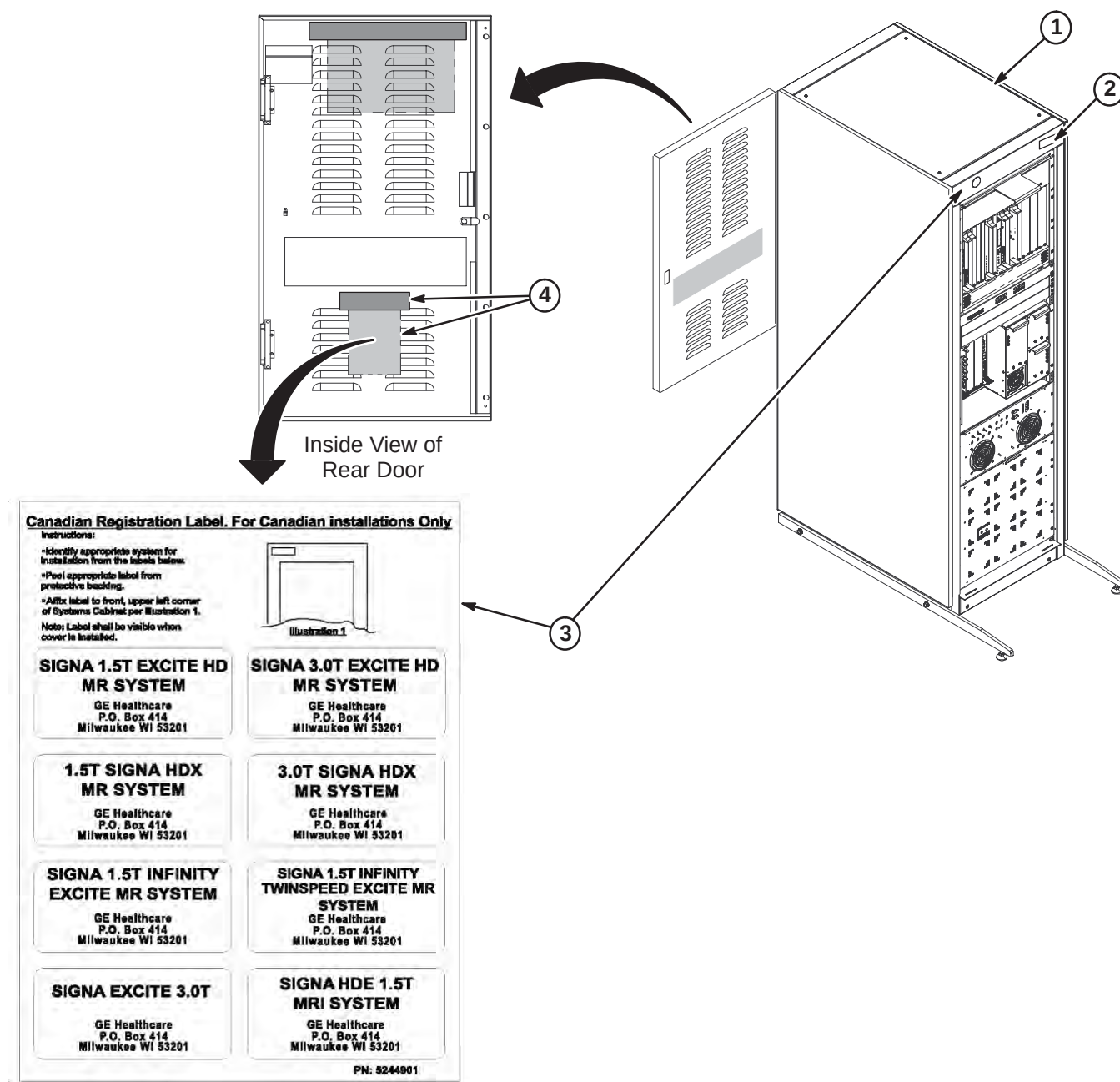
1. Move cabinet to final position as indicated by site layout drawings.
2. Anti-tip bars for the HFD Cabinet are shipped in the RFS Cabinet. Remove from RFS Cabinet for this procedure.
3. Attach bars to cabinet.
4. Adjust leveling pads to level cabinet.
5. From the Cabinet ID Label Sheet (5126718), locate the "GRADIENT/PDU" cabinet magnet and attach to front of cabinet as shown.



3-3 POSITION RFS CABINET AND ATTACH LABEL(S)

Comply with UL requirements or local codes for securing the cabinet to the floor with locally supplied brackets and floor anchors.

1. Position RFS cabinet to final position as indicated by site layout drawings.
2. From the Cabinet ID Label Sheet (5126718), locate the "RF/SYSTEMS CABINET" magnet and attach to front of cabinet as shown.
3. FOR INSTALLATION IN CANADA ONLY: Remove Registration Label sheet (5244901) from plastic sleeve on inside of rear door. Follow instructions on sheet for attaching appropriate label to cabinet.
4. For all non-Canada installations, remove plastic bag with labels and dispose. Label is not required.



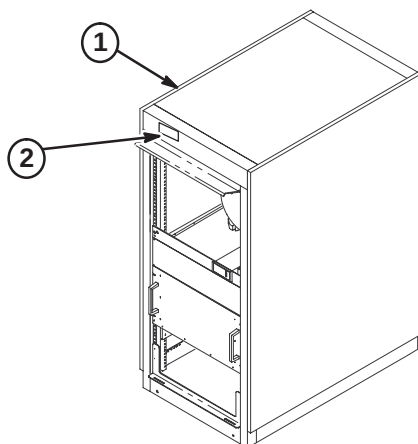
3-4 POSITION TAC CABINET AND ATTACH LABEL

Comply with UL requirements or local codes for securing the cabinet to the floor with locally supplied brackets and floor anchors.

Note

If supplied in cabinet, Run 975 is a service tool. It will not be part of the installation.

1. Position TAC cabinet as indicated by site layout drawings.
2. From the Cabinet ID Label Sheet (5126718), locate the "TAC CABINET" magnet and attach to front of cabinet as shown.



SECTION 4 – HFD CABINET SET-UP

TABLE OF CONTENTS

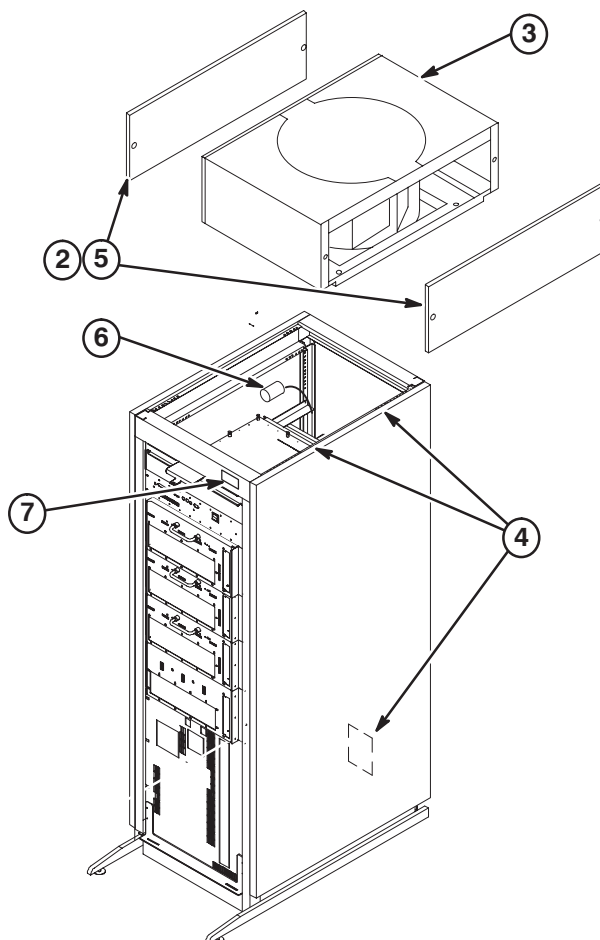
<u>SECTION</u>	<u>PAGE</u>
4-1 INSTALL FAN ASSEMBLY AND ATTACH INSITE MAGNET	4-2
4-2 INPUT VOLTAGE SELECTION	4-3
4-3 CIRCUIT BREAKER DIP SWITCH SETTINGS	4-3



LETHAL VOLTAGES ARE PRESENT WITHIN THE PDU. MAKE SURE THAT POWER AT THE MAIN DISCONNECT IS OFF, LOCKED, AND TAGGED BEFORE PROCEEDING.

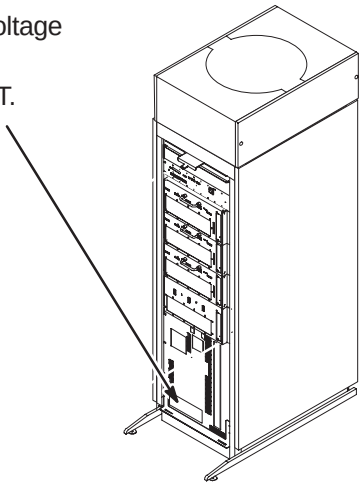
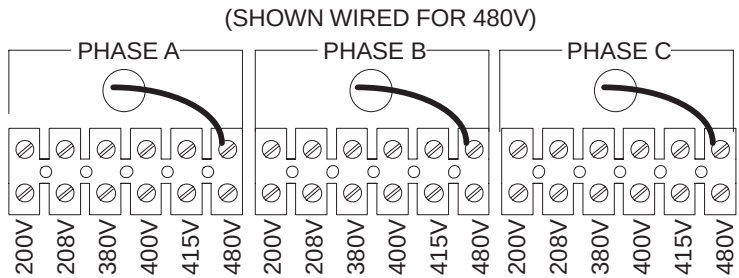
4-1 INSTALL FAN ASSEMBLY AND ATTACH INSITE MAGNET

1. Remove Fan Assembly from shipping container.
2. Remove two side trim panels (two screws from each).
3. Requiring two people, place the Fan Assembly on top of cabinet with the cable connector facing the rear of the cabinet.
4. Secure Fan Assembly in place with (4) four screws and washers, two on each side of cabinet. The fasteners are provided in a plastic bag located in PDU module. Rear PDU covers will need to be removed for access to plastic bags with fasteners.
5. Install two side trim panels with previously removed screws.
6. Connect cable from HFA Power Supply to the fan connector.
7. Make sure the "GRADIENT/PDU" cabinet magnet, part of the Cabinet ID Label Sheet (5126718), is attached to the front of cabinet as shown.

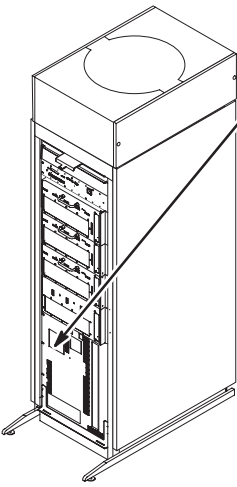


4-2 INPUT VOLTAGE SELECTION

- ① Using a voltmeter or other voltage indicating device, check to be sure that no voltage is being applied to the input of the PDU.
- ② Remove the plate from the front of the PDU marked INPUT VOLTAGE SELECT.
- ③ For each of the three phases, be sure the wire is connected to the proper terminal for the actual input voltage at the site and that the terminal screw is tight to make a good connection. Re-install plate.



4-3 CIRCUIT BREAKER DIP SWITCH SETTINGS



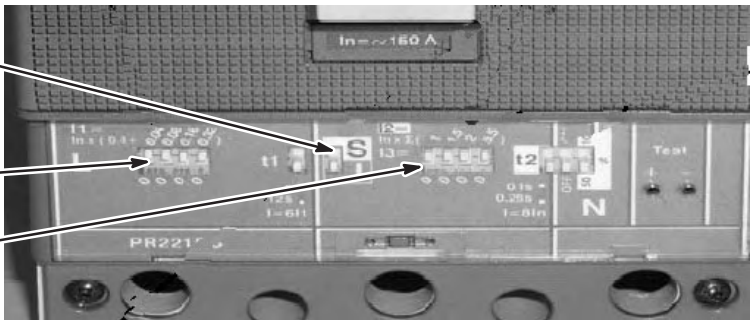
- ① Remove the plate from the front of the PDU marked CURRENT ADJUST.
- ② Table below shows the position of the DIP switches for each input voltage selection. Set DIP switches according to input voltage. The white portion indicates the switch.

INPUT VOLTAGE	200	208	380	400	415	480
"L" Switch Setting						
"I" Switch Setting						

- ③ The "S/I" switch must be in the "I", down position

DIP SWITCH
"L" SETTING

DIP SWITCH
"I" SETTING



SECTION 5 – COOLING SYSTEM INSTALLATION

TABLE OF CONTENTS

<u>SECTION</u>	<u>PAGE</u>
5-1 ROUTE/CONNECT COOLANT HOSES	5-2
5-2 INSTALL GROUND CLAMPS ON TEES	5-2
5-3 CONNECT COOLANT HOSES TO TEES	5-3
5-4 INSTALL CONNECTOR WRAPS	5-3
5-5 ROUTE/CONNECT GROUND WIRES TO MAGNET	5-4
5-6 COOLANT VALVE INSTALLATION	5-5
5-7 COOLANT CABINET OPERATING INSTRUCTIONS	5-5

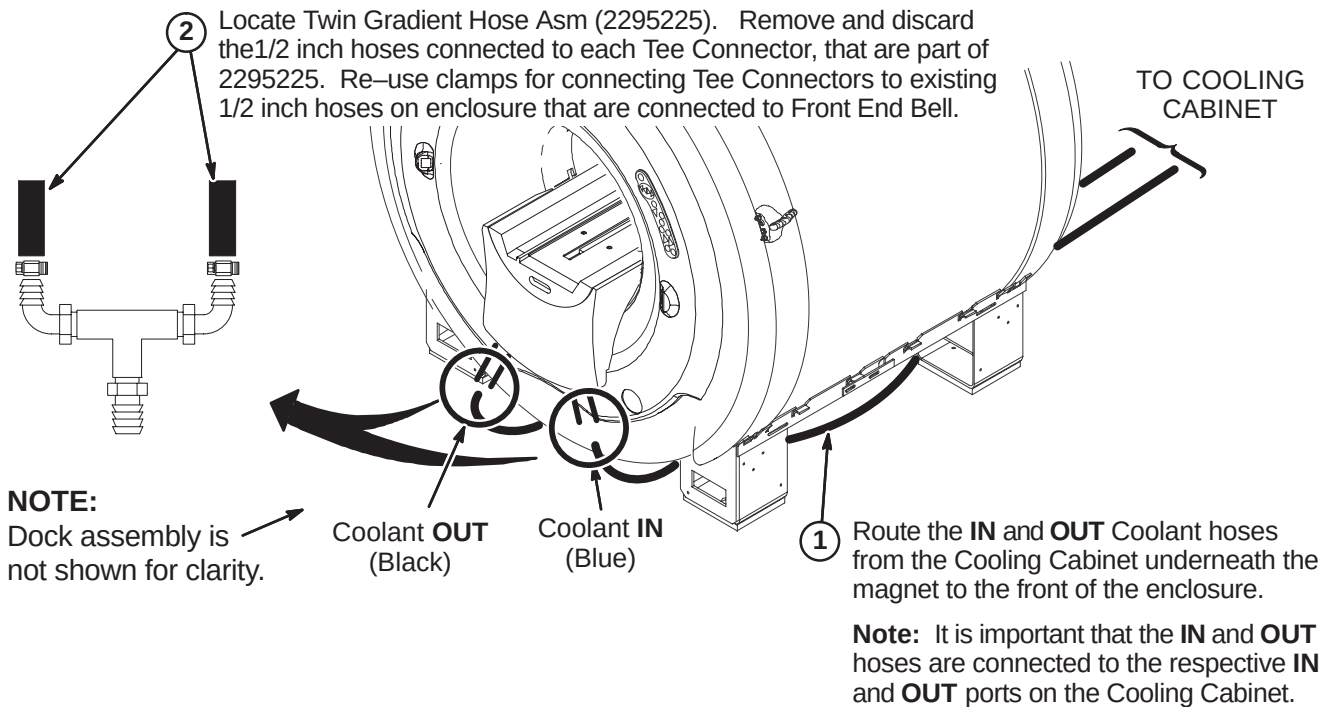
5-1 CONNECT MAGNET COOLANT HOSES TO TEES

Note

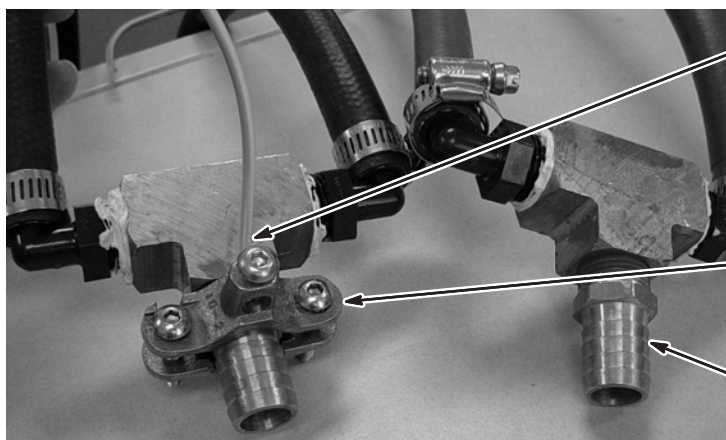
Mobile vans will connect to provided heat exchanger.

The Cooling Cabinet must be installed and electrically powered prior to the start of this procedure.

Magnet is delivered to site with four 1/2 inch hoses connected to the Front End Bell. The Twin Gradient Hose Assembly (2295225) is delivered to site and also has four duplicate 1/2" hoses. **Discard the 1/2 inch black (2293610-2) and blue (2293610-3) hoses that are part of 2295225.** Connect the IN and OUT coolant hoses from the Cooling Cabinet to the 1/2 inch hoses attached to the Front End Bell using the Tee Connectors and clamps shown below.

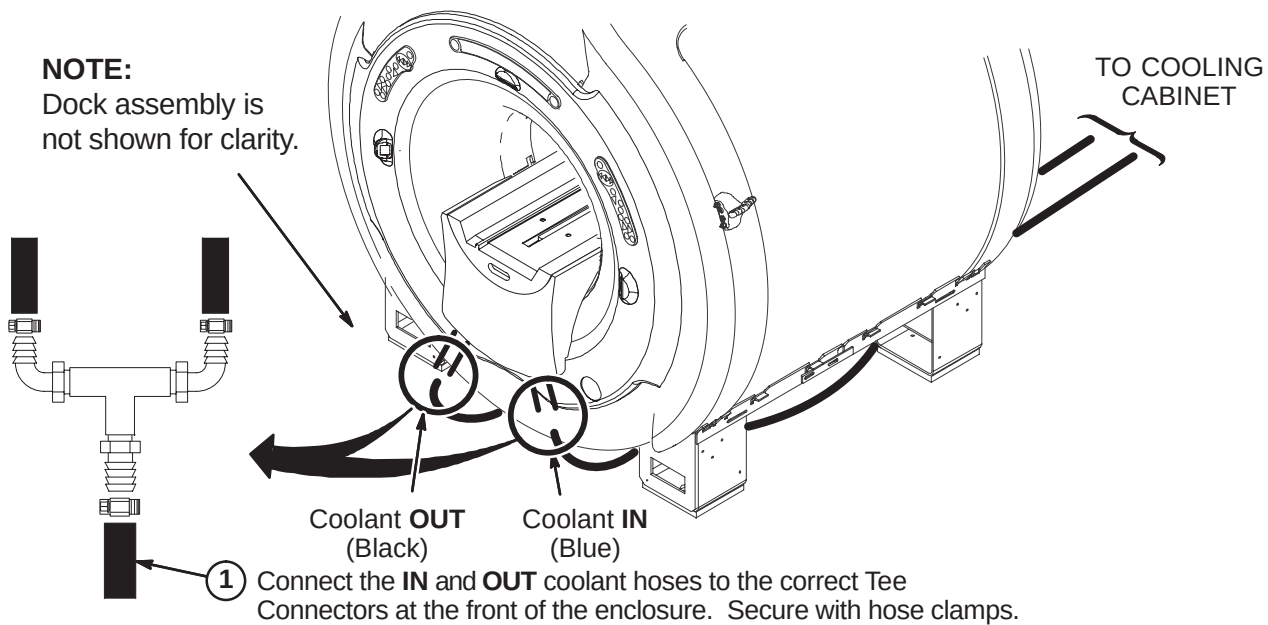


5-2 INSTALL GROUND CLAMPS ON TEES



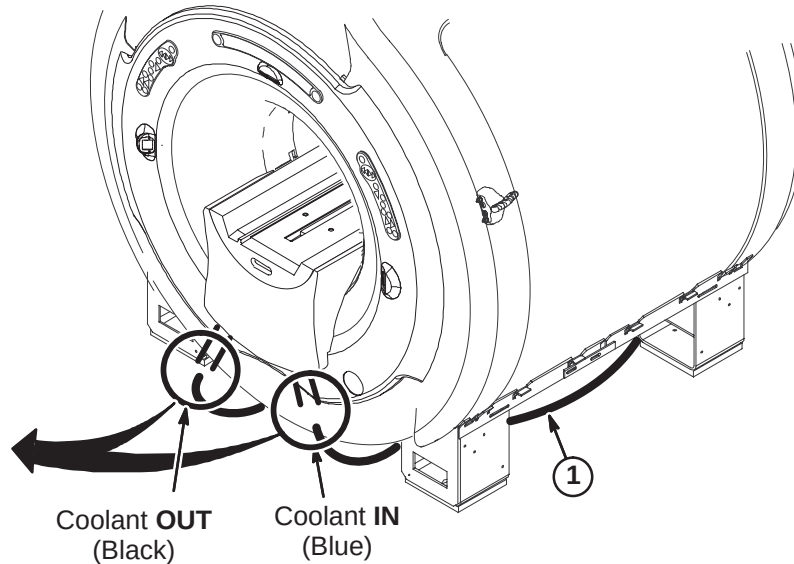
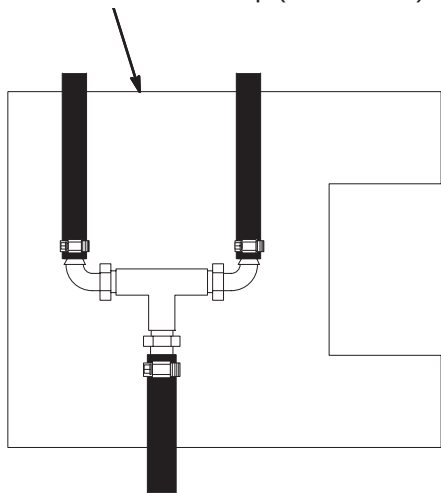
- 1 Verify that one end of each Jumper Wire Asm (5126241) is stripped bare of insulation for 1/2 inch (12.7mm) at one end and insert stripped end into hole of clamp. Tighten the wire securing screw.
- 2 Place Ground Clamp (5126242) over hex portion of 3/4 inch hose barb and tighten securely.
- 3 Repeat steps 1 and 2 for other coolant tee.

5-3 CONNECT COOLANT HOSES TO TEES



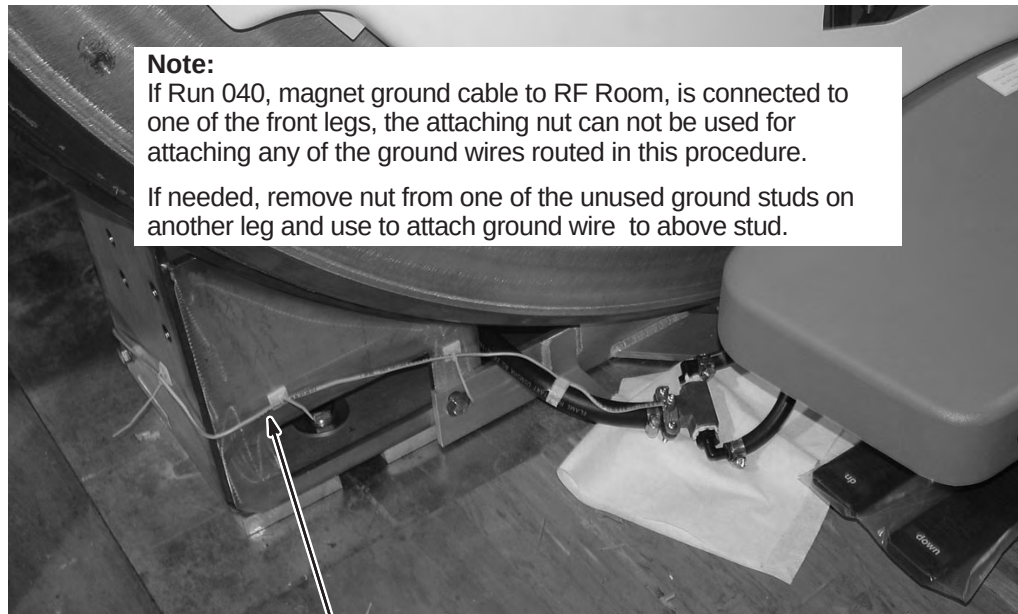
5-4 INSTALL CONNECTOR WRAPS

- ① Wrap each Tee Connector with 177 x 200 Connector Wrap (2236130-2)



5-5 ROUTE/CONNECT GROUND WIRES TO MAGNET**IMPORTANT !!!**

Make sure Jumper Wire Asm 5126241 is secure from motion along its entire length.



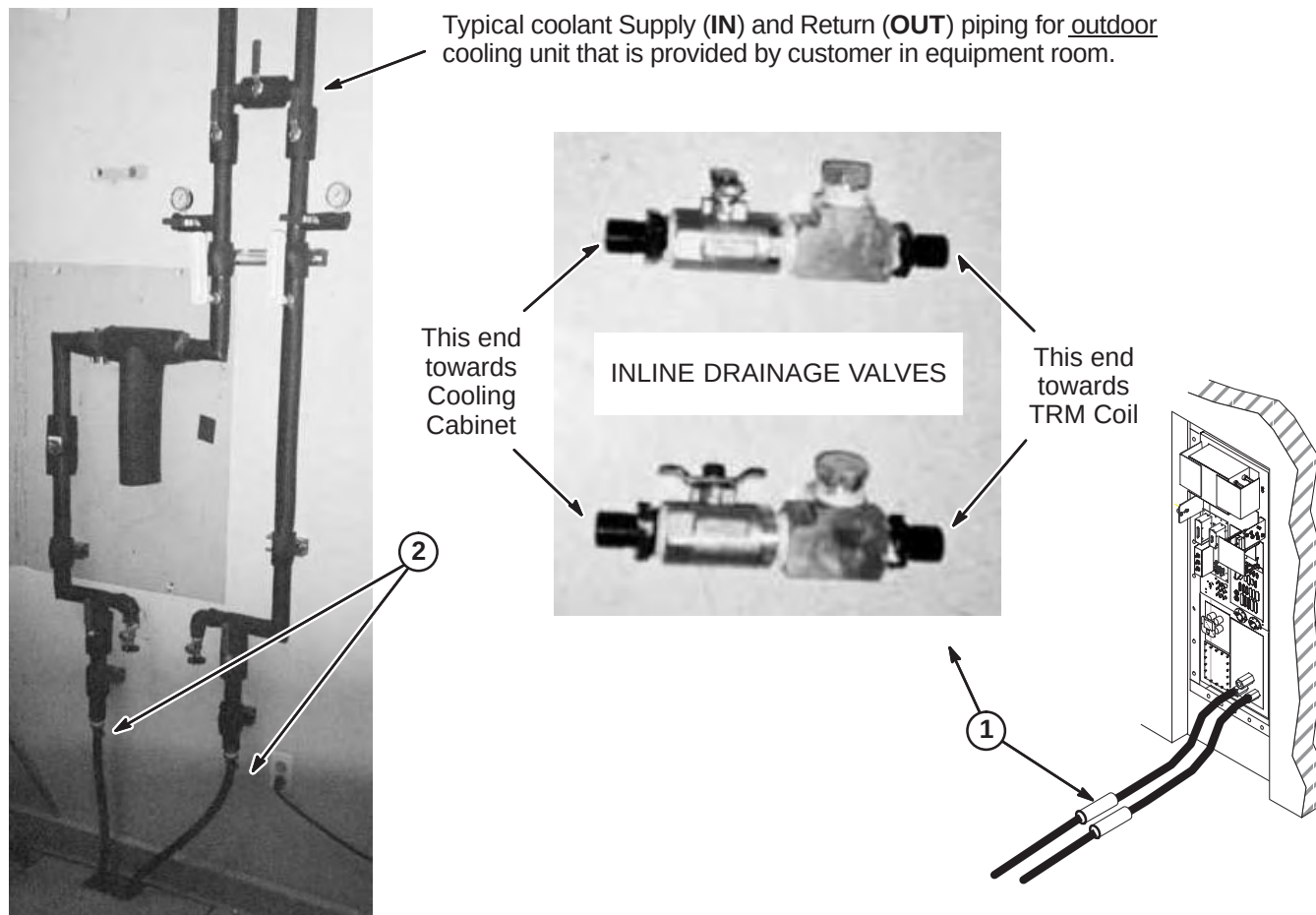
- ① Route one of the ground wires around the left leg to ground stud on inside of leg. Secure wire to leg with adhesive backed Tie Wrap Base (46-208747P1) and Tie Wraps (5112522). Securely fasten the ring terminal to ground stud.

- ② Route other ground wire to right leg ground stud. Secure and attach ground wire as shown.



5-6 COOLANT VALVE INSTALLATION

1. The Inline Drainage Valves shown above are installed in coolant hoses near the Penetration Panel on the Equipment Room side. The valve end must be positioned towards the Cooling Cabinet and the Tee end towards the TRM coil.
2. Connect **IN** and **OUT** coolant hoses to respective piping hose barbs, on outdoor or indoor unit.



5-7 COOLANT CABINET OPERATING INSTRUCTIONS

Consult vendor supplied manual that was delivered with Cooling Cabinet for installation, start-up, and operating instructions and procedures.

SECTION 6 – CABLE & SYSTEM INSTALLATION

TABLE OF CONTENTS

<u>SECTION</u>	<u>PAGE</u>
6-1 CABLE INTERCONNECT DOCUMENTATION	6-2
6-1-1 CABLE INTERCONNECT MAPS	6-2
6-1-2 SORTING AND ROUTING CABLES	6-7
6-2 EQUIPMENT ROOM CABLE INSTALLATION	6-8
6-2-1 HFD CABINET CABLE CONNECTIONS	6-8
6-2-2 EQUIPMENT ROOM CABINET POWER CABLE CONNECTIONS	6-11
6-2-3 GRADIENT CABLE INSTALLATION IN EQUIPMENT ROOM	6-12
6-2-4 INSTALLATION OF MAGNET MONITORING CABLES	6-21
6-2-5 INSTALL RUN 701 – RF DOOR SWITCH	6-21
6-2-6 INSTALLATION OF FIBER OPTIC CABLES	6-22
6-2-7 INSTALLATION OF REMAINING EQUIPMENT ROOM CABLES	6-26
6-3 MAGNET ROOM CABLE INSTALLATION	6-28
6-3-1 GRADIENT CABLE CLAMP BLOCK INSTALLATION	6-29
6-3-2 GRADIENT CABLE INSTALLATION IN MAGNET ROOM	6-30
6-3-3 INSTALLATION OF REAR PEDESTAL	6-39
6-3-4 INSTALLATION OF DRIVE BELT	6-43
6-3-5 ROUTE AND CONNECT CABLES AT REAR PEDESTAL	6-46
6-3-6 CONNECTION OF MAGNET GROUND, RUN 040	6-49
6-3-7 INSTALLATION OF BORE LAMP ASSEMBLY	6-49
6-3-8 MAGNET ROOM FIBER OPTIC PAC/SRI INSTALLATION	6-50
6-3-9 200 GAUSS ZONE LIMIT FOR MOVEMENT OF BLOWER BOX	6-52
6-3-10 INSTALLATION OF BLOWER BOX & AIR HOSE TO MAGNET FROM BOX	6-52
6-3-11 ROUTE AND CONNECT CABLES AT FRONT OF MAGNET ENCLOSURE	6-54
6-3-12 ROUTE AND CONNECT CABLES TO PAC/SRI PLATFORM	6-55
6-3-13 DOCK ADJUSTMENT FOR PATIENT TRANSPORT GAP	6-55
6-3-14 ATTACH APPLICABLE CONTROL PANEL OVERLAYS	6-56
6-4 INSTALLATION OF CABLES BETWEEN EQUIPMENT ROOM AND MAGNET ROOM	6-57
6-4-1 INSTALLATION OF CABLES AT PENETRATION PANEL J200, J201, and J202 ...	6-57
6-4-2 TIMES MICROWAVE RF CABLES INSTALLATION	6-58
6-4-3 VACUUM HOSE INSTALLATION	6-62
6-4-4 PNEUMATIC PATIENT ALERT INSTALLATION	6-64
6-4-5 EMERGENCY OFF CONNECTIONS	6-72

6-1 CABLE INTERCONNECT DOCUMENTATION

The cable maps illustrate the Signal HDx 1.5T TwinSpeed system interconnect for all supplied cables and interface with customer supplied wiring.

- Interconnects for additional cables supplied with other options are illustrated in the installation manual shipped with the option.
 - Instructions for connections for optional laser cameras are supplied with option.
 - Connection procedures for other options are covered in the installation manual shipped with the option.
 - Spectroscopy option cabling can be found on page 6-6.
- Procedures for connections made at the Magnet Enclosure, Operator Workspace, cabinets, etc. are detailed in the respective installation section within this manual.
- Interconnect details for magnet subsystem are covered in the applicable magnet subsystem manual.

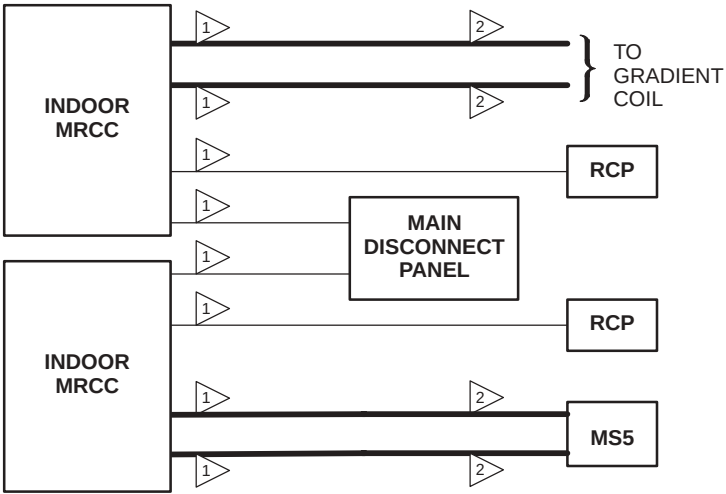
6-1-1 CABLE INTERCONNECT MAPS

For the convenience of the mechanical installers, a duplicate System Interconnect is provided. This will allow the install team to post one interconnect map in the Magnet Room and the other map in the Equipment/Operator Room. This will improve the efficiency of the installation when marking off cables when their connections are complete.

Fold-out HDx 1.5T TwinSpeed System Interconnect Cable Map is provided on the following pages:

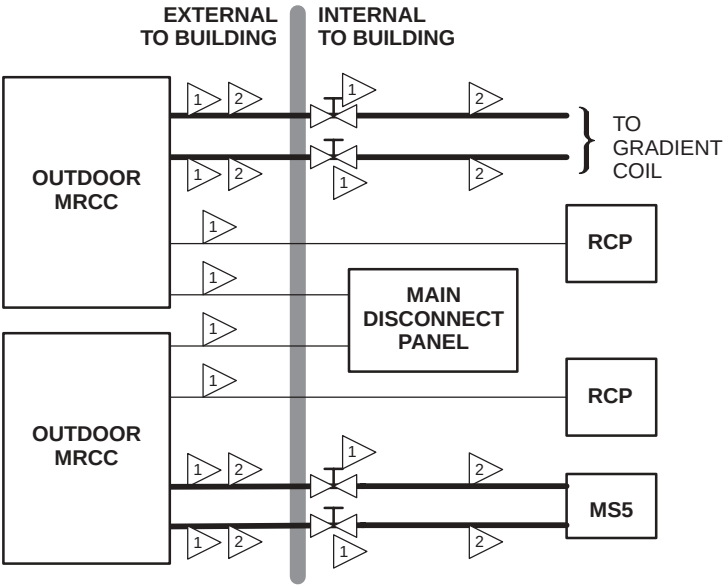
Description	Page
– Signal HDx 1.5T TwinSpeed System Interconnect (Fixed Site)	6-3
– Signal HDx 1.5T TwinSpeed System Cooling Cabinet Interconnect Options	6-4
– Signal HDx 1.5T TwinSpeed System Interconnect (Fixed Site) (DUPLICATE)	6-5
– Signal HDx 1.5T TwinSpeed System Interconnect for 2KW MNS Option	6-6

Tape maps in a convenient location in the Magnet Room and the Equipment/Operator Room to check off routed cables. Additional information on each cable is located in the Appendix. Upon completion of connecting cables, return fold-out maps to the binder for future reference.



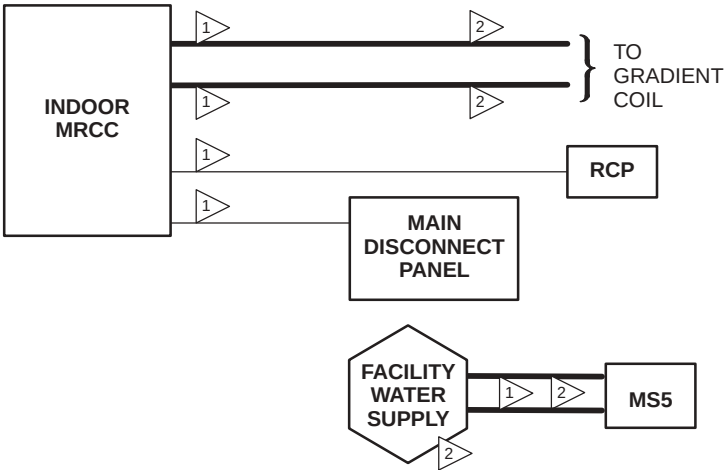
- NOTES:**
- 1 CUSTOMER FURNISHED.
 - 2 THIS GROUP CONTAINS WATER LINES WHICH SHALL BE ROUTED SEPARATE FROM ELECTRICAL LINES (I.E. POWER & SIGNAL).
- NOTE:**
- ONLY INTERCONNECTS SPECIFIC TO MRCC SUBSYSTEM EQUIPMENT SHOWN HERE.

DUAL INDOOR MRCC SUBSYSTEM GROUP INTERCONNECT DIAGRAM



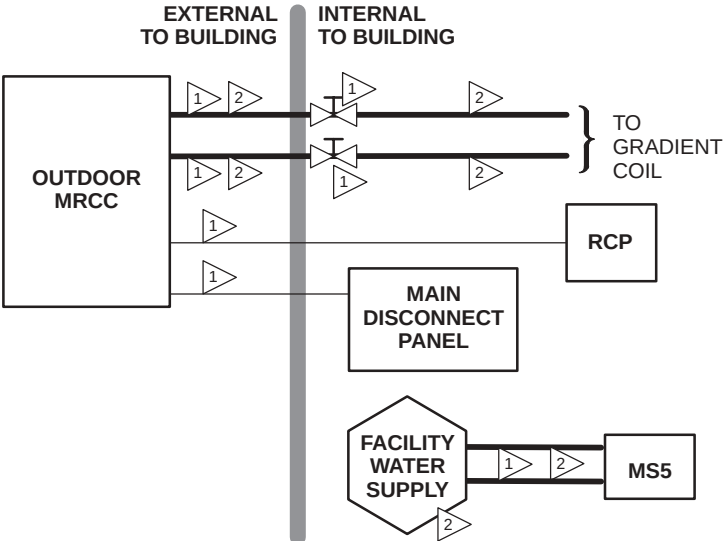
- NOTES:**
- 1 CUSTOMER FURNISHED.
 - 2 THIS GROUP CONTAINS WATER LINES WHICH SHALL BE ROUTED SEPARATE FROM ELECTRICAL LINES (I.E. POWER & SIGNAL).
- NOTE:**
- ONLY INTERCONNECTS SPECIFIC TO MRCC SUBSYSTEM EQUIPMENT SHOWN HERE.

DUAL OUTDOOR MRCC SUBSYSTEM GROUP INTERCONNECT DIAGRAM



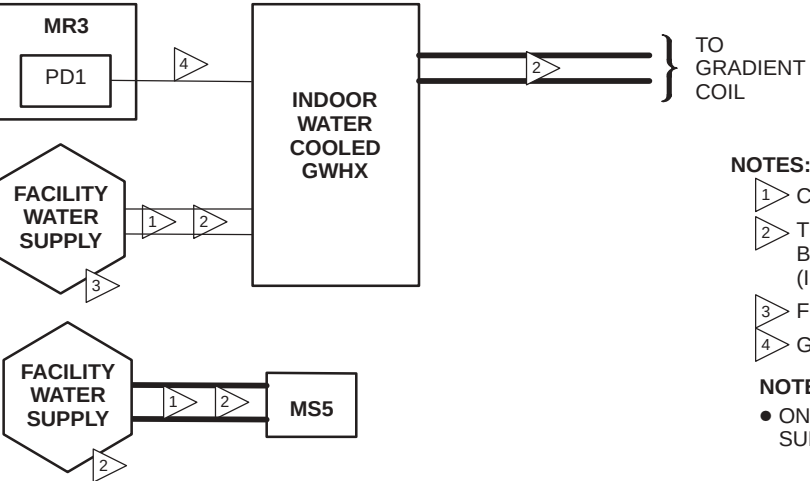
- NOTES:**
- 1 CUSTOMER FURNISHED.
 - 2 THIS GROUP CONTAINS WATER LINES WHICH SHALL BE ROUTED SEPARATE FROM ELECTRICAL LINES (I.E. POWER & SIGNAL).
- NOTE:**
- ONLY INTERCONNECTS SPECIFIC TO MRCC SUBSYSTEM EQUIPMENT SHOWN HERE.

INDOOR MRCC SUBSYSTEM GROUP INTERCONNECT DIAGRAM



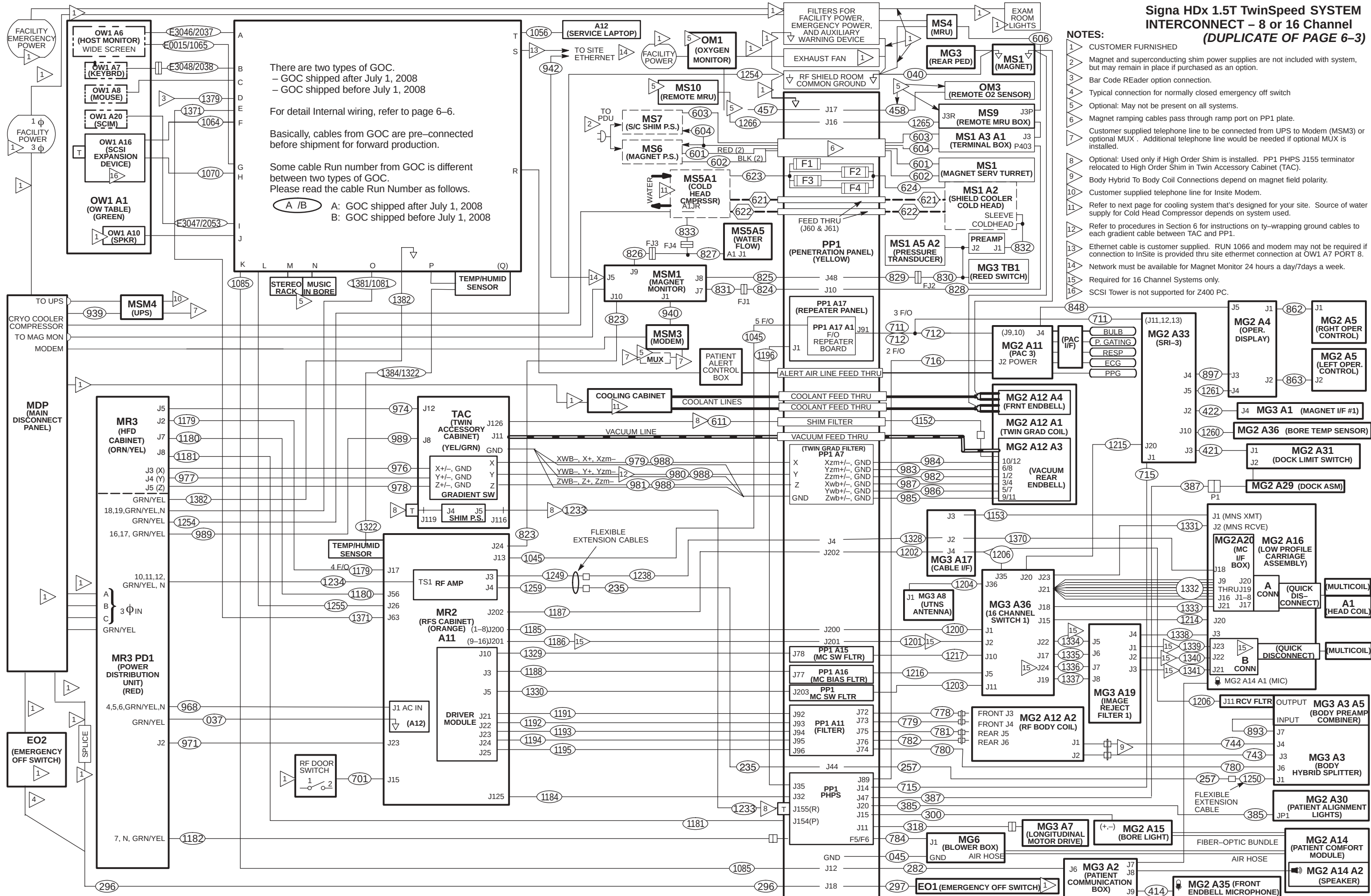
- NOTES:**
- 1 CUSTOMER FURNISHED.
 - 2 THIS GROUP CONTAINS WATER LINES WHICH SHALL BE ROUTED SEPARATE FROM ELECTRICAL LINES (I.E. POWER & SIGNAL).
- NOTE:**
- ONLY INTERCONNECTS SPECIFIC TO MRCC SUBSYSTEM EQUIPMENT SHOWN HERE.

OUTDOOR MRCC SUBSYSTEM GROUP INTERCONNECT DIAGRAM

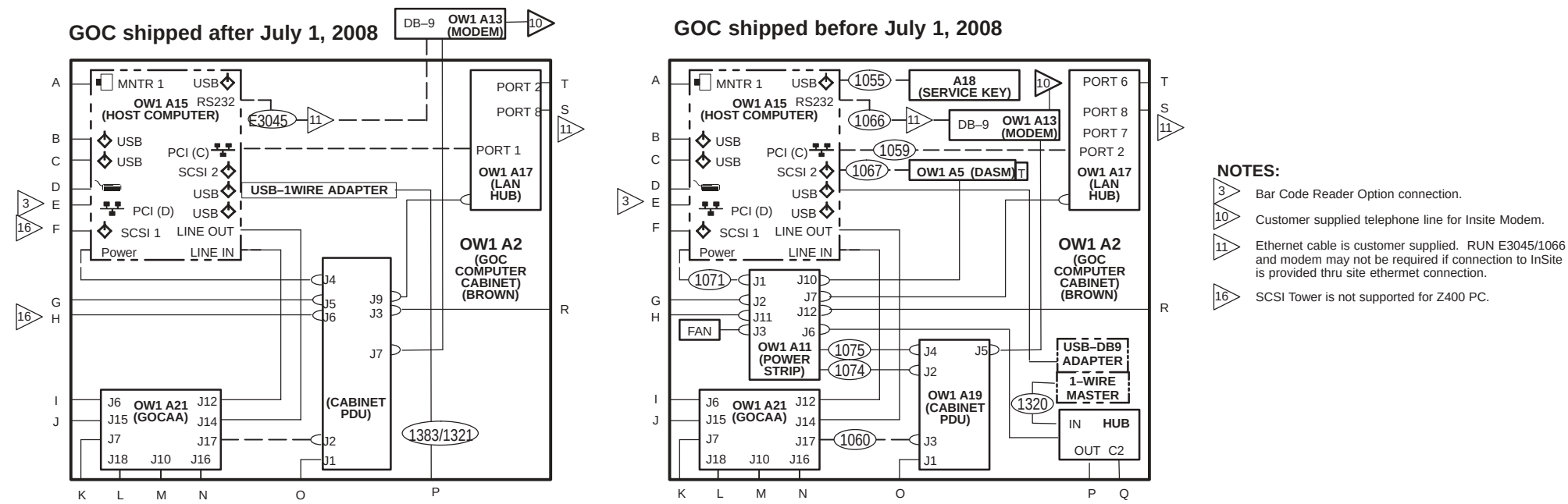


- NOTES:**
- 1 CUSTOMER FURNISHED.
 - 2 THIS GROUP CONTAINS WATER LINES WHICH SHALL BE ROUTED SEPARATE FROM ELECTRICAL LINES (I.E. POWER & SIGNAL).
 - 3 FOR INDOOR WATER COOLED GWHX ONLY.
 - 4 GWHX PROVIDED CABLE.
- NOTE:**
- ONLY INTERCONNECTS SPECIFIC TO GWHX SUBSYSTEM EQUIPMENT SHOWN HERE.

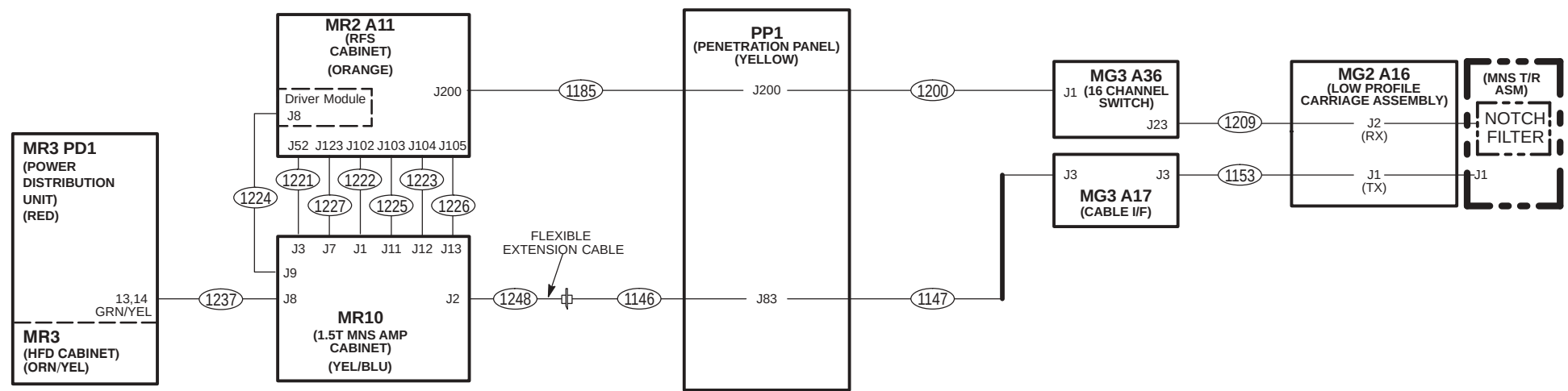
INDOOR GWHX SUBSYSTEM GROUP INTERCONNECT DIAGRAM



1.5T HDx Series TwinSpeed
GOC Configurations



Signa HDx 1.5T TwinSpeed, Release 14.0: 2KW MNS OPTION
SYSTEM INTERCONNECT (FIXED SITE)



6-1-2 SORTING AND ROUTING CABLES



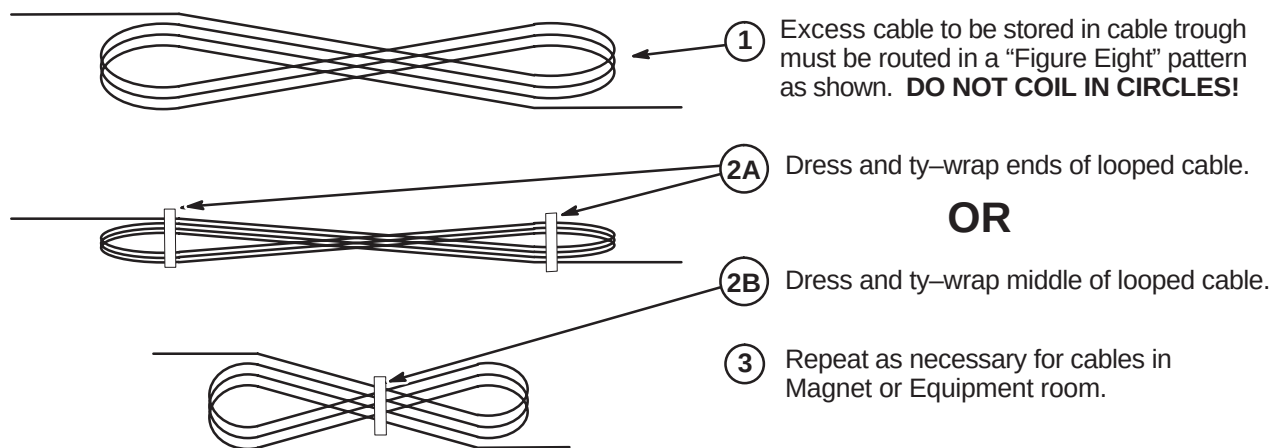
Ty-wraps must be cut flush with no protruding sharp edges or points. Failure to do so can result in numerous laceration hazards when servicing the equipment.

System power, ground, and data cables are color coded at each end by destination per colors shown on Cable Map.

1. Sort cables by destination. Normally, it is best to sort cables originating from the farthest cabinet or component from the PDU first. This will make it easier to pull cables through the troughs.

Note: Carefully review architectural drawings to insure that all ducts are properly installed and that all signal and power cables are accounted for before starting to route cable runs.

2. Unroll coiled cables along the intended route to insure that they will lay freely without twists or kinks. Route cables in accordance with the applicable System Interconnect Diagram.
3. Plan for storage location of coiled cable excess length if cables are to remain at delivered length. See Illustration 2-1. Some cables are usually cut to length such as gradient cables, power cables, Head and Body RF, and Fiber-optic cables.



PROPER STORAGE OF EXCESS CABLES

ILLUSTRATION 2-1

4. Place cables in provided troughs, conduit, or ducts. Leave sufficient slack for later connection when cabinets are in final position according to site architectural plans. Route cables inside exam room to Rear Pedestal area with sufficient length remaining to complete routing and connecting within Magnet Enclosure.

6-2 EQUIPMENT ROOM CABLE INSTALLATION

The equipment room cables and cabinets can be installed simultaneously. Cable locations depend on where cabinets are installed.

6-2-1 HFD CABINET CABLE CONNECTIONS



MAKE SURE ALL POWER TO CABINET IS OFF, LOCKED, & TAGGED BEFORE CONNECTING WIRES.

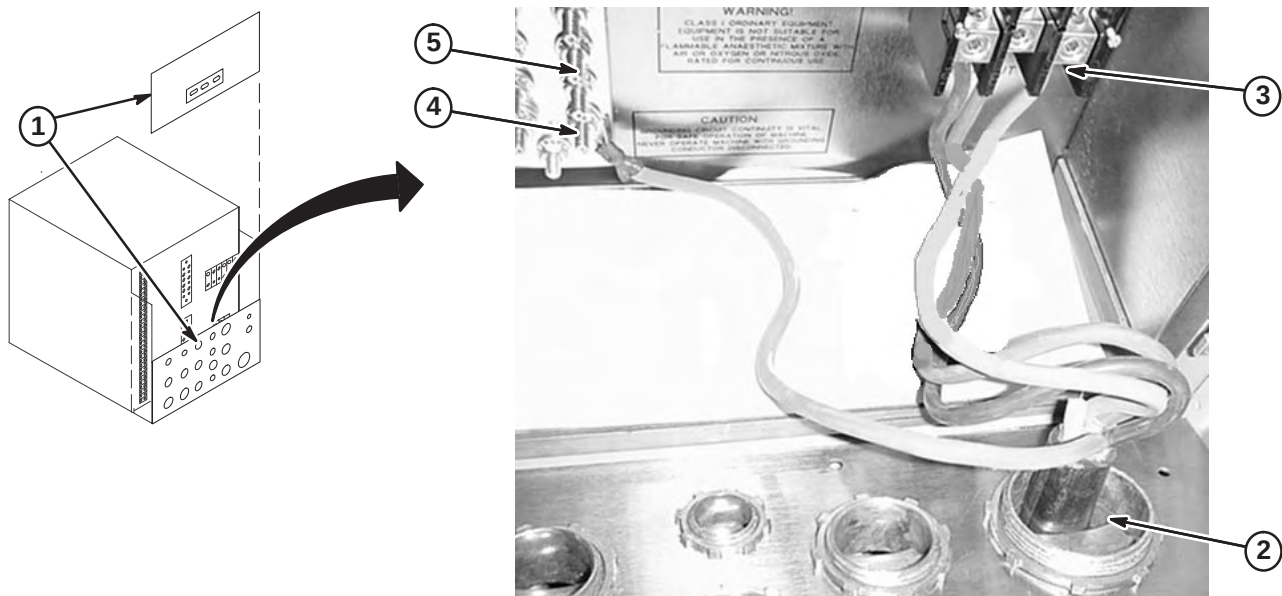
A black rectangular box with the word "WARNING!" in bold, white, sans-serif capital letters.

WARNING!

IF 3 PHASE WYE WITH NEUTRAL AND GROUND (5 WIRE SYSTEM) INPUT IS USED THE NEUTRAL MUST BE TERMINATED INSIDE THE MAIN DISCONNECT CONTROL AND NOT BROUGHT TO THE HFD CABINET.

6-2-1-A HFD Cabinet Power and Ground Cable Installation

1. Remove rear PDU module covers.
2. Route facility input power cable and ground wire thru interface panel.
3. Connect L1-L3 conductors to terminal block.
4. Connect facility input dedicated ground conductor to Ground (GRN/YEL) Bus Bar.
5. Connect RF screen room ground conductor to Ground Bus Bar.

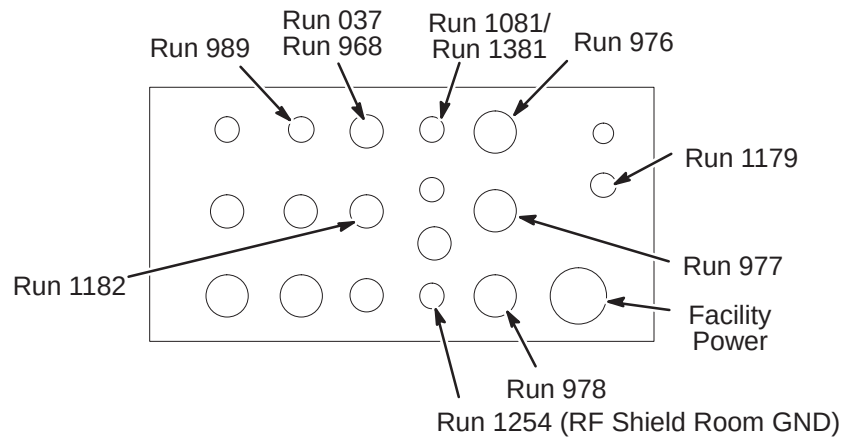


6-2-1-B Route Power Cables Thru Interface Panel

1. Route Runs thru interface panel as marked.

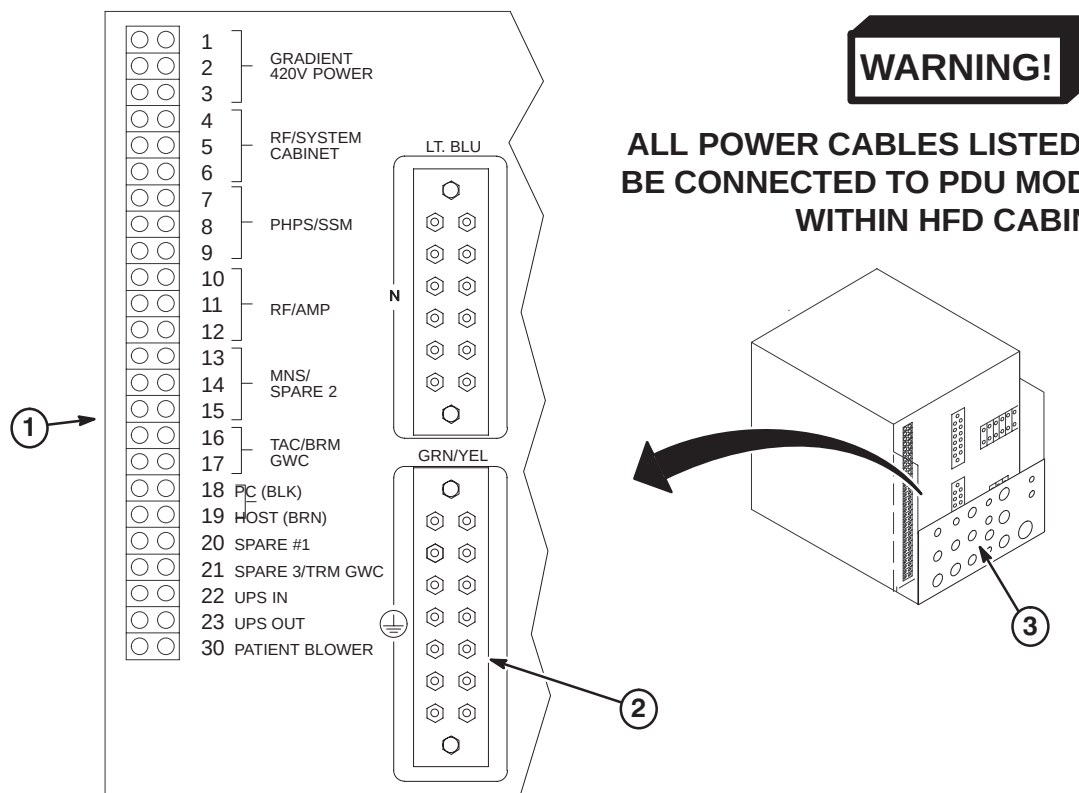
Note: The GRN/YEL ground wire, Run 037, **MUST** be routed with Run 968 thru the same port.

2. Route any option Runs as marked.



6-2-1-C Connect Power Cables to PDU Module

1. Route/connect power cables to pressure connectors and studs on back of PDU module as shown in Table below. Relable PDU connection panel as shown below.
2. Trim and connect ground wire cables (Runs 037 and 1254) to ground bus connectors.
3. Tighten clamps of interface panel to secure cables in place.



CONNECT TO	RUN	PHASE	DESTINATION
4,5,6	968	ϕ A, B, C	RFS Cabinet
7	1182	ϕ A	PHPS Module
10,11,12	1113	ϕ A, B, C	NB RF Amp Cabinet
16,17	989	ϕ A,C	TAC Cabinet
18,19	1081	ϕ C,A	Operator Workspace (Old GOC)
19	1381	ϕ A	Operator Workspace (New GOC)
(MDP)* or 21 (PDU)		ϕ A or ϕ A	AirSys Water Cooled Cabinet or AirSys Heat Exchanger
GRN/YEL	1382		GOC Ground Wire (new GOC)
GRN/YEL	037		System Cabinet Ground Wire
GRN/YEL	1254		RF Shield Room Ground (Customer Supplied)

COLOR CODE: for remainder of cables:
 3 ϕ = Black, Red, Orange Neutral = Light blue Gnd = Grn/Yel
 Single ϕ = Brown, Black

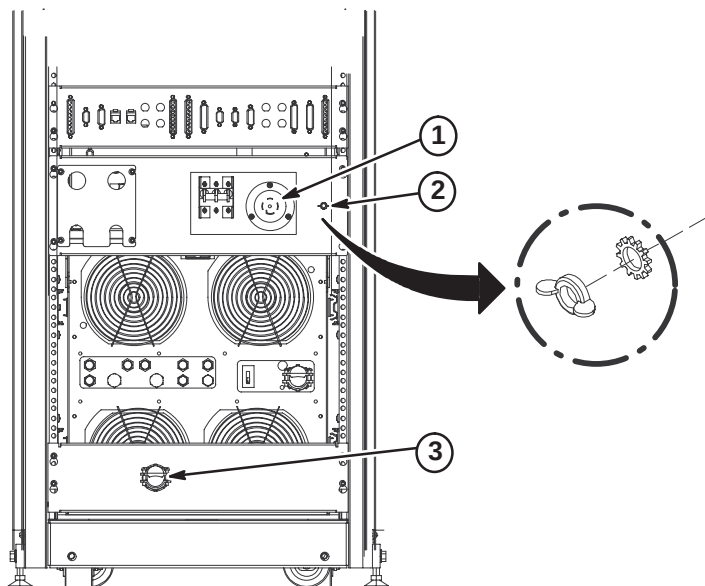
* Connection to Main Disconnect Panel to be completed by customer electrician.

6-2-2 EQUIPMENT ROOM CABINET POWER CABLE CONNECTIONS

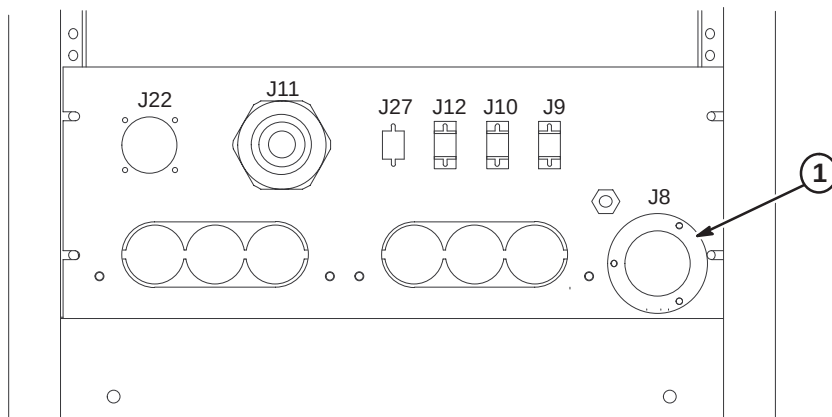
CABINET POWER AND GROUND CABLES MUST BE CONNECTED TO PDU MODULE LOCATED WITHIN HFD CABINET.

6-2-2-A RF/System Cabinet Power and Ground Cable Connections

1. Connect Run 968 cable plug to "J1". Be sure to push and turn plug for secure connection.
2. Connect Run 037 GND cable to ground stud.
3. Route Run 1234 thru lower I/F panel and clamp in place. Connect as marked to RF Amp module.

**6-2-2-B TAC Cabinet Power Cable Connection**

1. Connect Run 989 cable plug to J8.



6-2-3 GRADIENT CABLE INSTALLATION IN EQUIPMENT ROOM

Nine cables have been delivered to site. All have three lead wires with terminals and attached cable labels at one end. The leads on this cable end are connected to the Twin Gradient Filter on the Penetration Panel. The other end is cut flush with no labels or terminals. The nine cables with supplied labels are as follows:

RUN	PART #	LABEL	LENGTH	SUPPLIED LABELS
979	2345400-10	PP1-A7-X	59.5 ft.	(1) RUN 976, (1) RUN 979, (2) X+, (1) X-, (1) GND, (1) XZM-, (1) XWB-
980	2345400-11	PP1-A7-Y	59.5 ft.	(1) RUN 977, (1) RUN 980, (2) Y+, (1) Y-, (1) GND, (1) YZM- (1) YWB-
981	2345400-12	PP1-A7-Z	59.5 ft.	(1) RUN 978, (1) RUN 981, (2) Z+, (1) Z-, (1) GND, (1) ZZM- (1) ZWB-
982	2345400-18	PP1-A7-Z _{ZM}	26.5 ft.	(1) RUN 982, (1) ZZM- (1) ZZM+
983	2345400-17	PP1-A7-Y _{ZM}	26.5 ft.	(1) RUN 983, (1) YZM- (1) YZM+
984	2345400-16	PP1-A7-X _{ZM}	26.5 ft.	(1) RUN 984, (1) XZM- (1) XZM+
985	2345400-24	PP1-A7-Z _{WB}	26.5 ft.	(1) RUN 985, (1) ZWB- (1) ZWB+
986	2345400-23	PP1-A7-Y _{WB}	26.5 ft.	(1) RUN 986, (1) YWB- (1) YWB+
987	2345400-22	PP1-A7-X _{WB}	26.5 ft.	(1) RUN 987, (1) XWB- (1) XWB+

Refer to Section **6-3-2** for installation of Gradient cables in Magnet Room.

6-2-3-A General Steps for Cutting Cables to Length and Routing in Equipment Room

1. Move Run 979, 980, and 981 cables to equipment room. The end of cable with terminals will be connected to Penetration Panel.
2. Route cables from Gradient Filter of Penetration Panel to right side of interface panel at rear of Twin Accessory Cabinet (TAC).
3. Verify sufficient slack of at least two feet (610mm) is provided for termination and connection at TAC Cabinet. Mark location for cutting. Cut cables to length and label cut ends with applicable run number.
4. Refer to Procedure **6-2-3-E** to terminate and label TAC end cable Runs 979, 980, and 981
5. Refer to Procedure **6-2-3-F** for connection of Runs 979, 980, and 981 to TAC cabinet.
6. Route remainder of cables from rear of TAC to rear of HFD cabinet. Make sure enough slack is provided for termination and connection to cabinet. Mark location for cutting. Cut cables to length.
7. Refer to Procedure **6-2-3-H** for connection of Runs 976, 977, and 978 to TAC cabinet.
8. Verify sufficient slack of at least two feet (610mm) is provided for termination and connection at HFD Cabinet. Mark location for cutting. Cut cables to length and label cut ends with applicable run number.
9. Refer to Procedure **6-2-3-K** for connection of Runs 976, 977, and 978 to HFD cabinet.

6-2-3-B Connect Runs 979, 980, and 981 to Penetration Panel

1. Remove gradient filter terminal covers.
2. Route Run 981 (Z) to left side and Runs 979 (X) and 980 (Y) to right side of gradient filter on Penetration Panel.

Note: The three X and three Y terminals are routed to the right side and three Z terminals to the left side.

3. Connect +, ZM- (cable marked as TWO), & WB- (cable marked as THREE) wires to gradient filter terminals as marked on each side of gradient filter.

Note: Red tape indicates positive (+) wires.

4. Align cables to obtain maximum spacing from other cables and conductive metal objects.

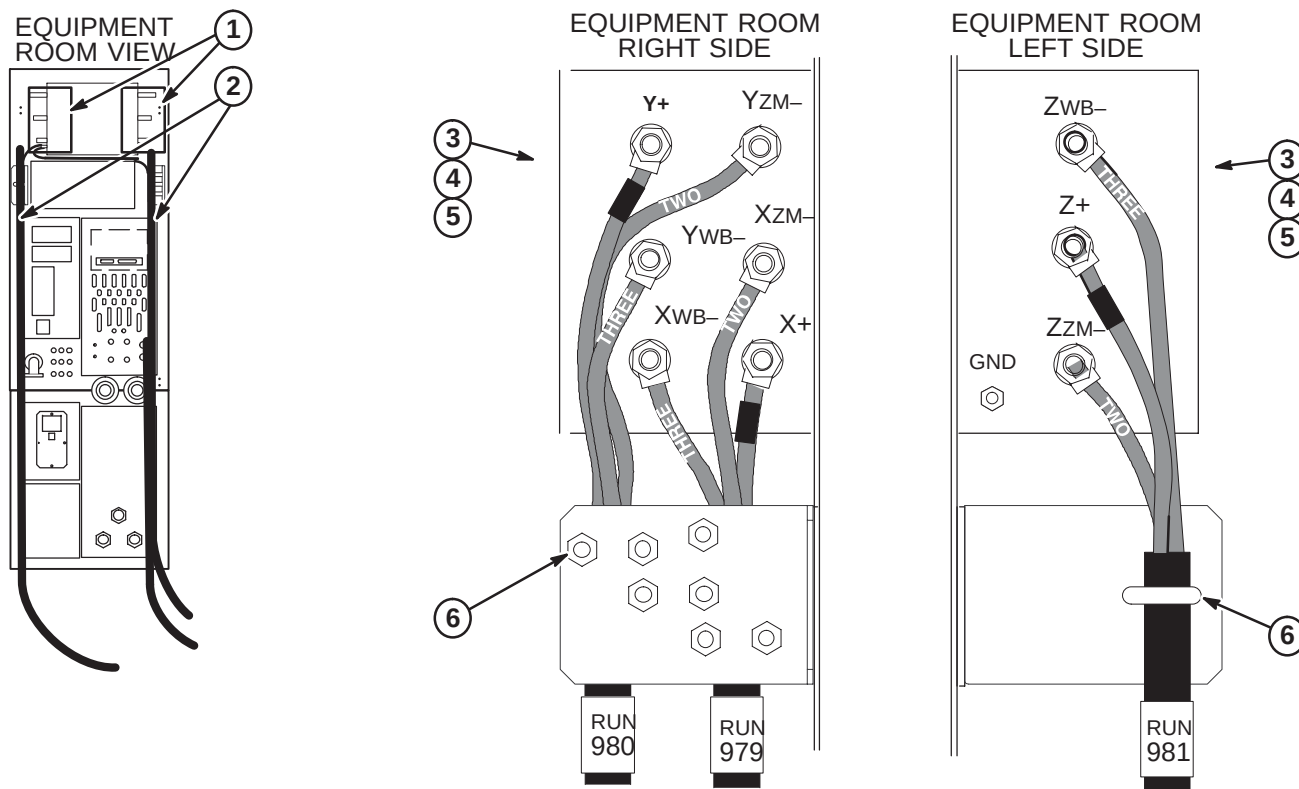
Note: When tightening nuts, it is important to hold screw body.

5. Tighten nuts to a torque of 35 ft-lbs (67–74Nm) or after tightening, verify that nothing is loose.

Note: Final check is by hand. Make sure screw is fixed, nut is tight, and cable is not detached from terminal lug.

6. Route and secure each cable Run to strain-relief bracket with U-bolts.

Note: Do not over-torque the strain relief. Cables may be damaged. Tighten enough to apply friction to cable, but not enough to “crimp” the insulation.

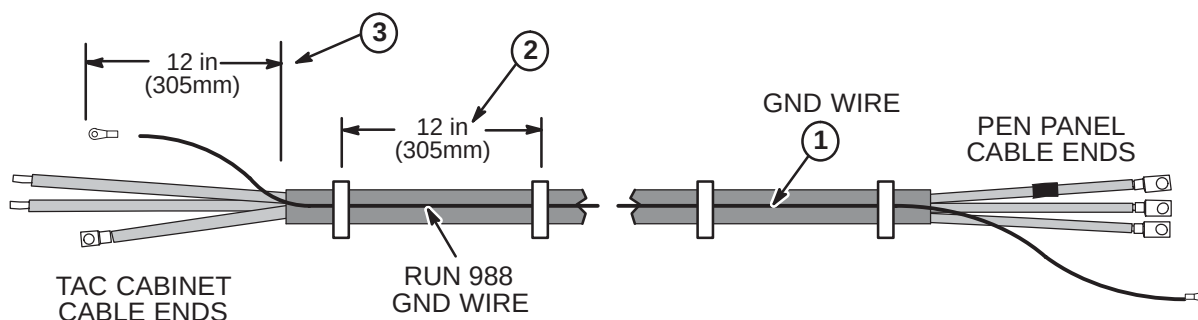


6-2-3-C Attach Ground Cables to Runs 979, 980, and 981

For each gradient cable routed between the Penetration Panel and TAC Cabinet, a ground wire, Run 988 (2283297), has to be ty-wrapped to each cable.

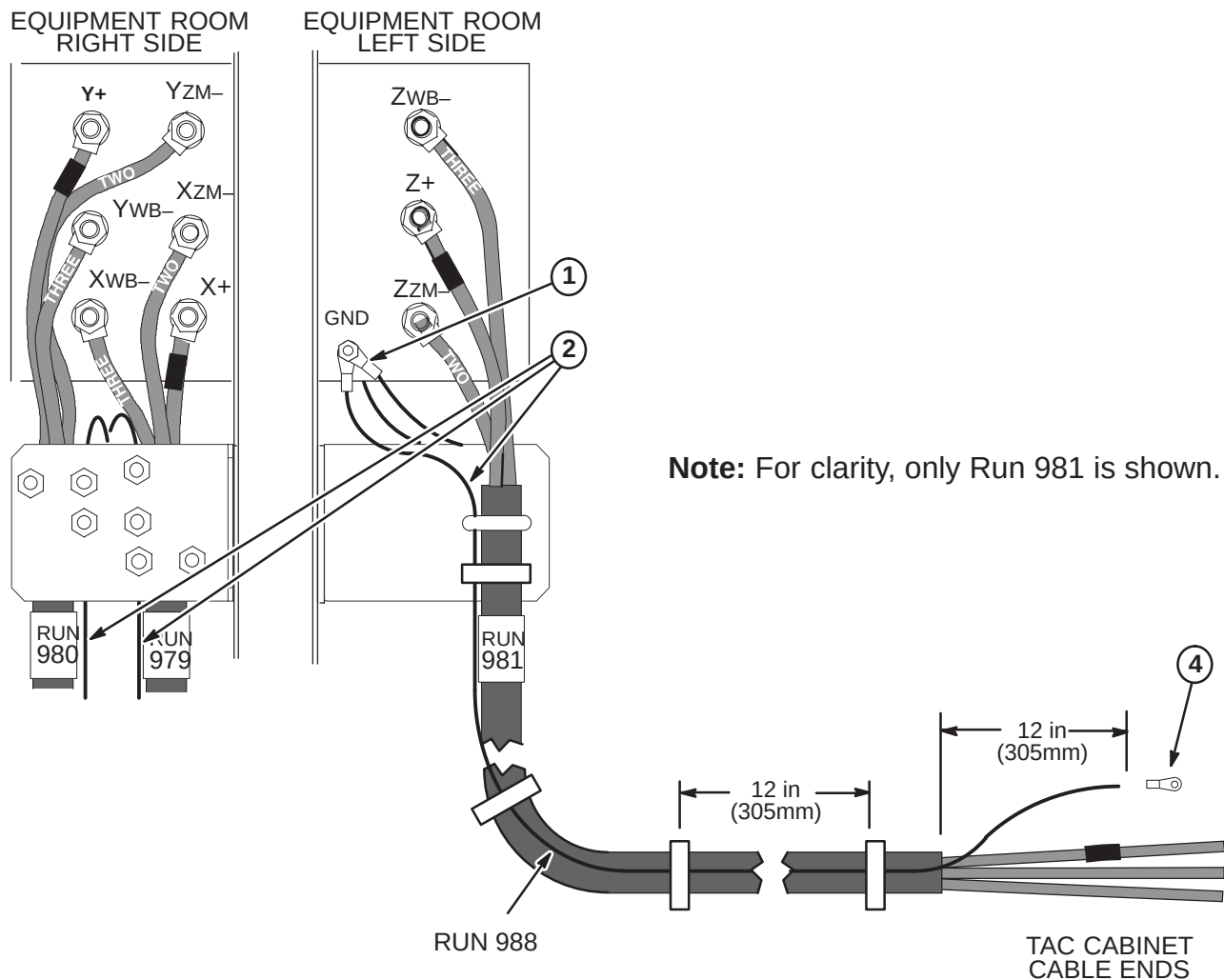
1. One Ground Wire will be routed along each of the Gradient Cables, Runs 979, 980, and 981.
2. Each Run 988 must be ty-wrapped about every 12 inches along entire length of each gradient cable.
Note: Run 988 must be routed parallel to the gradient cable. It **MUST NOT** be routed spirally around the gradient cable.
3. End of Run 988 at TAC cabinet should extend about 12 inches (305mm) past end of outer jacket after trimming.
Note: Refer to Procedure **6-2-3-E** for length of gradient cable jacket material to be removed.

Excess ground cable can not be coiled. It must be cut to length and reterminated.



6-2-3-D Connect Run 988 Cables Ground Wires to Gradient Filter

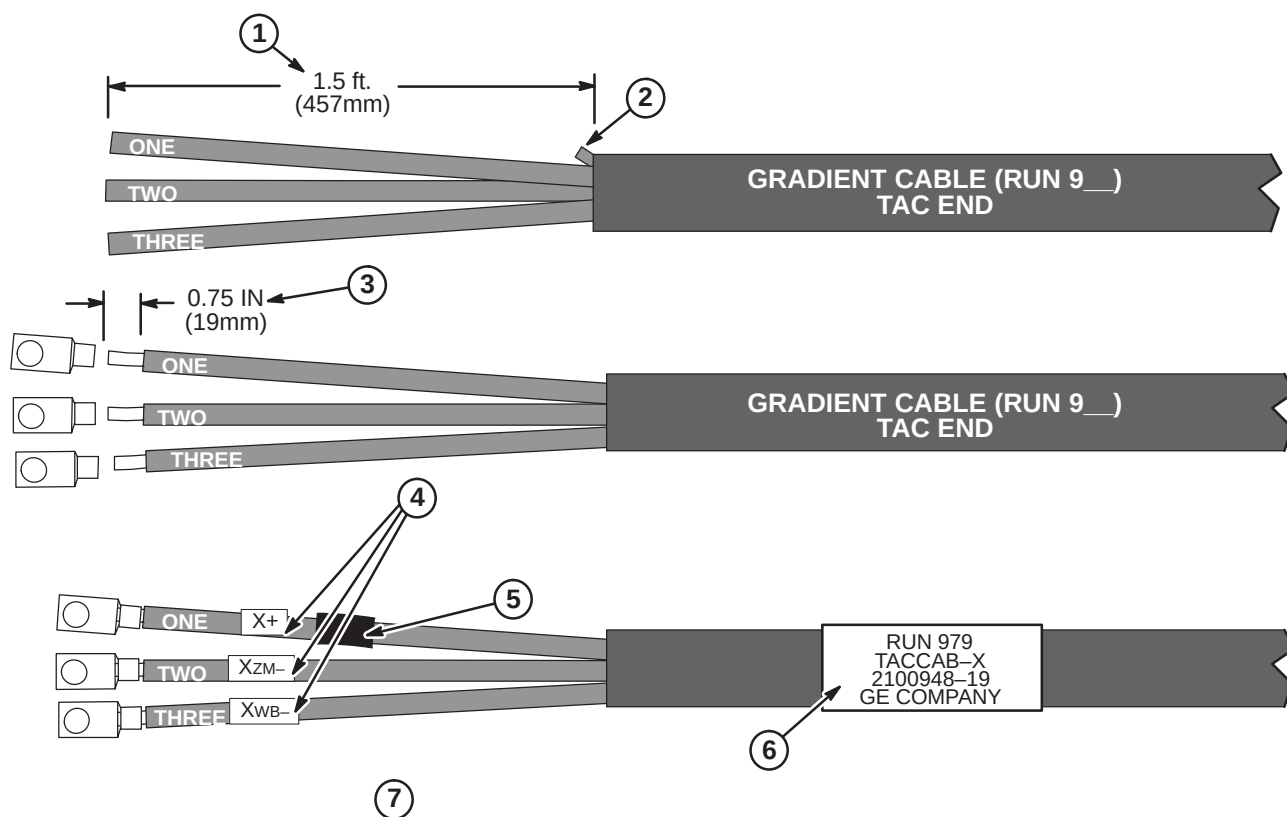
1. Connect the three Run 988 Gound Wires to GND on left side of Gradient Filter.
2. Each Run 988 must be ty-wrapped about every 12 inches along entire length of each gradient cable.
Note: Run 988 must be routed parallel to the gradient cable. It **MUST NOT** be routed spirally around the gradient cable.
3. Re-install gradient filter terminal covers.
4. Cut each Run 988 to length, reterminate with 3/8 ring terminal for connection to GND stud on TAC Cabinet.



6-2-3-E Terminate Runs 979, 980, and 981 for TAC Cabinet Connection

The following steps are required to terminate the end of the cable that was cut in Procedure **6-2-3-A**. The other end of the cable is connected to the Penetration Panel.

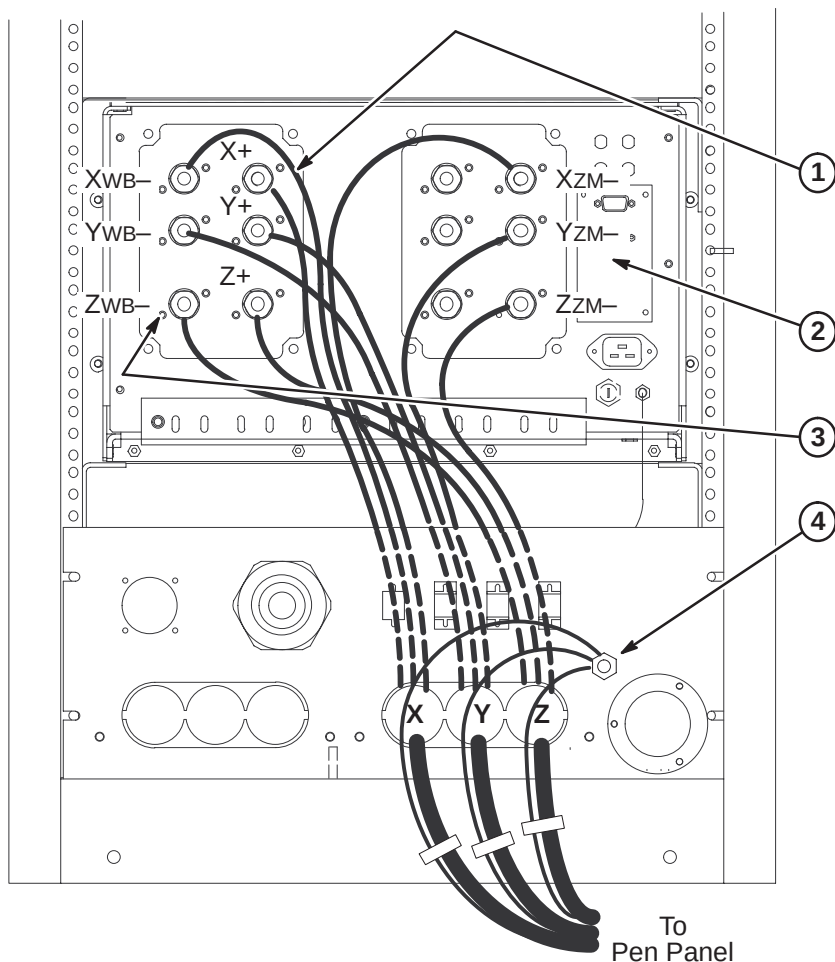
1. Remove 18 inches of jacket from end of cable Runs 979 (X), 980 (Y), and 981 (Z).
2. Trim rubber cable filler strands flush with end of jacket.
3. Strip 0.75 in. (19mm) of insulation from end of wires "ONE", "TWO" and "THREE" and crimp a furnished T & B 1/2" (G2-3-12) terminal to each using Ratcheting Crimper with #2 die.
4. Place furnished "X+" label on wire "ONE".
Place furnished "XZM-" label on wire "TWO".
Place furnished "XWB-" label on wire "THREE".
5. Wrap red tape around wire "ONE".
6. Place furnished "RUN 979" label on cable.
7. Steps above show completion of Run 979. Repeat same process for Runs 980 (Y+, YZM-, YWB-) and 981 (Z+, ZZM-, ZWB-).



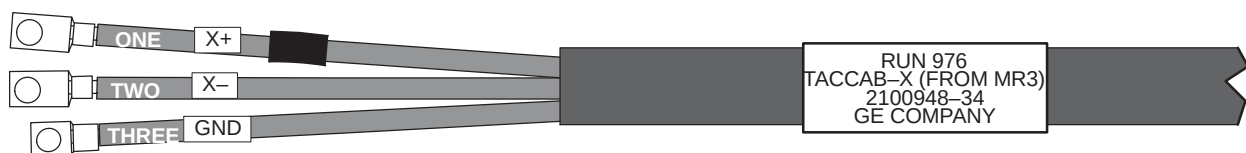
6-2-3-F Connect Runs 979, 980, and 981 to TAC Cabinet

To make installation easier, route and connect Run 981 (Z terminals) first, then Run 980 (Y), and last Run 979 (X).

1. Connect X+, Y+, and Z+ cables as marked. Cables should have red tape applied towards end of cable.
2. Connect XZM-, YZM-, and ZZM- cables as marked.
3. Connect XWB-, YWB-, and ZWB- cables as marked.
4. Route and connect the three Run 988 Ground wires to GND stud on interface panel.

**6-2-3-G Pre-terminated cable ends for Runs 976, 977, and 978**

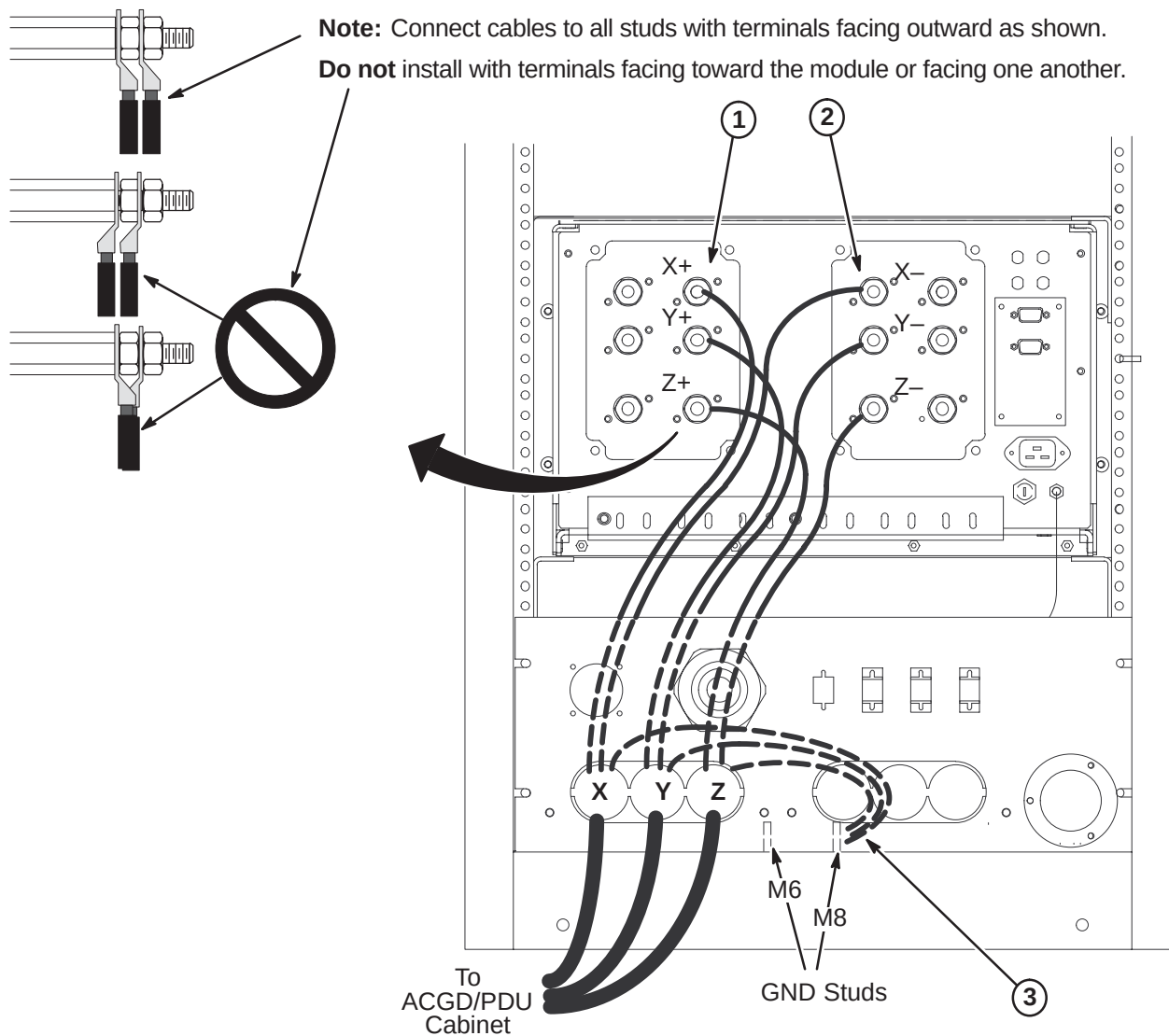
Cables delivered to site will be pre-terminated for the TAC Cabinet connection of Runs 976, 977, and 978. Refer to Procedure **6-2-3-A** for cutting cables to length for connection to HFD Cabinet. A typical pre-terminated end for each cable is shown below.



6-2-3-H Connect Runs 976, 977, and 978 to TAC Cabinet

To make installation easier, route and connect Run 978 (Z terminals) first, then Run 977 (Y), and last Run 976 (X).

1. Connect X+, Y+, and Z+ cables as marked. Cables should have red tape applied towards end of cable.
2. Connect X-, Y-, and Z- cables as marked.
3. Connect Ground leads from each cable to M8 GND stud located at bottom of cabinet.



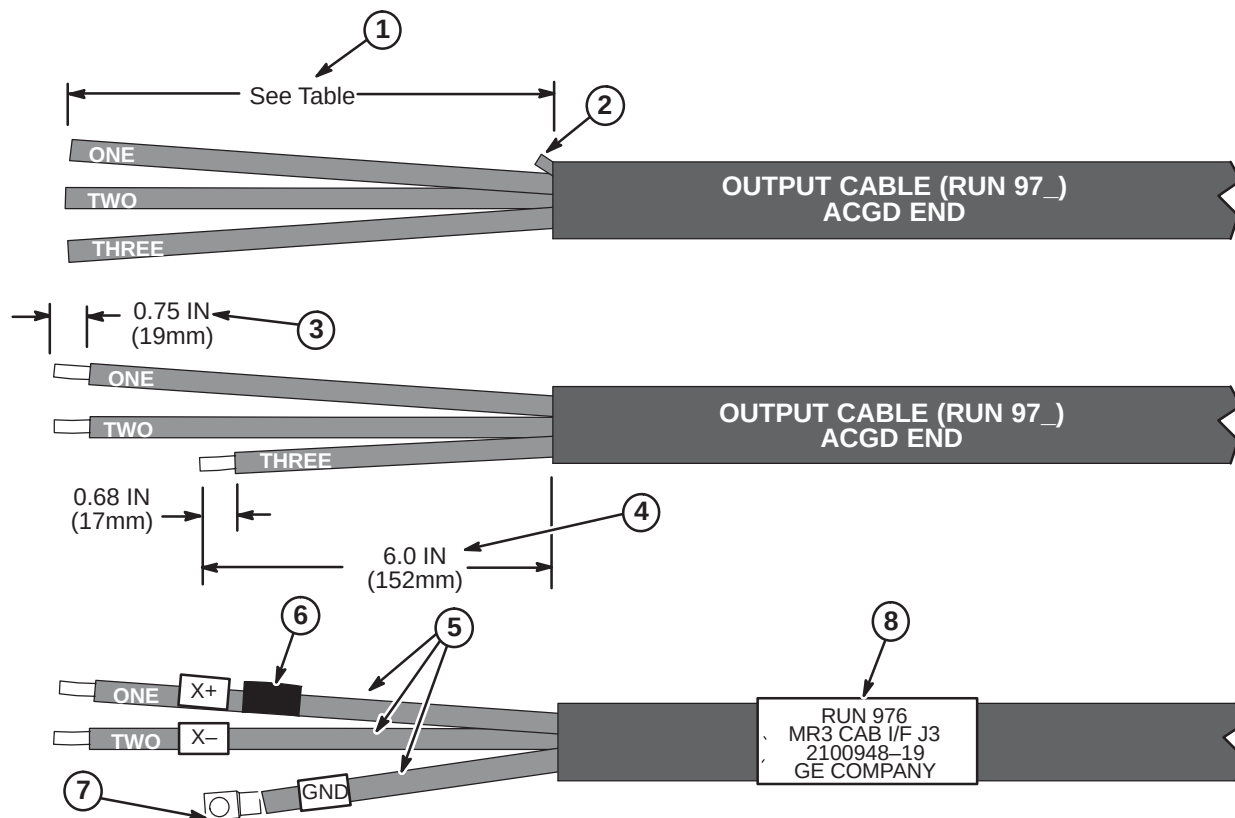
6-2-3-I Terminate Runs 976, 977, and 978 for Connection to HFD Cabinet

The following steps are required to terminate the end of the cable that is to be connected to the HFD Cabinet. The other end of the cable is connected to the TAC Cabinet.

	Run 976	Run 977	Run 978
Jacket Length to be Removed	8.0 IN (203mm)	11.0 IN (280mm)	14.0 IN (356mm)

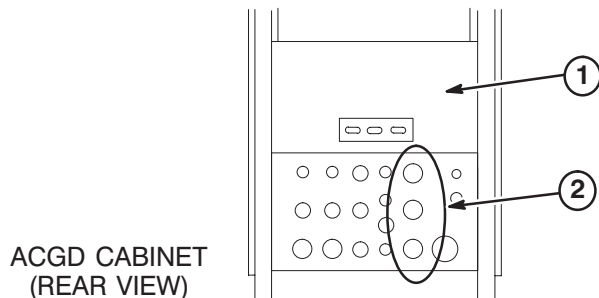
1. Refer to above table to determine length of jacket and foil to be removed from end of cable Runs 976 (X), 977 (Y), and 978 (Z).
2. Trim rubber cable filler strands flush with end of jacket.
3. Strip 0.75 in. (19mm) of insulation from end of wires "ONE" and "TWO".
4. Cut wire "THREE" to 6.0 inches (152mm).
5. Place furnished "X+" label on wire "ONE".
Place furnished "X-" label on wire "TWO".
Place furnished "GND" label on wire "THREE".
6. Wrap red tape around wire "ONE".
7. Crimp a furnished T&B #G2-38 terminal on GND wire end using a Ratcheting Crimper with #2 die.
8. Place furnished "RUN 976" label on cable.

Steps in illustration below show completion of Run 976. Repeat same process for Runs 977 (Y+, Y-) and 978 (Z+, Z-). **Red tape, Y+ (RUN 977) and Z+ (RUN 978) labels are attached to wire "ONE".**

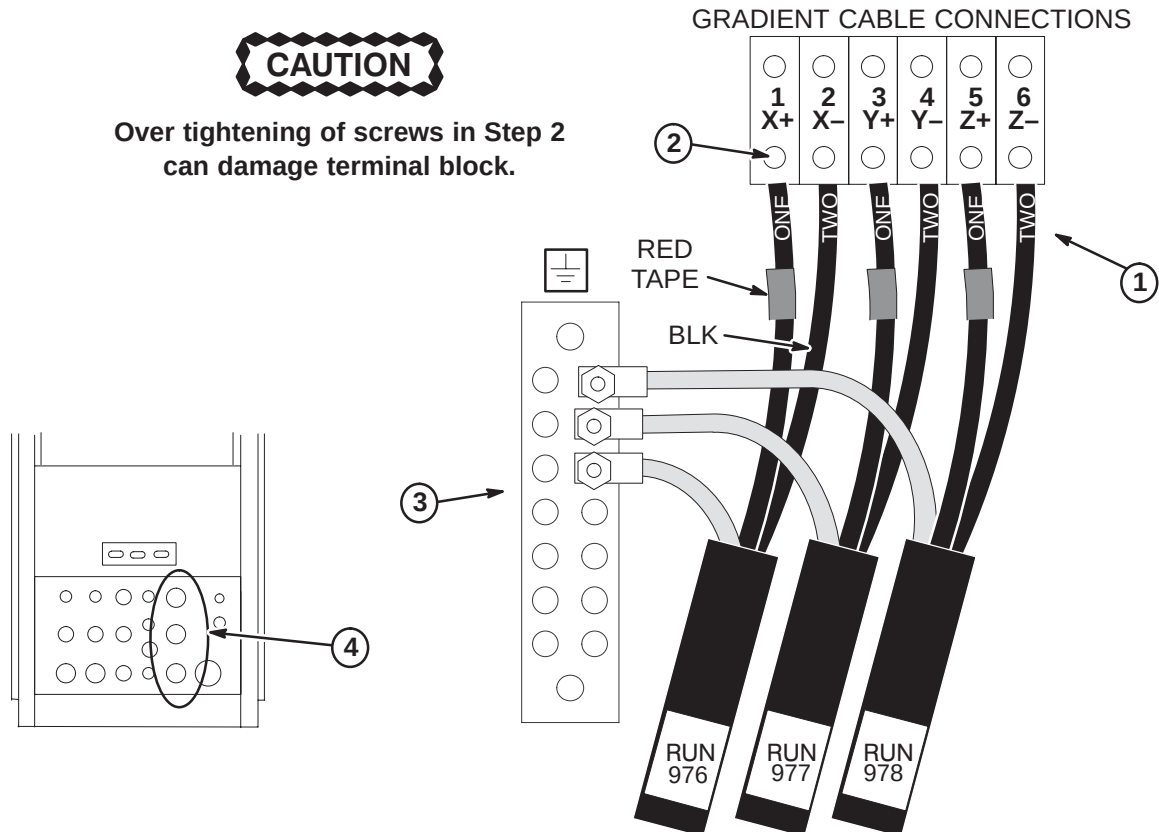


6-2-3-J Gradient Output Cable Interface Panel Access at HFD Cabinet

1. Remove Rear Cover Plate by removing seven screws that hold it in place.
2. Loosen screws on clamps for gradient output cables. Insert ends of gradient output cables into Cabinet Interface as marked. (Runs 976 into J3, Run 977 into J4, and Run 978 into J5)

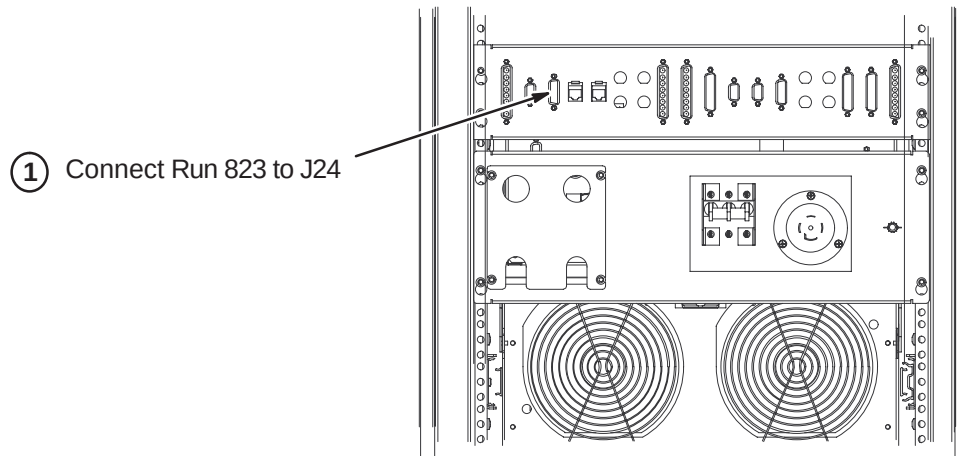
**6-2-3-K Gradient Output Cable Routing and Connection in HFD Cabinet**

1. Connect gradient output cables with red-taped (+) #1 wires to + output terminals and black (-) #2 wires to - output terminals.
2. Tighten six screws to 52-65 inch-lbs (5.88-7.34 N-M). This is about what can be applied using only your thumb and little finger on wrench. Loosen 1/2 turn and re-tighten to the same torque. Repeat approximately 6 times or until wrench returns to the same spot after torquing.
3. Connect the GND leads of each run to the Ground Bus Bar studs.
4. Tighten clamps on Interface Panel to secure gradient output cables.



6-2-4 INSTALLATION OF MAGNET MONITORING CABLES

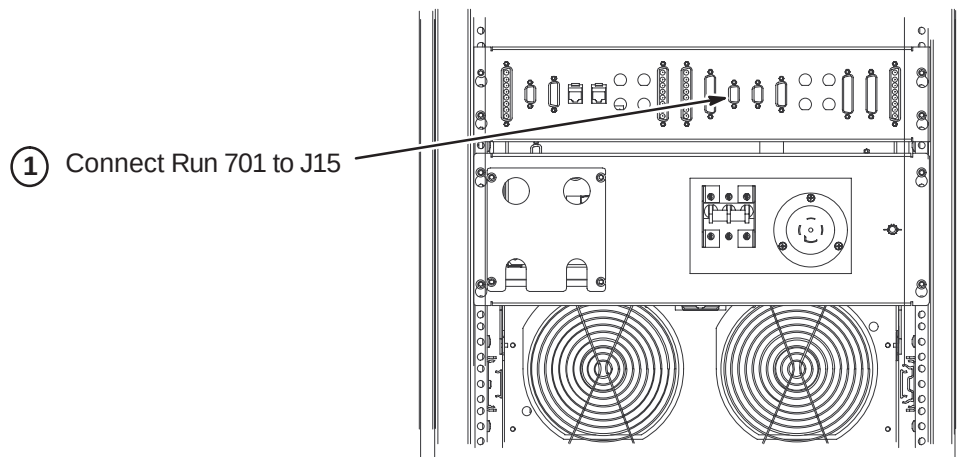
Refer to Direction 2230681, *Magnet Monitor Hardware Installation Manual*, for instructions on connecting cables in the System Cabinet. This manual should be found in the box containing the parts for Magnet Monitoring.



6-2-5 INSTALL RUN 701 – RF DOOR SWITCH

Run 701 is a 100 ft cable with a 9-pin subminiature “D” plug on one end. The other end is cut off with the black and red leads from pair #1 dressed for connecting to a switch. Should it be desired to cut off excess length, make sure that the black and red leads have continuity to pins 1 and 6 on the 9 pin subminiature D plug.

1. Connect Run #701 to J15 on System Cabinet and route to RF Door Switch.
2. Connect black lead on Run #701 to RF Door Switch COM (common) contact.
3. If RF door switch is normally open, proceed to step 4. If normally closed, proceed to step 5.
4. Connect red lead on Run #701 to RF Door Switch N.O. (normally open) contact.
5. If RF door switch is not normally open, connect red lead on Run #701 to RF Door Switch N.C. (normally closed) contact.

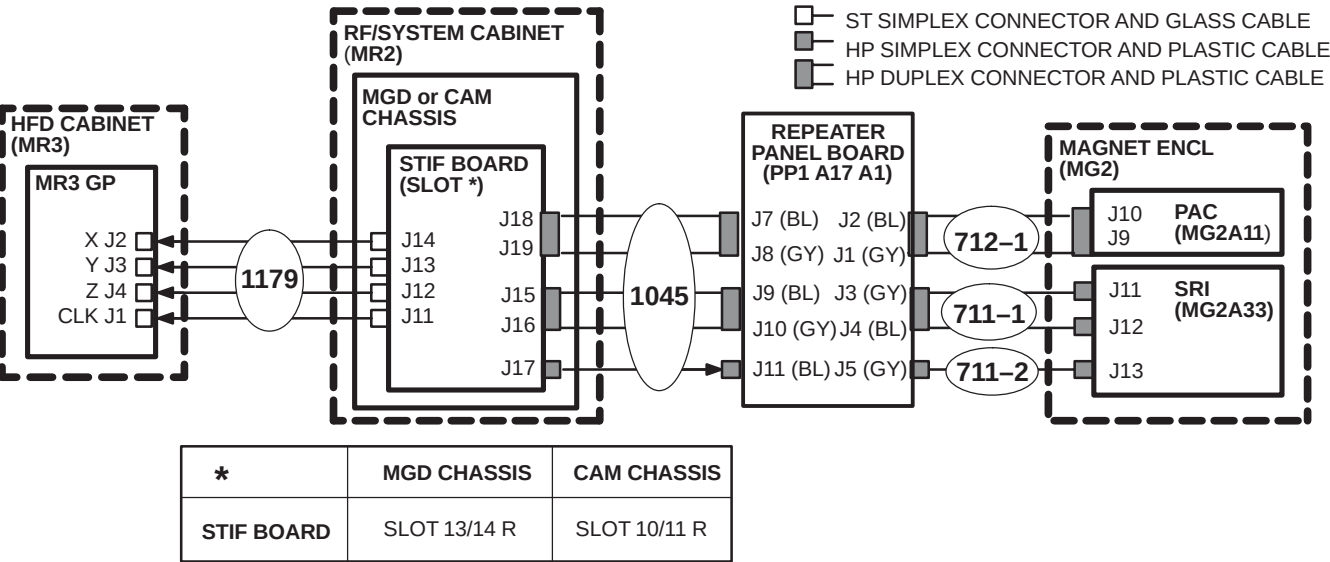


6-2-6 INSTALLATION OF FIBER OPTIC CABLES

Plan for storage of excess Run 1034 conduit of as it cannot be reterminated.

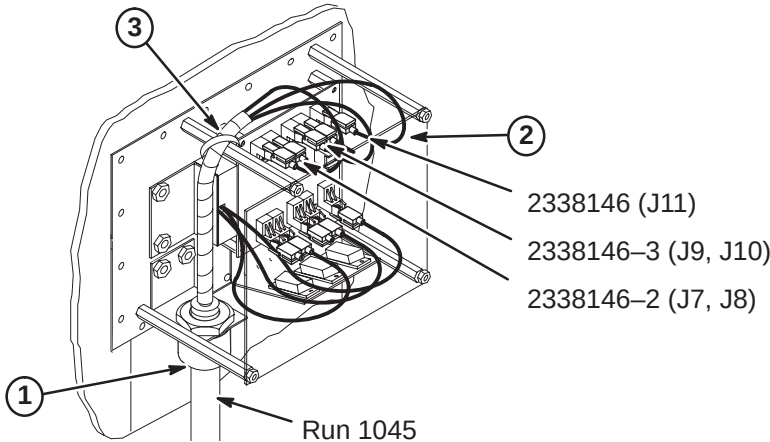
6-2-6-A Routing of Fiber Optic Cables

- 1. Unroll coiled fiber optic conduits for Runs 711/712, 1045, and 1179 along intended routes.
- 2. Connection of Runs is shown in diagram below.



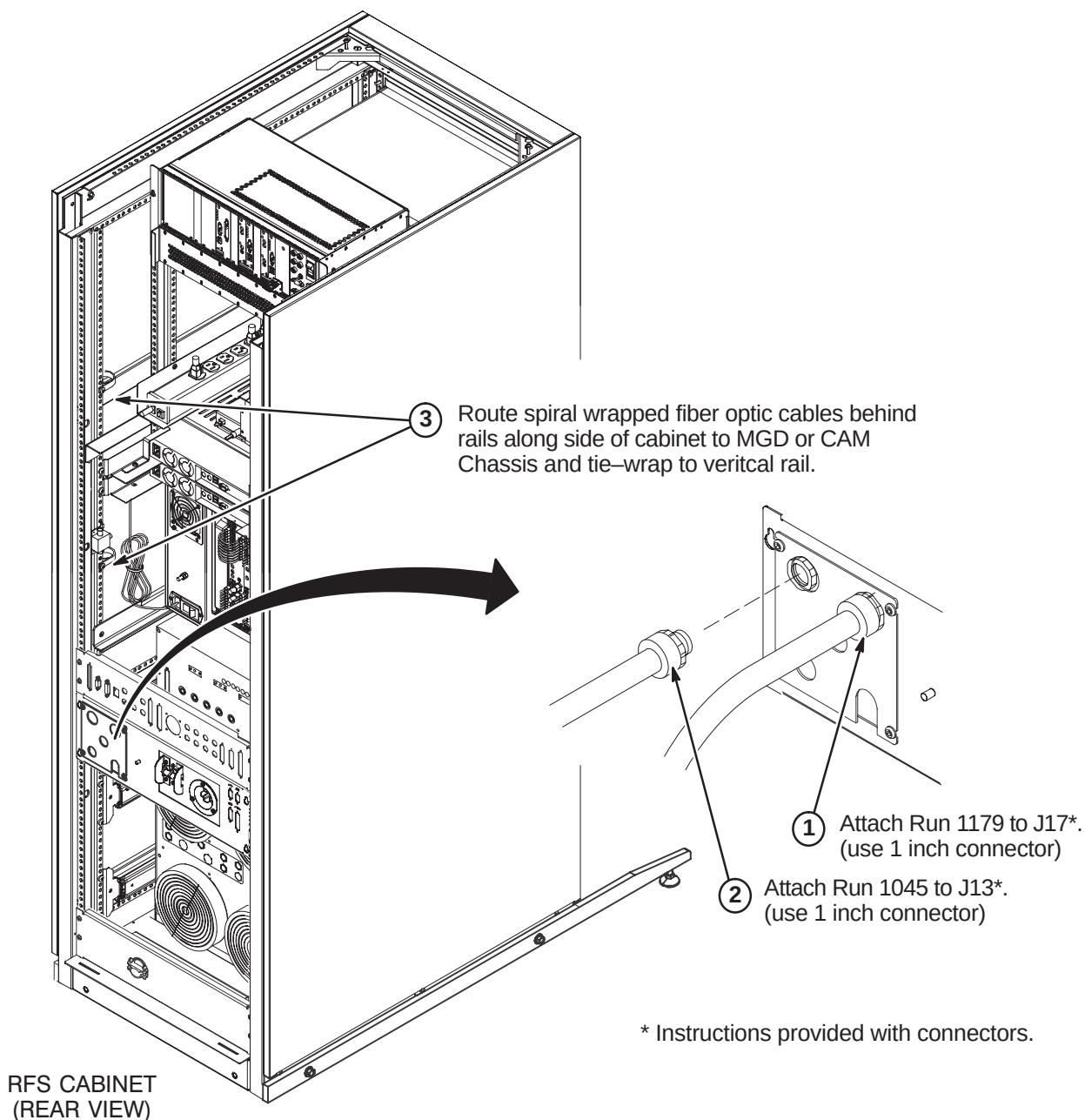
6-2-6-B Connect Run 1045 to Penetration Panel

- 1. Move Run 1045 to the Equipment Room side and attach "PP1-A17" end to access bracket on Penetration Panel. Refer to Direction 2123149 (packed with bag of FO Conduit connectors) for connector installation instructions.
- 2. Adjust route of individual fiber optic runs to maintain large radius and connect Run 1045 as shown.
- 3. Adjust position of protective spiral wrap and secure with tie-wrap.
- 4. Route other end of Run 1045 to System Cabinet.



6-2-6-C Route Fiber Optic Runs 1045, 1083, and 1179 in RF/System Cabinet**CAUTION**

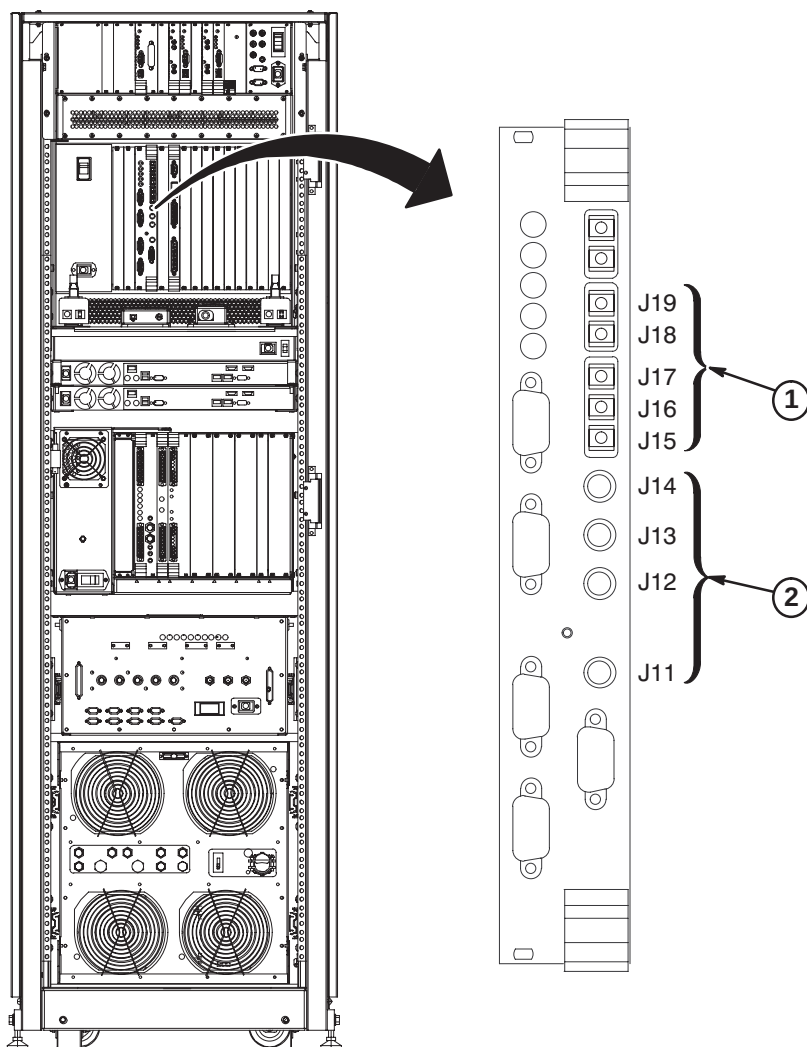
Handle fiber optic cables carefully. Do not bend fiber optic cables to radius smaller than two inches (50mm). Avoid scratching connector ends. Keep connectors protected until ready to connect.



6-2-6-D Route/Connect Fiber Optic Runs1079 and 1045 to STIF Board**CAUTION**

Handle fiber optic cables carefully. Do not bend fiber optic cables to radius smaller than two inches (50mm). Avoid scratching connector ends. Keep connectors protected until ready to connect.

1. Route/Connect Run 1045 fiber optic cables to J15, J16, J17, J18 and J19 on the STIF Board.
2. Route/Connect Run 1034 fiber optic cables to J11, J12, J13, and J14 on the STIF Board.

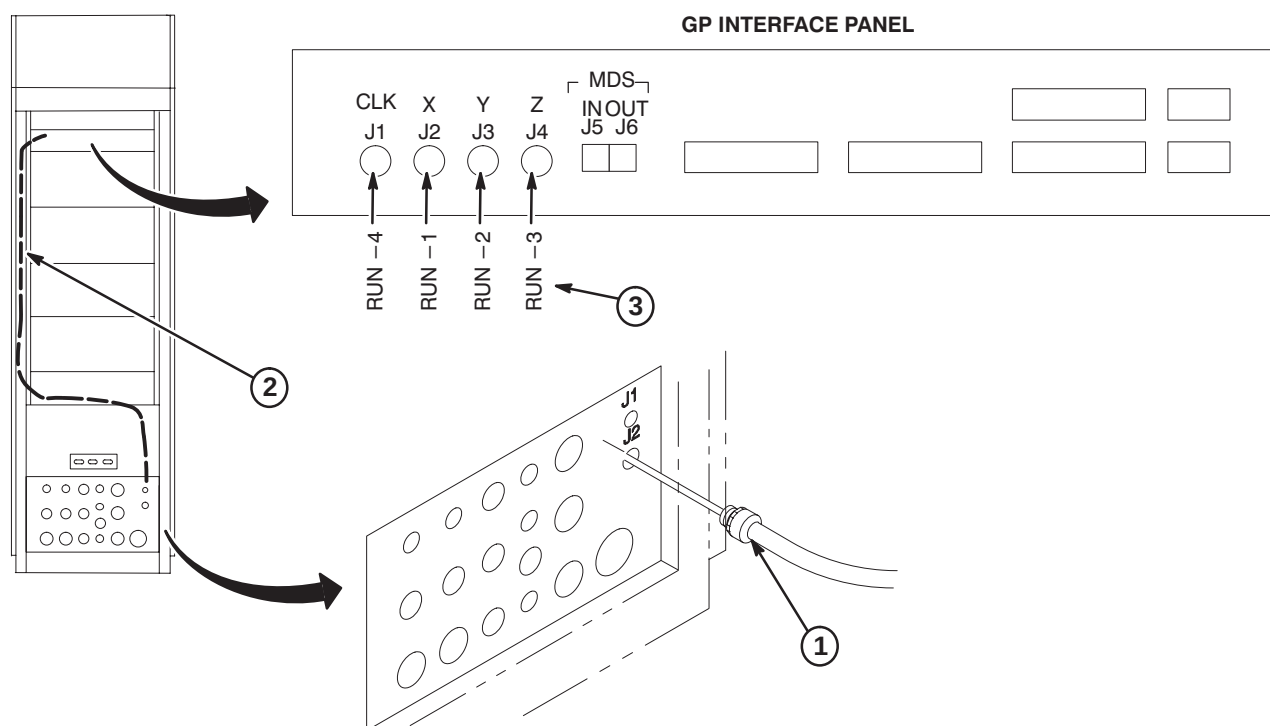
**Note:**

Use tie-wraps to secure fiber optic cables to MGD Chassis cables.

6-2-6-E Route/Connect Fiber Optic Runs 709 and 1034 to GP3 Module in HFD**CAUTION**

Handle fiber optic cables carefully. Do not bend fiber optic cables to radius smaller than two inches (50mm). Avoid scratching connector ends. Keep connectors protected until ready to connect. Routing of fiber optic cables must be done with care to prevent damage to optical fibers.

1. Attach Run 1179 to I/F J2.
2. Route fiber optic cables to GP3 Module. Leave enough slack for extending GP module forward.
3. Connect fiber optics to GP3 module according to table.

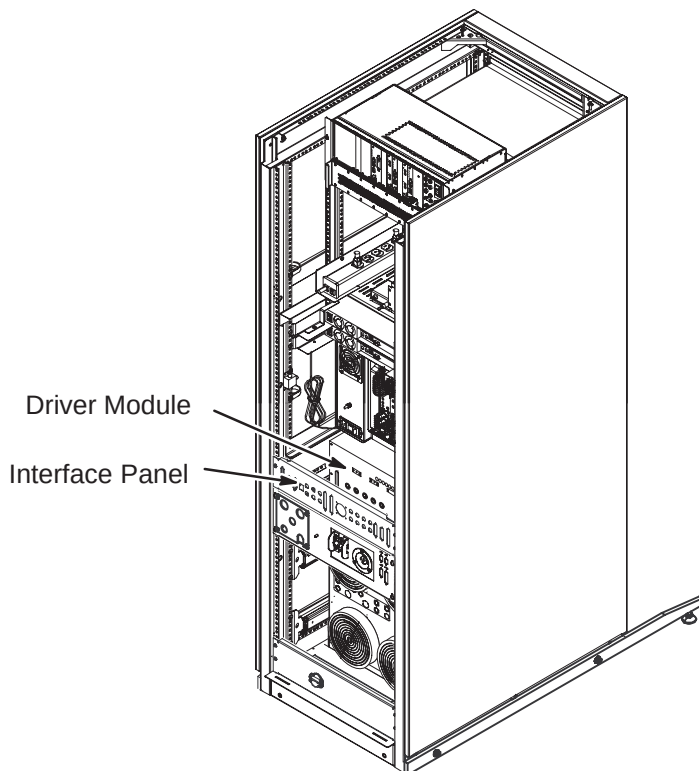


6-2-7 INSTALLATION OF REMAINING EQUIPMENT ROOM CABLES

Connect remaining Runs within equipment room. Dress and secure all cables after connections are made. Reference interconnect map on page 6-3 and Cable Table in Appendix for to/from designator information and Run number reference.

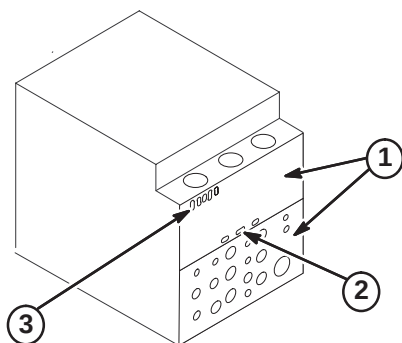
6-2-7-A RF/System Cabinet Connections

1. Connect routed cables to Interface Panels at bottom and middle of cabinet and to the Driver Module as marked. Refer to cable map.
2. Dress and secure cables. Adjust as necessary.



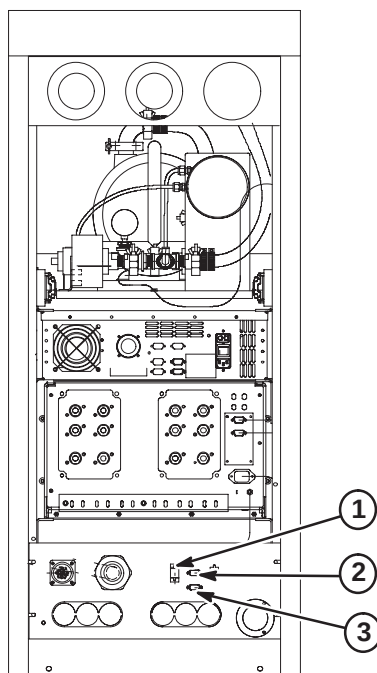
6-2-7-B HFD Cabinet PDU Module Connections

1. Attach rear PDU module covers removed earlier.
2. Connect Run 1179 to J2.
3. Connect Runs 974, 1180, and 1181 to J5, J7, and J8.



6-2-7-C TAC Cabinet Connections

1. Connect Run 974 to J12 on Interface panel.
2. Install Can Terminator on J119 on Interface Panel.
3. Connect Run1233 to J116 on Interface Panel.



6-3 MAGNET ROOM CABLE INSTALLATION

The magnet enclosure and magnet room cables must be installed simultaneously. Cable locations depend on how enclosures are installed.

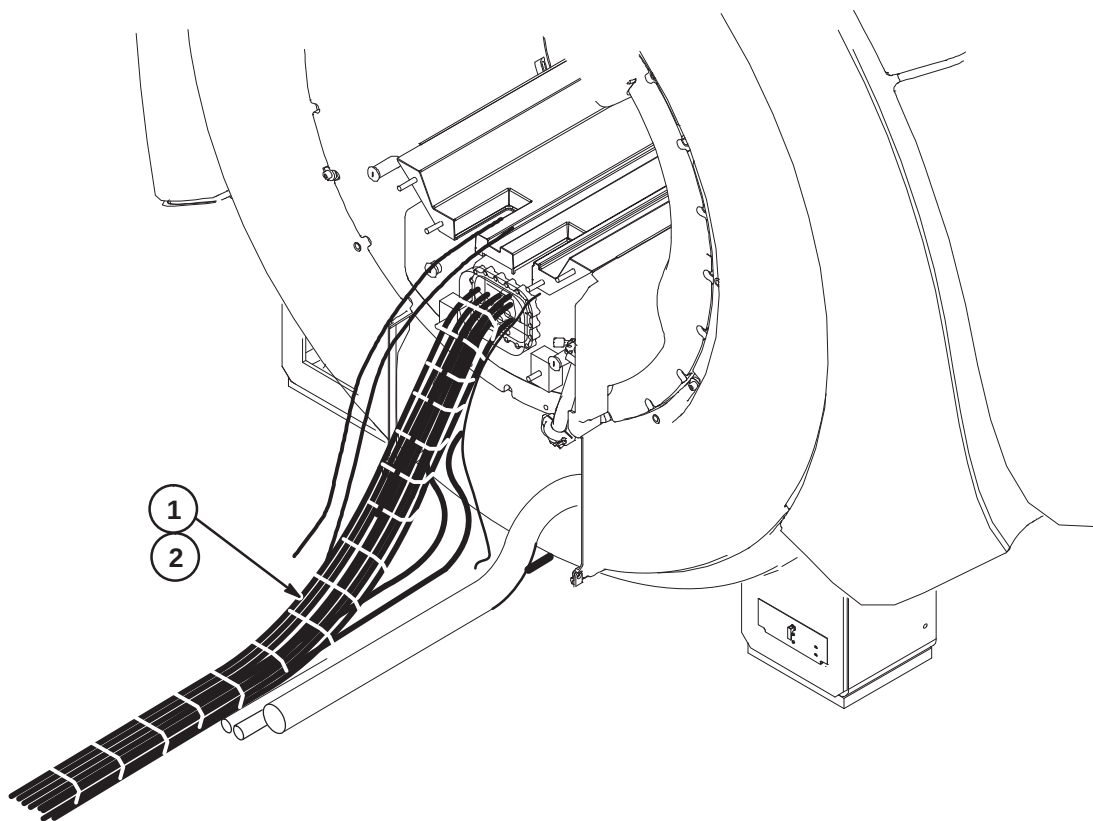
WARNING!

FERROUS MATERIAL HAZARD! THE RATCHETING CRIMP TOOL REQUIRED FOR THIS PROCEDURE CONTAINS FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME A DANGEROUS PROJECTILE. KEEP ALL FERROUS TOOLS AS FAR FROM THE MAGNET AS POSSIBLE.

IF MAGNET IS AT FULL FIELD – CUT GRADIENT CABLE TO LENGTH AND REMOVE FROM MAGNET ROOM TO PERFORM CABLE TERMINATION PROCEDURES.

Gradient cable leads from coil will arrive bundled with tie-wraps installed.

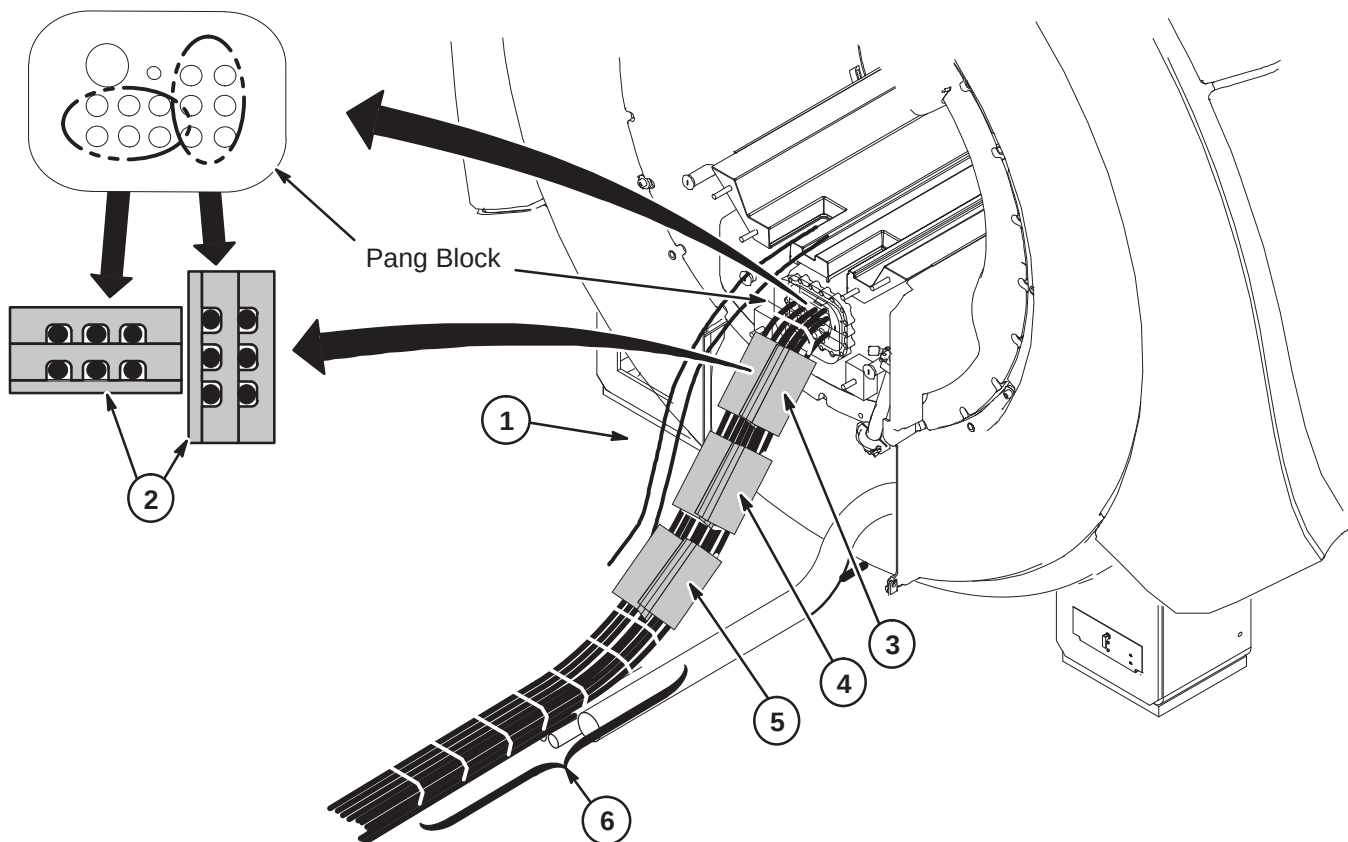
1. If not already done at time of Magnet delivery and installation, carefully unroll all cables and hoses from bore at rear of magnet.
2. Straighten and dress cables.



6-3-1 GRADIENT CABLE CLAMP BLOCK INSTALLATION

Note: Twelve blocks (5126956) and six covers (5126957) will be used, leaving six unused covers.

1. Remove and discard all existing foam tubes surrounding gradient cables and/or tie-wraps. [Approximatley the first 30 inches (750mm) of free cable exiting the magnet]
2. Install 2 Blocks (5126956) and a Cover (5126957) around each set of 6 cables.
Note: Orientate the Cable Clamp Blocks to cables exiting the Pang Block as shown.
3. Install the upper block assemblies 75mm to 100mm (3 to 4 inches) from the end of the Pang Block.
4. Install the middle block assemblies 25mm to 50mm (1 to 2 inches) from the end of the Pang Block.
5. Install the lower block assemblies 75mm to 100mm (3 to 4 inches) from the end of the Pang Block.
6. Install tie-wraps, about 3 to 4 inches apart (75mm to 100mm) around the remaining lengths of the gradient cables.



6-3-2 GRADIENT CABLE INSTALLATION IN MAGNET ROOM

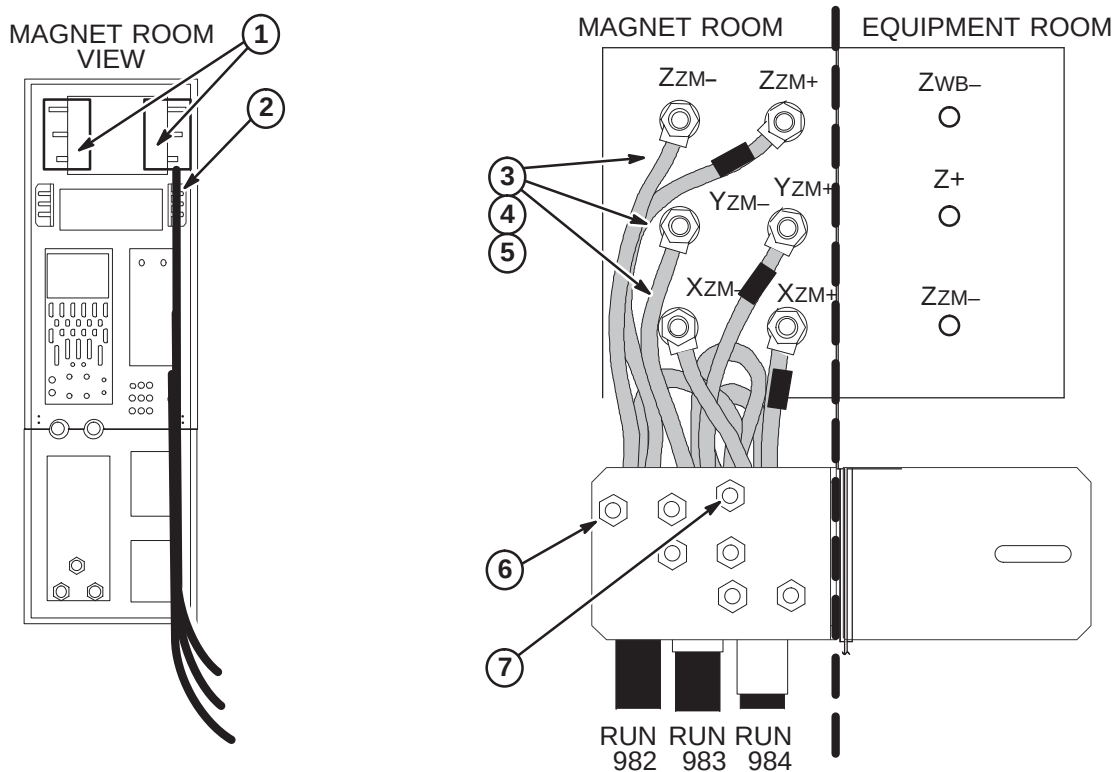
Nine cables have been delivered to site. All have three lead wires with terminals and attached cable labels at one end. The leads on this cable end are connected to the Twin Gradient Filter on the Penetration Panel. The other end is cut flush with no labels or terminals. The nine cables with supplied labels are as follows:

RUN	PART #	LABEL	LENGTH	SUPPLIED LABELS
979	2345400-10	PP1-A7-X	59.5 ft.	(1) RUN 976, (1) RUN 979, (2) X+, (1) X-, (1) GND, (1) XZM-, (1) XWB-
980	2345400-11	PP1-A7-Y	59.5 ft.	(1) RUN 977, (1) RUN 980, (2) Y+, (1) Y-, (1) GND, (1) YZM- (1) YWB-
981	2345400-12	PP1-A7-Z	59.5 ft.	(1) RUN 978, (1) RUN 981, (2) Z+, (1) Z-, (1) GND, (1) ZZM- (1) ZWB-
982	2345400-18	PP1-A7-Z _{ZM}	26.5 ft.	(1) RUN 982, (1) ZZM- (1) ZZM+
983	2345400-17	PP1-A7-Y _{ZM}	26.5 ft.	(1) RUN 983, (1) YZM- (1) YZM+
984	2345400-16	PP1-A7-X _{ZM}	26.5 ft.	(1) RUN 984, (1) XZM- (1) XZM+
985	2345400-24	PP1-A7-Z _{WB}	26.5 ft.	(1) RUN 985, (1) ZWB- (1) ZWB+
986	2345400-23	PP1-A7-Y _{WB}	26.5 ft.	(1) RUN 986, (1) YWB- (1) YWB+
987	2345400-22	PP1-A7-X _{WB}	26.5 ft.	(1) RUN 987, (1) XWB- (1) XWB+

Refer to Section **6-2-3** for installation of Gradient cables in Equipment Room.

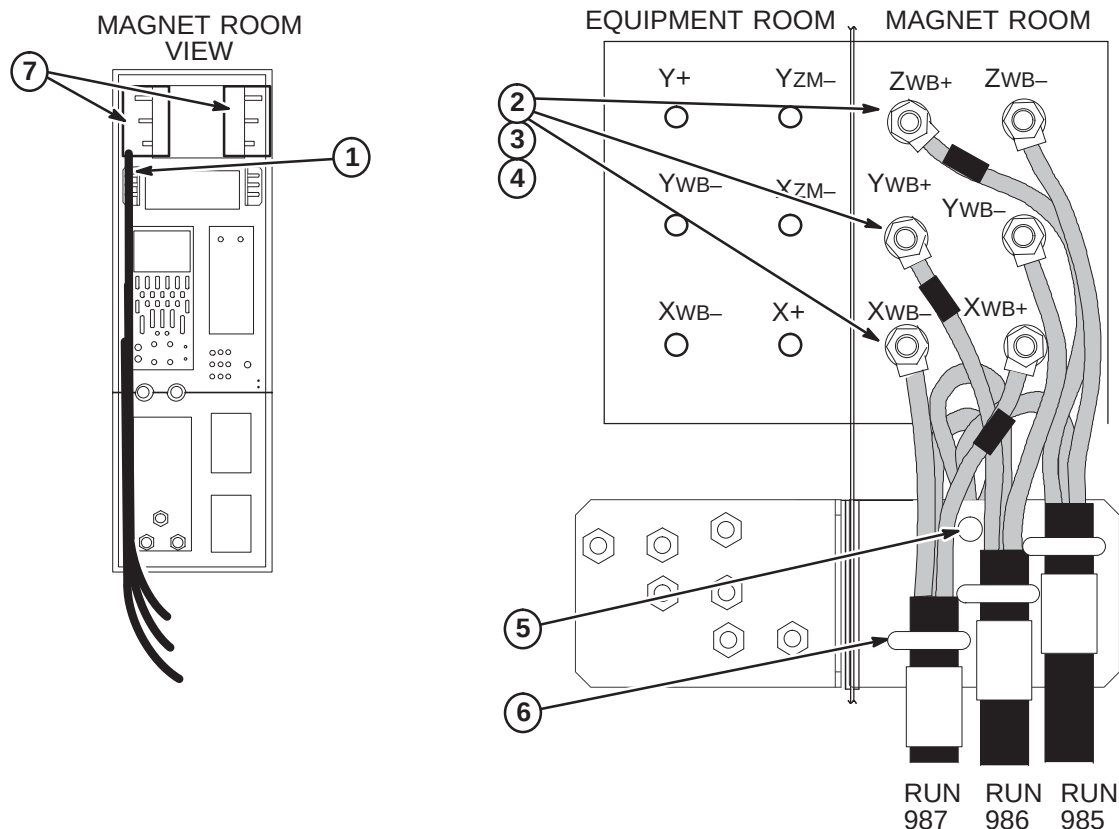
6-3-2-A Connect ZM (Zoom) Cables to Penetration Panel

1. Remove gradient filter terminal covers.
2. Route Run 982, 983, and 984 to right side of gradient filter on Penetration Panel.
Note: The end of the cable with terminals will be connected to Penetration Panel.
3. Connect six ZM- and ZM+ wires to gradient filter terminals as marked.
Note: Red tape indicates positive (+) wires.
4. Align cables to obtain maximum spacing from other cables and conductive metal objects.
Note: When tightening nuts, it is important to hold screw body.
5. Tighten nuts to a torque of 35 ft-lbs (67–74Nm) or after tightening, verify that nothing is loose.
Note: Final check is by hand. Make sure screw is fixed, nut is tight, and cable is not detached from terminal lug.
6. Route and secure each cable Run to strain-relief bracket with U-bolts.
Note: Do not over-torque the strain relief. Cables may be damaged. Tighten enough to apply friction to cable, but not enough to “crimp” the insulation.
7. Secure GND lead wire on Ground Stud for each cable Run.



6-3-2-B Connect WB (Whole Body) Cables to Penetration Panel

1. Route Run 985, 986, and 987 to left side of gradient filter on Penetration Panel.
Note: The end of the cable with terminals will be connected to Penetration Panel.
2. Connect six WB- and WB+ wires to gradient filter terminals as marked.
Note: Red tape indicates positive (+) wires.
3. Align cables to obtain maximum spacing from other cables and conductive metal objects.
Note: When tightening nuts, it is important to hold screw body.
4. Tighten nuts to a torque of 35 ft-lbs (67–74Nm) or after tightening, verify that nothing is loose.
Note: Final check is by hand. Make sure screw is fixed, nut is tight, and cable is not detached from terminal lug.
5. Secure GND lead wire on Ground Stud for each cable Run.
6. Route and secure each cable Run to strain-relief bracket with U-bolts.
Note: Do not over-torque the strain relief. Cables may be damaged. Tighten enough to apply friction to cable, but not enough to “crimp” the insulation.
7. Replace gradient filter terminal covers.



6-3-2-C Route Gradient Coil Output Cables Thru Rear Pedestal**WARNING!**

FERROUS MATERIAL HAZARD! If magnet is ramped, the wheel assemblies attached to the Rear Pedestal for shipping and delivery **MUST** be removed before moving Rear Pedestal into Magnet Room.

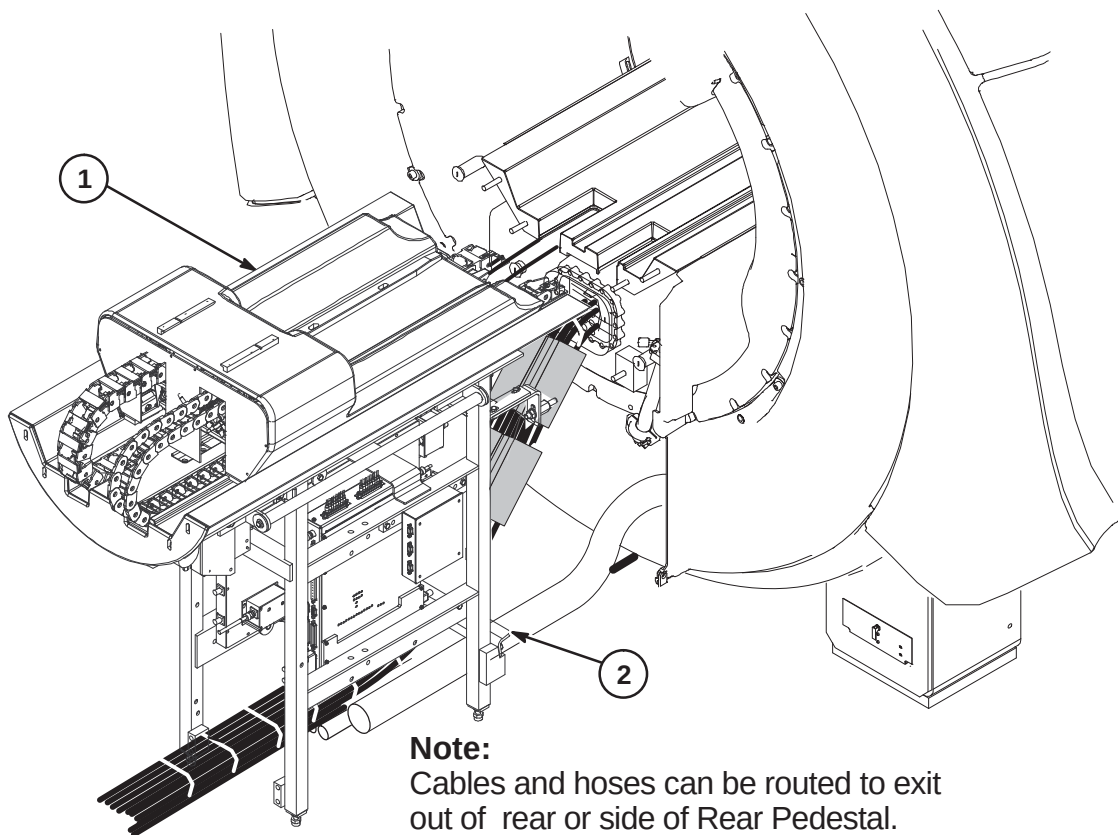
CAUTION

The weight of the Rear Pedestal requires at least **THREE** people to move Rear Pedestal into position at rear of magnet.

1. Move Rear Pedestal close to its final position at rear of magnet.

Note: Routing the Gradient Cables above rather than below the cross brace will reduce white pixels.

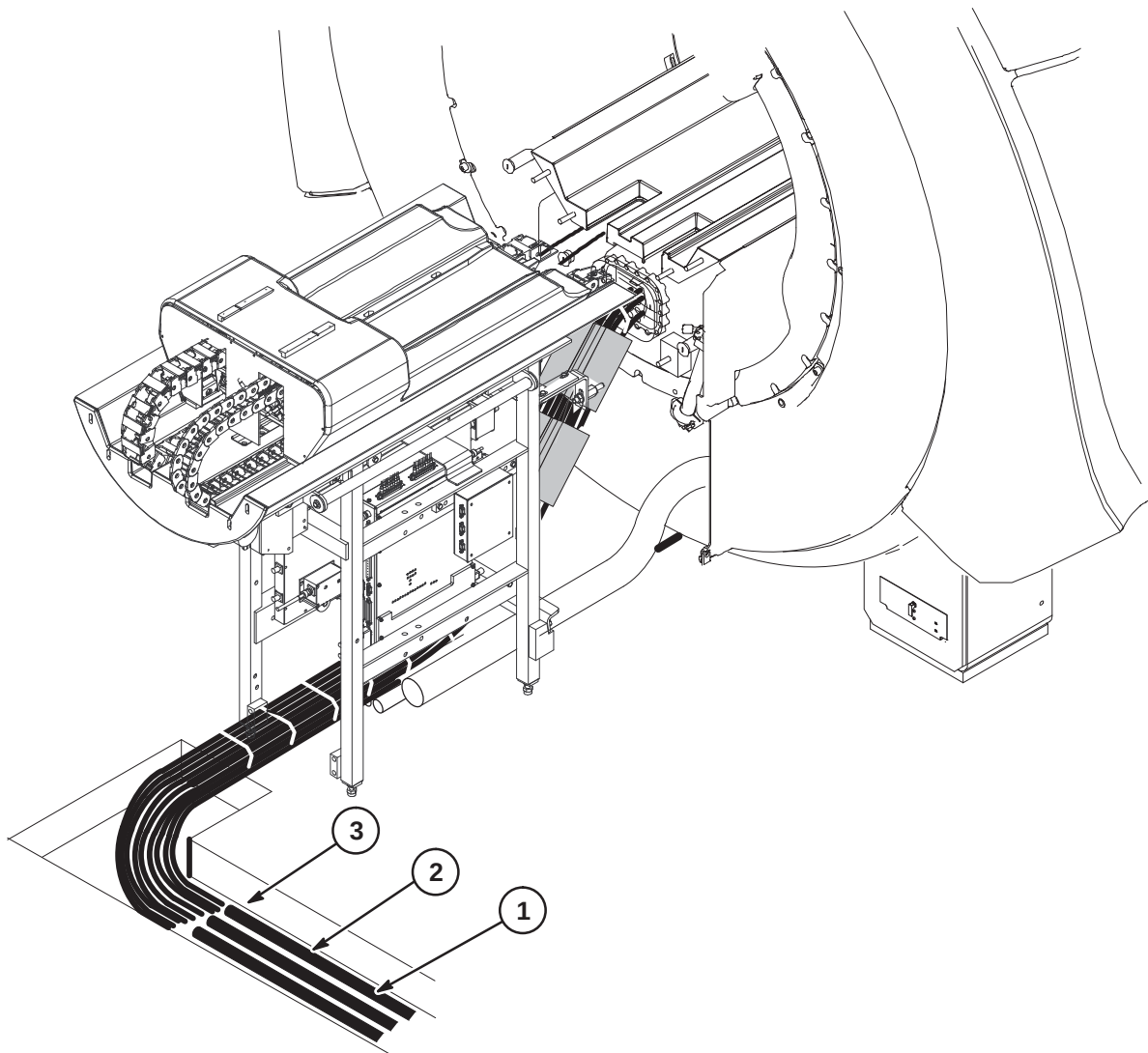
2. Route the bundled Gradient Cables above lower cross brace at front of Rear Pedestal.



6-3-2-D Cut Cables to Length and Route in Magnet Room

Do not cut TRM Body Coil gradient cables. Only Gradient Output cables from Penetration Panel are to be cut to length.

1. Route Gradient Output cables from Gradient Filter of Penetration Panel thru designated cable trough to rear of magnet. The TRM Body Coil gradient cables should have been routed to area in cable trough at rear of magnet.
2. Verify sufficient slack of at least two feet (610mm) is allowed for termination and connection. Mark location for cutting.
3. Cut cables to length and label cut ends with applicable run number.



6-3-2-E Terminate Runs 982, 983, And 984 for Magnet Room

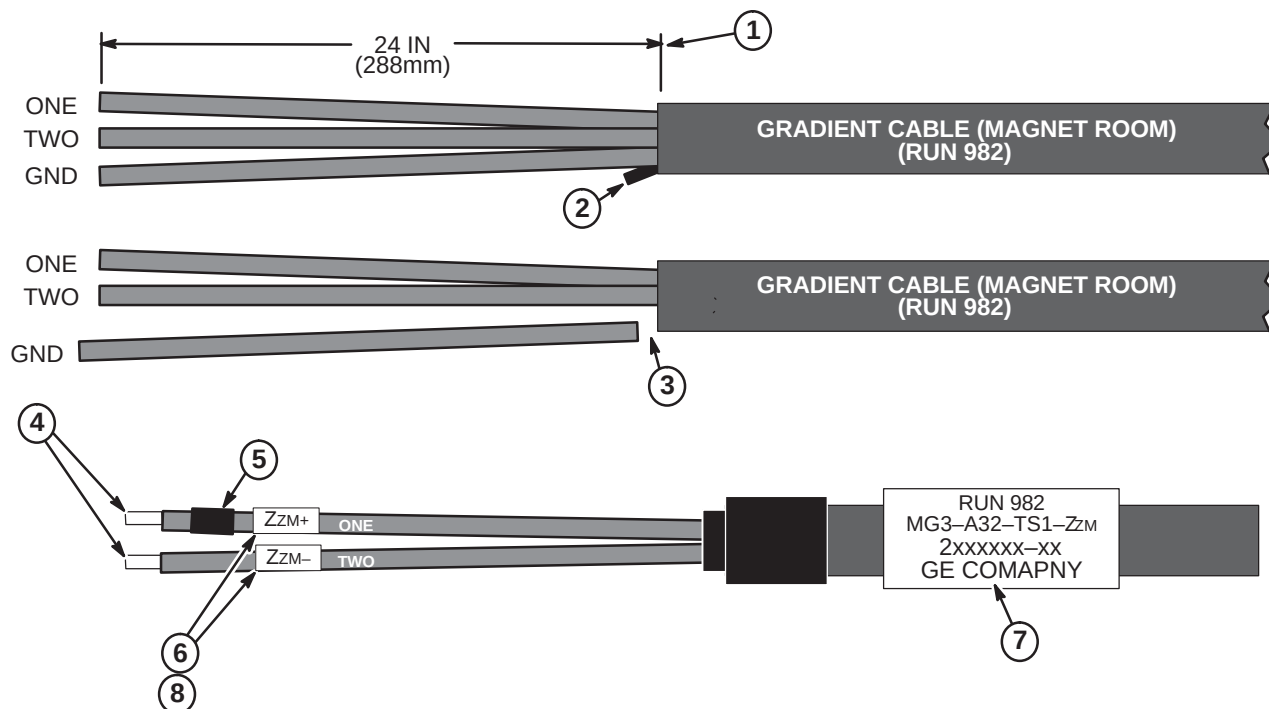


Rear End Bell **MUST** be installed on magnet with gradient coil cables routed thru block before connecting gradient coil cables to gradient cables routed from Penetration Panel with splices in cable trough. Failure to do so will result in inability to install Rear End Bell.

Note: The following steps are required to terminate the end of the cable that was cut in Step 6-3-2-D. The other end of the cable is connected to the Penetration Panel.

1. Remove 24 in. (288 mm) of jacket from end of cable Run 982.
2. Trim rubber cable filler strands flush with end of jacket.
3. Cut and remove GROUND wire.
4. Strip 0.75 in. (19 mm) of insulation from end of each wire.
5. Wrap red tape (from Pen Panel end of cable) around wire "ONE".
6. Place furnished "Zzm+" label on wire "ONE". Place furnished "Zzm-" label on wire "TWO".
7. Place furnished "Run 982" label on cable.
8. Repeat Steps 1 through 7 for Runs 983 and 984.

Note: Red tape, Yzm+ (RUN 983) and Xzm+ (RUN 984) labels are attached to wire "ONE".

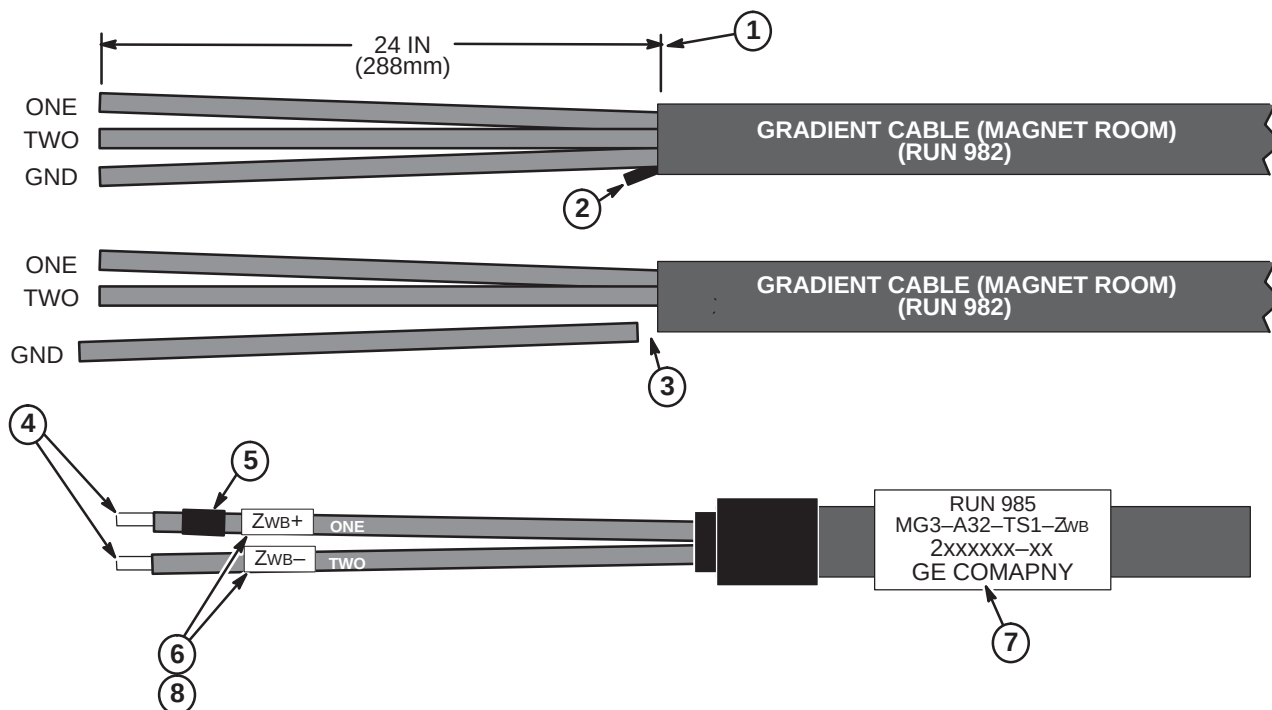


6-3-2-F Terminate Runs 985, 986, And 987 for Magnet Room

Note: The following steps are required to terminate the end of the cable that was cut in Step **6-3-2-D**. The other end of the cable is connected to the Penetration Panel.

1. Remove 24 in. (288 mm) of jacket from end of cable Run 982.
2. Trim rubber cable filler strands flush with end of jacket.
3. Cut and remove GROUND wire.
4. Strip 0.75 in. (19 mm) of insulation from end of each wire.
5. Wrap red tape (from Pen Panel end of cable) around wire "ONE".
6. Place furnished "ZWB+" label on wire "ONE". Place furnished "ZWB-" label on wire "TWO".
7. Place furnished "Run 985" label on cable.
8. Repeat above process for Runs 986 and 987.

Note: Red tape, Ywb+ (RUN 986) and Xwb+ (RUN 987) labels are attached to wire "ONE".



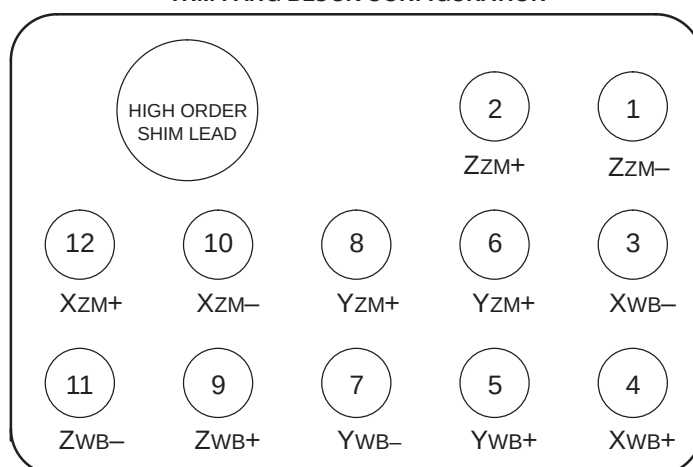
6-3-2-G Connection of Body Coil Gradient Cables

WARNING!

THE RATCHETING CRIMP TOOL REQUIRED FOR THIS PROCEDURE CONTAINS FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET WHICH COULD CAUSE INJURY OR DAMAGE TO THE EQUIPMENT.

The illustration of the Pang Block below and the table provide information for connection and crimping of Body Coil gradient cable instructions on the next page.

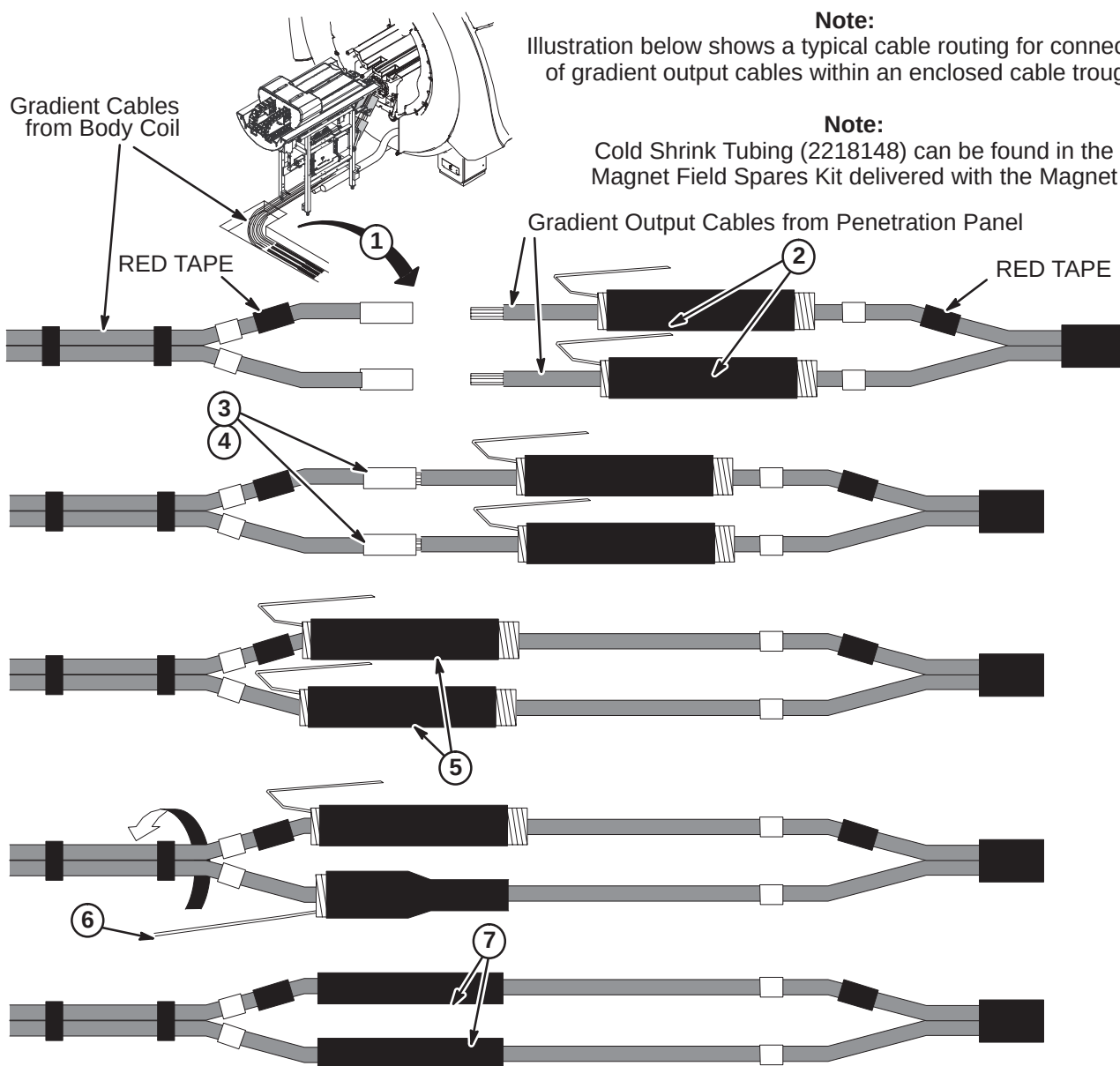
TRM PANG BLOCK CONFIGURATION



TRM COIL CABLE LABELS			OUTPUT CABLE LABELS		
WIRE #	LABEL	DESCRIPTION	RUN #	LABEL	WIRE #
1	Z ZM-	Z6 In (-)	982	ZM-	2
2	Z ZM+	Z5 Out (+)	982	ZM+	1
3	X WB-	X8 180 (-)	987	WB-	2
4	X WB+	X7 0 (+)	987	WB+	1
5	Y WB+	Y9 270 (+)	986	WB+	1
6	Y ZM-	Y4 90 (-)	983	ZM-	2
7	Y WB-	Y10 90 (-)	986	WB-	2
8	Y ZM+	Y3 270 (+)	983	ZM+	1
9	Z WB+	Z11 Out (+)	985	WB+	1
10	X ZM-	X2 180 (-)	984	ZM-	2
11	Z WB-	Z12 In (-)	985	WB-	2
12	X ZM+	X1 0 (+)	984	ZM+	1

6-3-2-G Connection of Body Coil Gradient Cables (Continued)

1. Cable connection process shown below is for Run 987. Repeat process for Runs 982, 983, 984, 985 and 986.
Note: Red tape, XWB+ (RUN 987), YWB+ (RUN 986), ZWB+ (RUN 985), XZM+ (RUN 984), YZM+ (RUN 983), and ZZM+ (RUN 982) labels are attached to wire "ONE".
2. Position Cold Shrink tubing (2218148) on output cables **prior to connecting**.
3. Insert ends of Run 987 into correct butt splice (2218149) on Body Coil gradient cable.
4. Crimp butt splices using a T&B WT 3185 Gradient Cable Crimper (2144091) with #2 die.
5. Locate Cold Shrink tubing at desired starting point. Position tubing 1-1/2 to 2 inches (38 to 50mm) beyond end of connector.
6. Pull loose tab gently while unwinding in a counter-clockwise direction. Repeat process for other cable.
7. Connection of cables and installation of Cold Shrink tubing is now complete.



6-3-3 INSTALLATION OF REAR PEDESTAL

6-3-3-A Position Rear Pedestal and Connect Front & Rear Split Bridge

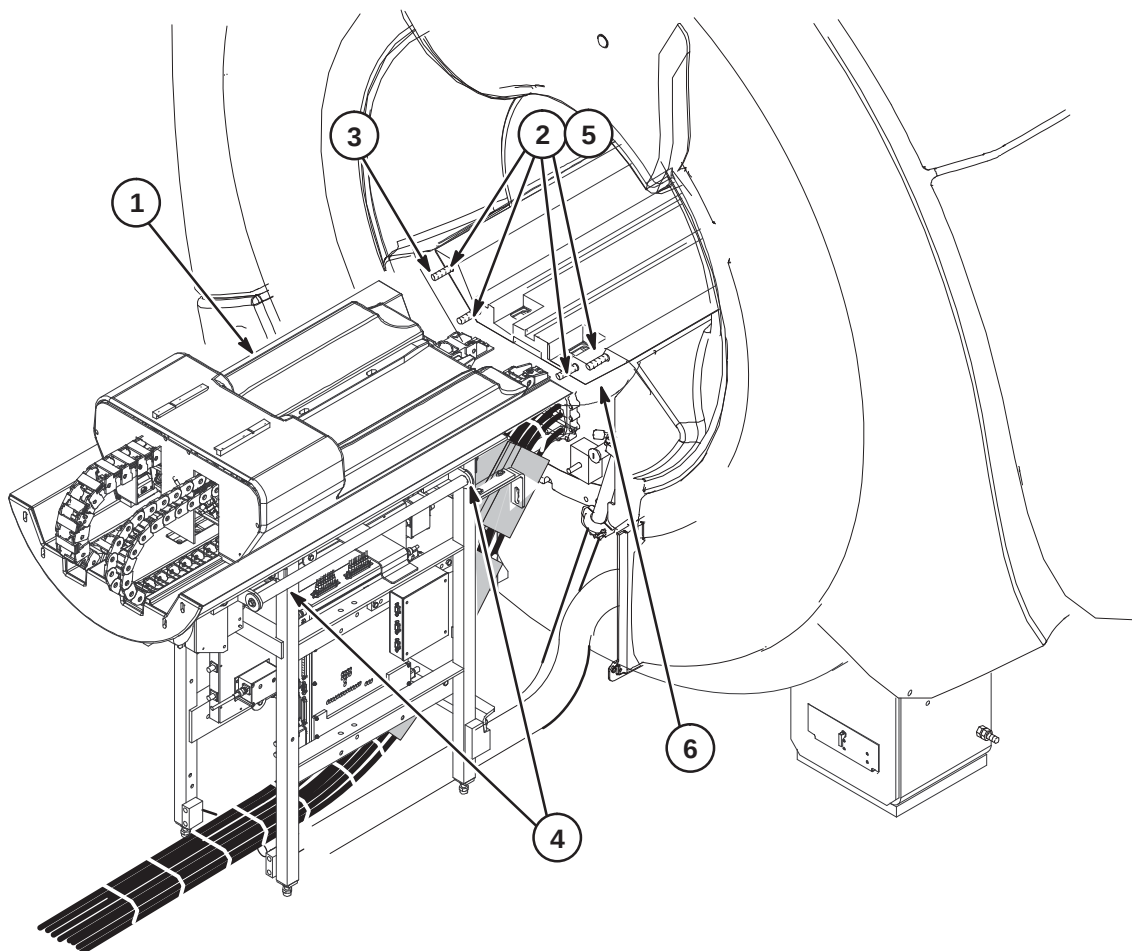


The weight of the Rear Pedestal requires at least THREE people to move Rear Pedestal into position at rear of magnet.

1. Move Rear Pedestal into position at rear on magnet as shown below.

Note: Dress and align all cables and hoses routed thru Rear Pedestal while moving Rear Pedestal closer to magnet.

2. Remove four M10 hex nuts (2109875-4) and washers (2228717-2) from end of Bridge in magnet. These will be used for attaching the Rear Split Bridge.
3. Move Rear Pedestal forward and engage front split bridge studs.
4. Adjust the Rear Pedestal height to align the Rear Split Bridge with the Front Split Bridge for connection.
5. Fasten, BUT DO NOT TIGHTEN, the Front and Rear Split Bridge with four nuts and washers that were removed from end of Front Split Bridge.
6. Remove foam and tape that was used to hold Front Split Bridge in place while in transit to site.



6-3-3-B Height Adjustment at Front of Bridge (If Needed)

1. Verify that the top of the Bridge is flush to 1mm below the Front End Bell surfaces.
2. To obtain this alignment, adjust the nuts on the left and right height adjusters at bottom of legs.
3. Tighten the mounting bolts.

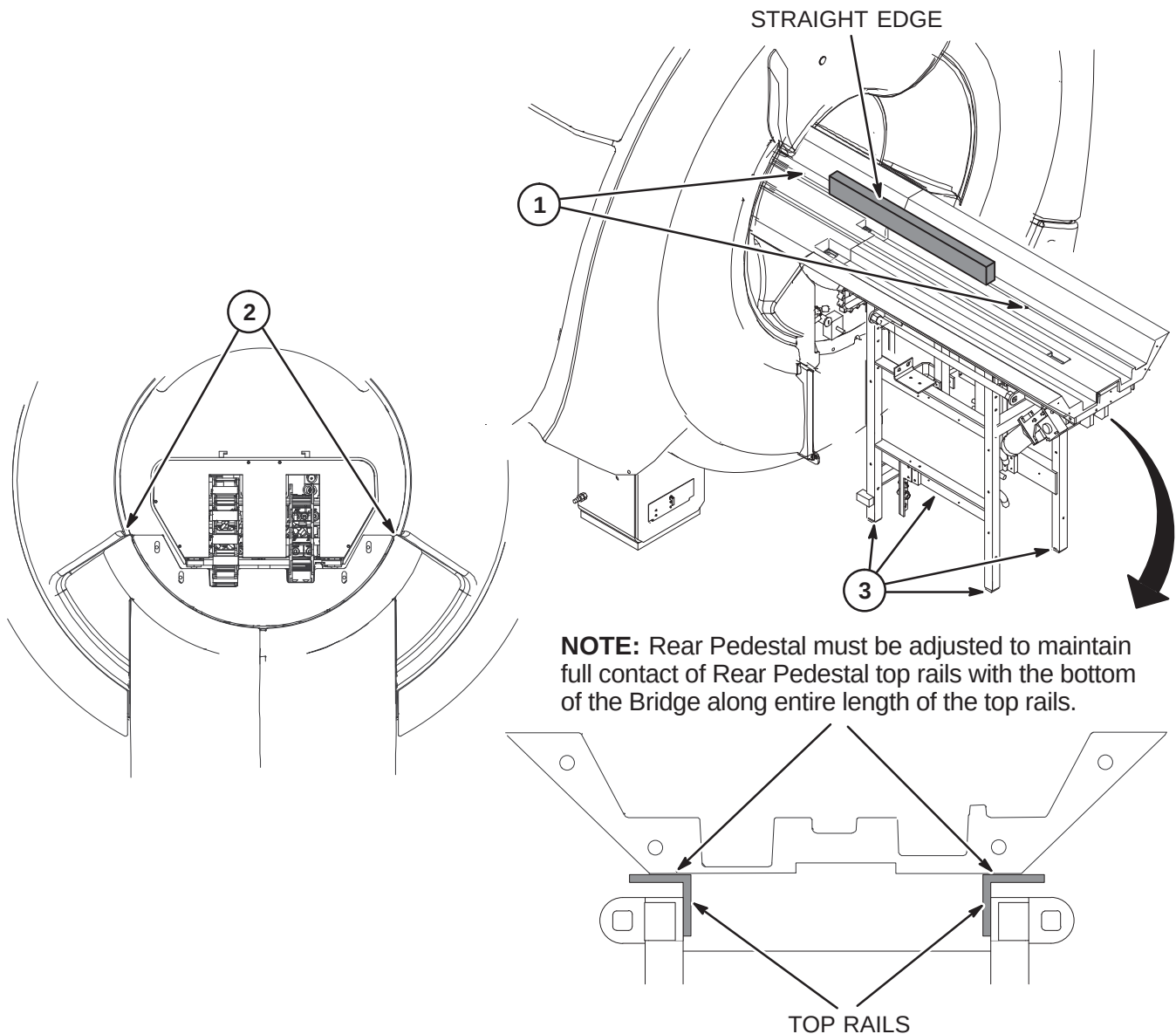


6-3-3-C Height Adjustment at Rear Bridge

Note: If Bridge is not adjusted to be flush to 1mm below front and rear End Bells, the Rear Pedestal covers will not fit properly. The Bridge does not need to be “level” to Earth in achieving the alignment with the end bells.

Therefore, you do **NOT** need a level at this point of the installation, you **DO NEED** a straight edge. (Though you can use a level as a straight edge)

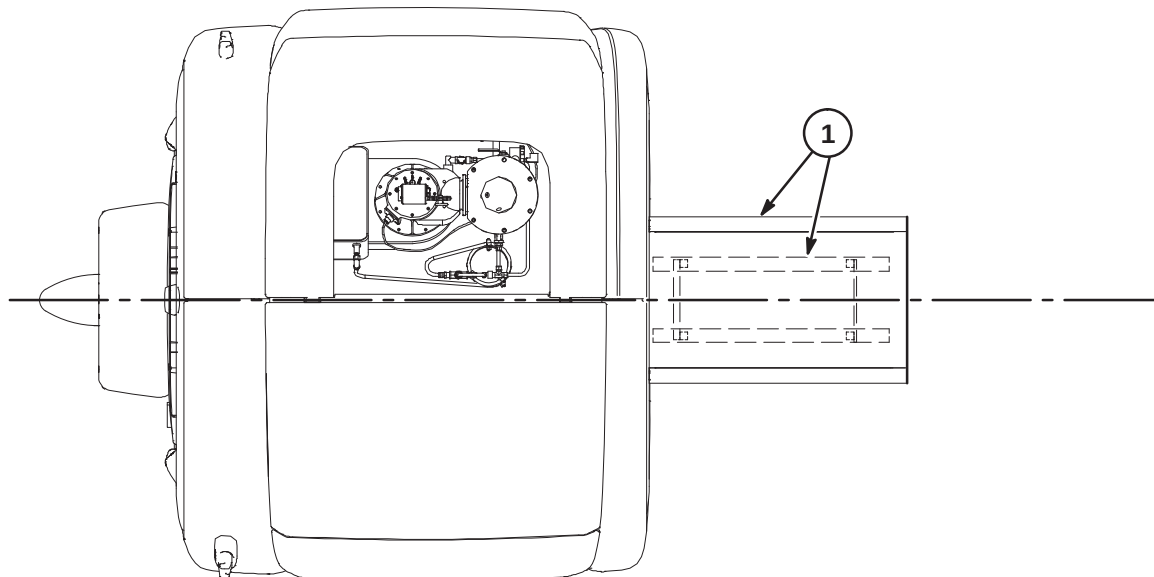
1. Use a Straight Edge to make sure that the front and rear Bridge surfaces, on both sides of Bridge where the LPCA wheels roll, are even. This will provide smooth travel for the Carriage wheels. If uneven, refer to Step 3.
2. Make sure the Bridge surface is flush to 1mm below the surfaces of the rear End Bell. If uneven, refer to Step 3.
3. Adjust height using feet of Rear Pedestal so the Bridge surface is flush with the rear End Bell surface to 1mm below the rear End Bell surface; and that the Front Split Bridge is even with Rear Split Bridge.



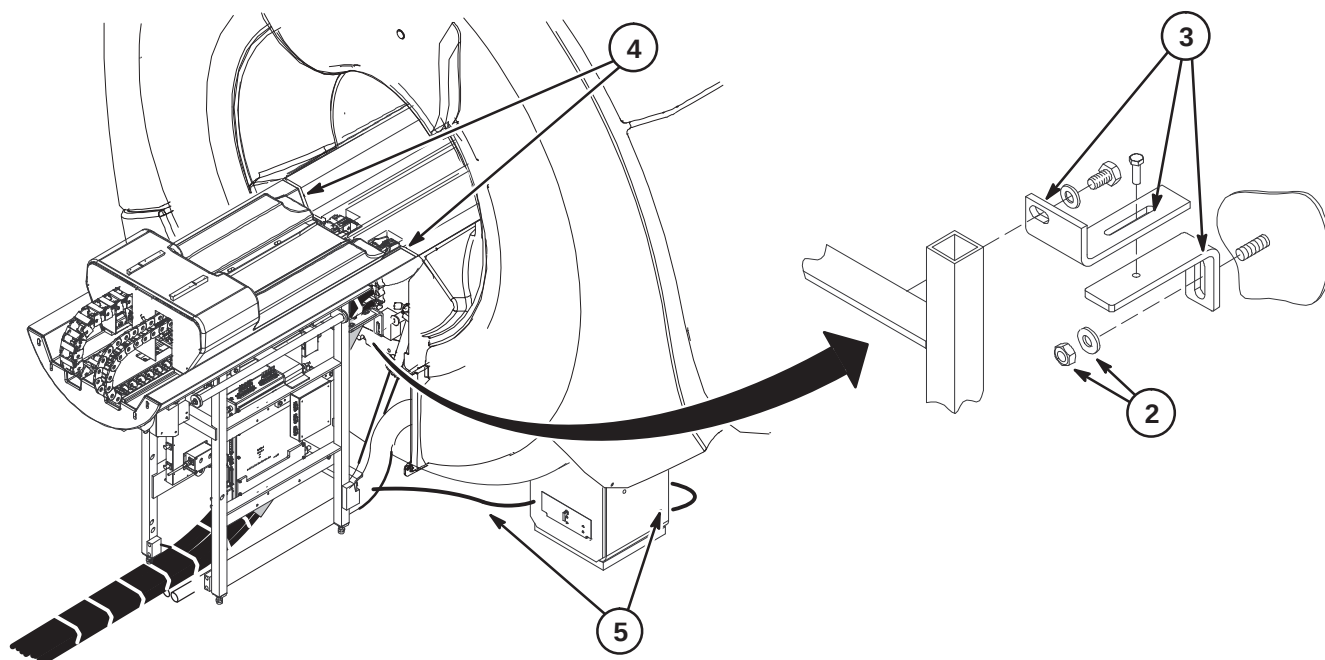
6-3-3-D Connect Rear Pedestal to Magnet and Tighten Bridge Fasteners

Note: Tongue to Bridge alignment is critical to smooth belt tracking and operation. Make sure Tongue is square to Bridge and Rear Pedestal.

1. Align Rear Pedestal to Rear Split Bridge and center Rear Split Bridge to Rear Endbell.

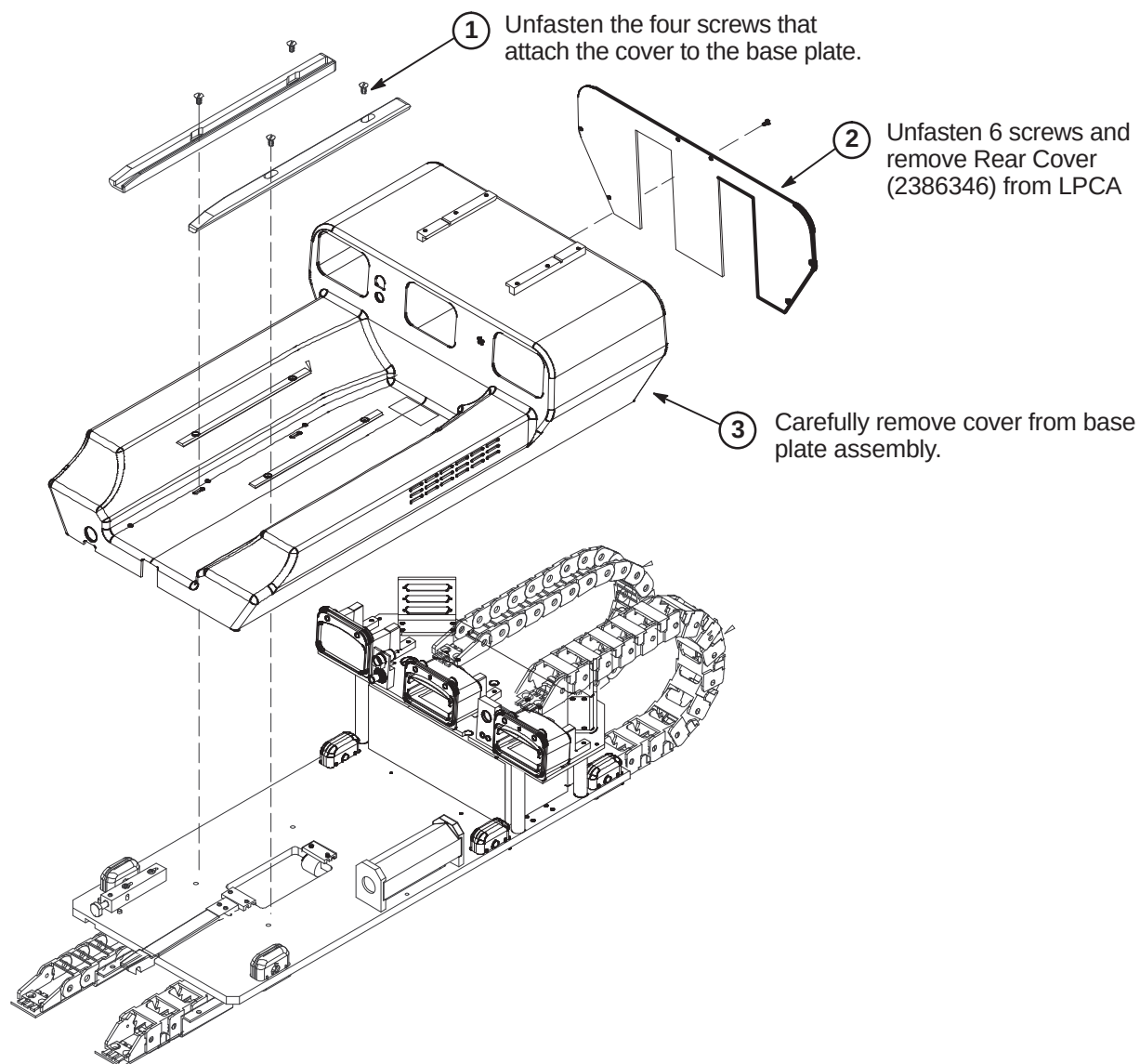


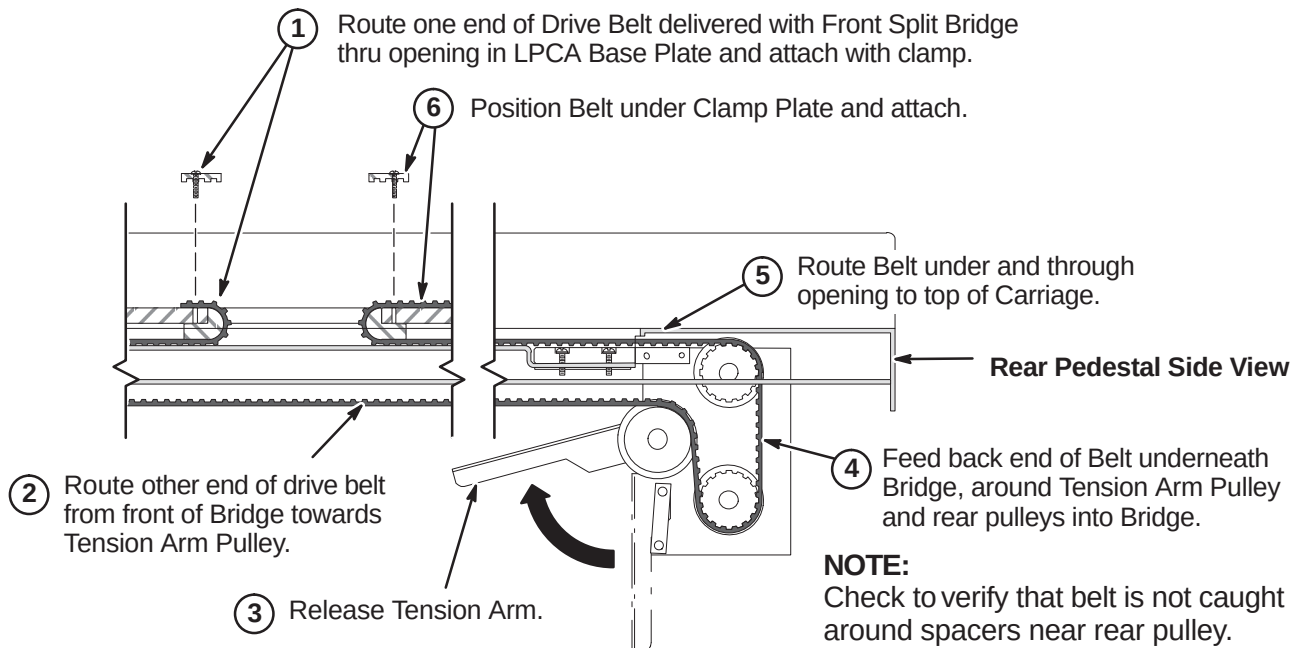
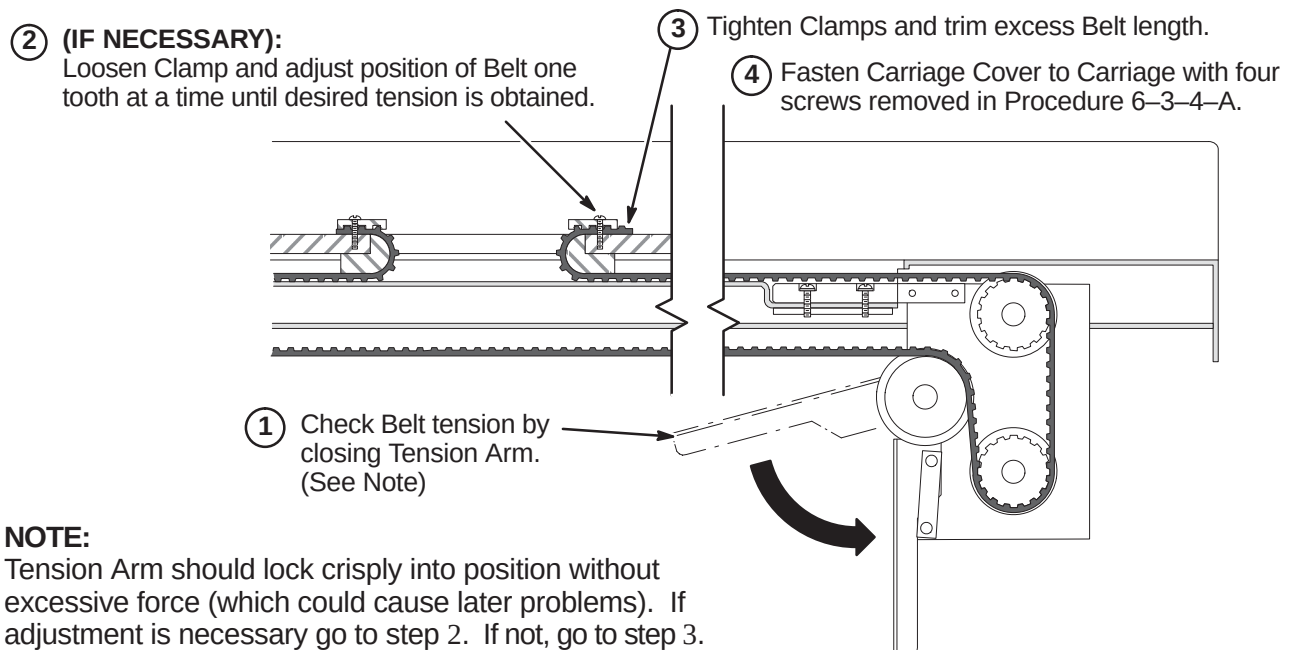
2. Attach Rear Pedestal to Support Studs using M10 Nut (2109875-4) and Flat Washer (2109878-4) on each stud.
3. Adjustment of Rear Pedestal location can be made at these slotted hole locations when connecting the Split Bridge.
4. Tighten the four M10 nuts fastening the Rear Split Bridge to the Front Split Bridge.
5. Route Ground cable from Rear Pedestal under magnet and connect to Ground Stud at inside of magnet leg.



6-3-4 INSTALLATION OF DRIVE BELT

6-3-4-A Remove LPCA Cover



6-3-4-B Route and Attach Drive Belt**6-3-4-C Final Adjustment of Drive Belt**

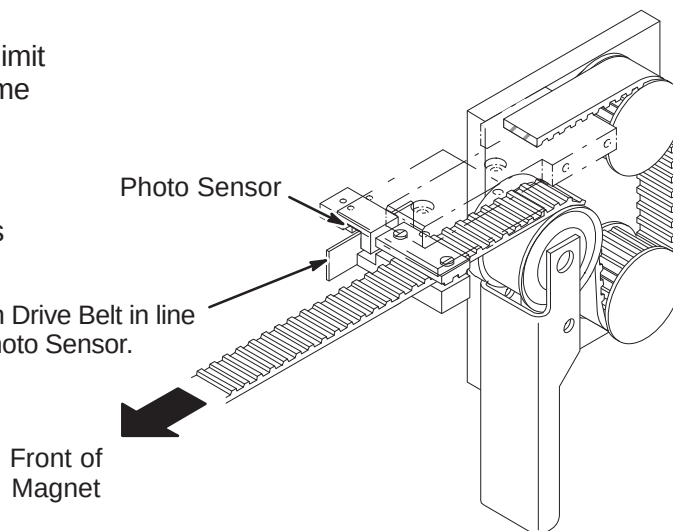
6-3-4-D Installation of Sensor Flag**NOTE:**

Function of the Flag is to interrupt the reflected limit switch sensor beam when the Carriage is in home position all the way forward in the bore.

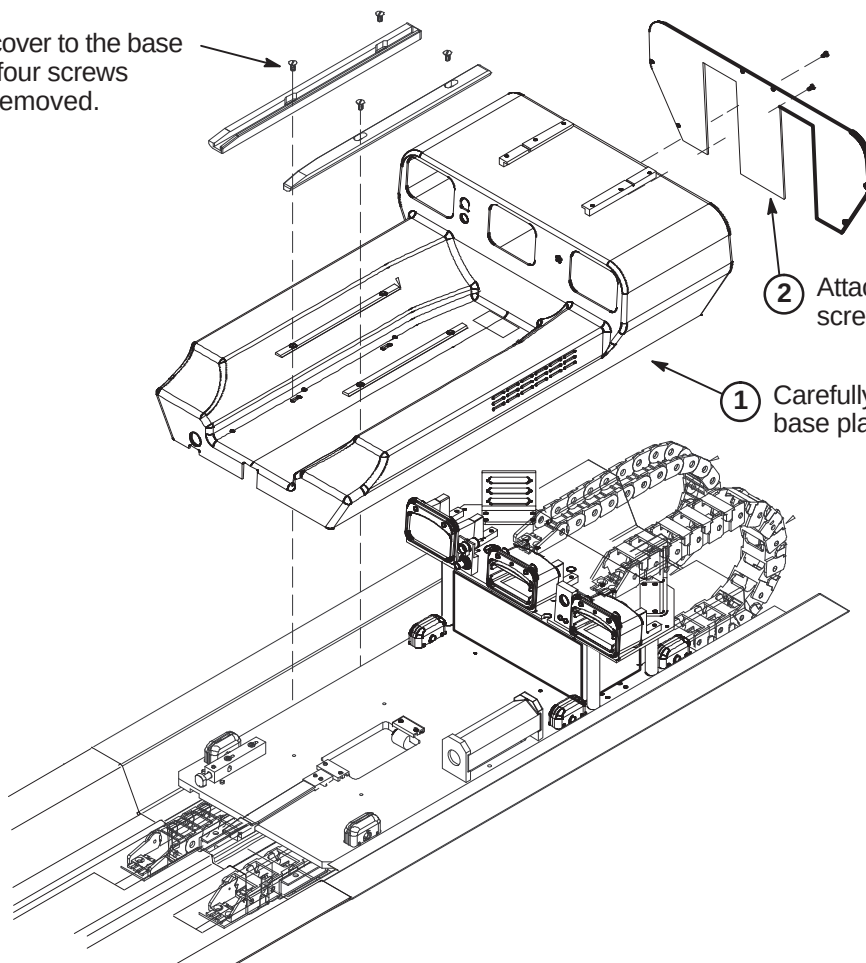
NOTE:

The following Step is needed if flag was removed during routing of drive belt.

- ① Install Flag Assembly on Drive Belt in line with Bridge mounted Photo Sensor.

**6-3-4-E Installation of LPCA Cover**

- ③ Attach the cover to the base plate using four screws previously removed.

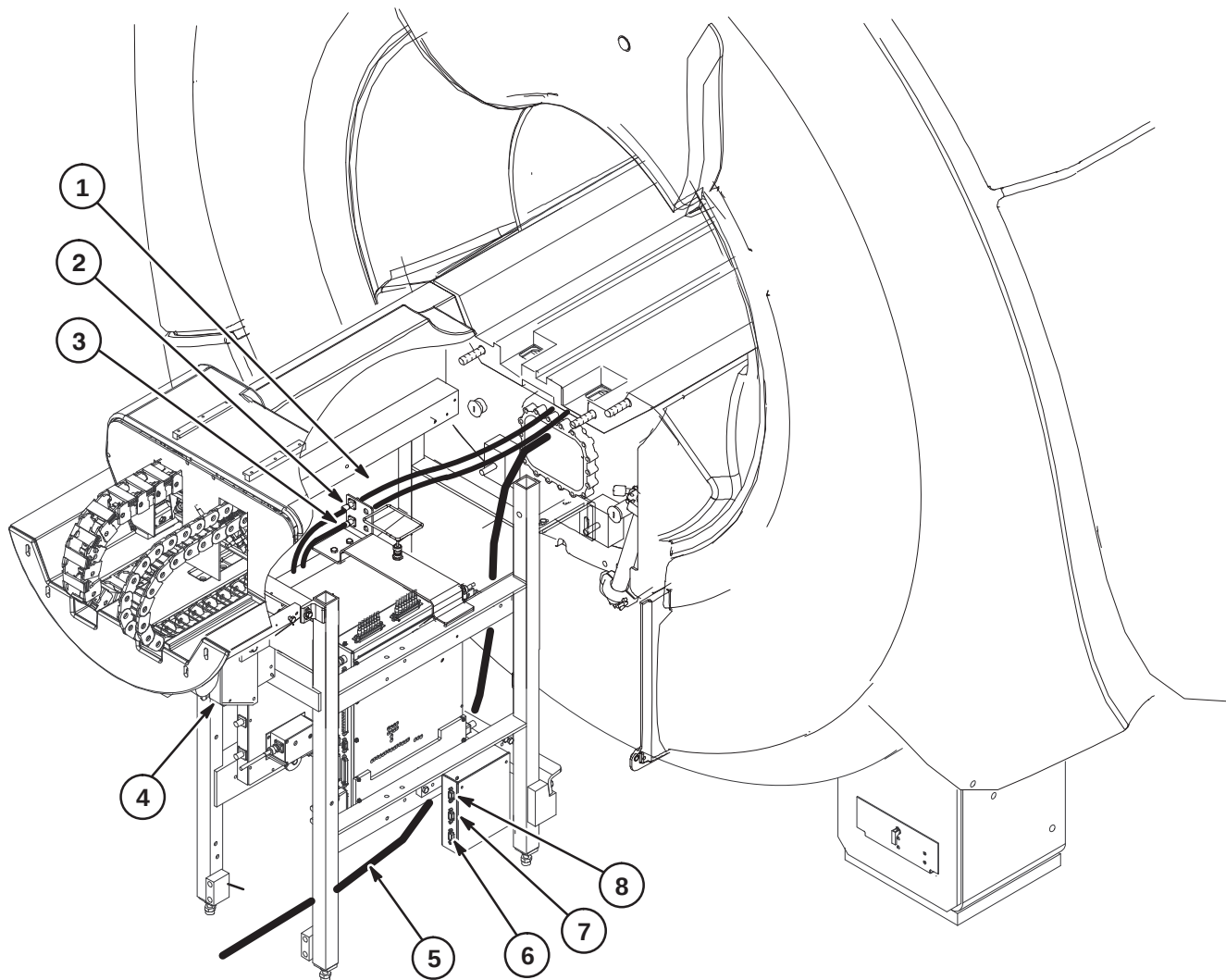


6-3-5 ROUTE AND CONNECT CABLES AT REAR PEDESTAL**Note**

For Mobiles, follow van manufacturers process for cable installation.

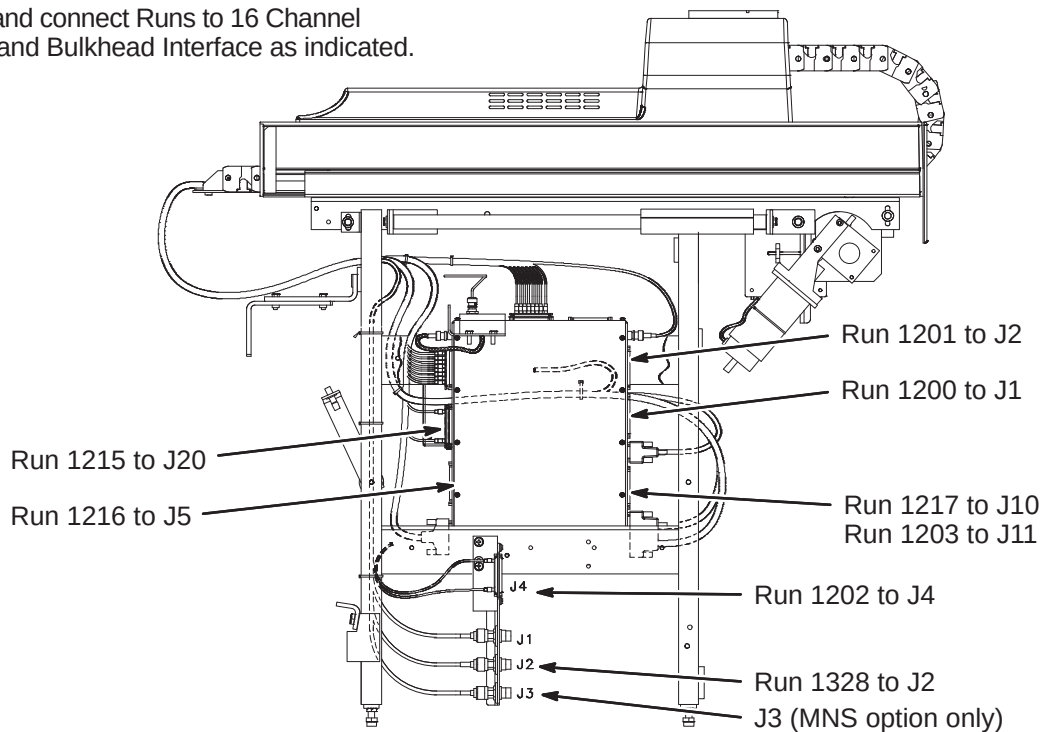
6-3-5-A Route/Connect Runs 282, 414, 422, 781, 782, 1057, Multi-coil, & Front End Bell Speaker Cable

1. Route and connect RF Bias cables from RF Coil to side of UTNS bracket..
2. Route and connect Run 781 from Bias cable J5 to J75 on the Penetration Panel.
3. Route and connect Run 782 from Bias cable J6 to J76 on the Penetration Panel.
4. Route/connect Run 422 from J2 on SRI to J4 on Magnet Interface #1.
5. Route/connect Run 1152 from Shim Filter on Penetration Panel to Shim Filter cable thru Rear End Bell gasket.
6. Route/connect Front End Bell microphone cable Run 414 to J9.
7. Route/connect speaker cable from Patient Comfort Module to J8.
8. Route/connect Run 282 from J12 on Pen Panel to J6.

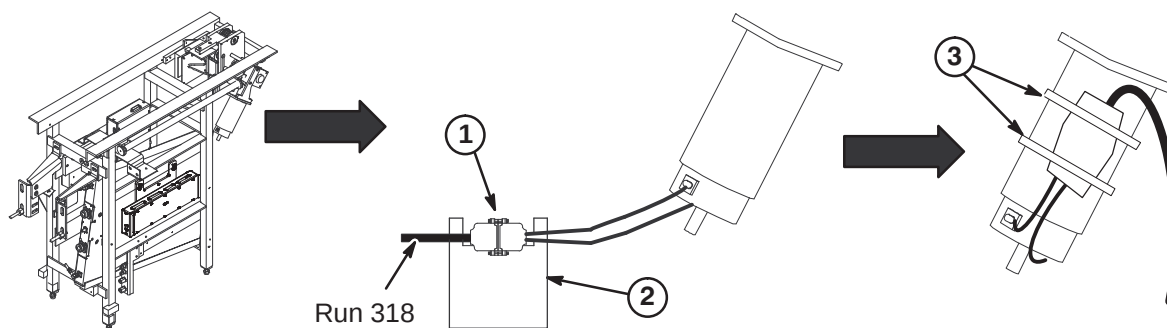


6-3-5-B CONNECT REMAINING CABLES TO REAR PEDESTAL

- ① Route and connect Runs to 16 Channel Switch and Bulkhead Interface as indicated.

**6-3-5-C Route/Connect Run 318 to Longitudinal Drive Motor**

1. Route Run 318 from J11 on Penetration Panel and connect to Longitudinal Drive Motor cable.
2. Wrap both connectors with 177 x 200 Connector Wrap (2236130-2).
3. Ty-wrap cable to motor to prevent movement.



6-3-5-D Connect Cables to Hybrid Splitter

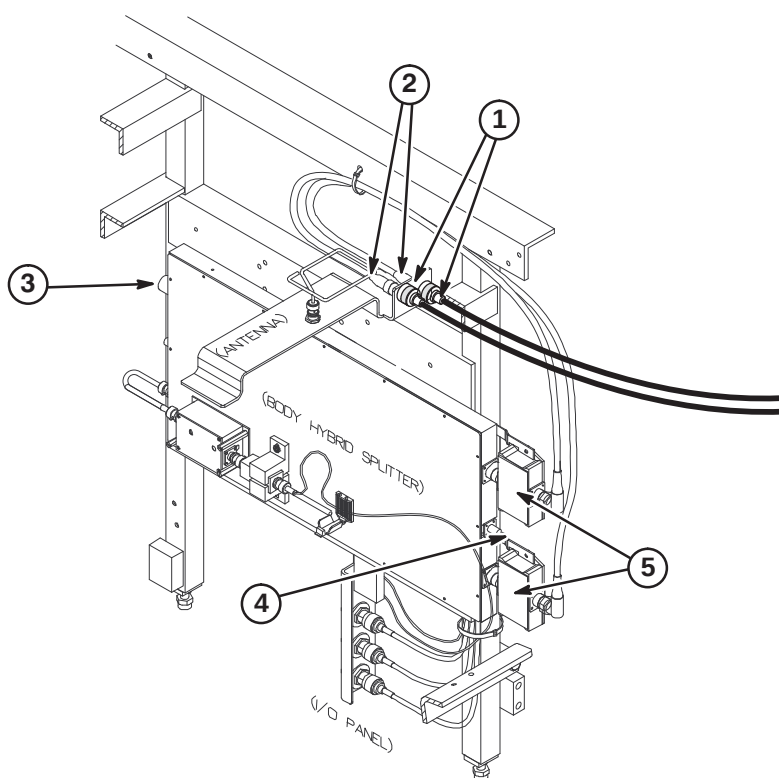
The RF Drive Input cables utilize a corrugated solid RF shield. If mishandled, the cable can be kinked and subsequently broken.

Also, use care when connecting these cables to the RF Splitter connectors J3 and J4. Cross threading or inadequate engagement of the center pin can cause arcing and damage the cables.

1. Route and connect I and Q RF Drive input cables from RF Coil and connect as marked to J1 and J2 on bracket.

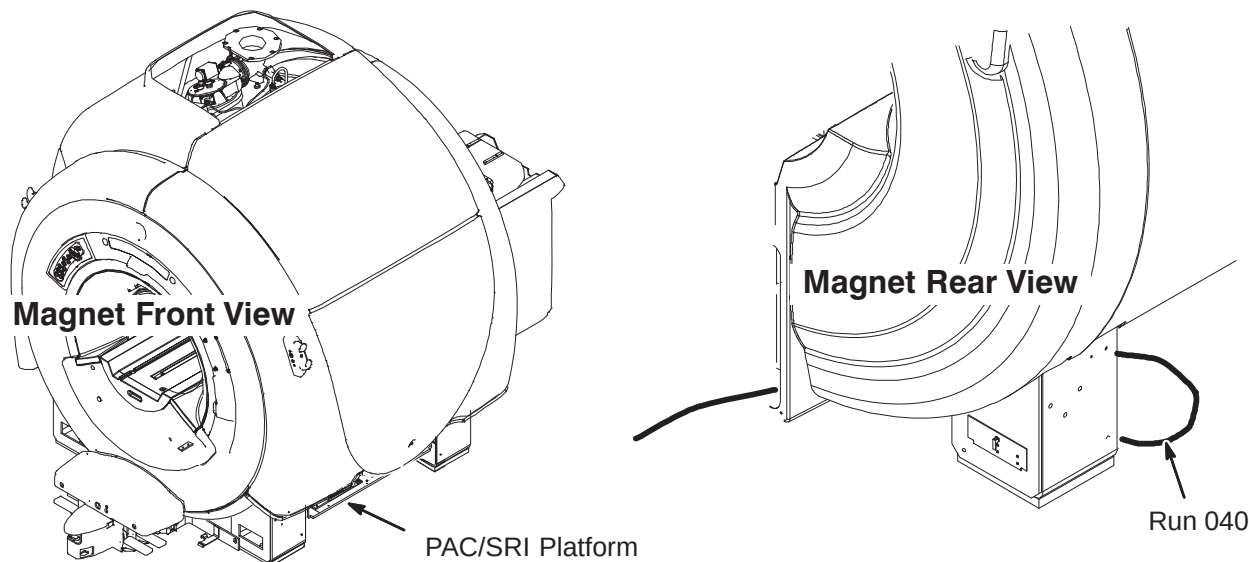
Note: Runs 743 and 744, Body Coil to Hybrid Splitter cable connections, may be reversed depending on magnet field polarity. Consult local Field Engineer for proper routing.

2. Connect Run 743 to J2 and Run 744 to J1 on bracket.
3. Connect RF Flexible cable Run 1160 to J1. Refer to Procedure **6-4-2-B** for instructions on installing the Times Microwave cables. Other end is connected to Run 257.
4. Route and connect Run 780 from J74 on Penetration panel to J6 at front of Hybrid Splitter.
5. If present, **remove** the two Ground Isolation Filter Boxes (5118952) that are installed on end of Hybrid Splitter and connected to J3 and J4. Reconnect Runs 743 and 744 to J3 and J4 on Hybrid Splitter.



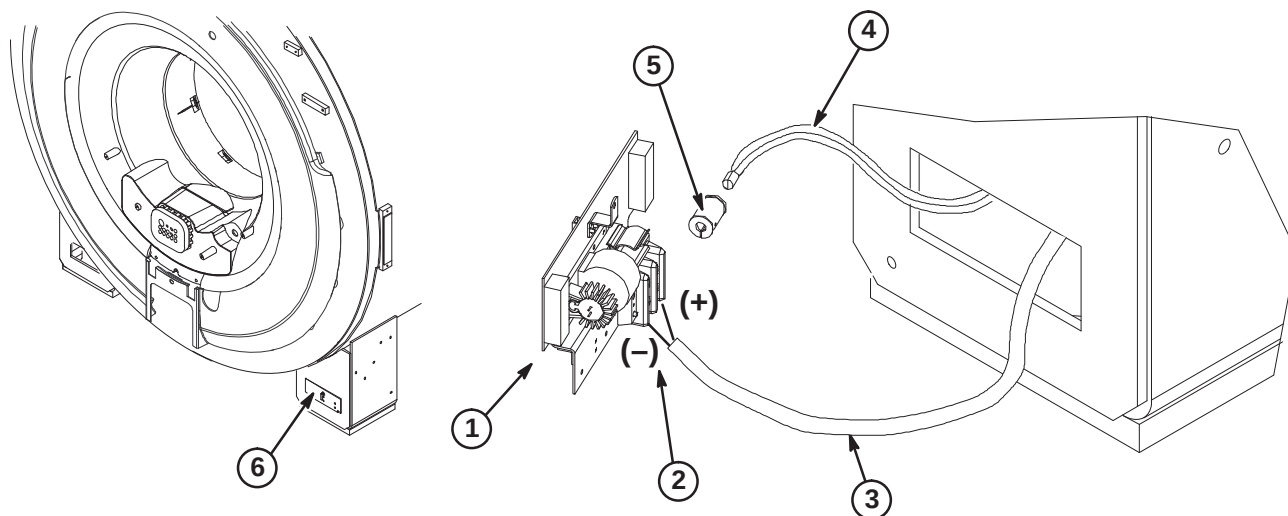
6-3-6 CONNECTION OF MAGNET GROUND, RUN 040

The PAC/SRI Platform is located on right side of magnet. The Magnet ground cable, RUN 040, has to be connected to the inside of the opposite side rear left leg as shown below. The other end of Run 040 is connected to the RF Screen Room common ground.



6-3-7 INSTALLATION OF BORE LAMP ASSEMBLY

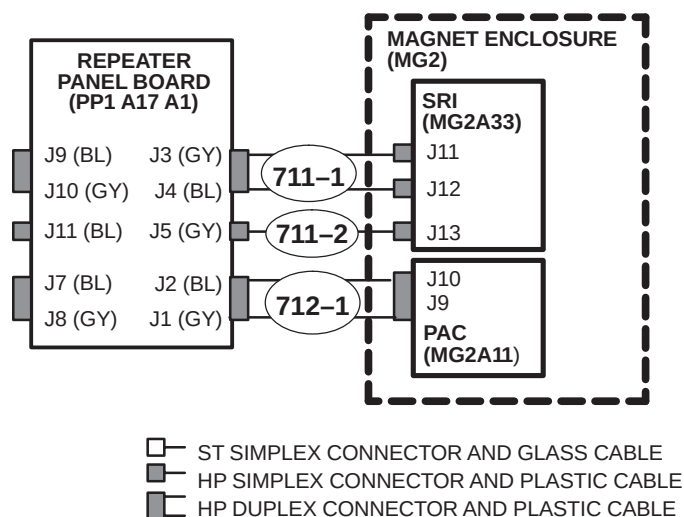
1. Move Bore light Assembly (2222547) to area of left rear magnet leg.
2. Connect the (+) and (-) wires of Run 300 to the terminal strip as shown.
3. Route Run 300 thru opening in rear leg, under Magnet and Rear Pedestal and connect to J15 on Penetration Panel.
4. Route fiber bundle from Rear End Bell thru leg opening and insert into bushing. Fasten to bushing with set screws.
5. Insert into clip on Bore light Assembly. Bushing flanges to be inside clip.
6. Peel off protective paper on velcro and install assembly in opening in rear leg.



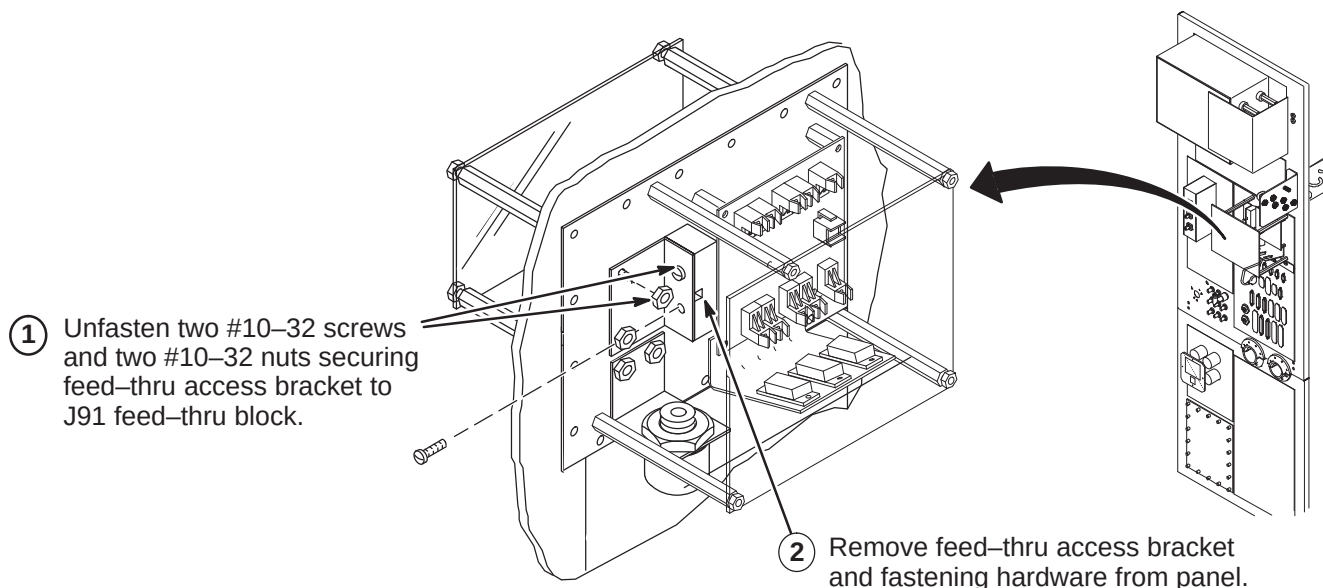
6-3-8 MAGNET ROOM FIBER OPTIC PAC/SRI INSTALLATION

Fiber Optic Run 711/712 is delivered to site connected to the SRI and PAC modules on the platform assembly mounted to the right side of the magnet between the legs. The conduit is stored for shipment in the bore of the magnet. Unroll conduit along designated path in cable trough to Penetration Panel.

Shown below are the connections for the fiber optics cables routed in the magnet room from the Penetration Panel, thru the cable trough to the Rear Pedestal, and under the magnet to the PAC/SRI platform.

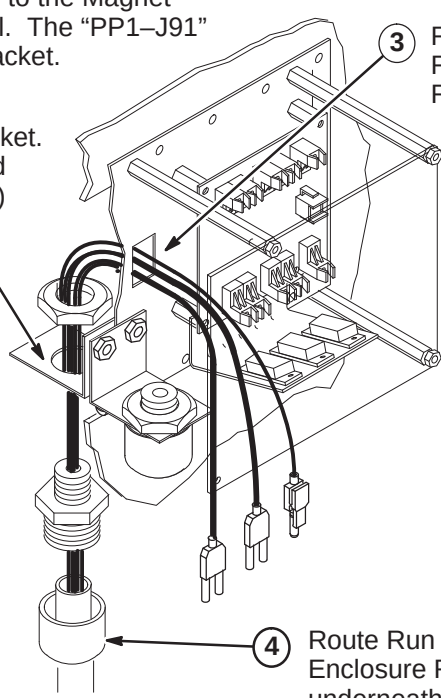


6-3-8-A Repeater Panel Feed-thru Bracket Removal



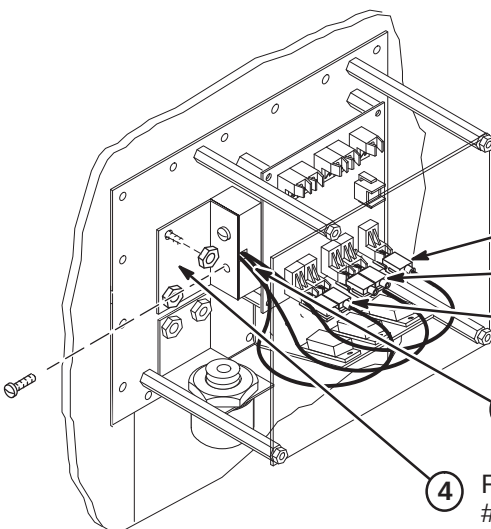
6-3-8-B Routing Run 711/712 in Magnet Room

Note: If Run 711/712 excess conduit is a problem in the magnet room, excess may be cut and ends terminated with the Fiber Optic Connector Repair Kit. Refer to Appendix for instructions. **If cables need to be shortened, mark each cable prior to cutting. This will aid in identifying the cables when reterminating.**

- ① Move Run 711/712 (46-328079G9) to the Magnet Room side of the Penetration Panel. The "PP1-J91" end will be attached to interface bracket.
 - ② Attach Run 711/712 to access bracket. Refer to Direction 2123149 (packed with bag of FO Conduit connectors) for connection installation.
 - ③ Route ends of individual fiber-optic Runs through J91 opening to the Repeater Board (PP1 A17 A1)
 - ④ Route Run 711/712 to Magnet Enclosure Rear Pedestal and underneath magnet.
- 

6-3-8-C Connecting Fiber Optic Run 711/712 to Repeater Panel

Note: Take care to maintain the minimum 2 inch bend radius.

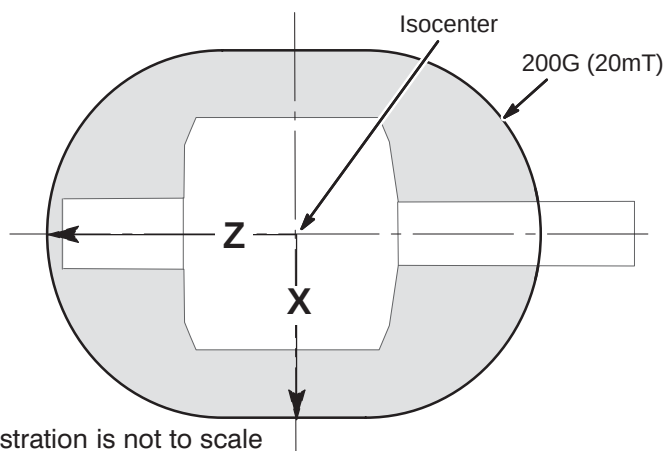
- ① Adjust route of individual fiber optic runs to maintain large radius.
 - ② Connect Runs 712-1, 711-1, and 711-2 as shown.
 - ③ Gather fiber optic runs and route through wave guide notch in feed-thru block.
 - ④ Replace feed-thru access bracket and fasten with two #10-32 screws and two #10-32 nuts removed in step A1.
- 
- Run 711-2 (J5)
Run 711-1 (J3, J4)
Run 712-1 (J1, J2)

6-3-9 200 GAUSS ZONE LIMIT FOR MOVEMENT OF BLOWER BOX

Refer to the illustration and table below for applicable 200 Gauss zone for magnet at site.



THE BLOWER BOX IS HIGHLY FERROUS! IF MAGNET IS RAMPED, ANY MOVEMENT OF THE BLOWER BOX REQUIRES TWO PEOPLE AND MUST BE OUTSIDE THE 200 GAUSS ZONE.



Approximate Distance from isocenter to 200G Line		
	X	Z
1.5T	150 cm (59 in) ¹	205 cm (81 in) ¹
Unshielded Magnet	Measure ²	
Note 1: These distances represent maximum distances for the magnet strength listed. Use gauss meter if needed. Note 2: For unshielded magnets, the magnetic field must be measured using a gauss meter.		

6-3-10 INSTALLATION OF BLOWER BOX AND AIR HOSE FROM MAGNET TO BLOWER BOX

WARNING!

RF SHIELD INTEGRITY MUST BE MAINTAINED FOR MOUNTING BLOWER BOX IN THE MAGNET ROOM

IMPORTANT!

Location of Blower Box and type of fastener used for securing Blower Box outside of minimum service area should be specified in the Site Design (architectural)/ Construction drawings. If proper mounting area is not specified, contact the Project Manager of Installation (PMI).

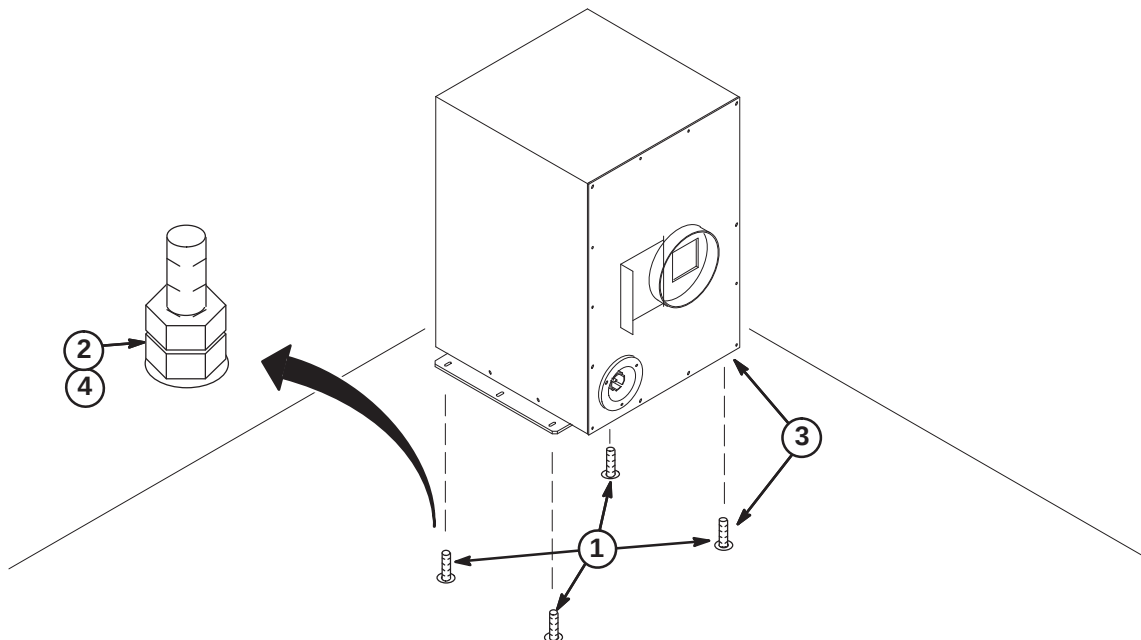
CAUTION

THE BLOWER BOX CONTAINS FERROUS MATERIAL. The proper mounting of the Blower Box is critical to safety. It MUST be mounted per these instructions.

- Check for correct installation of the anchors:
 - Confirm that the anchors are physically secured to the building structure, not to a removable floor, raised floor, or false floor.
 - Confirm there are at least four (4) anchors.
 - If any or the anchors appears wobbly or loose, stop the installation and contact the PMI to fix anchoring.

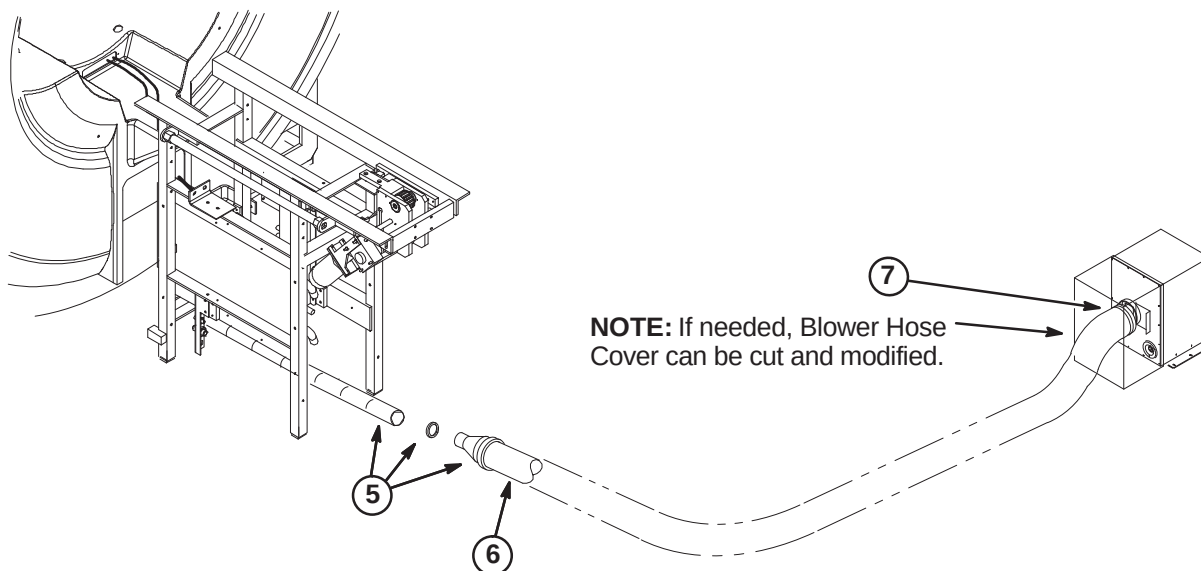
6-3-10 INSTALLATION OF BLOWER BOX & AIR HOSE FROM MAGNET TO BLOWER BOX (Continued)

2. Remove any mounting hardware from the anchors.
3. Position the blower box on top of anchors.
4. Secure the blower box using the hardware provided by the RF shield room vendor to attach.



The following items are part of Blower Hose Kit, 2129412-5

5. Connect 2 inch air hose from Rear End Bell to Reducer (2175143) with 2 inch clamp (46-208765P28). **Do not cut or reduce the length of the 2 inch hose.**
6. Connect 4 inch air hose (46-282666P11) to reducer with 4 inch clamp (46-208765P34). **If necessary, the 4 inch hose may need to be cut to reduce length.**
7. Route 4 inch hose to Blower Cabinet. Trim to appropriate length and attach to vent opening with 4 inch Hose Clamp (46-208765P34).



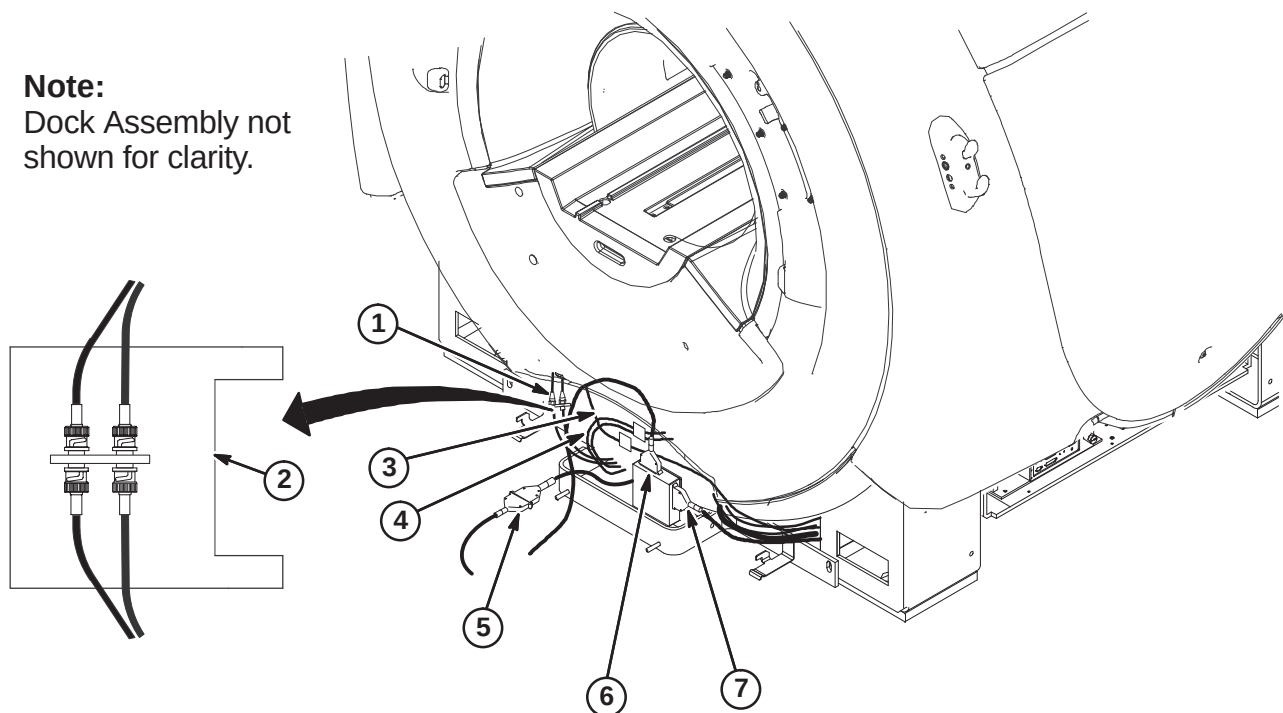
6-3-11 ROUTE AND CONNECT CABLES AT FRONT OF MAGNET ENCLOSURE**CAUTION**

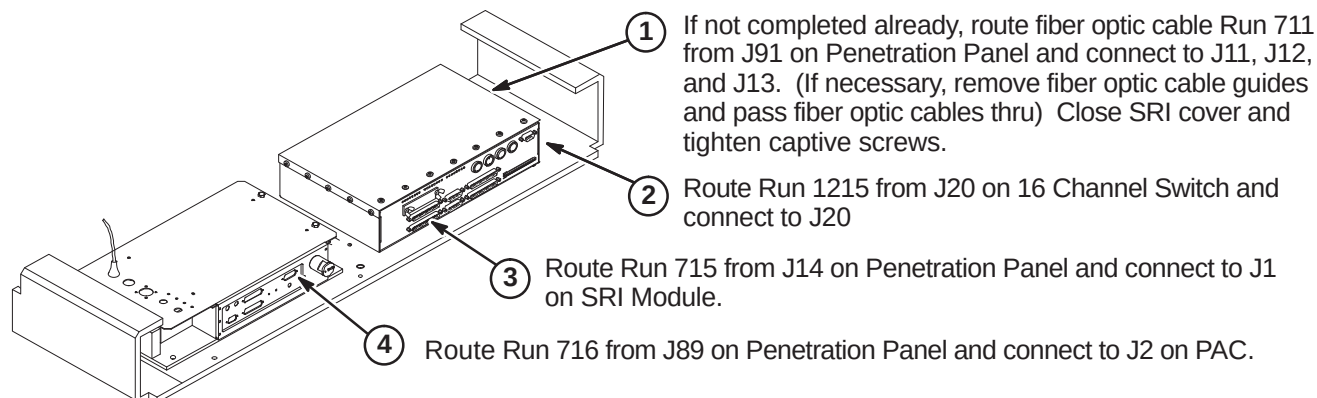
**No metal to metal contact is allowed for connectors to other areas.
i.e. connector touching the magnet.**

1. Route Runs 778 and 779 from J72 and J73 on Penetration Panel and connect to front RF Bias cables J3 & J4.
2. Wrap the RF Bias cable connections with one 177 x 200 Connector Wrap (2236130-2).
3. Route Run 1260, Bore Temperature Sensor, from Front End Bell and connect at J10 on SRI.
4. Route Front End Bell Microphone and Speaker cables underneath magnet to Rear Pedestal and connect to Patient Communication Box as marked.
5. Connect Run 387 to Dock Cable (MG2 A29 P1). If needed to prevent contact with other metal parts, cover connectors with provided velcro jacket or tape.
6. Connect Dock Cable to J2 on Dock Limit Box
7. Route/Connect Run 421 from SRI J3 and connect to J1 on Dock Limit Box.

Note:

Dock Assembly not shown for clarity.

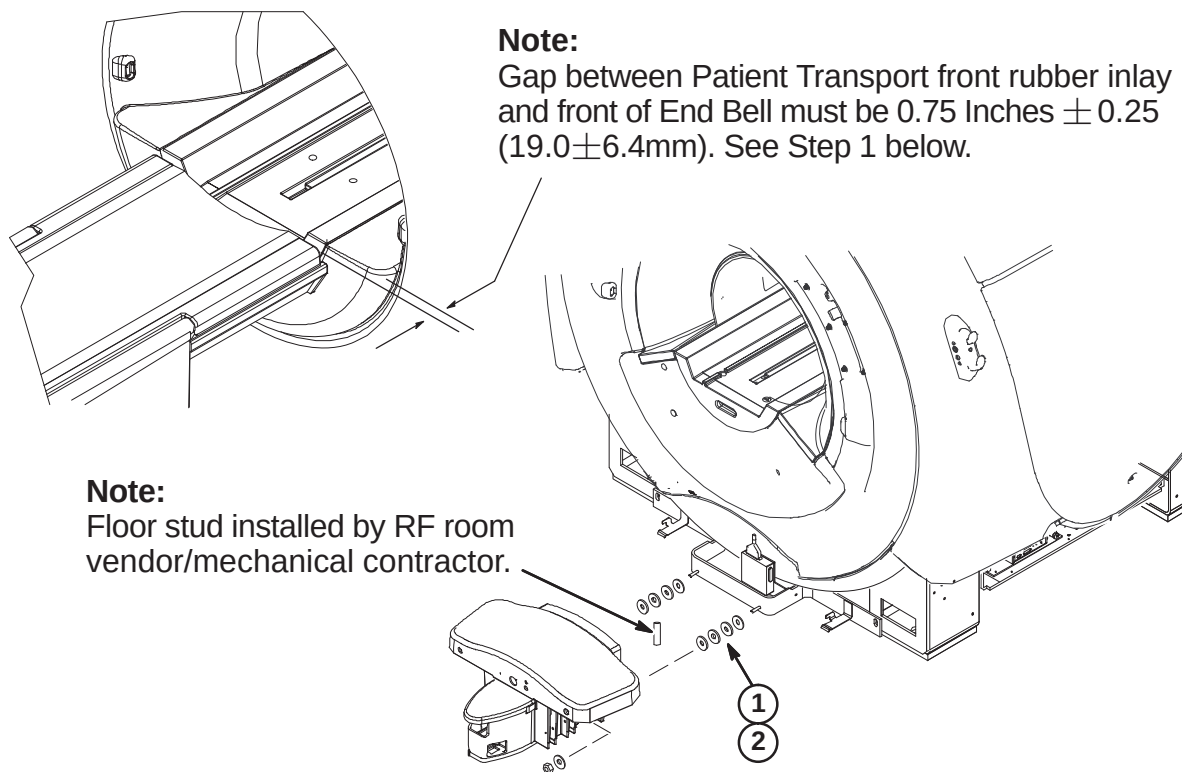


6-3-12 ROUTE AND CONNECT CABLES TO PAC/SRI PLATFORM**6-3-13 DOCK ADJUSTMENT FOR PATIENT TRANSPORT GAP**

WARNING!

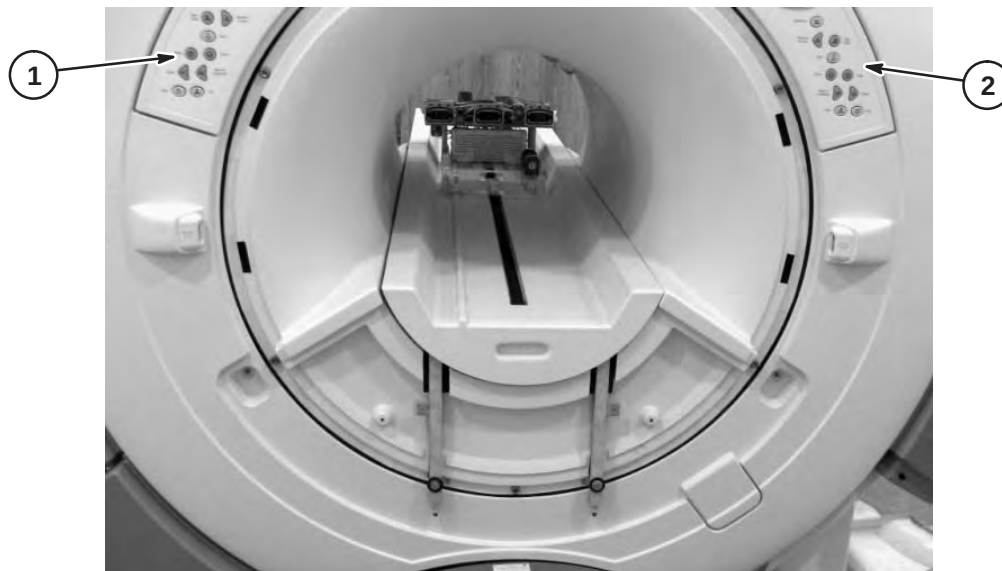
DOCK CONTAINS FERROUS MATERIAL. HANDLE WITH CARE DURING INSTALLATION.

1. Start with two (2) 11mm washers on each side between Dock and Dock Support.
2. If adjustment is required, add or remove washers until proper gap is achieved.



6-3-14 ATTACH APPLICABLE CONTROL PANEL OVERLAYS

1. Select the correct language overlay from 2215716-60 and position on Left Control Panel.
2. Select the correct language overlay from 2215716-70 and position on Right Control Panel.

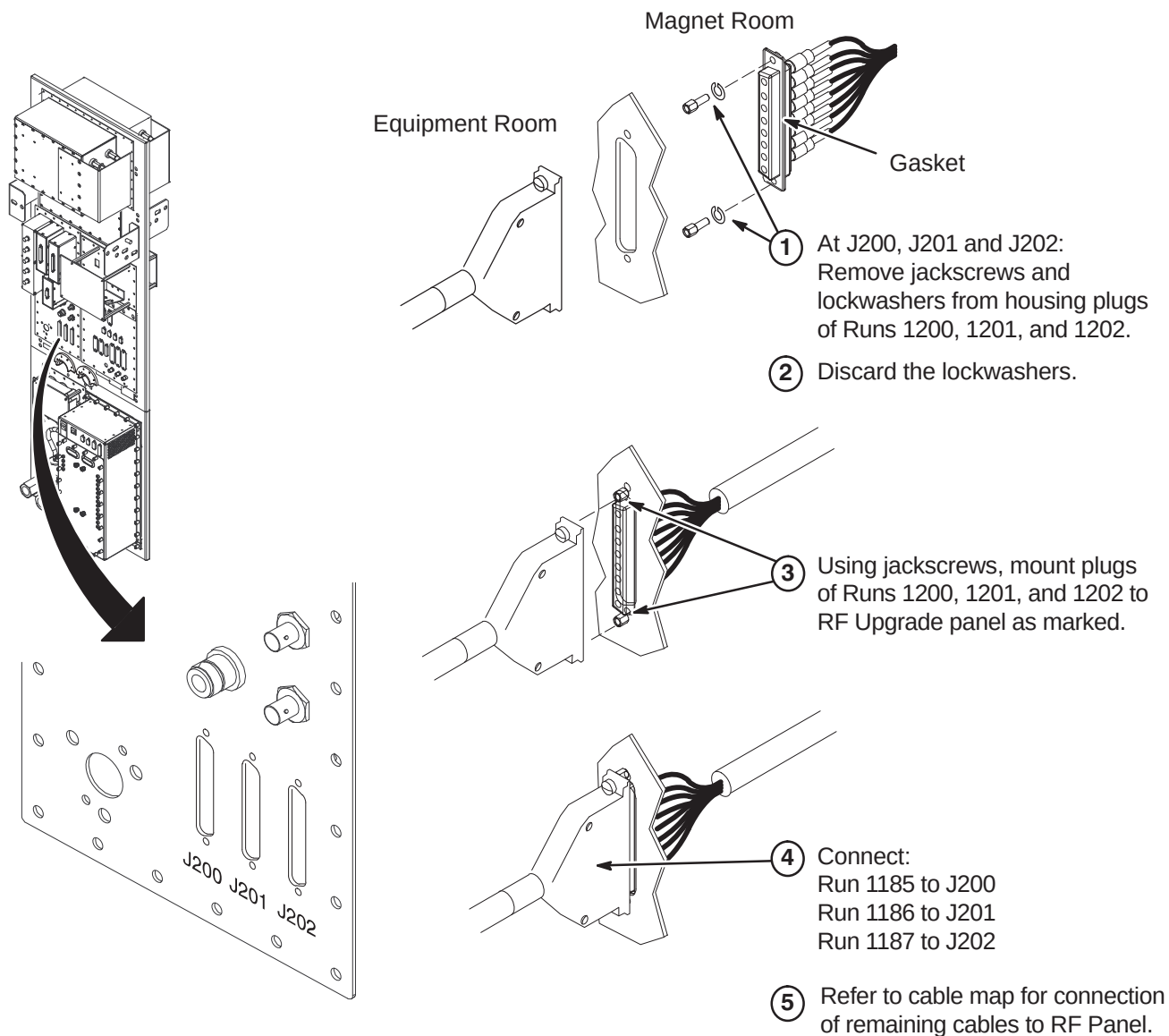


6-4 INSTALLATION OF CABLES THAT RUN BETWEEN EQUIPMENT ROOM AND MAGNET ROOM

6-4-1 INSTALLATION OF CABLES AT PENETRATION PANEL J200, J201, and J202

Notes:

- Two people are required, one on each side of the Penetration panel, to install the cables in Step 1.
- Runs 1201 and 1186 are not included in 8 Channel system. They are only installed for 16 or 32 Channel systems.



6-4-2 TIMES MICROWAVE RF CABLES INSTALLATION

6-4-2-A Cable Information

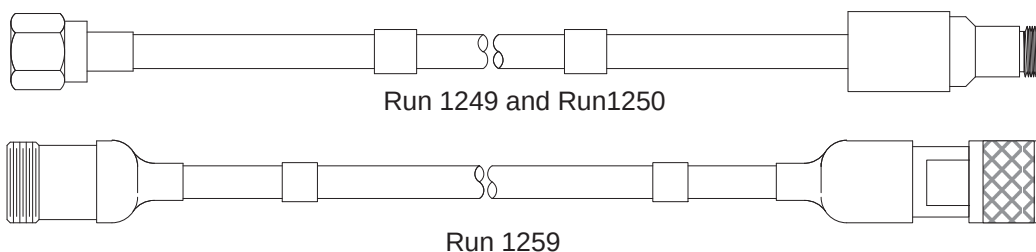
Five cables, three extension cables, and other associated parts to be installed are provided for this procedure.

The Times Microwave stripping and cutting tools are not delivered with the cables. If the tools are not on site, order the RF Transmit Cable Service Kit, 5131924. The kit contains stripping tools for LMR 900 (2384858), LMR 600 (2352193), and LMR1200 (284858), cables and the Times Microwave cutting tool (5111565).

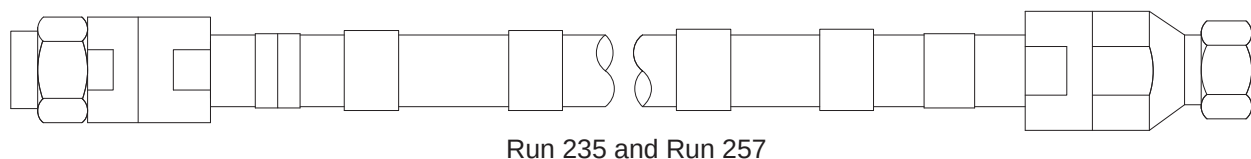


Do not subject LMR-600FR cable (Head) to a bend radius less than 3 inches (75mm). Do not subject LMR-1200FR cable (Body) to a bend radius less than 6.5 inches (165mm). The cables will be permanently damaged if it is kinked by a bend radius less than the specified minimum.

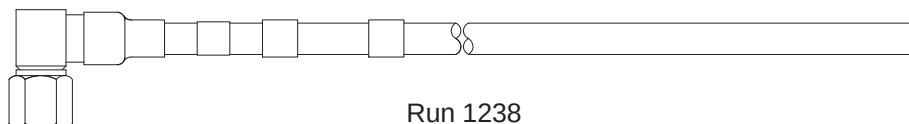
Two extension cables, 5116213-2, Run 1249 (Head) and 5121269, Run 1159 (Body) are connected to the RF Amplifier module in the RF/System Cabinet. The other extension cable, Run 1250, 5121245, is connected to the Body Hybrid Splitter in the Rear Pedestal.



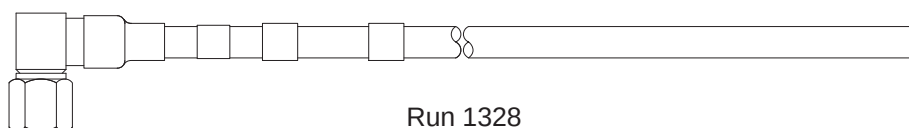
Cable 2316584-2 (Body, Runs 235 & 257), will be cut to length to make two cables and terminated for routing in the equipment and scan rooms.



Cable 5115916 (Head, Run 1238), will be cut to length and terminated for routing in the equipment room.



Cable 5160826 (Head, Run 1328), will be cut to length and terminated for routing in the magnet room.

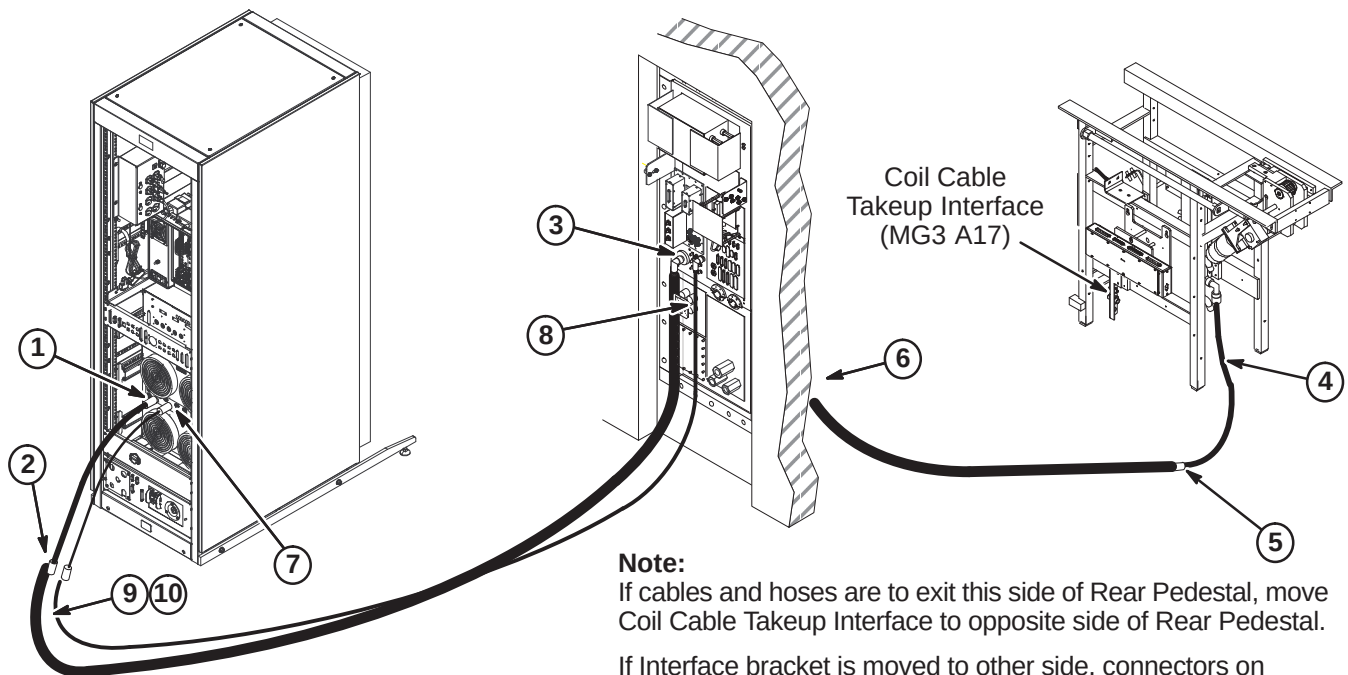


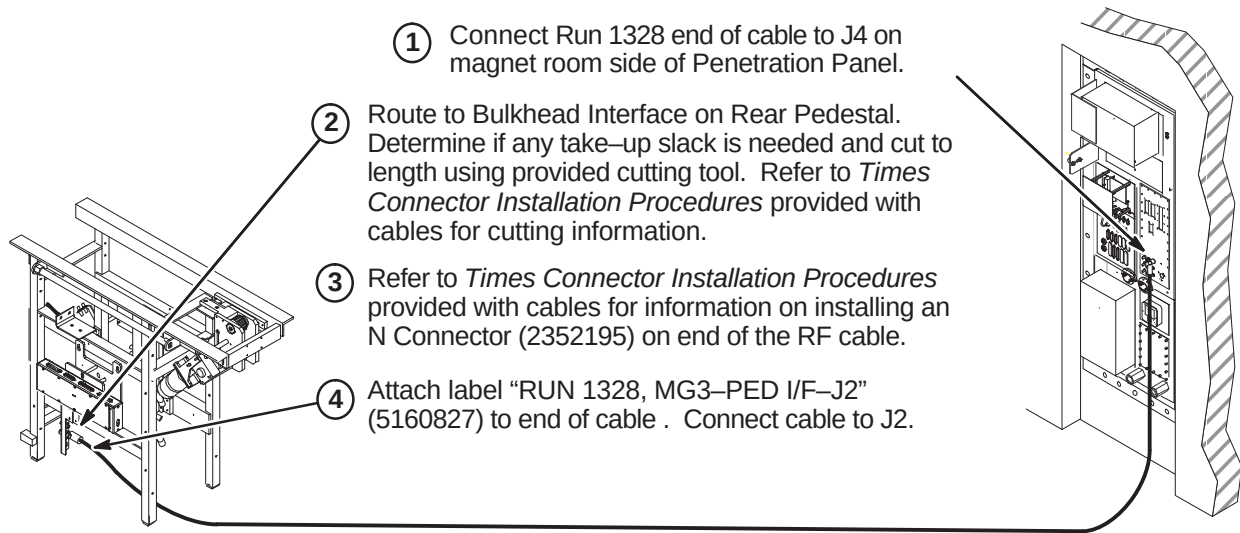
The remaining connectors, fasteners, and labels will be installed as directed in the following steps.

6-4-2-B Install Runs 235, 257, 1238, 1249, 1250, and 1259

RF cables will be routed and connected as indicated in the following steps.

1. Connect extension cable Run 1259 to J4 Body Output at rear of RF Amp in RFS cabinet.
2. Connect end of Times RF Body cable marked "(RUN 235) FJ5" to Run 1259, Body extension cable. Attach label "RUN 235, Connect to Run 1259" (5121244) over existing label.
3. Route cable to J44 at Penetration Panel. Determine if any take-up slack is needed and cut to length using provided cutting tool. Refer to *Times Connector Installation Procedures* provided with cables for cutting information. Refer to Procedure **6-4-2-D** for termination procedure of cut end.
4. Connect Run 1250 to J1 on Hybrid Splitter.
5. Move remainder of RF Body cable to Scan Room and connect other end of cable marked "(RUN 257) MG3A3-J1" to Run 1250 at Rear Pedestal as shown below. Attach label "RUN 257, Connect to Run 1250" (5121244-2) over existing label.
6. Route remainder of cable to J44 at Penetration Panel. Determine if any take-up slack is needed and cut to length. Refer to *Times Connector Installation Procedures* provided with cables for cutting information. Refer to Procedure **6-4-2-D** for termination instructions of cut end.
7. Connect extension cable Run 1249 to J3 Head Output at rear of RF Amp IN RFS cabinet.
8. Connect end of Times RF Head cable marked "(RUN 1238) PP1-J4" to J4 at Penetration Panel. Route to RF Head extension cable at rear of SRF cabinet. Determine if any take-up slack is needed and cut to length using provided cutting tool. Refer to *Times Connector Installation Procedures* provided with cables for cutting information.
9. Attach label "RUN 1238, CONNECT TO RUN 1249" (5116214) to end of cable after termination.
10. Connect cable to Run 1249.



6-4-2-C Install Run 1328

6-4-2-D Terminate and Connect RF Body Cables at Penetration Panel

Perform the following steps for terminating and connecting the RF Body cables at the Penetration Panel J44.

1. Refer to *Times Connector Installation Procedures* provided with cables for information on installing an EIA Right Angle Connector (2384859) on each end of the RF cables. Use provided cutting tool (5111565) and the LMR1200 Stripping Tool (2384858).

Note

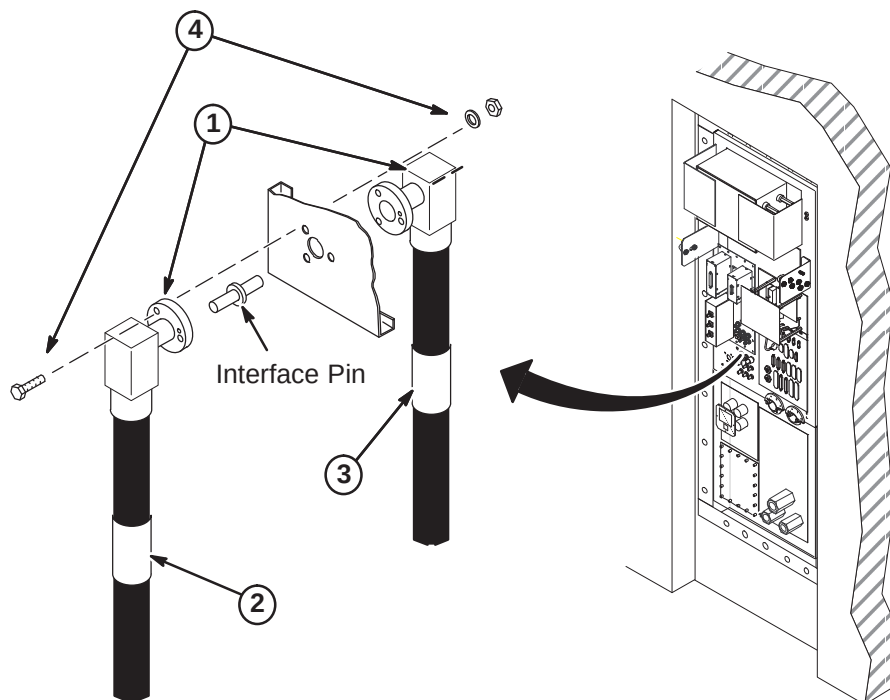
Make sure the fastener holes in the connector flanges align with the holes in the Penetration Panel so the cables will route toward floor when installed.

2. Attach label "RUN 235, PP1-J44" (2357230) to cable as shown on Equipment room side of Penetration Panel.
3. Attach label "RUN 257, PP1-J44" (2357230-2) to cable as shown on Scan room side of Penetration Panel.

CAUTION

Do NOT install O-Ring or Grease that may be provided with Cable Termination Kit. Installation of these parts will prevent the ground from making contact resulting in RF leaks.

4. Insert Interface Pin into one of the connectors. At J44, align connectors on each side of Penetration Panel and push together. Fasten together using three of each of the following:
 - 1/4-20 x 1.25 Hex Head Bolt (brass)
 - 1/4 Screw Size Washer (phos bronze)
 - 1/4-20, 7/16" Hex x 3/16" thk Nut (brass)



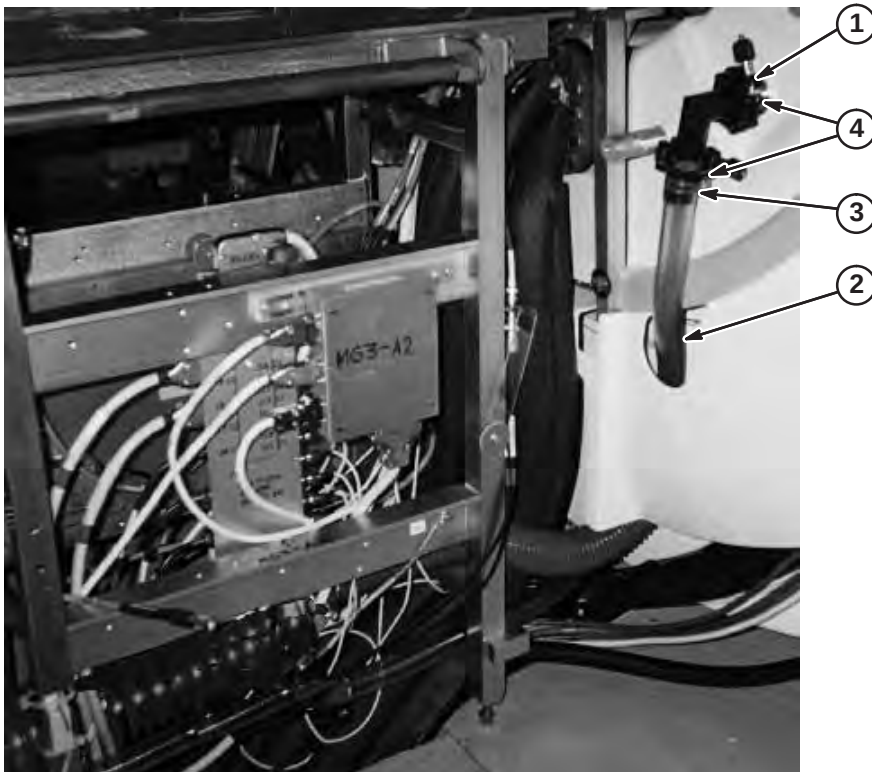
6-4-3 VACUUM HOSE INSTALLATION

The parts for installing the vacuum hose are supplied with the Twin Vacuum Hose Assembly (2295226).

Note: Vacuum Grease must be applied to all Centering Rings (2293608) and outside edge of vacuum connector flanges.

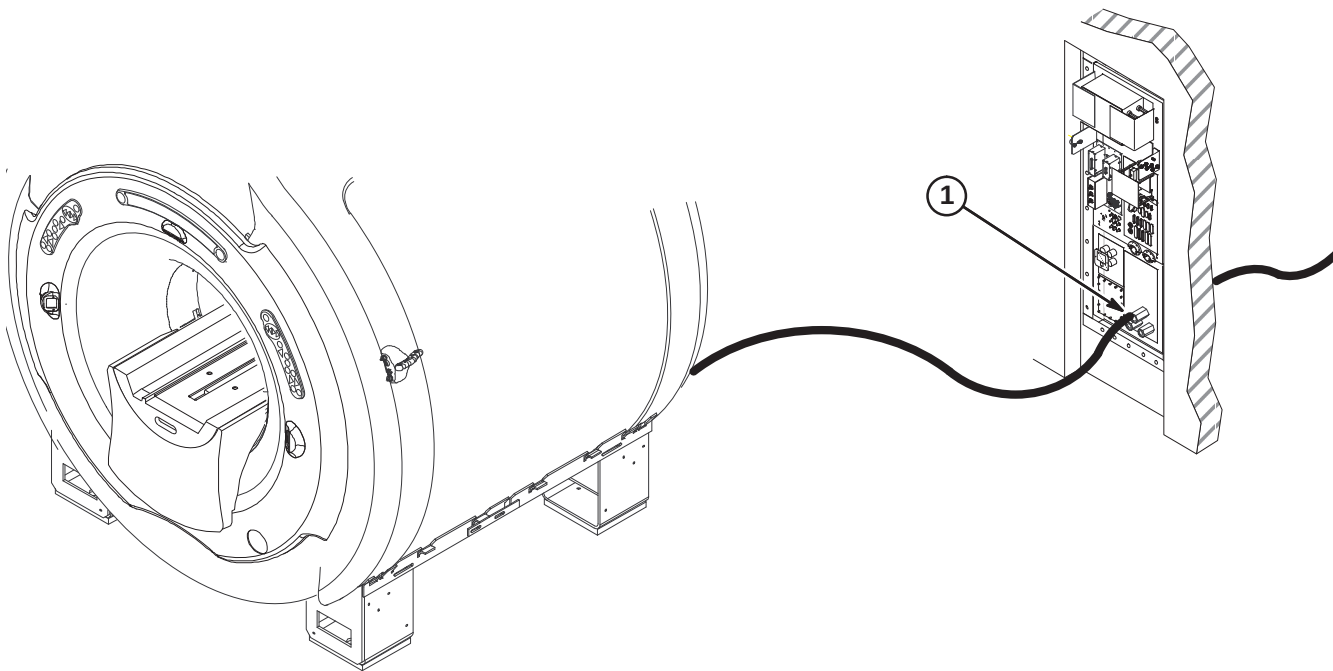
6-4-3-A Connect Vacuum Hose to Magnet

1. Install Elbow (2293609) to End Bell using centering ring (2293608) and chain clamp (2293618-2). Tighten chain clamp to “finger tight”.
2. Route vacuum hose (2293607) underneath Rear Pedestal, under magnet end cover, and thru hole to Elbow on End Bell.
3. At end of vacuum hose insert Hose Connector (2293616), apply vacuum grease to outside edge of connector and fasten with two (2) Clamps (2294010-3).
4. Apply vacuum grease to outside edge of Hose Connector flanges and respective connector flanges on End Bell.

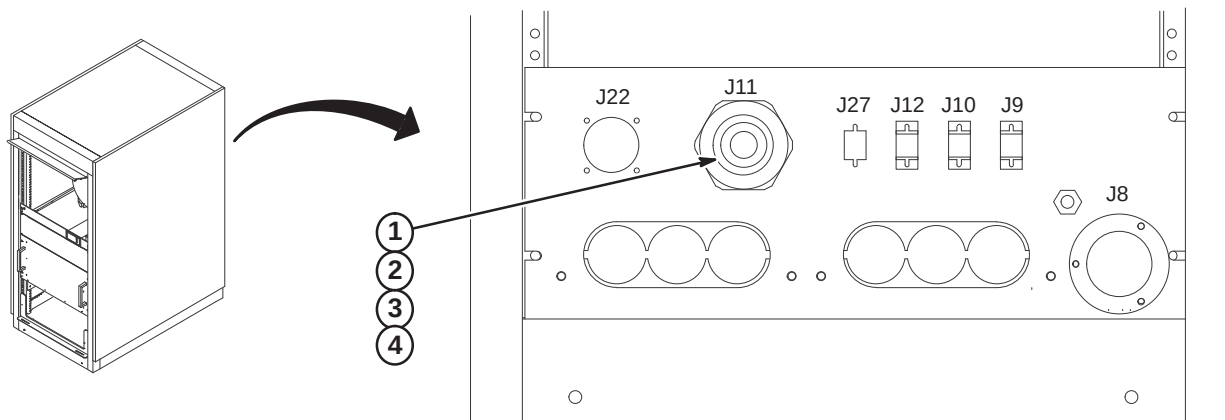


6-4-3-B Route Vacuum Hose

1. Route Vacuum Hose thru wave guide in Penetration Panel to TAC Cabinet.

**6-4-3-C Connect Vacuum Hose to TAC Cabinet**

1. Route vacuum tube to J11 at back of cabinet and cut to length.
2. Install 1" Hose Connector (2293616) in the end of hose and attach with worm-drive clamp (2294010-3).
3. Apply vacuum grease to centering ring (2293608) and connect vacuum tube to J11 using supplied Chain Clamp (2293618-2).
4. Apply vacuum grease to outside edge of hose connector flanges and respective connector flanges on TAC Cabinet.



6-4-4 PNEUMATIC PATIENT ALERT INSTALLATION

The customer and users should be involved in the following decisions:

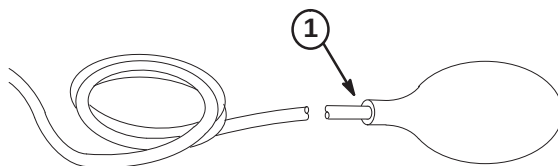
- Routing of Pneumatic Tubing
- Location of Control Box

CAUTION

Do not substitute other types of tubing for the pneumatic tubing supplied with this Installation Kit. Substitution of other types of tubing may cause a control box malfunction.

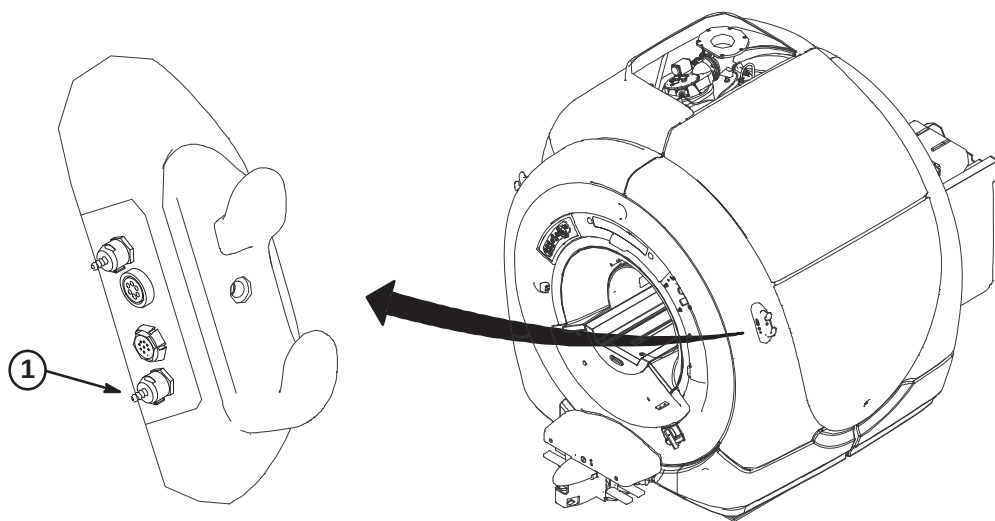
6-4-4-A SQUEEZE BULB INSTALLATION

1. Insert end of Pneumatic Tubing into Squeeze Bulb approximately 1/2 inch (13mm). For ease of pneumatic tubing insertion, pinch pneumatic tubing in one hand while squeezing squeeze bulb and pushing pneumatic tubing in with the other hand. This will create a positive pressure and expand squeeze bulb opening.
2. Before cutting pneumatic tubing from roll to length required, verify that when the squeeze bulb is held by a patient being scanned, the tubing length can be routed to the "CALL" connector on the Remote PAC Interface Assembly without getting in the way of operator or patient during use.



6-4-4-B CONNECT TUBE TO REMOTE PAC INTERFACE ASSEMBLY

1. IConnect end of tubing to "CALL" connection on Remote PAC Interface Assembly.



6-4-4-C MAGNET ROOM ROUTING OF TUBING & WAVE GUIDE INSTALLATION

CAUTION

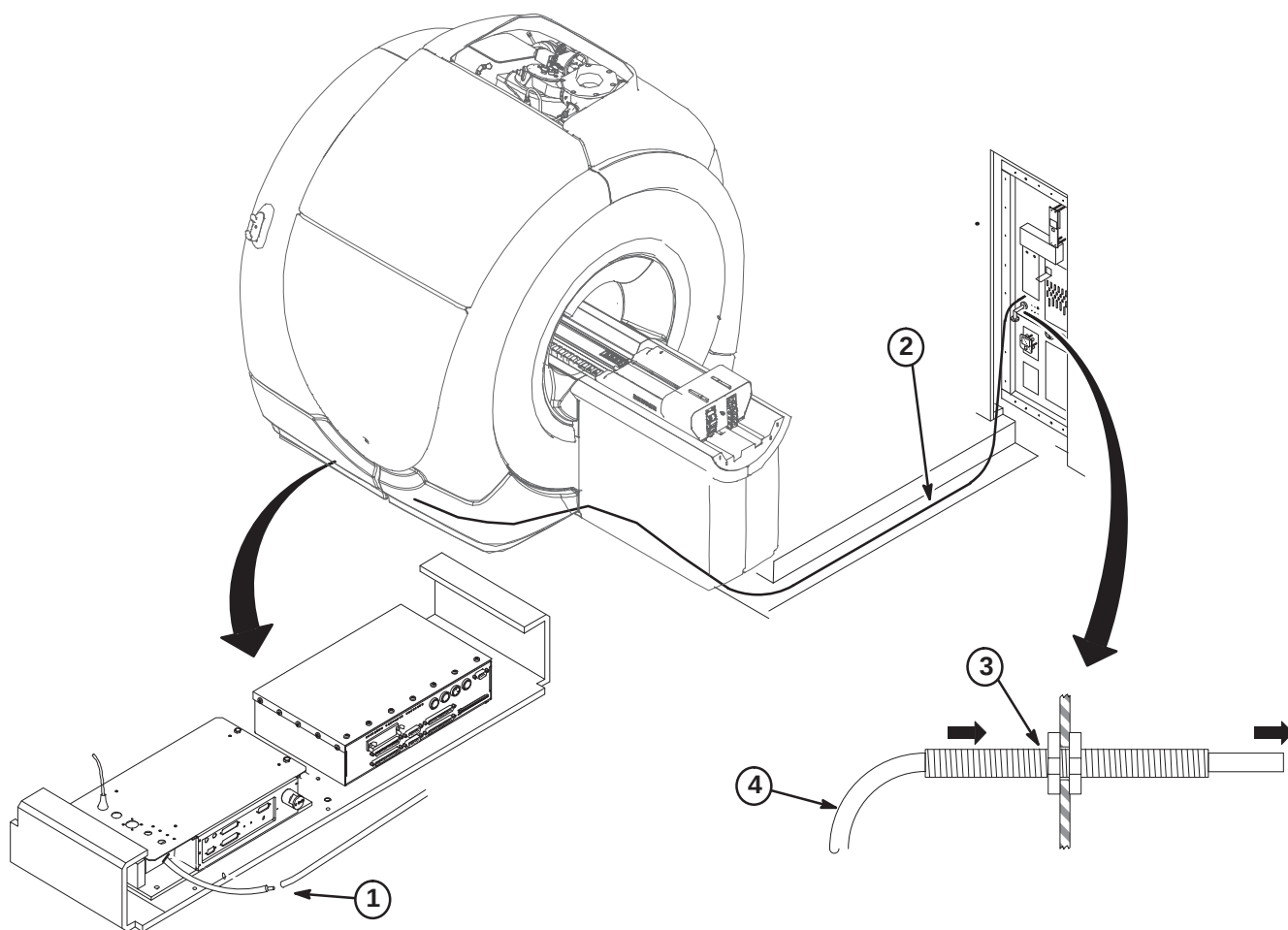
Make sure that pneumatic tubing is not routed where it can be stepped on, pinched, or have an object set on it. The control box alarm will not work if pneumatic tubing is pinched.

Also, make sure that the pneumatic tubing can be routed from the PAC connector through the Rear Pedestal, and to the Penetration panel without being pinched along the route by other cables.

1. Push end of pneumatic tubing onto PAC-II tubing connector.

Note: If the pneumatic tubing is to be pulled thru fiber-optic conduit, coat with baby powder, not grease, to allow easier pulling.

2. Route Pneumatic Tubing from Rear Pedestal to Penetration Panel with other system cables.
3. Insert Wave Guide thru "ALERT AIR LINE" hole and fasten to Penetration Panel with nut on each side.
4. Route end of pneumatic tubing from Magnet Room side of Penetration Panel thru Wave Guide into the Equipment Room where the Control Box will be installed.



6-4-4-D DETERMINING NEED OF EXTENDER BOX

If installation requires greater than 115 feet of pneumatic tubing between the squeeze bulb (hanging on Magnet Enclosure) and control box (near Operator's Console), Extender Kit, 46-317758P2, must be ordered.

- **If the pneumatic tubing is long enough to reach control box:**
The extender is not needed. Go to Procedure **6-4-4-F**, CONTROL BOX INSTALLATION.
- **If the pneumatic tubing is not long enough to reach control box:**
The extender is needed. Go to Procedure **6-4-4-E**, INSTALLATION OF EXTENDER KIT.

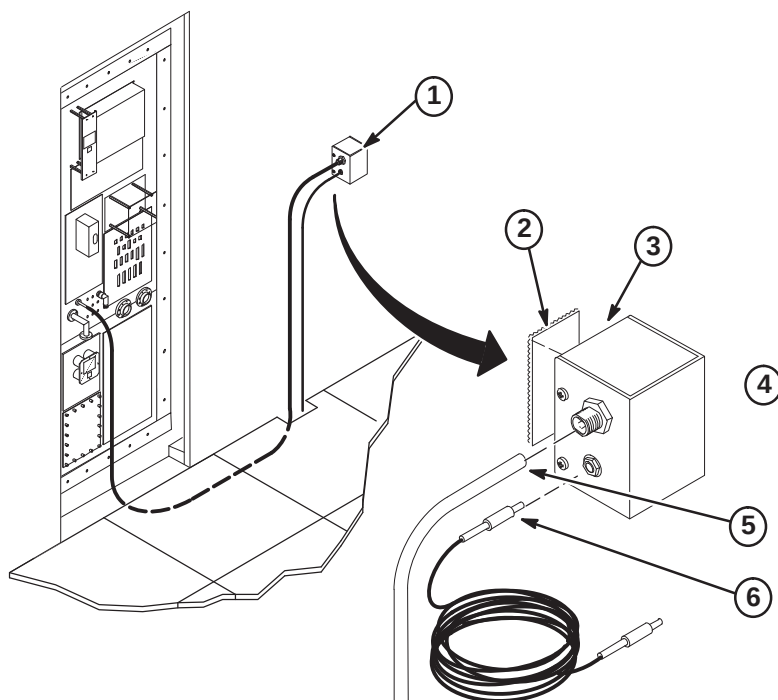
6-4-4-E INSTALLATION OF EXTENDER KIT (46-317758P2)

CAUTION

Do not mount Extender Box in Magnet Room because the box contains ferrous parts.

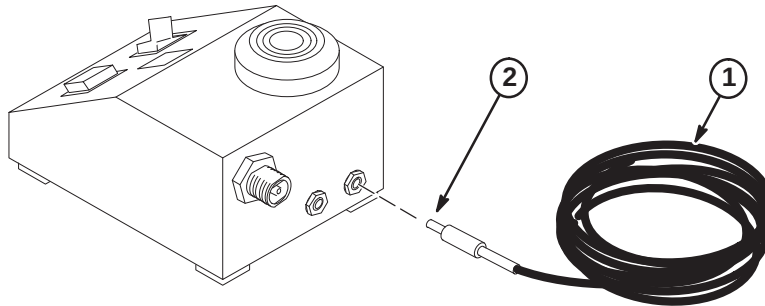
Install Extender Box

1. Choose the mounting location for extender box on equipment room wall next to Penetration Panel. Extender box will be mounted with pneumatic and electrical jacks on side of extender box.
2. Cut a two inch long piece of Velcro loops and a two inch long piece of Velcro hooks. Both are obtained from the Pneumatic Patient Alert Kit.
3. Clean the back of extender box, remove protective paper from back of Velcro loops, and attach Velcro to center of extender box back.
4. Clean wall in the area where extender box will be mounted, remove protective paper from back of Velcro hooks, and attach velcro to wall in same orientation as Velcro loops on extender box.
5. Push end of pneumatic tubing fully onto Feed-thru connector on Extender Box.
6. Push the jack on one end of the 65 foot Extender Wire into the plug on side of Extender Box.



6-4-4-E Installation of Extender Kit (46-317758P2) (Continued)**Connection of Extender Wire to Control Box**

1. Route the extender wire from extender box to control box. Control box will be installed near Operator Console, within five feet of an electrical outlet.
2. Push the jack at other end of extender wire into the plug on back of control box.

**6-4-4-F CONTROL BOX INSTALLATION OPTIONS**

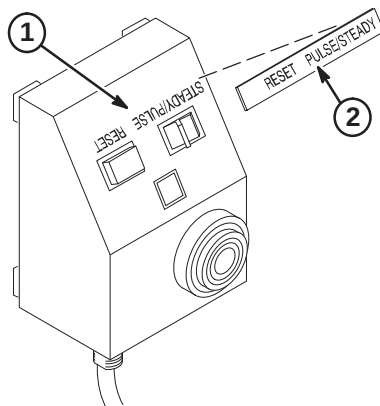
Consult with the operator in choosing the mounting orientation for the control box. Some key factors to consider are ease of use by operator, remaining within sight of operator, and remaining within five feet of an electrical outlet. Choose one of the following control box locations:

- Mount control box on a wall or other vertical surface per Procedure **6-4-4-G**, CONTROL BOX INSTALLATION ON WALL OR UNDER SHELF.
- Mount control box under shelf per Procedure **6-4-4-G**, CONTROL BOX INSTALLATION ON WALL OR UNDER SHELF.
- Place Control Box on a counter top, desk top, or other horizontal surface. Set the Control Box near Operator Console and within five feet of electrical outlet. Control Box should be placed within sight of operator. Go to Procedure **6-4-4-K**, FINAL CONNECTIONS TO CONTROL BOX.

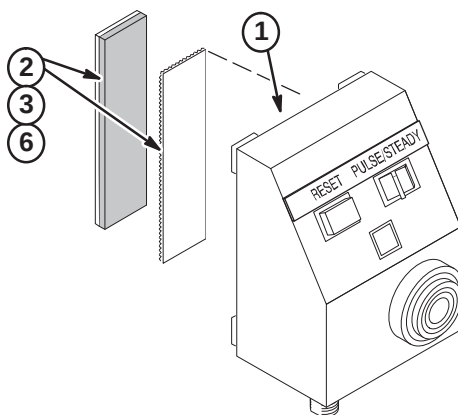
Note: The installer must provide additional mounting hardware if supplied screws or adhesive backed Velcro are not adequate for installation of the control box.

6-4-4-G CONTROL BOX INSTALLATION ON WALL OR UNDER SHELF

1. Clean Control Box surface.
2. Remove adhesive backing from metalized nameplate and apply over silk screened lettering as shown.
3. Select one of the following for installation of Control Box:
 - If control box will be mounted to WALL or SHELF with Velcro, go to Procedure **6-4-4-H**
 - If control box will be mounted to WALL with screws, go to Procedure **6-4-4-I**
 - If control box will be mounted to SHELF with screws, go to Procedure **6-4-4-J**

**6-4-4-H Control Box Mounted With Velcro**

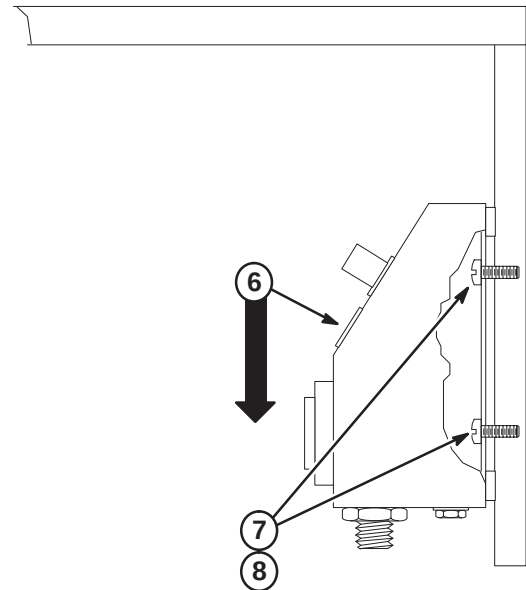
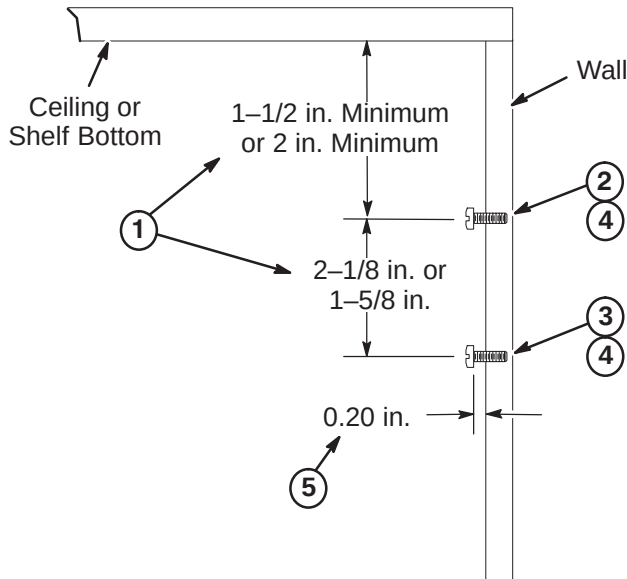
1. Clean the bottom surface of the Control Box.
2. From the Pneumatic Patient Alert Kit, cut a three inch long piece of Velcro loops and a three inch piece of Velcro hooks.
3. Remove protective paper from back of Velcro loops, and attach Velcro to center bottom of control box.
4. Choose the mounting location for control box on wall (or under shelf) next to operator console. Control box should be mounted within sight of operator.
5. Clean the area where control box will be mounted.
6. Remove protective paper from back of Velcro hooks and attach to mounting location in same orientation as Velcro on box.
7. jMount the control box by pressing Velcro on control box against Velcro on mounting location and twisting slightly.
8. Proceed to Procedure **6-4-4-K**, FINAL CONNECTIONS TO CONTROL BOX.



6-4-4-I Installing Control Box with Mounting Screws on Vertical Surface

Choose the mounting area near the Operator Console and install mounting screws (item 9). Control box should be mounted within sight of operator.

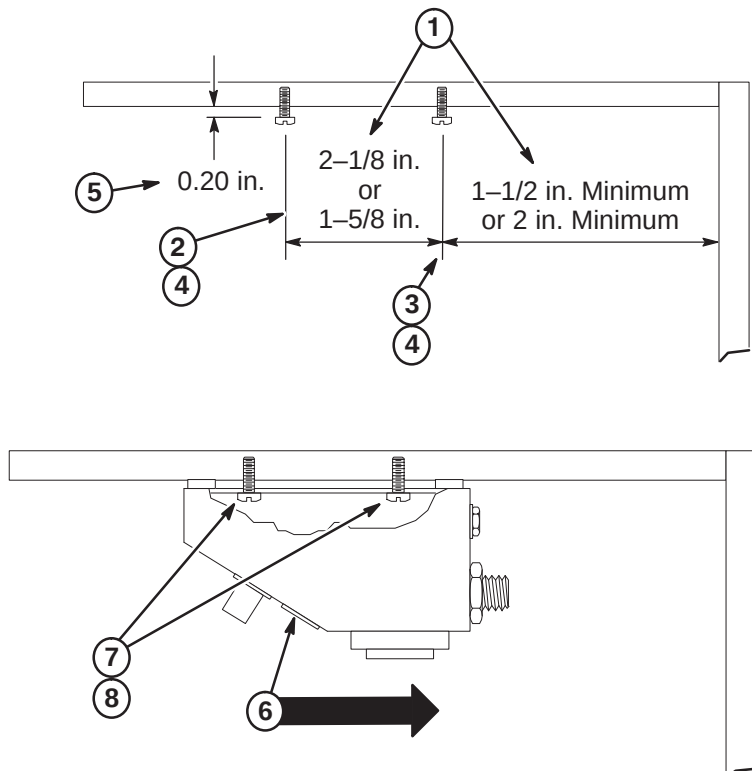
1. Determine which of the shown keyhole slot spacings are applicable to Control Box and select appropriate dimensions to use.
2. Mark top mounting hole position on wall.
3. Mark bottom mounting hole position on wall.
4. Drill a 1/16 in. pilot hole at locations if furnished screws are used.
5. Install screws to depth as shown.
6. Center box keyhole slots over screws, push in on box, and then push downward to seat screws within slots.
7. Tighten or loosen screws as necessary to obtain a snug fit of box against wall.
8. Proceed to Procedure **6-4-4-K, FINAL CONNECTIONS TO CONTROL BOX**.



6-4-4-J Installing Control Box with Mounting Screws Under Shelf

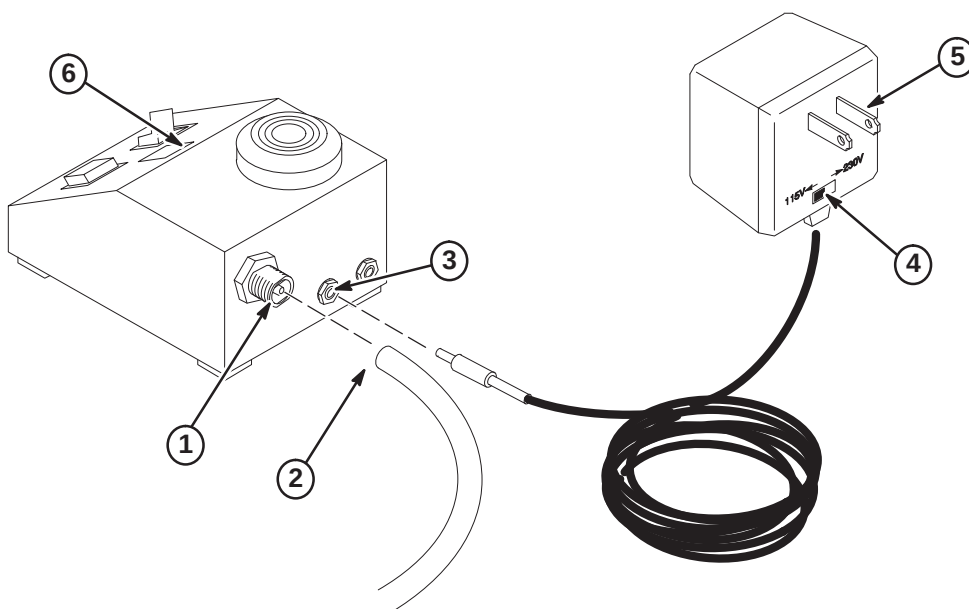
Choose the mounting area near the Operator Console and install mounting screws (item 9). Control box should be mounted within sight of operator.

1. Determine which of the shown keyhole slot spacings are applicable to Control Box and select appropriate dimensions to use.
2. Mark front mounting hole position on underside of shelf.
3. Mark rear mounting hole position on underside of shelf.
4. Drill a 1/16 in. pilot hole at locations if furnished screws are used.
5. Install screws to depth as shown.
6. Center box keyhole slots over screws, push in on box, and then push inward to seat screws within slots.
7. Tighten or loosen screws as necessary to obtain a snug fit of box against wall.
8. Proceed to Procedure **6-4-4-K, FINAL CONNECTIONS TO CONTROL BOX**.



6-4-4-K Final Connections to Control Box

1. Bring the pneumatic tubing to back of control box and mark length. Leave two inches of extra pneumatic tubing for connection.
2. Cut tubing and push fully onto feed-thru connector on back of control box.
3. jPlug Power Supply Jack into Control Box.
4. Set Power Supply to outlet voltage (115VAC or 230VAC).
5. Plug Power Supply into AC outlet.
6. Check that green LED is lighted.



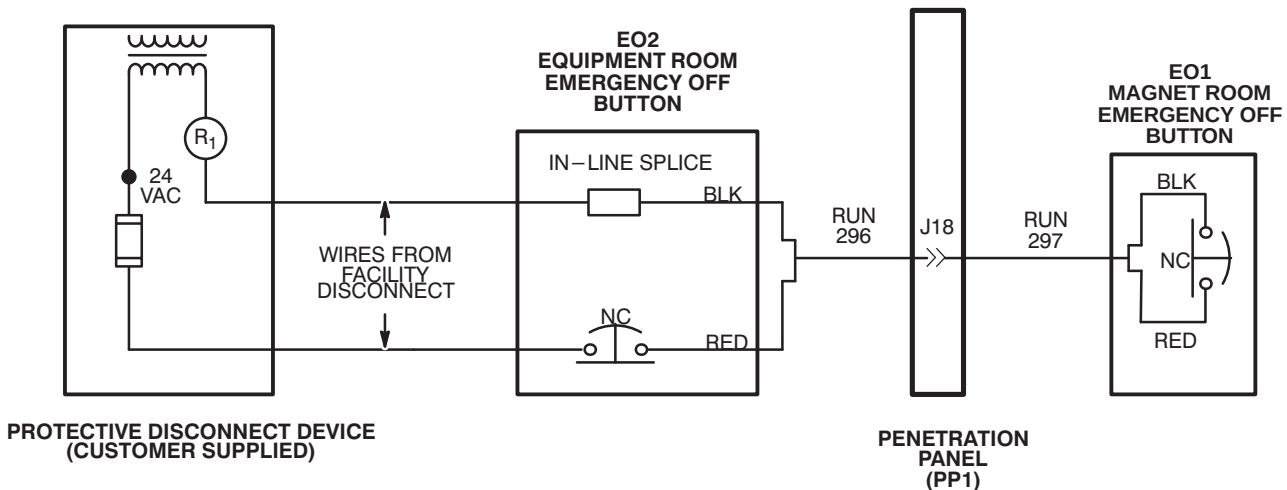
6-4-5 EMERGENCY OFF CONNECTIONS

Run 296 is routed to EO2 (Equipment Room Emergency Off) and connected to EO2 and wire from Facility Disconnect. Run 297 is routed from Penetration Panel to EO1 (Magnet Room Emergency Off Button). Refer to applicable *Pre-Installation*, Section 5, Protective Disconnect Device, for detailed wiring diagram.

1. Complete routing of Run 296 to EO2 (Equipment Room Emergency Off) location, and trim cable to length. Locate red and black pair of wires and prepare ends for applicable terminals.

Note: In Illustration below, black and red wires are used for connections in Runs 295 and 297. Actually any pair of wires on these runs could be used so long as both ends are consistent with one another. Runs 296 and 297 are actually nine wire cables.

2. At equipment room room "Emergency Off" (EO2) location, terminate red wire at end of Run 296 with local supplied terminal and connect to customer supplied Equipment Room Emergency Off Button (EO2).
3. Terminate customer supplied wire (from fuse in Protective Disconnect Device) with local supplied applicable terminal and connect to customer supplied Emergency Off Button (EO2).
4. Terminate customer supplied wire from R1 in Protective Disconnect Device with local supplied push-on terminal.
5. Terminate black wire at end of Run 296 with local supplied "push-on" terminal.
6. Connect black wire from end of Run 296 to customer supplied wire from protective disconnect device with local supplied in-line splice.
7. Complete routing of Run 297 to EO1 (Magnet Room Emergency Off) location, and trim cable to length. Locate red and black pair of wires and prepare ends for applicable terminals.
8. Terminate red and black wires at end of Run 297 with local supplied terminals and connect to customer supplied Magnet Room Emergency Off Button (EO1).



SECTION 7 – FINAL ENCLOSURE INSTALLATION

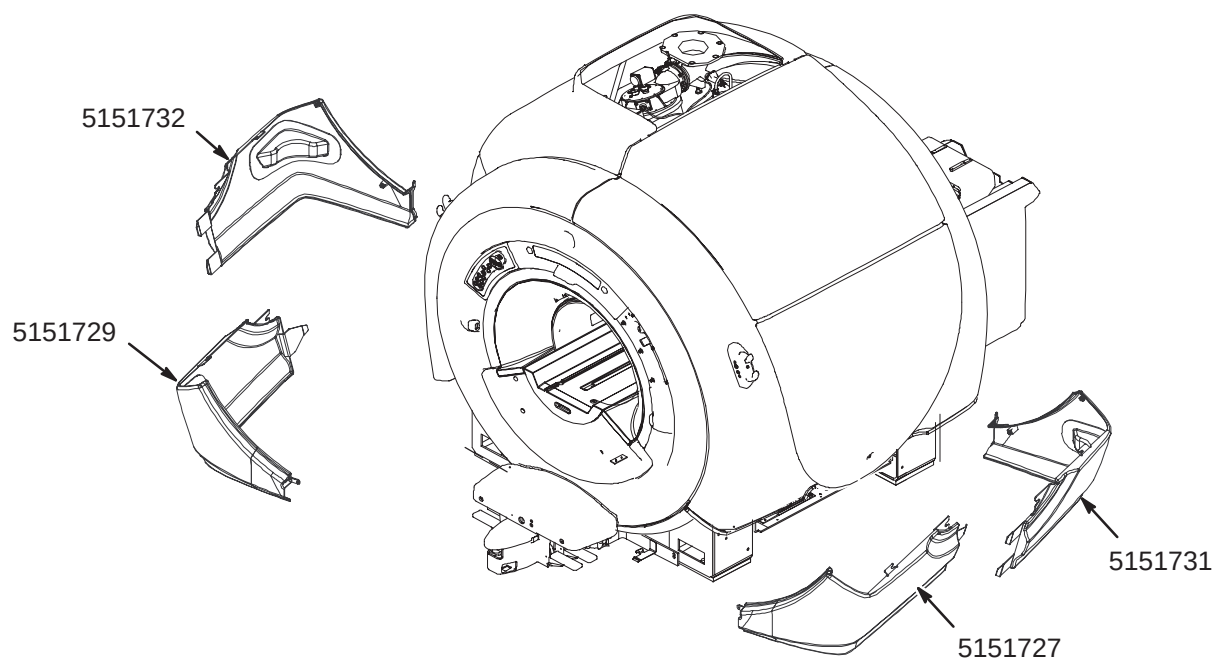
TABLE OF CONTENTS

<u>SECTION</u>	<u>PAGE</u>
7-1 INSTALL SKIRT COVERS	7-2
7-2 INSTALL LEFT SIDE COVERS	7-4
7-3 INSTALL REAR PEDESTAL COVERS	7-8
7-4 INSTALL LASER ALIGNMENT LIGHT WARNING LABELS	7-9
7-5 PATIENT TRANSPORT INSTALLATION	7-10
7-6 INSTALLATION OF HEARING LOSS WARNING SIGN	7-11

7-1 INSTALL SKIRT COVERS

Note

For Mobiles, follow van manufacturers process for magnet cover installation.



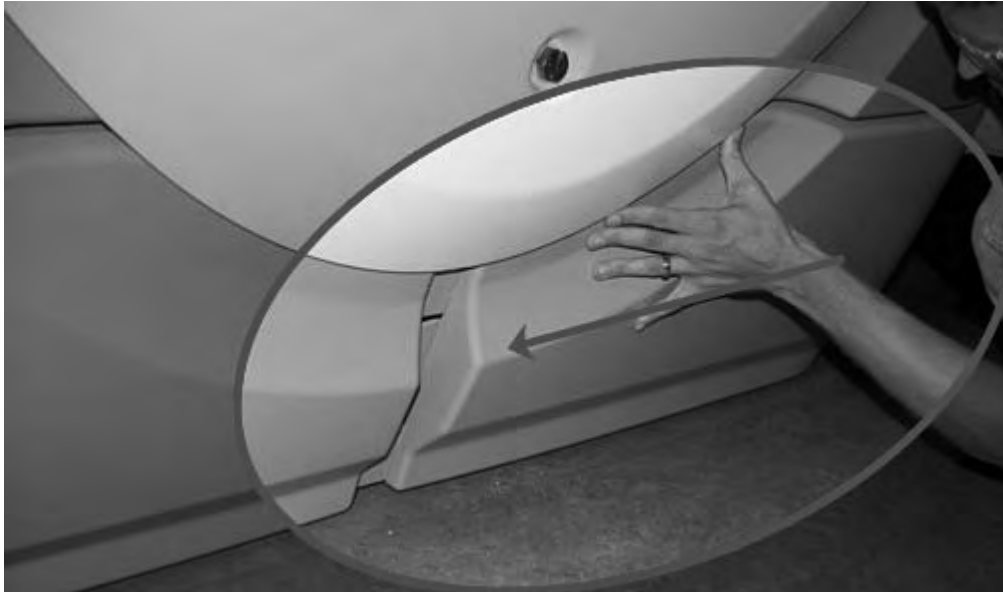
7-1-1 Align Skirt's Hook with Slot

Align fastener on skirt cover to the notch on the arc cover and slide cover towards the center of the magnet.



7-1-2 Align Skirt Tabs

Align tabs as to interlock with tabs on adjoining skirt cover.



7-1-3 Snap Covers in Place

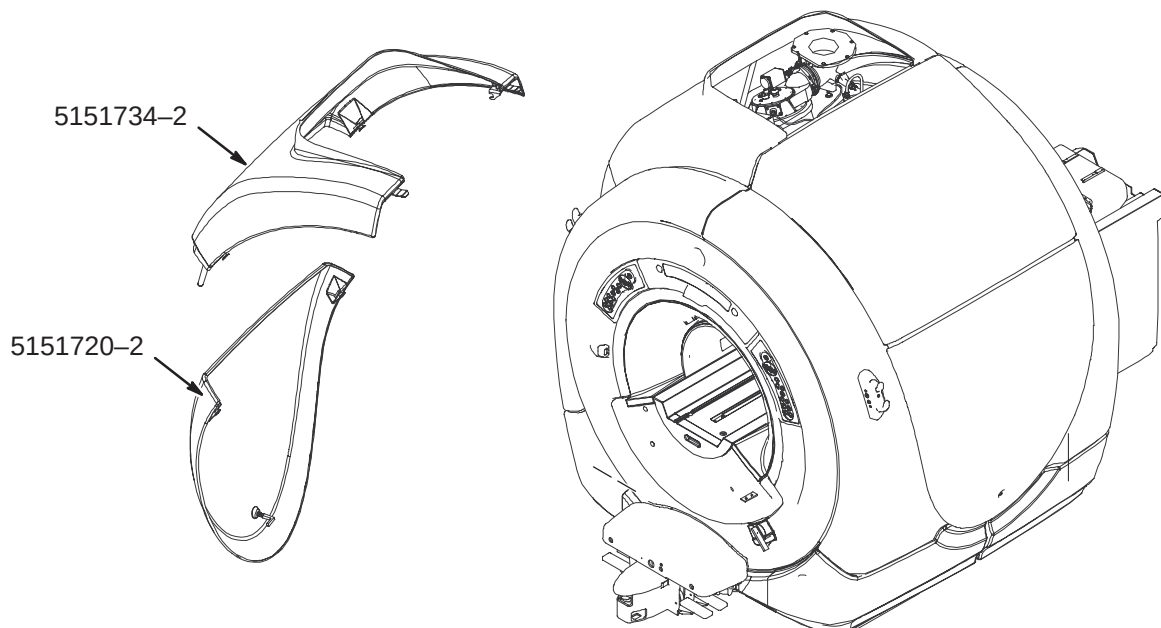
Snap poppers on the front end of arc cover into place.



7-2 INSTALL LEFT SIDE COVERS

Install the left side enclosure covers as shown in the following procedures.

Note: To install the left side covers, the right side cover must be installed



7-2-1 Using the Arc Cover as Guide Rails

Using the arc covers as guide rails, slide the side cover up to meet the right side cover at the top of the magnet.



7-2-2 Align Tabs at Left and Right Covers

Hold the handles away from the enclosure to ensure that the alignment tabs get properly slotted under the right side's slot.

**Note:**

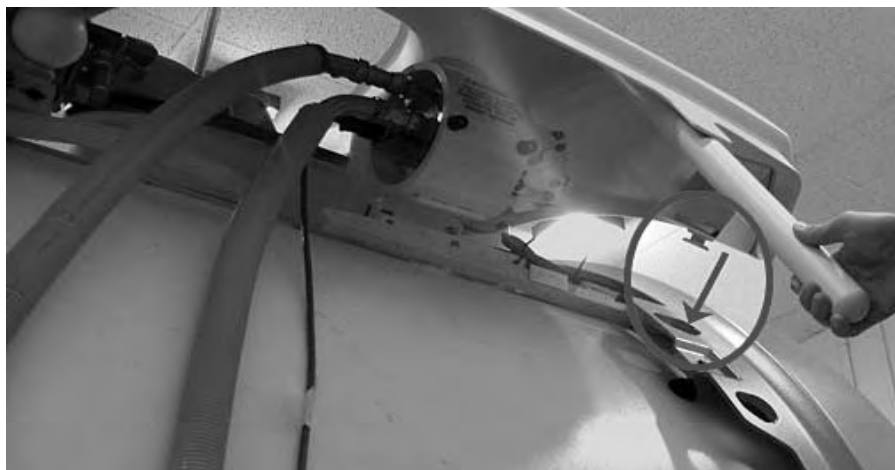
This is not easily accomplished since you cannot see all the alignment tabs while holding on to the cover's handles. Before proceeding to install the lower side cover, go over to the front and rear of the magnet and visually verify that the alignment tabs have been properly inserted and are not protruding over the top of the right side cover.



Make sure tab is properly placed.

7-2-3 Place Arc Cover Foot

Set the round foot on the enclosure side of the cover into the upper oval on the arc cover. Snap into place.

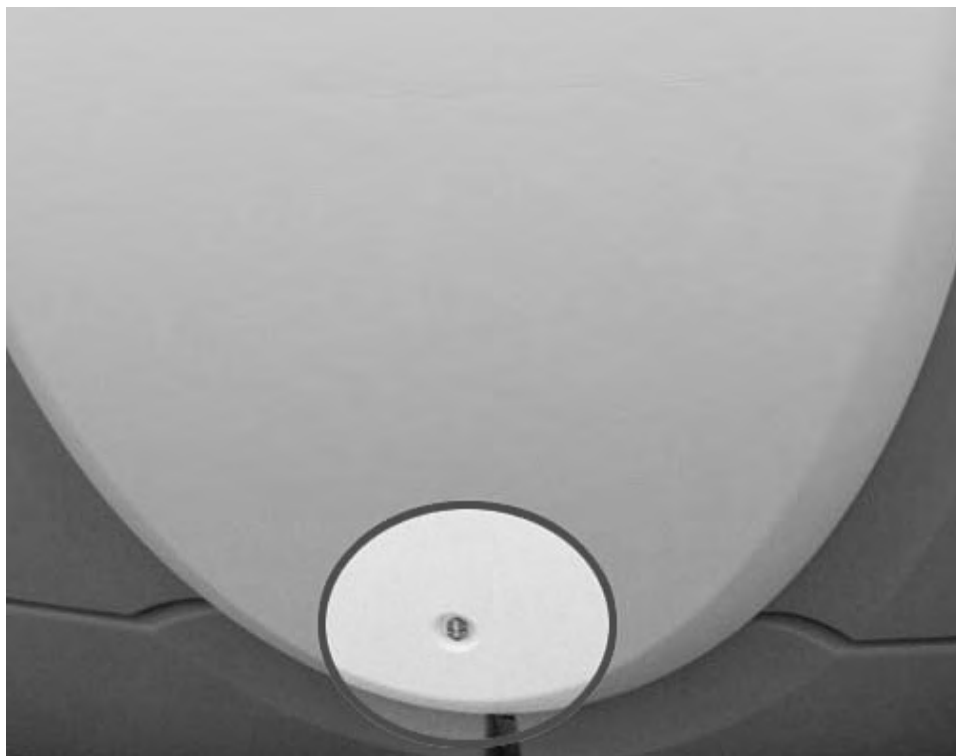
**7-2-4 Installing Left Side Cover**

Hold the sides of the Lower Cover and align the round feet on the enclosure—side of the lower cover with the arc cover's ovals. Snap into place.



7-2-5 Secure Side Cover

Latch the cover into place by tuning the screw clockwise so the latch is secure/

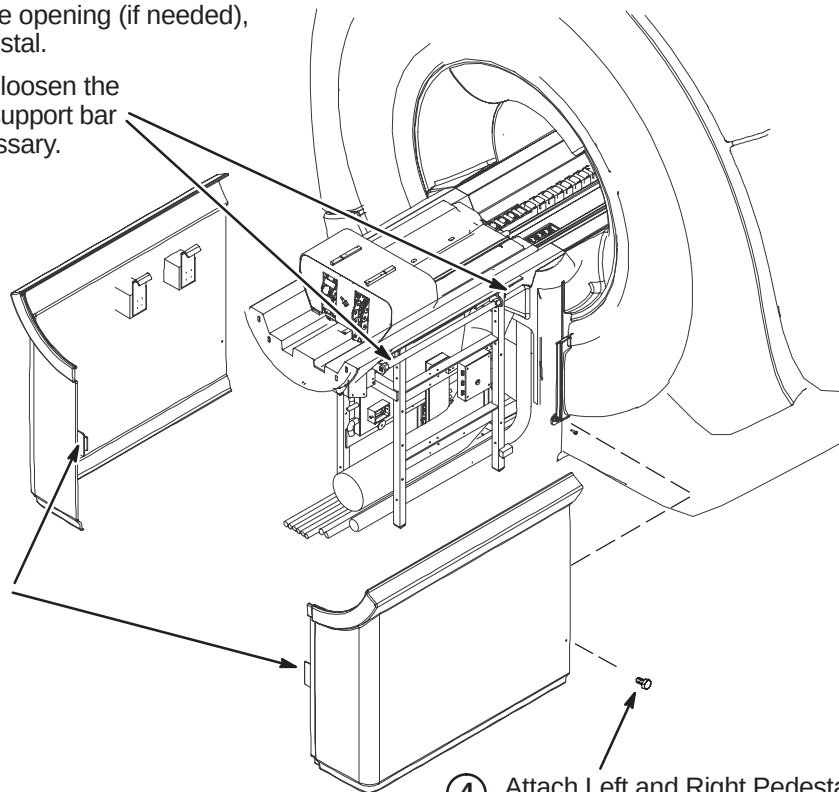


7-3 INSTALL REAR PEDESTAL COVERS

Note: Pedestal Covers may require an opening to be cut out depending on routing of cables and hoses. Covers are marked in three places (left side, right side, or rear) for cutting.

- ① After cutting out appropriate opening (if needed), hang covers on Rear Pedestal.
- ② If covers need adjustment, loosen the screws at each end of the support bar and reposition bar as necessary.

- ③ Snap covers together.



- ④ Attach Left and Right Pedestal Covers with supplied fastener on each cover.

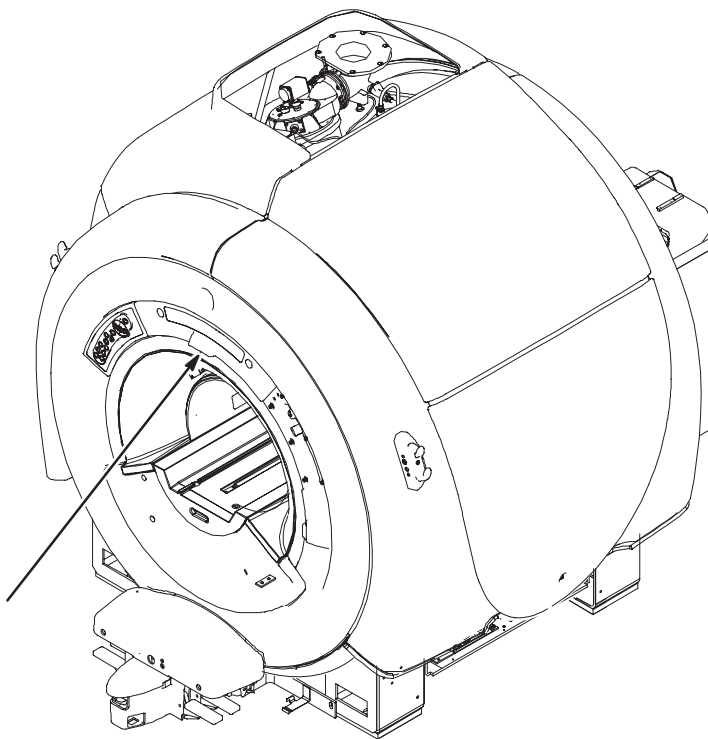
7-4 INSTALL LASER ALIGNMENT LIGHT WARNING LABELS

A sheet of Laser Alignment Light Warning Labels has been supplied with the magnet, attached to the magnet bridge.

The magnet enclosure is shipped with a label in English attached below the laser light.

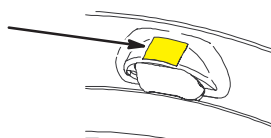


English



- ① For those sites which require a label other than English, peel off the appropriate label and affix over the English label at the laser light location.

- ② If enclosure does not have label, attach new labels as indicated.



French



German



Portuguese



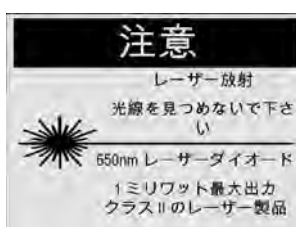
Spanish



Italian



Swedish



Japanese



Chinese

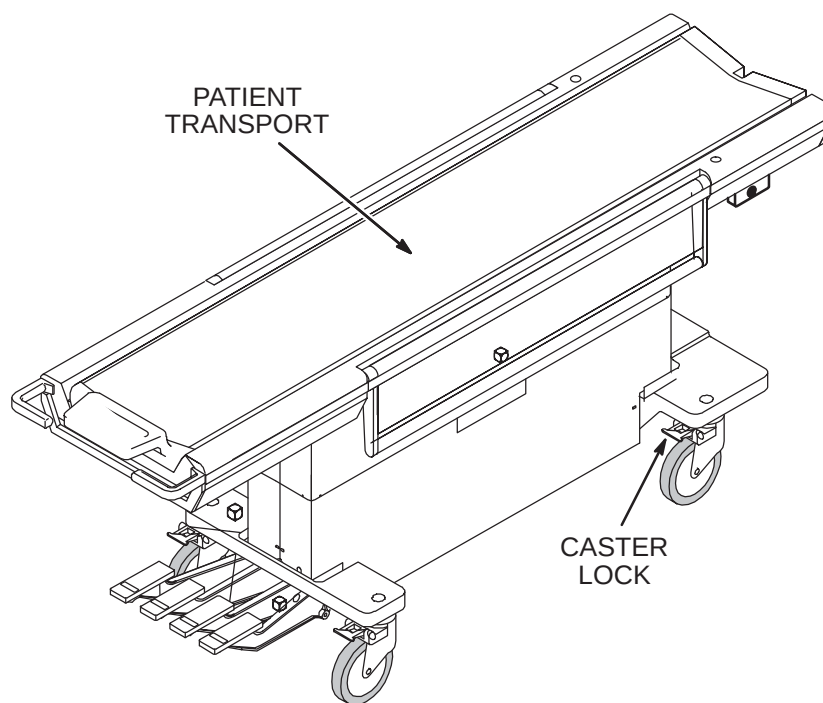
7-5 PATIENT TRANSPORT INSTALLATION

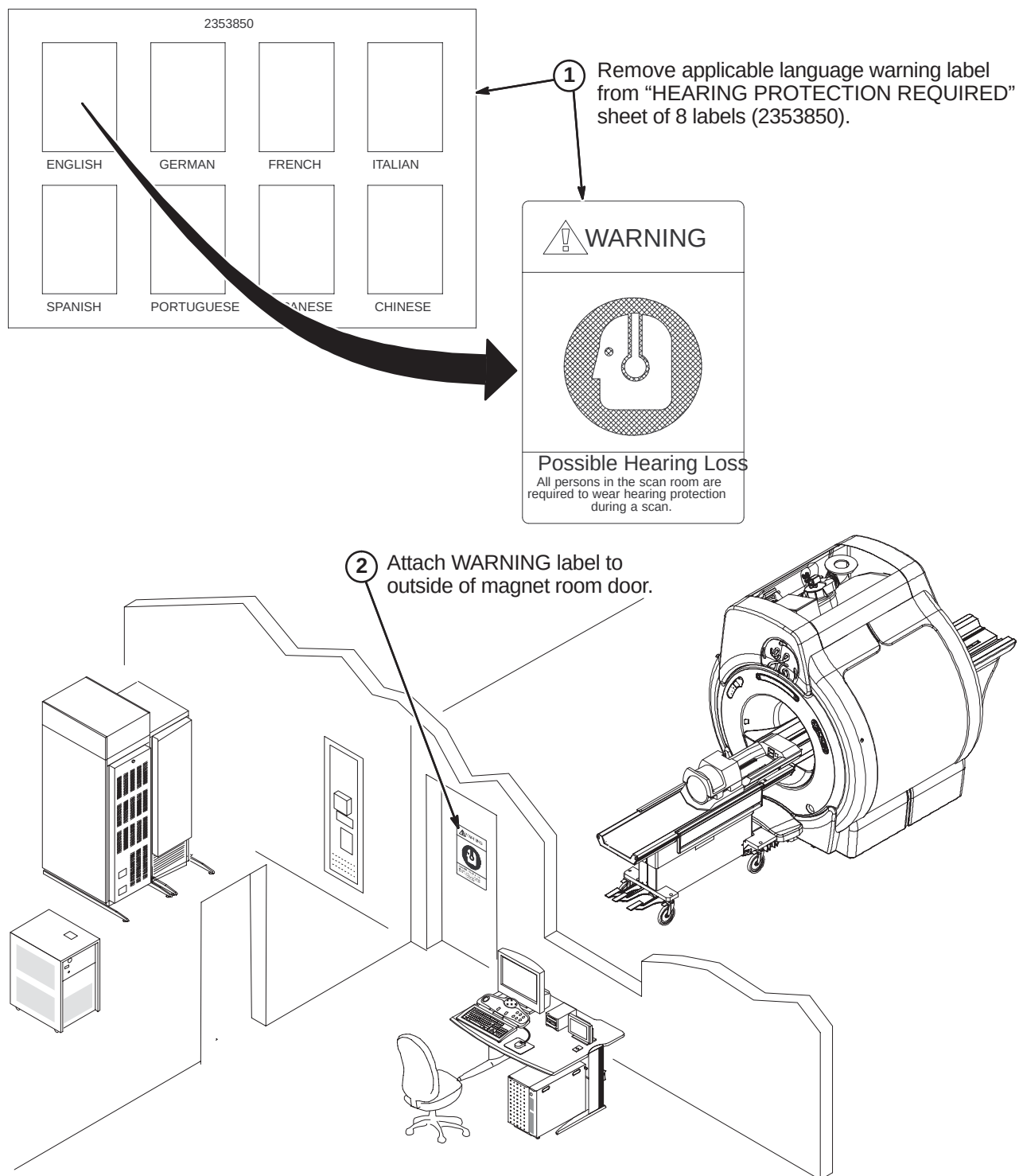
Patient Transport is shipped covered with protective cardboard which should remain in place until ready to be moved into the magnet room. Cover table with plastic previously removed when checks and adjustments are completed.

1. Unlock casters and move Patient Transport to magnet room for adjustments and function checks at the dock.
2. Remove protective plastic sheet from table.
3. Perform mechanical procedures in MR CD-ROM, Set up and Calibration: Patient Transport, LIGHTWEIGHT PATIENT TRANSPORT.

Note: If hardware fastening procedures for installation of the dock and support brackets were followed, the hardware used to fasten the dock and dock supports was left loose. The loose hardware permits a quick and easy dock centering adjustment with table docked.

4. If additional Patient Transports are supplied, refer to *Direction 15477, Second Patient Transport Option Installation*, (delivered with M1020BD option).



7-6 INSTALLATION OF HEARING LOSS WARNING SIGN

SECTION 8 – OPERATOR WORKSPACE

TABLE OF CONTENTS

<u>SECTION</u>	<u>PAGE</u>
8-1 INTRODUCTION	8-2
8-2 OPERATOR WORKSPACE UPGRADE FOR SYSTEMS SHIPPED AFTER JULY 1, 2008	8-3
8-2-1 OPERATOR WORKSPACE TABLE INSTALLATION (If Applicable)	8-3
8-2-2 POSITION NEW GOC COMPUTER CABINET	8-4
8-2-3 SEISMIC ANCHORING (If Required)	8-4
8-2-4 COMPUTER CABINET CABLE ROUTING AND CONNECTIONS	8-5
8-2-5 GOC CABINET COVER REMOVAL	8-6
8-2-6 INSTALL MODEM AND DASM IN GOC CABINET (If Applicable)	8-7
8-2-7 GOC CABLE CONNECTIONS	8-8
8-2-8 POWER CABLE CONNECTIONS	8-9
8-2-9 LINUX PC CABLE CONNECTIONS	8-10
8-2-10 INSTALL TEMPERATURE AND HUMIDITY SENSOR COMPONENTS	8-11
8-2-11 INSTALL MONITOR AND SCIM & KEYBOARD CABLES	8-14
8-2-12 INSTALL SCSI TOWER and MODEM (Option)	8-15
8-2-13 FINAL POSITIONING OF GOC CABINET	8-16
8-2-14 POWER ON INFORMATION	8-16
8-3 OPERATOR WORKSPACE UPGRADE FOR SYSTEMS SHIPPED BEFORE JULY 1, 2008 ..	8-17
8-3-1 OPERATOR WORKSPACE TABLE INSTALLATION (If Applicable)	8-17
8-3-2 POSITION NEW GOC COMPUTER CABINET	8-18
8-3-3 SEISMIC ANCHORING (If Required)	8-18
8-3-4 COMPUTER CABINET CABLE ROUTING AND CONNECTIONS	8-19
8-3-5 GOC CABINET COVER REMOVAL	8-20
8-3-6 INSTALL MODEM AND DASM IN GOC CABINET (If Applicable)	8-20
8-3-7 ROUTE CABLES FOR LOWER OPENING	8-21
8-3-8 POWER STRIP CONNECTIONS	8-21
8-3-9 LINUX PC CABLE CONNECTIONS	8-22
8-3-10 TEMPERATURE AND HUMIDITY SENSOR KIT INSTALLATION	8-23
8-3-11 INSTALL MONITOR AND SCIM & KEYBOARD CABLES	8-26
8-3-12 INSTALL SCIM/KEYBOARD	8-26
8-3-13 INSTALL SCSI TOWER	8-27
8-3-14 FINAL POSITIONING OF GOC CABINET	8-28
8-3-15 POWER ON INFORMATION	8-28

8-1 INTRODUCTION



POWER TO SYSTEM MUST BE OFF, LOCKED AND TAGGED, BEFORE STARTING ANY PROCEDURES.

Beginning July 1, 2008, a new GOC will be delivered to site. Instructions for installing the existing version of the GOC along with instructions for installing the new GOC are included in this section.

Select the applicable sub-section for the GOC delivered to the site being upgraded.

- Section 8-2, **OPERATOR WORKSPACE UPGRADE FOR SYSTEMS SHIPPED AFTER JULY 1, 2008**
or
- Section 8-3, **OPERATOR WORKSPACE UPGRADE FOR SYSTEMS SHIPPED BEFORE JULY 1, 2008**

8-2 OPERATOR WORKSPACE UPGRADE FOR SYSTEMS SHIPPED AFTER JULY 1, 2008

8-2-1 OPERATOR WORKSPACE TABLE INSTALLATION (If Needed)

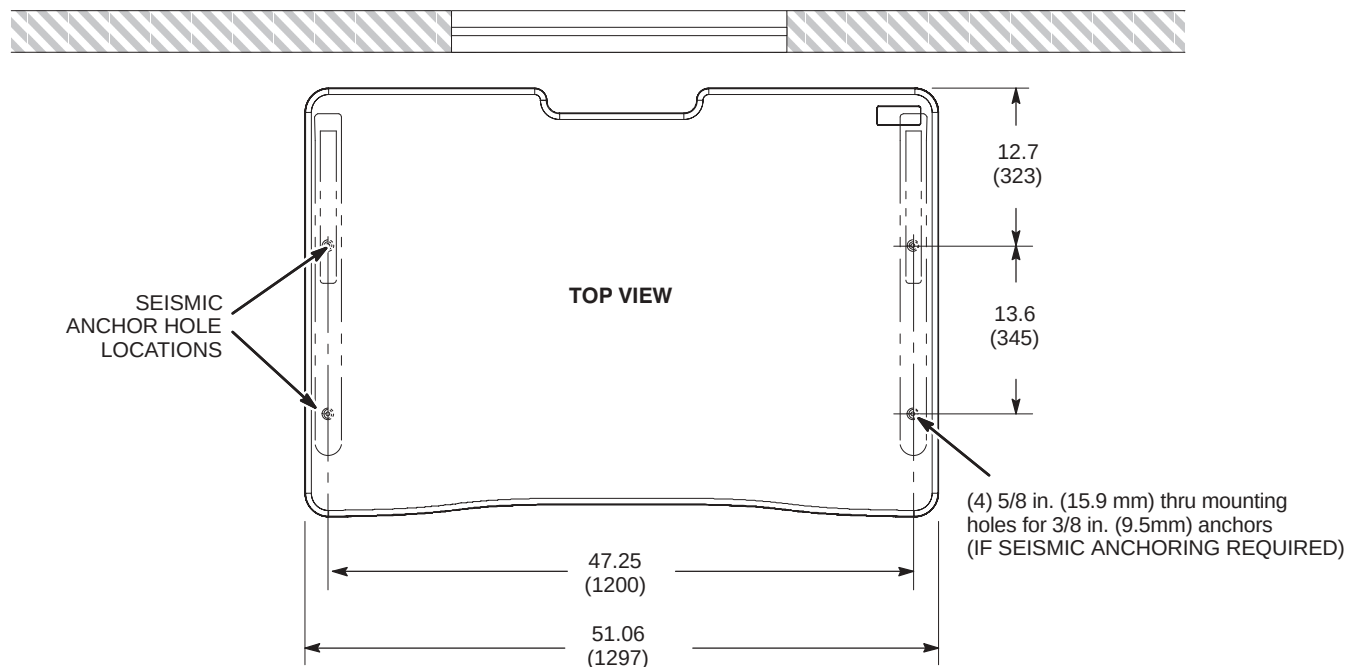
Refer to room layout plan and consult with Site customer for exact location of Operator Workspace Table.

1. Open palletized M1000MW box that contains the Operator Workspace table. Use furnished *Operator's Workspace Table Assembly Instructions* (5189137-50) to assemble the model 5189137 table.
2. Move table into designated position. Refer to room layout plan and/or consult with Site customer for exact location of Operator Workspace Table. Typical location with table in front of Magnet Room viewing window is shown below.

Note

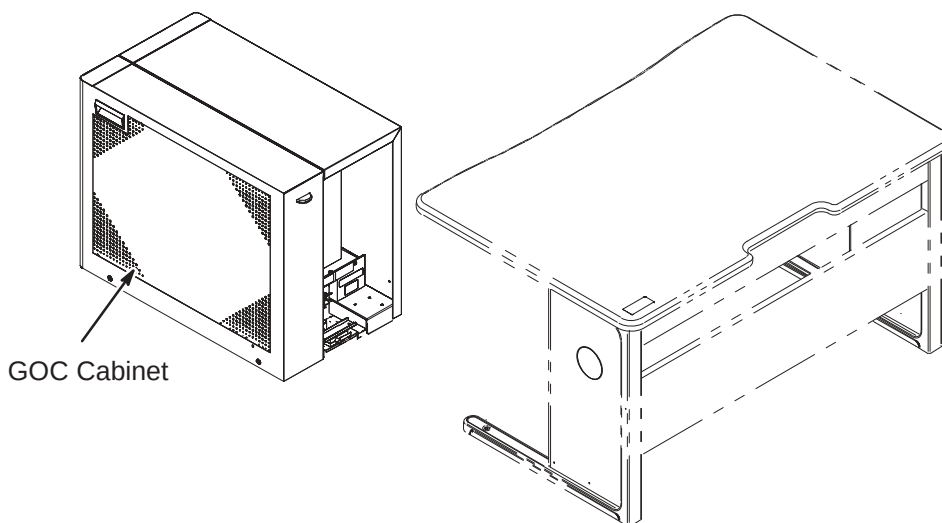
The design of the new model 5189137 table does not require anti-tip anchors. This table meets the ANSA/BIFMA anti-tip standards.

3. If seismic anchoring is required, refer to site architectural drawings and procure and install specified anchors.



8-2-2 POSITION COMPUTER CABINET

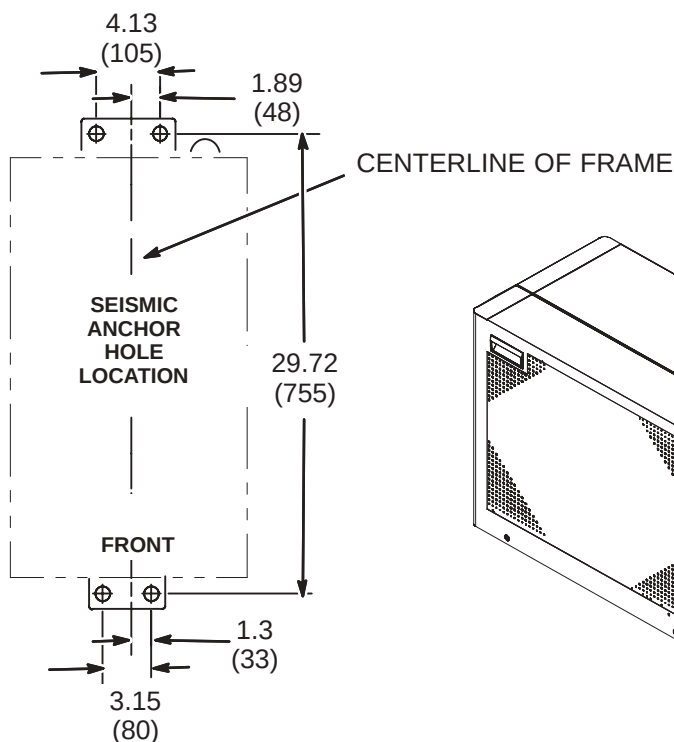
Consult with Site customer for exact location of Global Operator Console (GOC). The location could be underneath the left or right side of the Operator Workspace table, or on the outside left or right side of the table.



8-2-3 SEISMIC ANCHORING (IF REQUIRED)

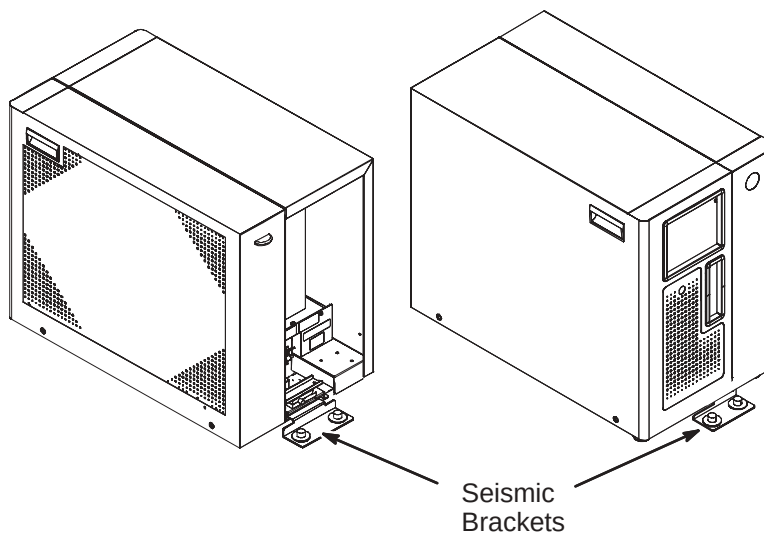
GOC Seismic Mount kit is shipped with GOC. Find kit inside of plastic bag.

If seismic anchoring is required, refer to site architectural drawings for fastener type details or applicable Pre-installation manual. Illustration below indicates hole location for installing brackets.



NOTE:

- ALL DIMENSIONS ARE IN INCHES. ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.



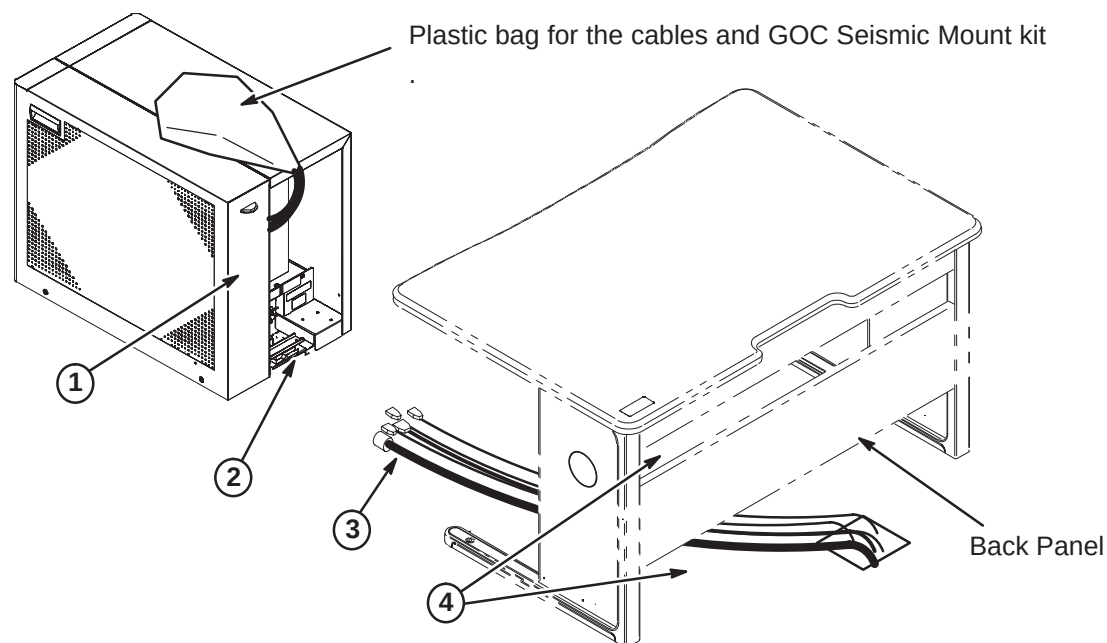
8-2-4 COMPUTER CABINET CABLE ROUTING AND CONNECTIONS

The GOC (Computer) Cabinet is shipped with cables connected to the computer that will be routed and connected to equipment that will be installed on the top of the OW Table. The cables are:

- Two power cords
- One video output cable
- Keyboard output cable
- SCIM output cable
- SCSI output cable
- Modem output cable

These cables are already connected to the computer and routed thru the opening located at the top of the rear of the GOC cabinet as shown in the illustration below. The remainder of the cables to be connected to designated modules within the GOC Cabinet will have to be routed thru the lower opening.

1. Runs 1064, 1070, E0015, E3045, E3046, E3047 and E3048 are pre-connected to computer and are routed at top of opening.
(These cables are inside of plastic bag)
2. Runs 1381, 1085, 1255, 1371, power cord for the Patient Alert system, and any other option cables will be routed thru lower opening.
3. Route all cables before connecting any cables. Refer to cable map for connection information.
4. If GOC Cabinet is to be located under OW Table, the cable coming out of the top opening of the cabinet will be routed above the Back Panel. The cables coming out of the lower opening will be routed below the Back Panel.



8-2-5 GOC CABINET COVER REMOVAL

The Computer Cabinet covers need to be removed to access the cable connection points.

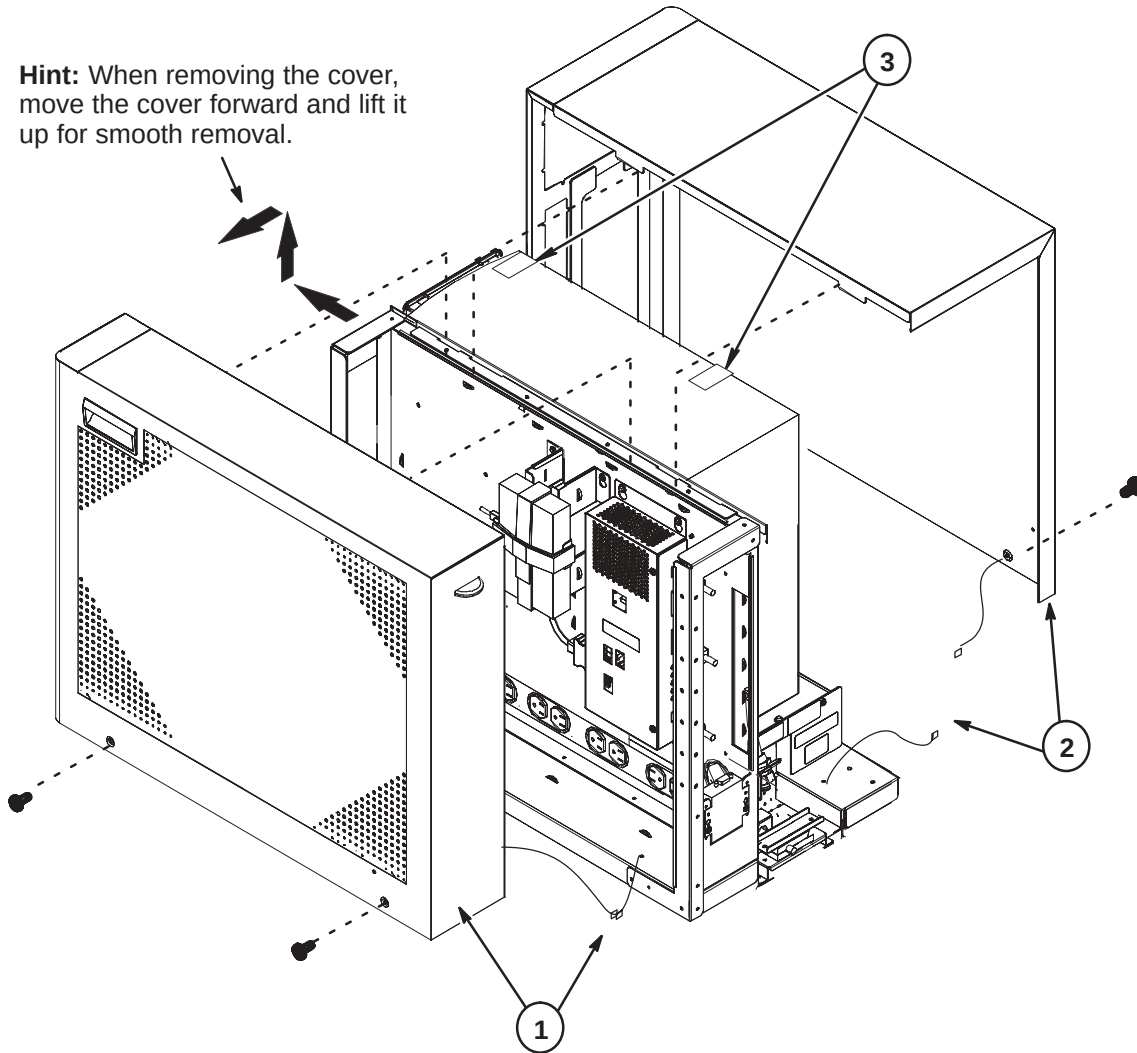
1. For right side cover, remove two screws and lift up. Disconnect Ground cable connector and remove the cover.
2. For left side cover, remove two screws and lift up. Disconnect Ground cable connector and remove the cover.

Note

A ground wire is connected to each side cover. Use care when removing side covers.

3. Peel the two shipping tapes off from PC.

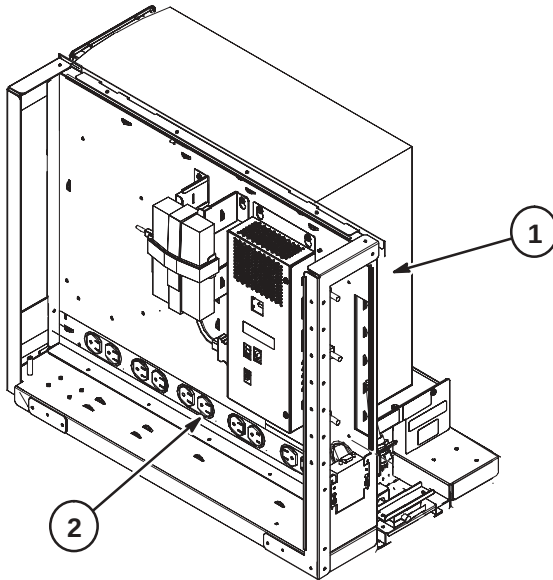
Hint: When removing the cover, move the cover forward and lift it up for smooth removal.



8-2-6 INSTALL MODEM CABLE TO GOC CABINET (If Applicable)

If Modem is part of system installation or upgrade, it will be placed on Operator Table or GOC.

1. Modem cable to the Linux PC is provided. Connect Run E3045 to Modem.
2. Modem power cord must be connected to J7 on GOC PDU Module. Route this cable to the rear top of the GOC Cabinet.



8-2-7 GOC CABLE CONNECTIONS

Refer to cable map for information on cables to be connected thru lower area of rear opening.

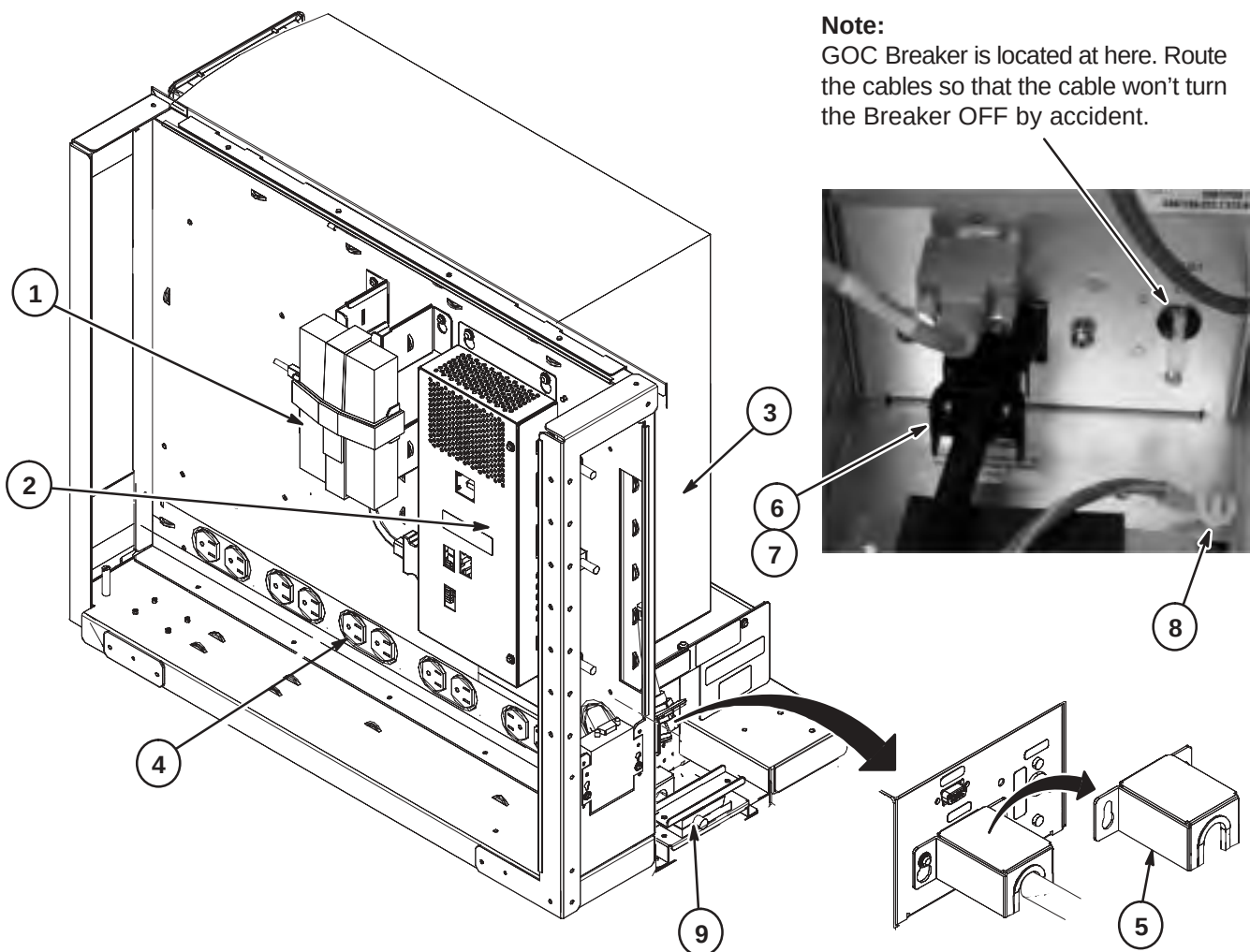
1. Route and connect Ethernet cables to Lan Hub.

Note Use ports 3, 7 or 8 for connecting site ethernet cables such as AW. Don not use any other ports for site ethernet cables.

2. Connect Run 1085 to J7 and Run 1255 to J18 on GOC AA.
3. Refer to Section 8-2-9 for all cables connected to Linux PC computer.
4. Refer to Section 8-2-8 for power cable connections to GOC PDU. Make sure Patient Alert power is connected to J3. If Modem is installed, power cable is connected to J7.
5. Before connecting GOC power cable 1381 to J1, loosen screws and remove cover.
6. Route and connect GOC power cable Run 1381 to J1 on PDU.
7. Confirm that GOC Power Cable is firmly connected to J1 by pulling the cable once. Re-install cover removed in Step 5 after connecting cable and tighten screws.
8. Connect Run 1382 to GND. Make sure nut securing GND cable is firmly tightened to GND stud.
9. Tighten clamp to secure cables in place.

Note:

GOC Breaker is located at here. Route the cables so that the cable won't turn the Breaker OFF by accident.

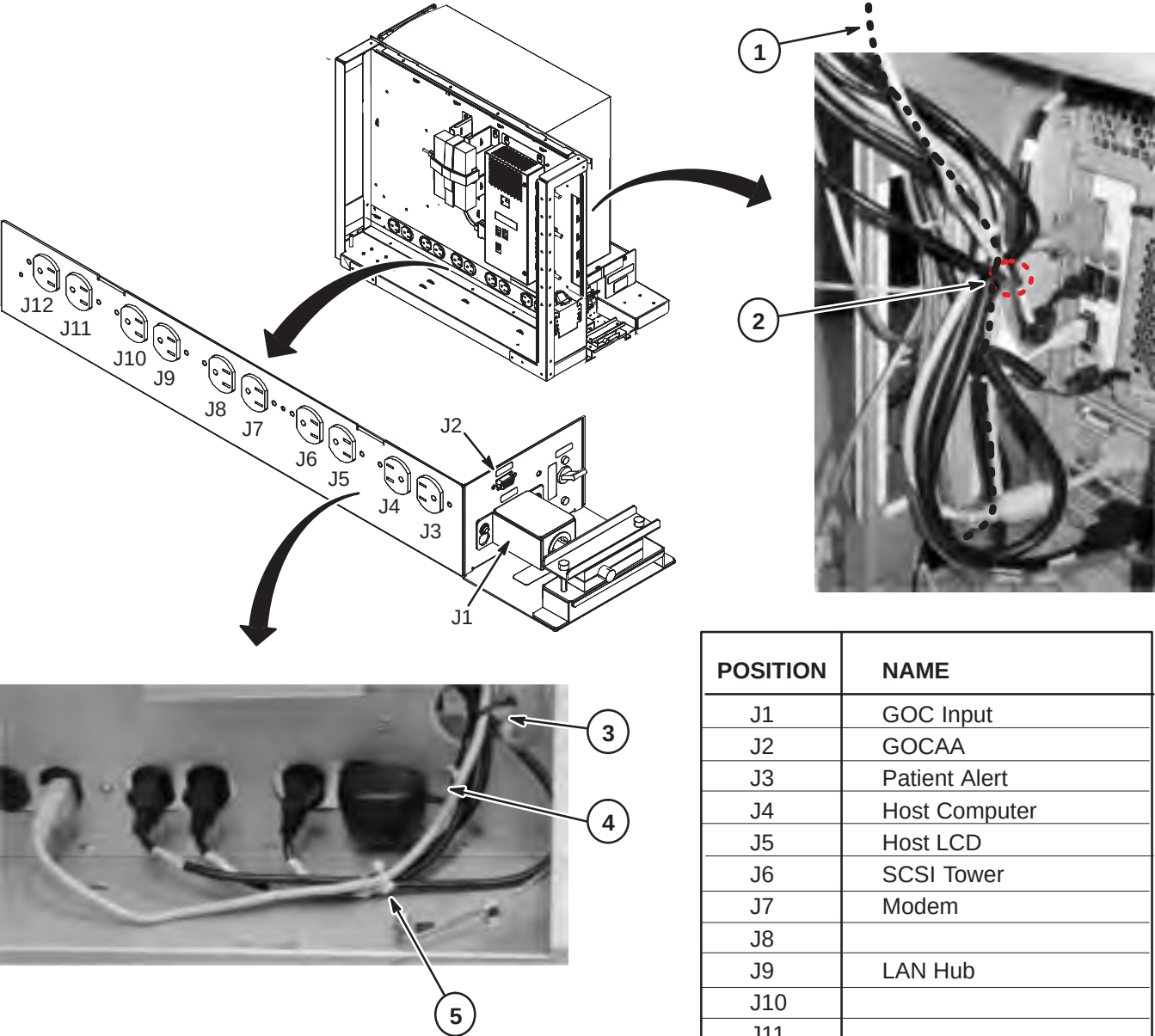


8-2-8 POWER CABLE CONNECTIONS

The illustration below show the power cable connection locations to the power strip in the GOC cabinet. Make sure power cables are attached as shown.

Also refer to cable map for information on power cable conections to OW1-PDM.

1. If Modem is installed, route power cable and connect to J7.
2. Tie-wrap point for Modem power cable.
3. Route power cables connected to J3 thru J7 through access hole.
4. route and connect Patient Alert power cable to J3 and tie-wrap at this point.
5. Tie-wrap power cables J4 to J7 at this point.

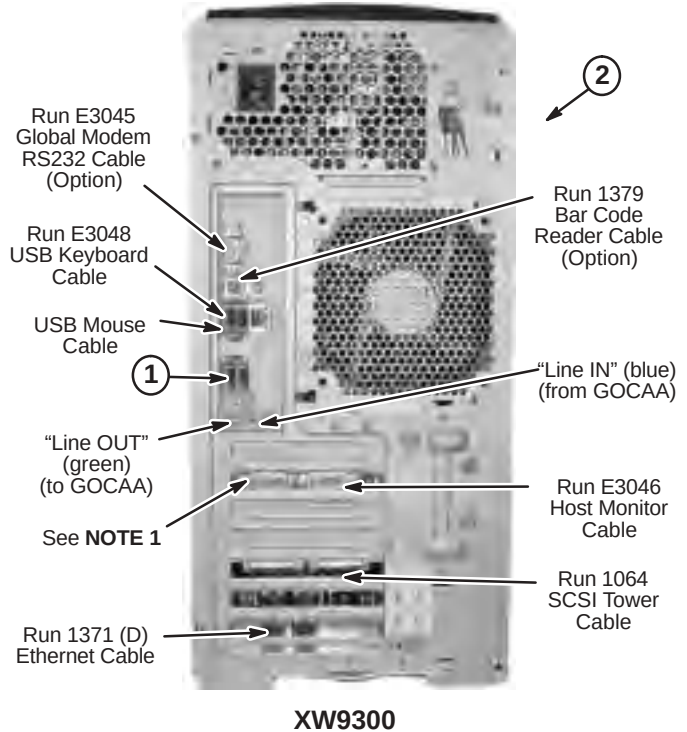
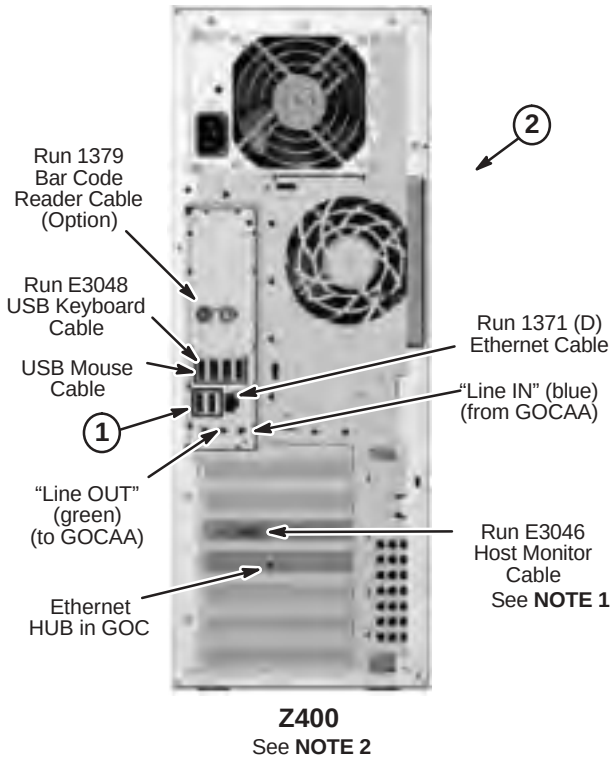


POSITION	NAME
J1	GOC Input
J2	GOCAA
J3	Patient Alert
J4	Host Computer
J5	Host LCD
J6	SCSI Tower
J7	Modem
J8	
J9	LAN Hub
J10	
J11	
J12	One Wire Hub

8-2-9 LINUX PC CABLE CONNECTIONS

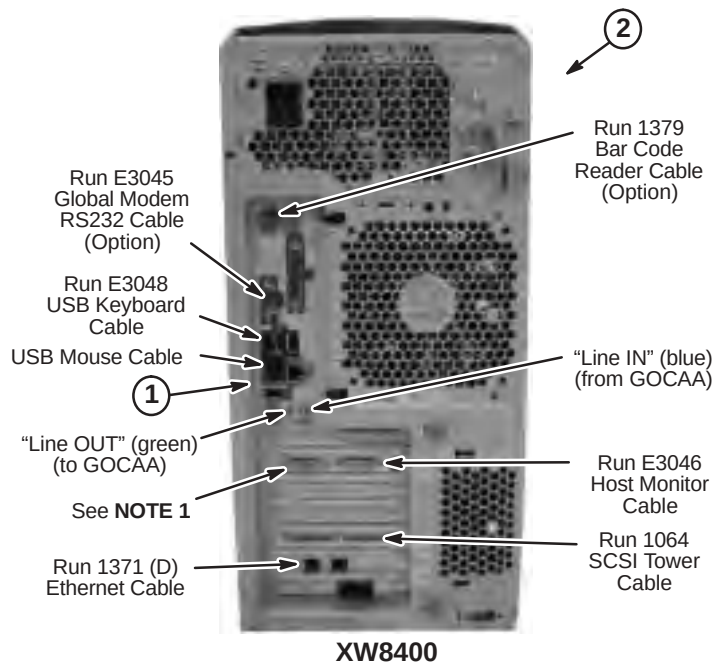
The connection points of cables routed to the Linux PC are described in the following two sections. All cables shown are preconnected except for Run 1371 and USB cable for USB–Serial DB9 Adaptor which has to be routed and connected at site. Refer to the applicable PC for your site.

1. Refer to Section 8-2-10 for installing the USB to 1–Wire Adapter (5307784) to USB port.
2. Connect remaining cables as marked at rear of computer.



NOTE 1:
Do not connect any additional monitors prior to loading operating system software. Doing so will cause boot-up to stop.

NOTE 2:
SCSI Tower is not supported for Z400PC.

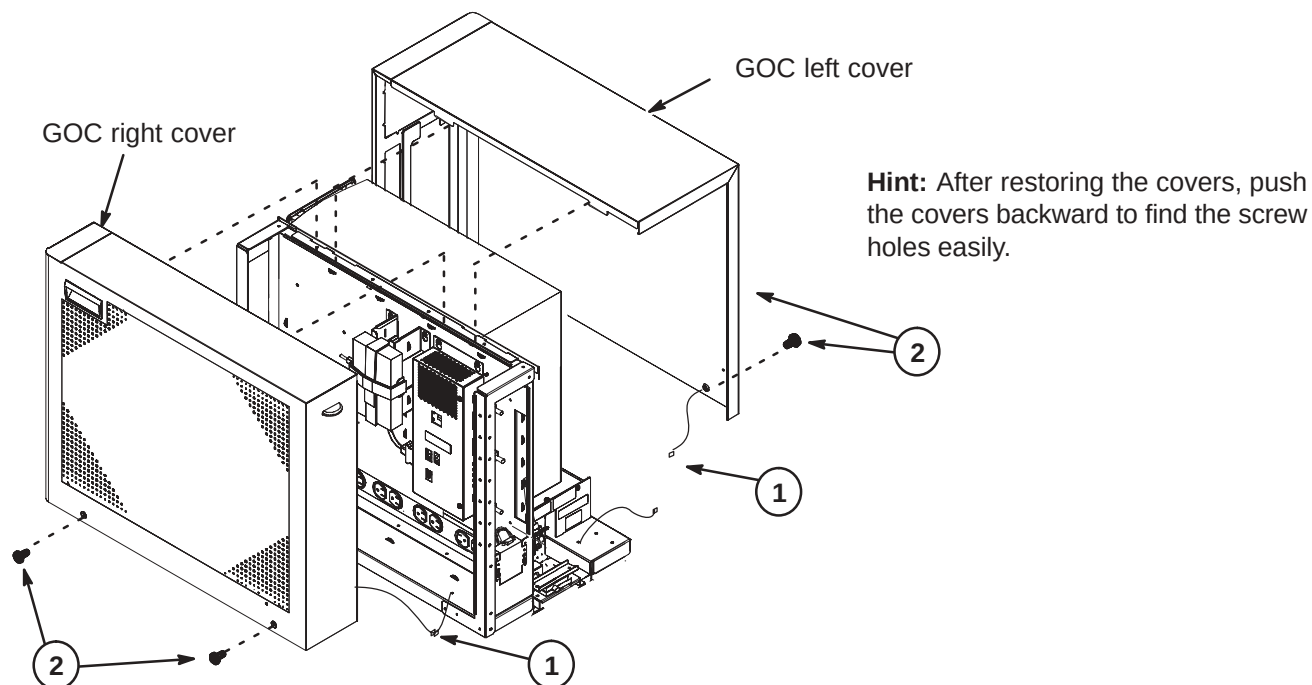


8-2-10 INSTALL TEMPERATURE AND HUMIDITY SENSOR COMPONENTS

The components of the Temperature and Humidity Sensor Kit (5147136-2) will be installed on the back of the GOC cabinet and on the back of the System Cabinet with one cable routed from the GOC and connected to it.

8-2-10-1 RESTORE GOC COVERS

1. Connect Ground cables and install both side covers by inserting tabs at top of covers into slots.
2. Install screws that were removed in earlier procedure

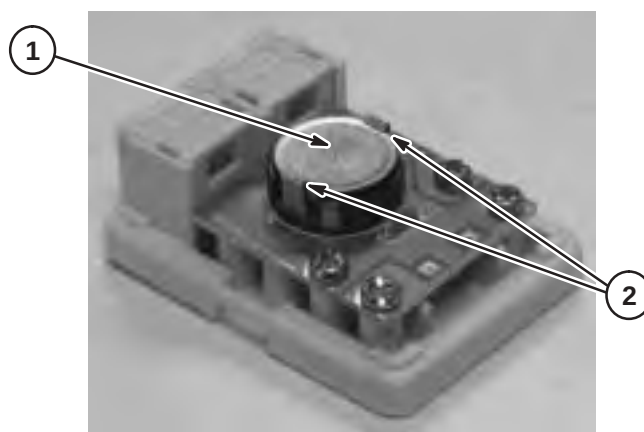
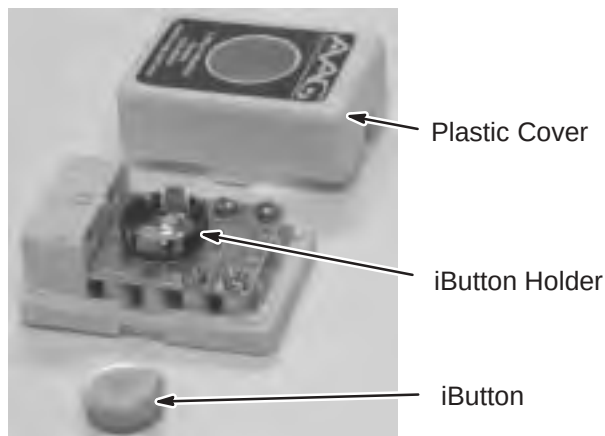


8-2-10-2 INSTALL iBUTTON INTO HOLDER

Install the iButton into the holder in the orientation shown in the illustration.

Note: The pictures below show the plastic cover removed although this is not necessary to install the iButton.

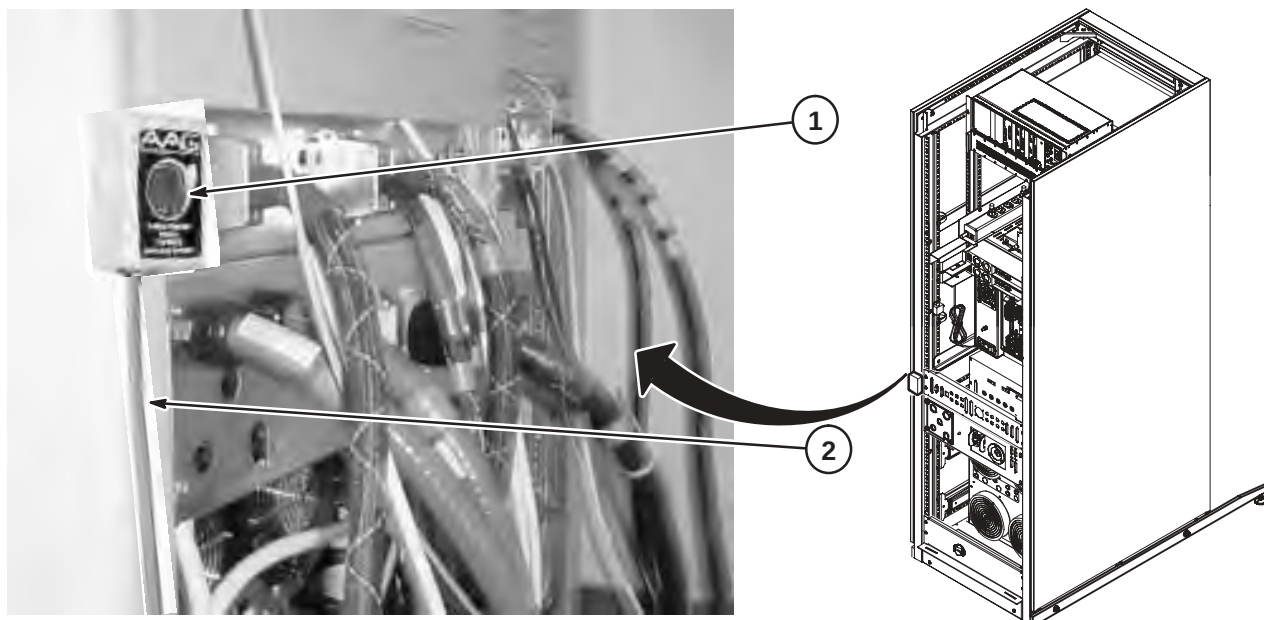
1. Install the iButton in the holder so that the solid part of the case is exposed.
2. Make sure the “fingers” latch over the top of the iButton and secure it in the holder.
3. If removed, reinstall the Plastic Cover.



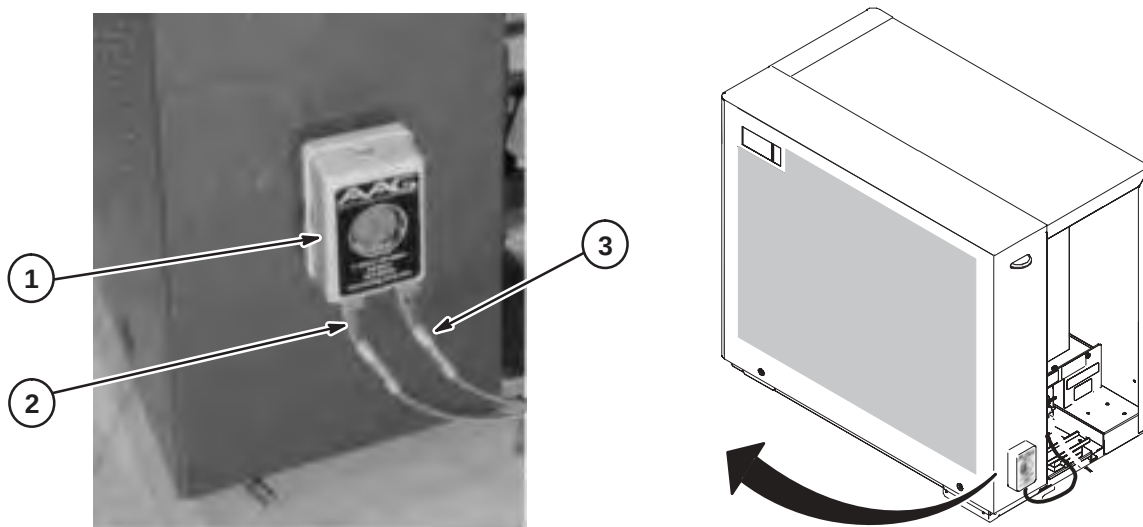
8-2-10-3 INSTALL TEMPERATURE & HUMIDITY SENSOR ON SYSTEM CABINET

Note: iButton Holders are unique to each location. Make sure to install 5141857-4 to System Cabinet and 5141857-5 to GOC.

1. Attach iButton Sensor assembly (Holder, 5141856, and Humidity iButton (5141857-4), to back of System/RFS Cabinet at this location using Dual Locking Velcro Tape (2341779 & 2341779-2).
2. Connect cable Run 1384 (5143794-2) to Sensor. Route and to iButton at rear of GOC.

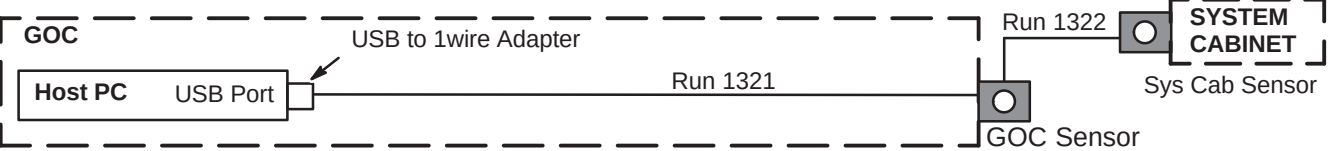
**8-2-10-4 INSTALL TEMPERATURE & HUMIDITY SENSOR ON GOC**

1. Attach iButton Sensor assembly (Holder, 5141856, and iButton 5141857-5), to back of GOC Cabinet at this location using Dual Locking Velcro Tape (2341779 & 2341779-2).
2. Route and connect Run 1322 (5143794-2) from System cabinet to Sensor at back of GOC.
3. Connect cable Run 1383 (5143792-3) from Sensor to USB to 1-Wire Adapter (5307784) installed in any open USB port at rear of PC. Refer to step 1 in Section 8-2-9.

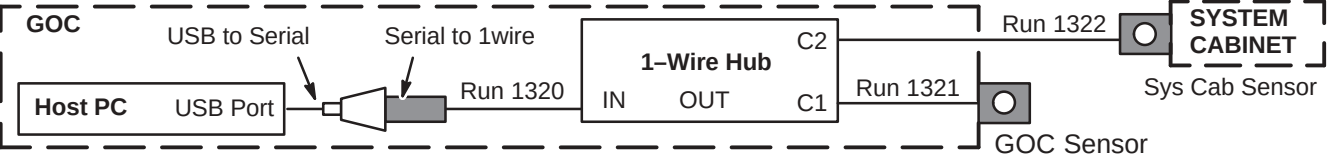


8-2-10-5 TEMPERATURE & HUMIDITY SENSOR WIRING SCHEMATICS

Environmental Sensors Block Diagram (Shipped after July 1, 2008)



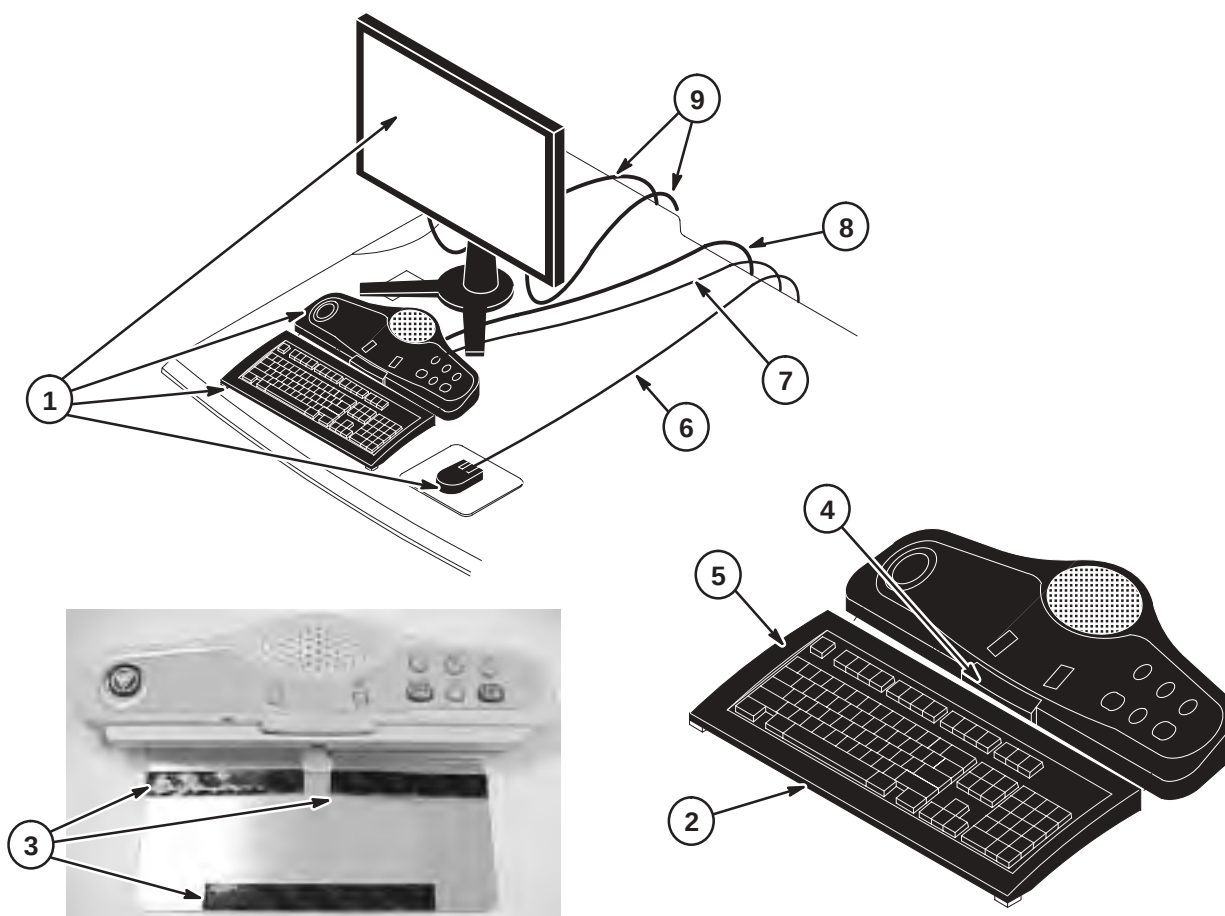
Environmental Sensors Block Diagram (Shipped before July 1, 2008)



8-2-11 INSTALL MONITOR AND SCIM & KEYBOARD CABLES

Consult with Site customer for exact location of SCIM/Keyboard and Monitor on Operator Workspace (OW) Table.

1. Position Host Display Monitor, SCIM, Mouse, and keyboard on table.
2. Place keyboard in front of SCIM, pull the keyboard forward, flip it, and place on top of the SCIM up side down.
3. Remove the plastic strips from the velcro located on SCIM plate, exposing the adhesive surface (see illustration on next page).
4. Guide the cable back into the center notched opening until all the cable is under the SCIM base.
5. Place the keyboard tight against the SCIM, Center it as best as possible, and drop into place. The Velcro pads will now be sticking to the bottom of the keyboard.
6. Connect Mouse cable to USB port at rear of computer in GOC.
7. Connect Run E3048 to USB Keyboard cable. Route and connect other end to computer in GOC.
8. Connect Run E3047 to SCIM. Route and connect other end to computer in GOC.
9. Route power cord Run E0015 and Run E3046 from GOC and connect to Display Monitor.

**Note**

The adhesive on the velcro takes 24 hours to completely adhere to the bottom of the keyboard. Allow the adhesive to dry completely before removing keyboard from SCIM base.

8-2-12 INSTALL SCSI TOWER and MODEM (Option)

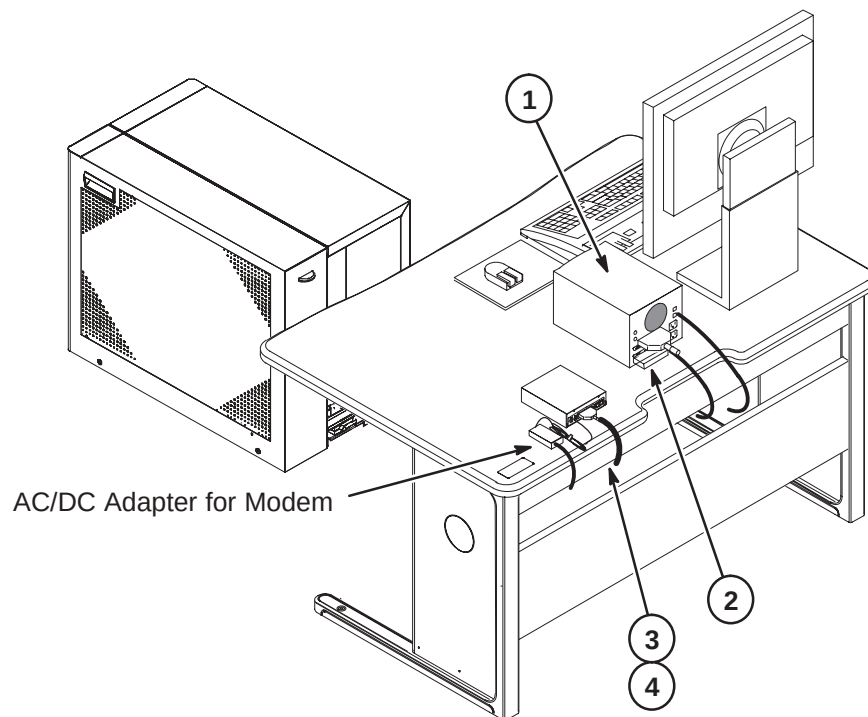
Note: SCSI Tower is an option with release 15.x or HDxt.

Note: SCSI Tower and Modem are not compatible with Z400 computer. The Z400 computer will be delivered and installed at sites around the 4th quarter of 2010.

1. Place the SCSI Tower on the OW table and connect Runs 1064 and 1070 from GOC Cabinet.
2. Install terminator on back of SCSI Tower.

The following steps are for the Modem option:

3. If applicable, place the Modem on the OW table and connect Runs E3045 and power cable from GOC Cabinet.
4. Connect customer supplied telephone line.



8-2-13 FINAL POSITIONING OF GOC CABINET

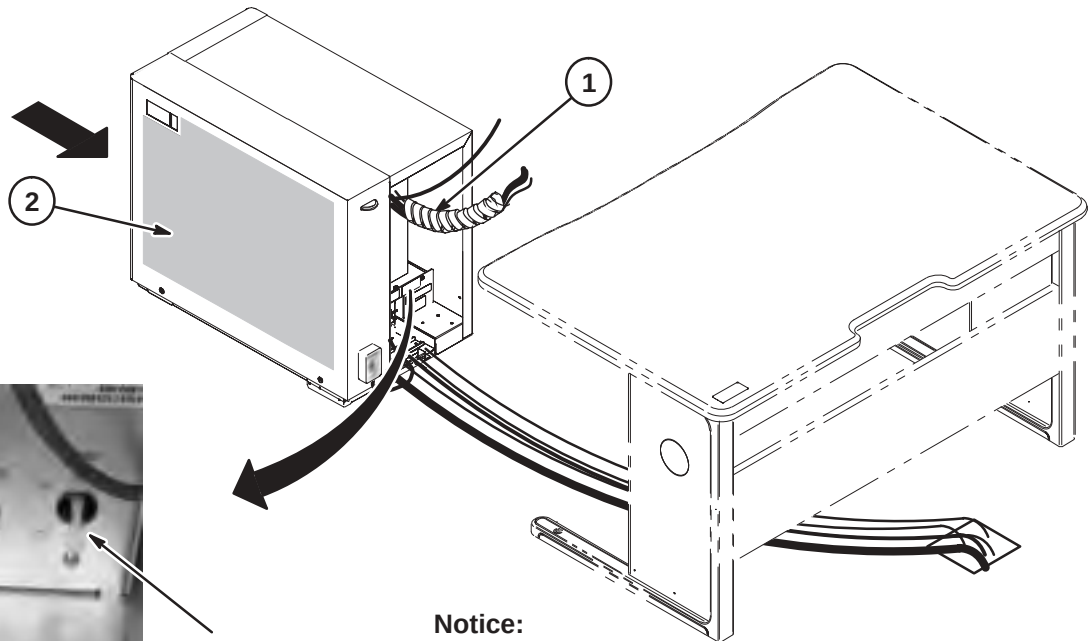
1. Use the spiral tube to bind the following cables. (Length of spiral tube is about 1 m) Spiral tube is included in the plastic bag which is shipped with GOC.

Notice

Do Not include Modem power cable in Spiral Tube.

- Three power cords (LCD Monitor ,SCSI Tower(if applicable))
- One video cable
- Keyboard cable
- SCIM cable
- SCSI Tower cable (if applicable)
- Modem cable (if applicable)

2. Place cabinet in final position as determined by customer.

**Notice:**

GOC circuit breaker location. To avoid turning the Breaker OFF by accident, do not place the cables inside of GOC Cabinet near the circuit breaker.

8-2-14 POWER ON INFORMATION**IMPORTANT**

When turning on the circuit breaker in the PDU, both the PC and Host circuit breakers must be energized.

Note

When powering up the GOC, there is a Main breaker located in the rear of the GOC.

8-3 OPERATOR WORKSPACE UPGRADE FOR SYSTEMS SHIPPED BEFORE JULY 1, 2008

8-3-1 OPERATOR WORKSPACE TABLE INSTALLATION (If Needed)

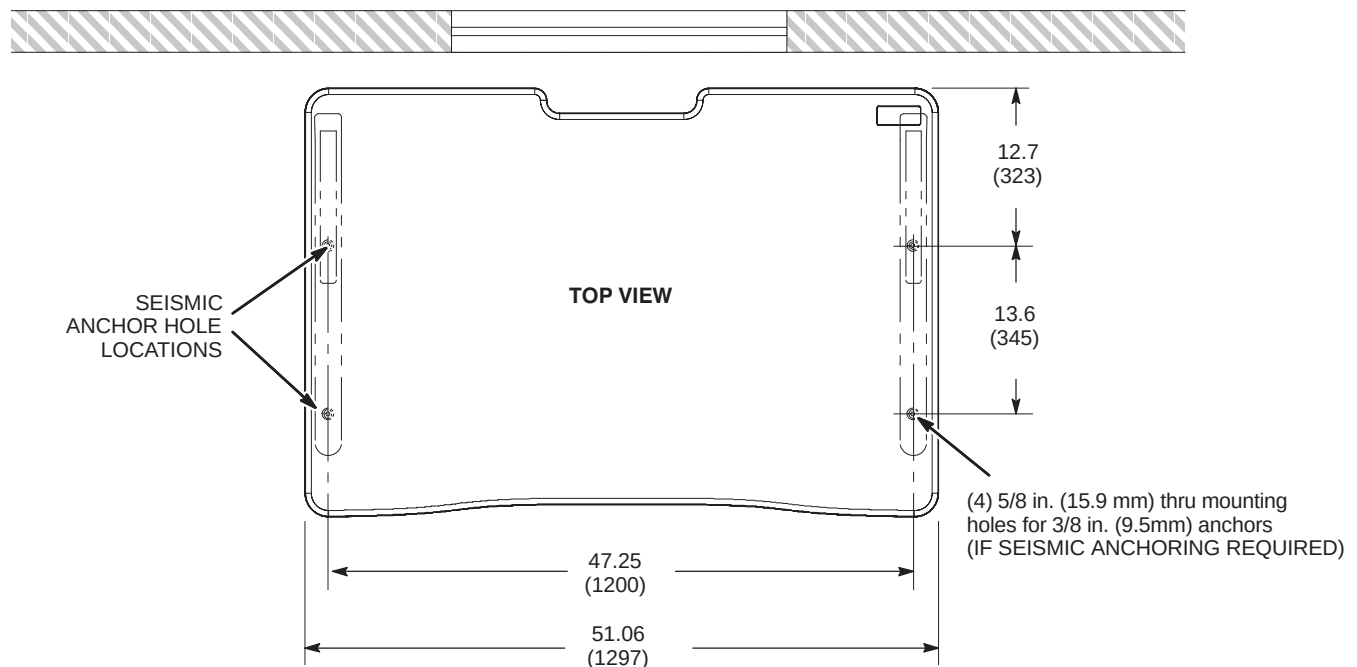
Refer to room layout plan and consult with Site customer for exact location of Operator Workspace Table.

1. Open palletized M1000MW box that contains the Operator Workspace table. Use furnished *Operator's Workspace Table Assembly Instructions* (5189137-50) to assemble the model 5189137 table.
2. Move table into designated position. Refer to room layout plan and/or consult with Site customer for exact location of Operator Workspace Table. Typical location with table in front of Magnet Room viewing window is shown below.

Note

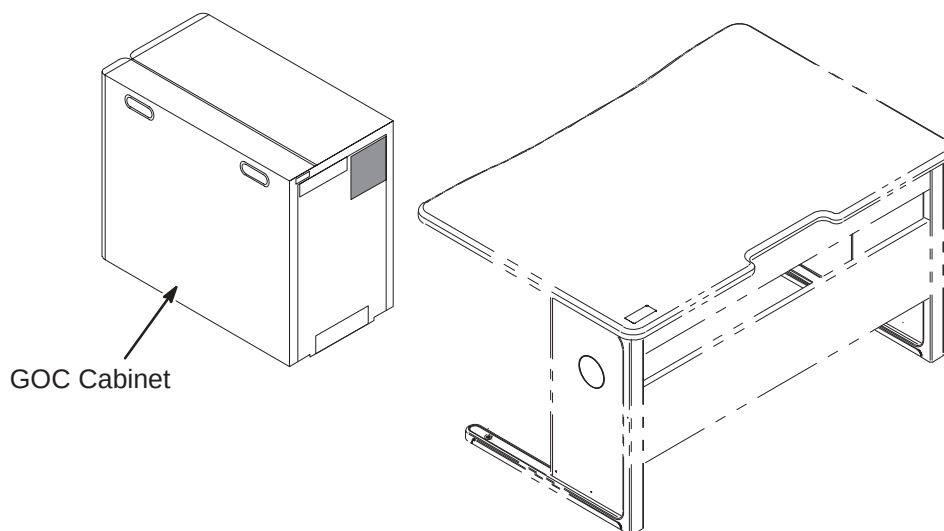
The design of the new model 5191031 table does not require anti-tip anchors. This table meets the ANSA/BIFMA anti-tip standards.

3. If seismic anchoring is required, refer to site architectural drawings and procure and install specified anchors.



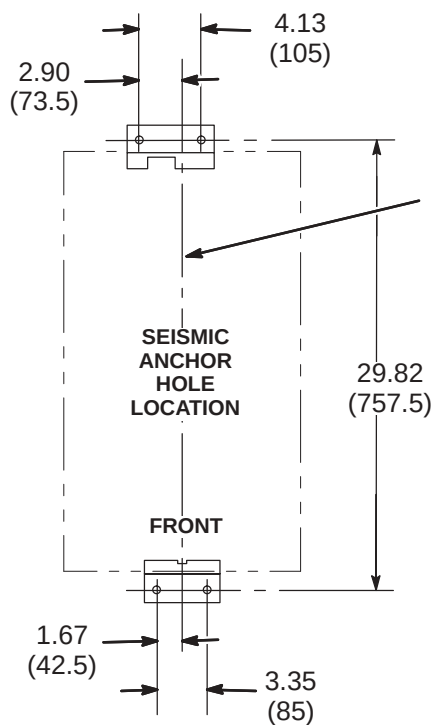
8-3-2 POSITION NEW GOC COMPUTER CABINET

Consult with Site customer for exact location of Global Operator Cabinet (GOC). The location could be underneath the left or right side of the Operator Workspace table, or on the outside left or right side of the table.



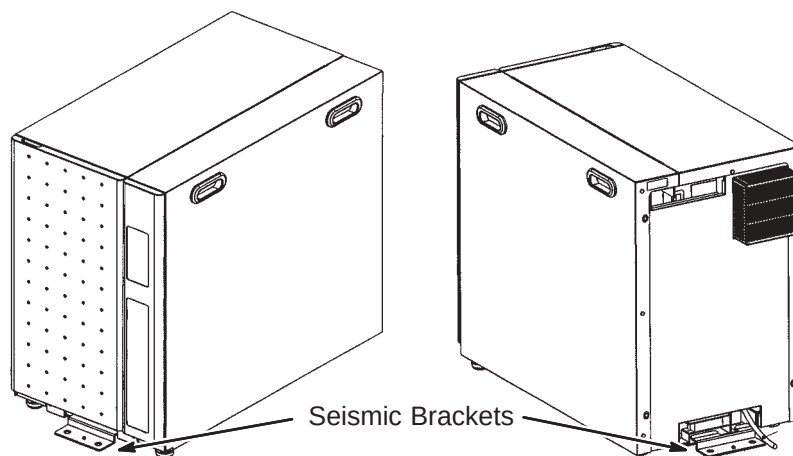
8-3-3 SEISMIC ANCHORING (IF REQUIRED)

If seismic anchoring is required, refer to site architectural drawings for fastener type details or applicable Pre-installation manual. Illustration below indicates hole location for installing brackets.



NOTE:

- ALL DIMENSIONS ARE IN INCHES. ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.



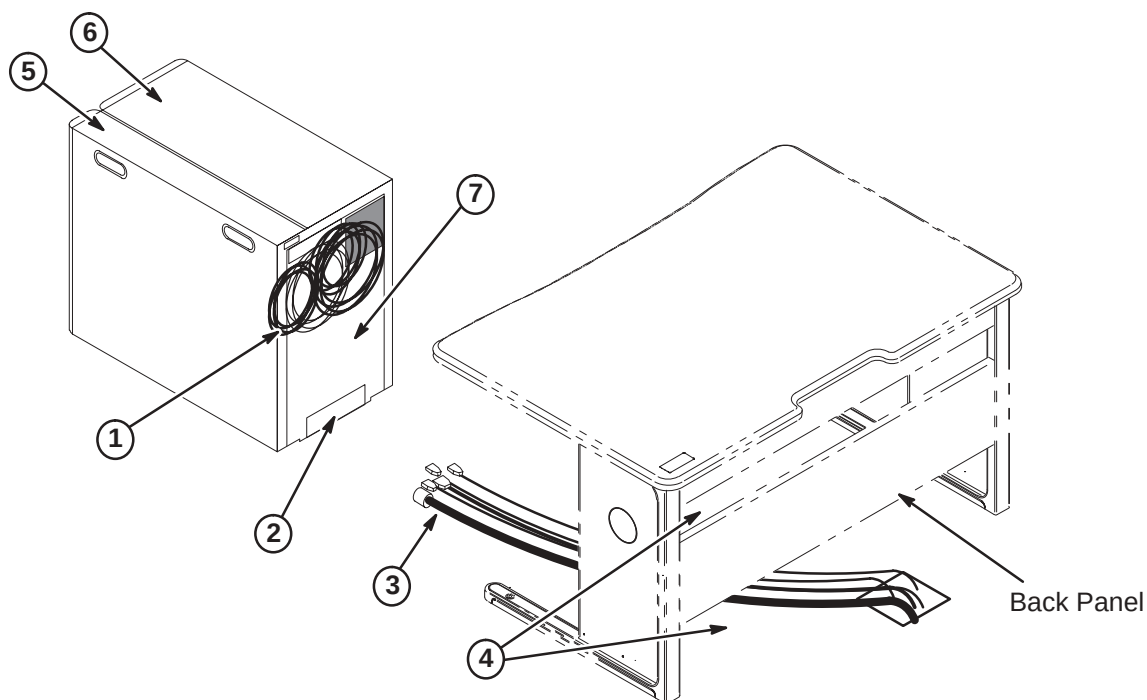
8-3-4 COMPUTER CABINET CABLE ROUTING AND CONNECTIONS

The GOC (Computer) Cabinet is shipped with seven cables connected to the computer that will be routed and connected to equipment that will be installed on the top of the OW Table. The cables are:

- Three power cords
- Two video output cables
- Keyboard output cable
- SCSI output cable

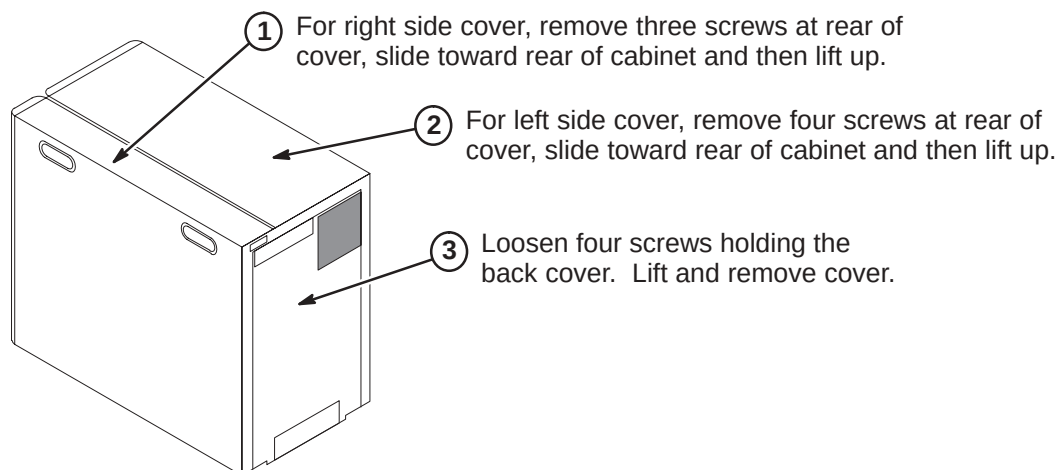
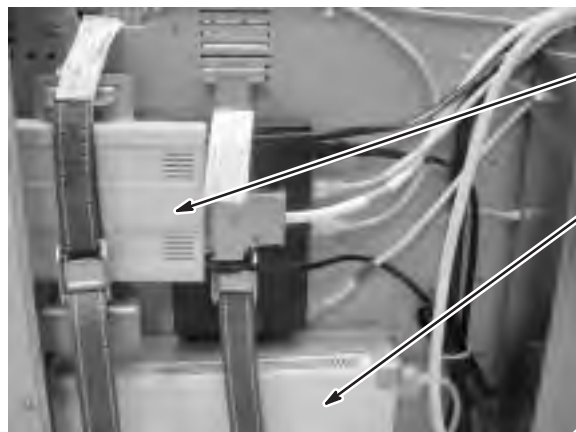
These cables are already connected to the computer and routed thru the opening located at the top of the rear of the GOC cabinet as shown in the illustration below. The remainder of the cables to be connected to designated modules within the GOC Cabinet will have to be routed thru the lower opening.

1. Runs 1064, 1065, 1070, 1327, 2037, 2038 and 2053 are pre-connected to computer and are routed thru top opening.
2. Runs 1081, 1085, 1255, 1371, power cord for the Patient Alert system, and any other option cables will be routed thru lower opening.
3. Route all cables before connecting any cables. Refer to cable map for connection information.
4. If GOC Cabinet is to be located under OW Table, the cable coming out of the top opening of the cabinet will be routed above the Back Panel. The cables coming out of the lower opening will be routed below the Back Panel.



The Computer Cabinet covers need to be removed to access the cable connection points.

5. For right side cover, remove three screws at rear of cover, slide toward rear of cabinet and then lift up.
6. For left side cover, remove four screws at rear of cover, slide toward rear of cabinet and then lift up.
7. Loosen four screws holding the back cover. Lift and remove cover.

8-3-5 GOC CABINET COVER REMOVAL**8-3-6 INSTALL MODEM AND DASM IN GOC CABINET (If Applicable)**

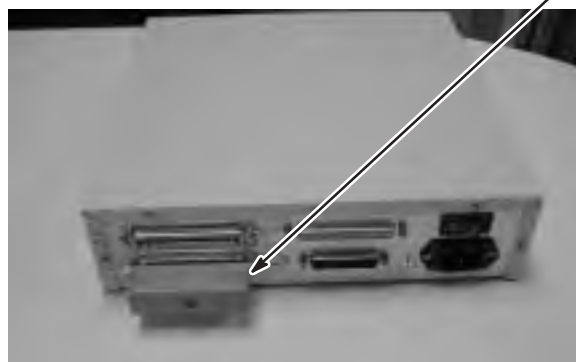
① Mount Modem to GOC so that the underside is facing out to cover. Route customer provided phone line and connect to Modem.

② Mount DASM at this location. Route and connect cable from filming device.

③ Cable routed between DASM or Modem to the Linux PC are provided. Connect Run 1066 to Modem and Run 1067 to DASM.

④ If applicable, install the SCSI Terminator onto the bottom port of the DASM.

⑤ Verify that the **Active SCSI Terminator** located on the bottom of the DASM is in the **OFF** position.

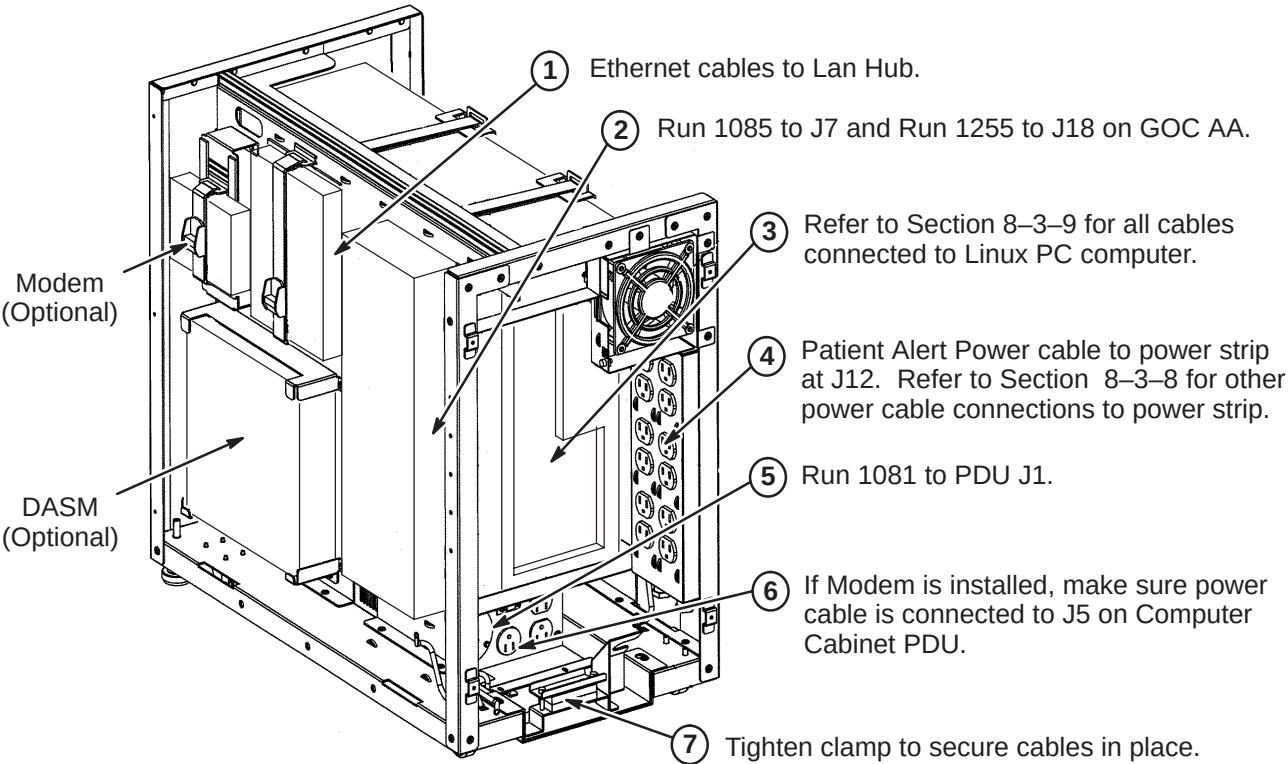


⑥ Modem power cord must be connected to J5 on GOC PDU Module



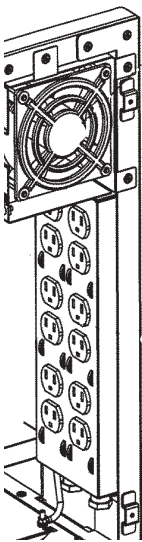
8-3-7 ROUTE CABLES FOR LOWER OPENING

Refer to cable map for information on cables to be connected thru lower opening.



8-3-8 POWER STRIP CONNECTIONS

The illustration below show the power cable connection locations to the power strip in the GOC cabinet. Make sure power cables are attached as shown.



J1		J7
J2		J8
J3		J9
J4		J10
J5		J11
J6		J12

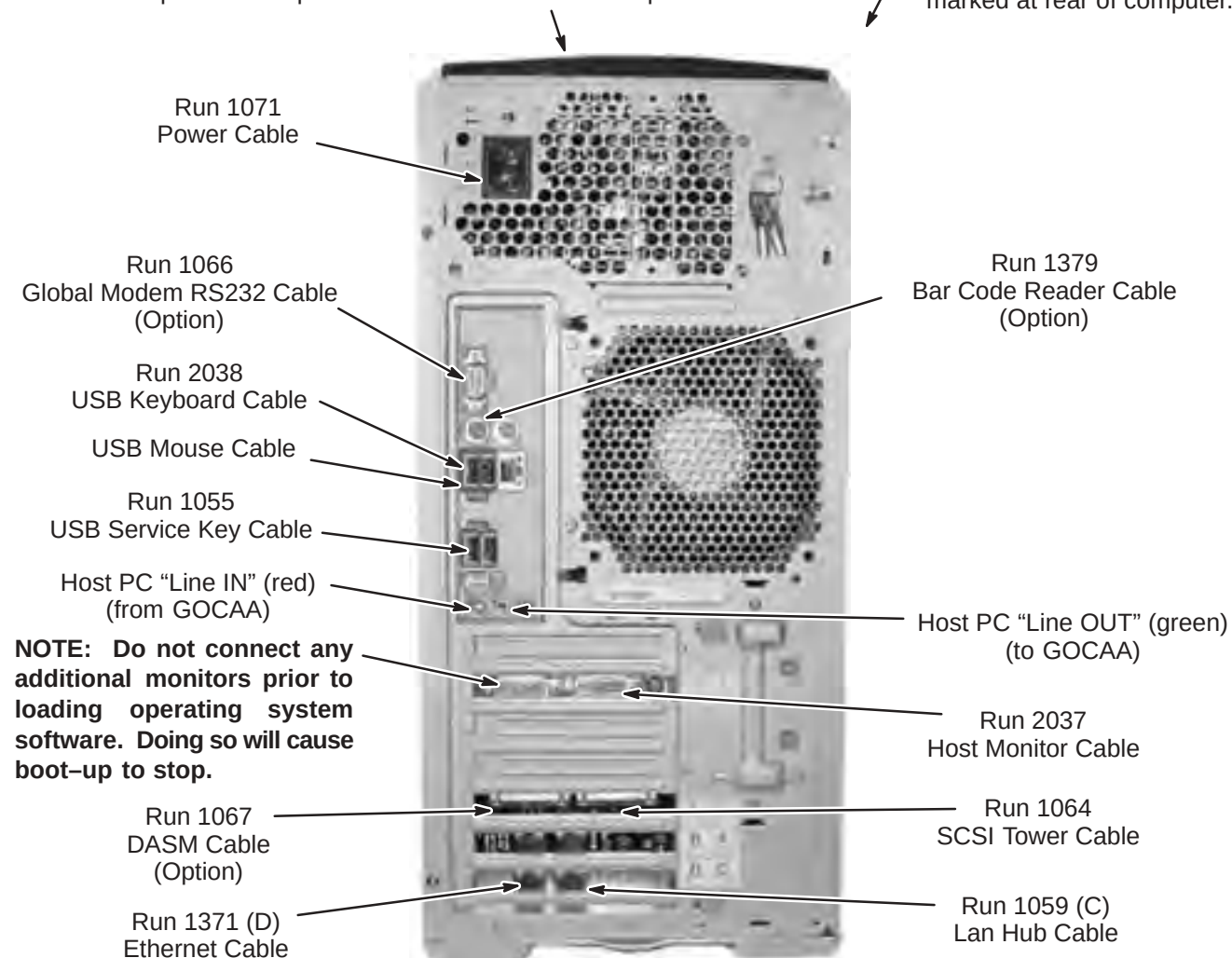
POSITION	NAME	MAX AMP	
		CIRCUIT 1	CIRCUIT 2
J1	Host Comp	6.4	
J2	Host LCD	0.8	
J3	Fan	0.2	
J4			
J5			
J6	1-Wire hub	0.3	
J7	LAN Hub		0.2
J8	2nd LCD		0.8
J9			
J10	DASM		0.6
J11	SCSI Tower		7.0
J12	Patient Alert		0.2
TOTAL		7.7	8.8

8-3-9 LINUX PC CABLE CONNECTIONS

The connection points of cables routed to the HP9300 PC are described in the following two sections. All cables shown are preconnected except for Run 1371 which has to be routed and connected at site. Refer to the applicable PC for your site.

- ① Route and connect USB end of cable from USB–Serial DB9 Adaptor to USB port at FRONT of HP 9300 computer.

- ② Connect remaining cables as marked at rear of computer.

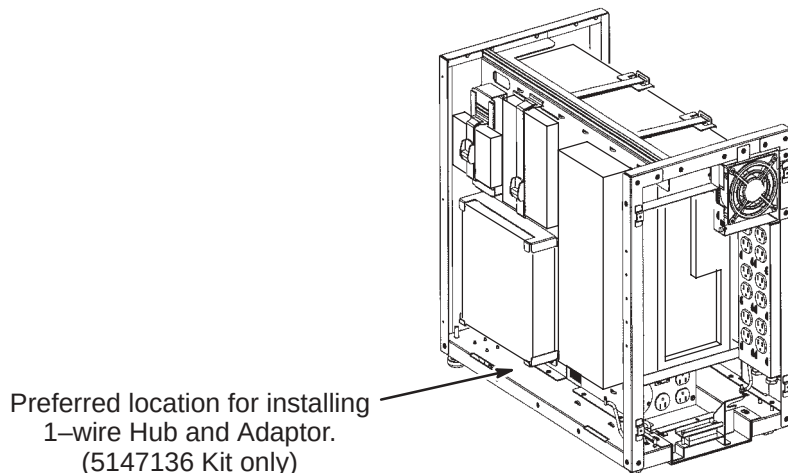


8-3-10 TEMPERATURE AND HUMIDITY SENSOR KIT INSTALLATION

One of two styles of Temperature and Humidity Sensor Kits has been delivered to the site for installation. The original kit, 5147136, or the newer kit, 5147136-2.

- Section 8-3-10 provides instructions for installing 5147136.
- ISection 8-2-10 provides instructions for installing 5147136-2.

The following instructions mount the 1-Wire Hub and Adaptor to the inside of the right GOC cover. The preferred location for installing the parts is at the bottom of the GOC as indicated below.

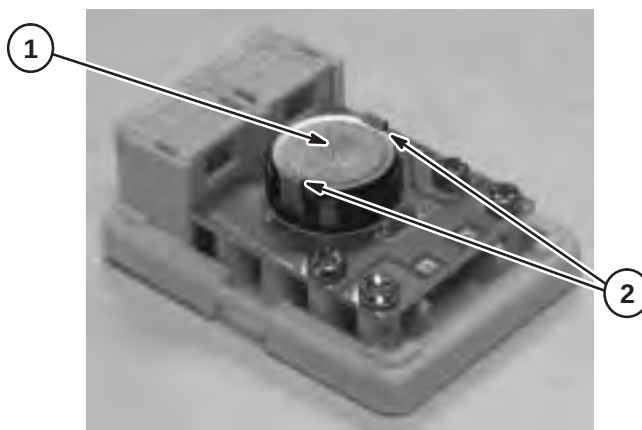
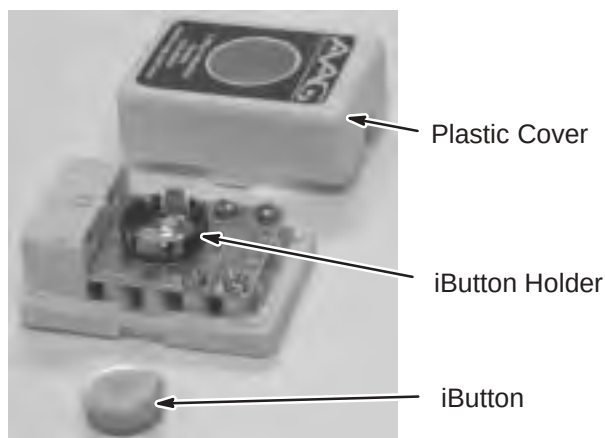


8-3-10-1 INSTALL iBUTTON INTO HOLDER

Install the iButton into the holder in the orientation shown in the illustration.

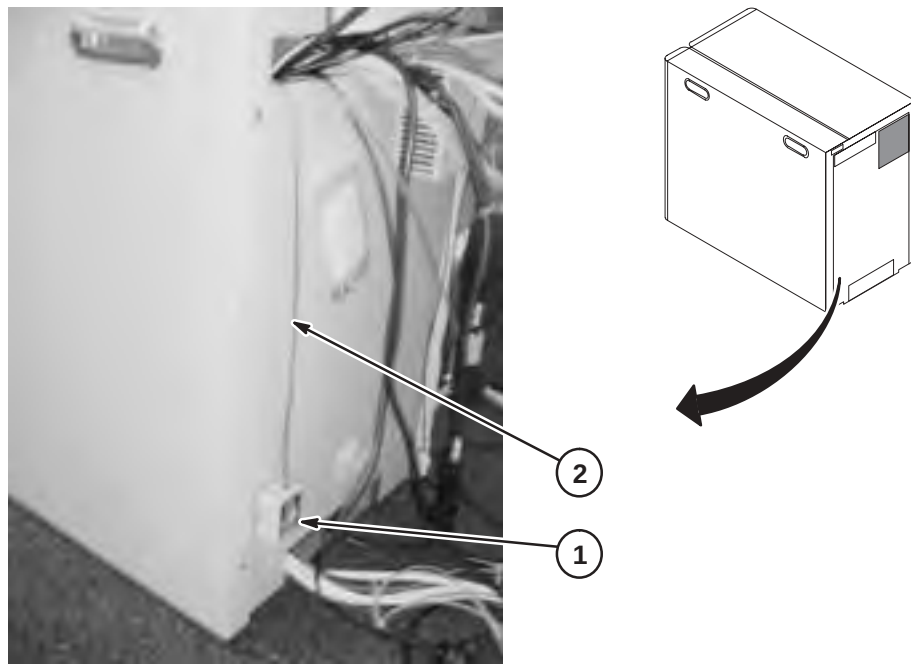
Note: The pictures below show the plastic cover removed although this is not necessary to install the iButton.

1. Install the iButton in the holder so that the solid part of the case is exposed.
2. Make sure the “fingers” latch over the top of the iButton and secure it in the holder.
3. If removed, reinstall the Plastic Cover.



8-3-10-2 INSTALL TEMPERATURE & HUMIDITY SENSOR ON GOC

1. Attach iButton Holder (5141856), with Humidity iButton (5141857), to back of GOC Cabinet at this location using Dual Locking Velcro Tape (2341779 & 2341779-2).
2. Connect cable (5143792-2) to Sensor and route to inside right GOC Cover thru top opening.

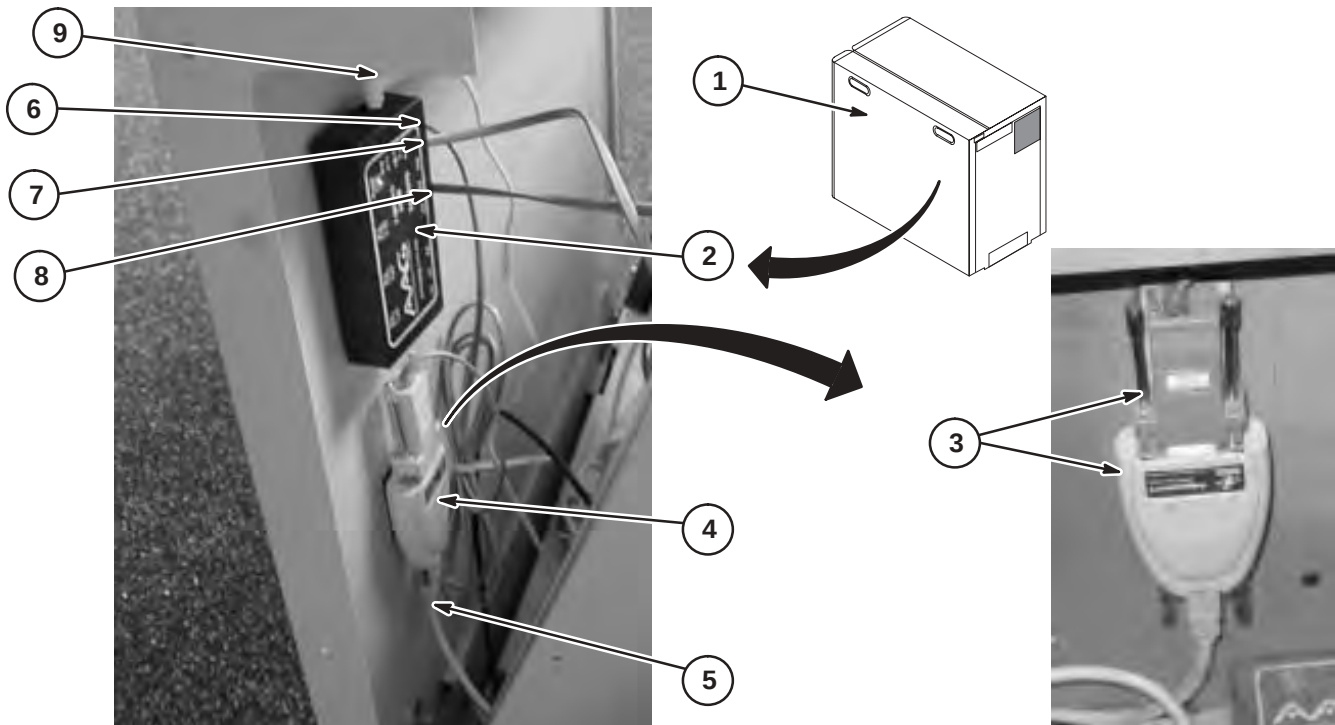
**8-3-10-3 INSTALL TEMPERATURE & HUMIDITY SENSOR ON SYSTEM CABINET**

1. Attach iButton Holder, with Humidity iButton, to back of System/RFS Cabinet at this location using Dual Locking Velcro Tape (2341779 & 2341779-2).
2. Connect cable Run 1322 (5143794) to Sensor and route to inside right GOC Cover thru top opening.

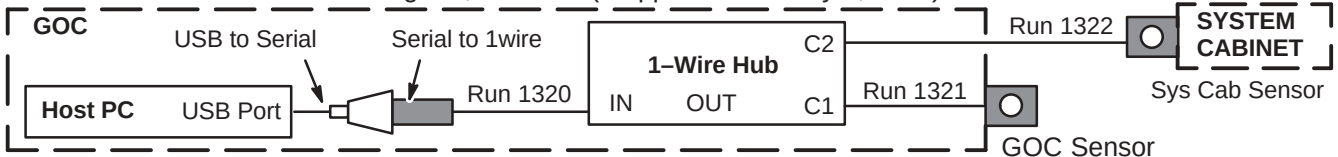


8-3-10-4 INSTALL TEMPERATURE AND SENSOR COMPONENTS INSIDE GOC

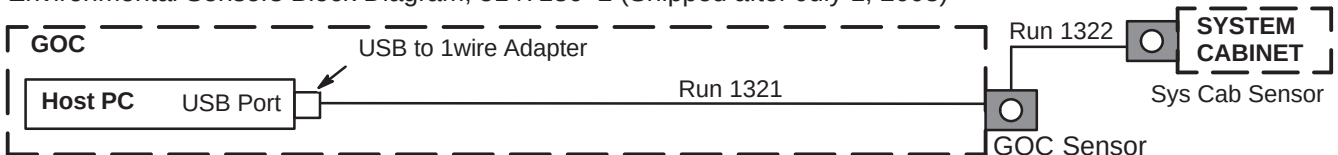
1. Open right side cover of GOC.
2. Attach 1-Wire Hub (5141855) to inside of right cover at this location using two Dual Locking Velcro Tapes (2341779 & 2341779-2).
3. Connect 1-Wire Serial Master (5141859) to USB-Serial DB9 Adaptor.
4. Attach USB-Serial DB9 Adaptor (5146678) to inside of cover at this location using Dual Locking Velcro Tapes (2341779 & 2341779-2).
5. Connect Type B USB (square) end of cable to Adaptor. Route and connect other end to USB port at FRONT of HP 9300 computer.
6. Route and connect Run 1320 (5143792) between 1-Wire Serial Master and 1-Wire Hub IN.
7. Route and connect Run 1321 (5143792-2) between OW Cabinet Sensor and 1-Wire Hub OUT.
8. Route and connect Run 1322 (5143794) between System Cabinet Sensor and 1-Wire Hub C2.
9. Connect power supply cord 5141858) to top of 1-Wire Hub. Connect other end of power cord to J6 on power strip inside GOC cabinet.



Environmental Sensors Block Diagram, 5147136 (Shipped before July 1, 2008)

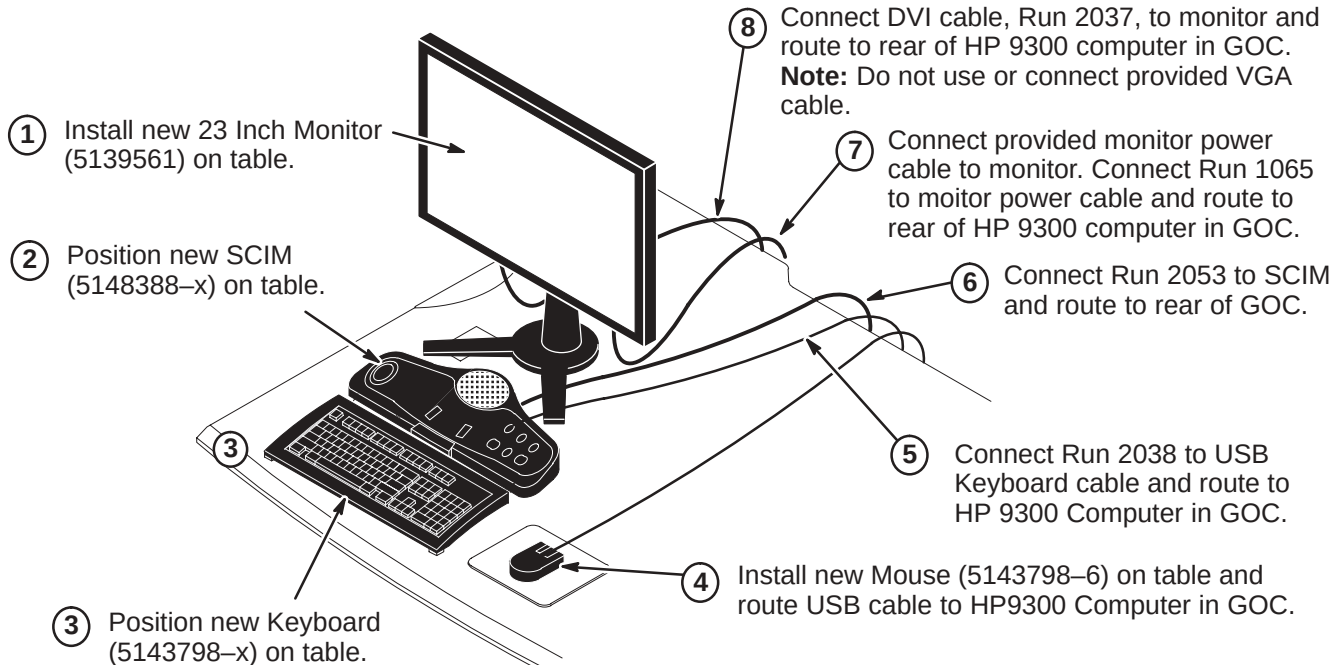


Environmental Sensors Block Diagram, 5147136-2 (Shipped after July 1, 2008)



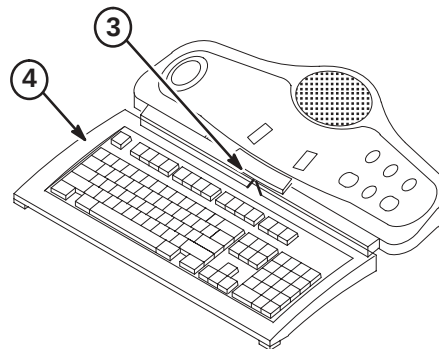
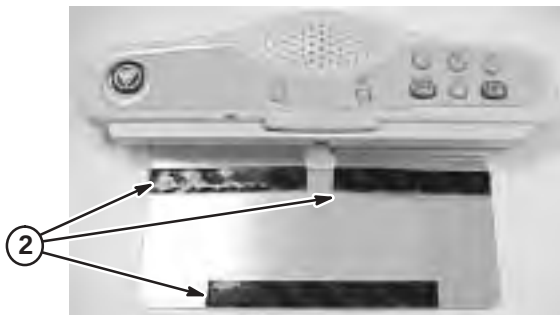
8-3-11 INSTALL MONITOR AND SCIM & KEYBOARD CABLES

The Scan-Control Intercom Module (SCIM) is the current keyboard on all Signa systems. Consult with Site customer for exact location of SCIM/Keyboard and Monitor on Operator Workspace (OW) Table.



8-3-12 INSTALL SCIM/KEYBOARD

1. Place keyboard in front of SCIM, pull the keyboard forward, flip it, and place on top of the SCIM up side down.
2. Remove the plastic strips from the velcro located on SCIM plate, exposing the adhesive surface (see illustration on next page).
3. Guide the cable back into the center notched opening until all the cable is under the SCIM base.
4. Place the keyboard tight against the SCIM, Center it as best as possible, and drop into place. The Velcro pads will now be sticking to the bottom of the keyboard.

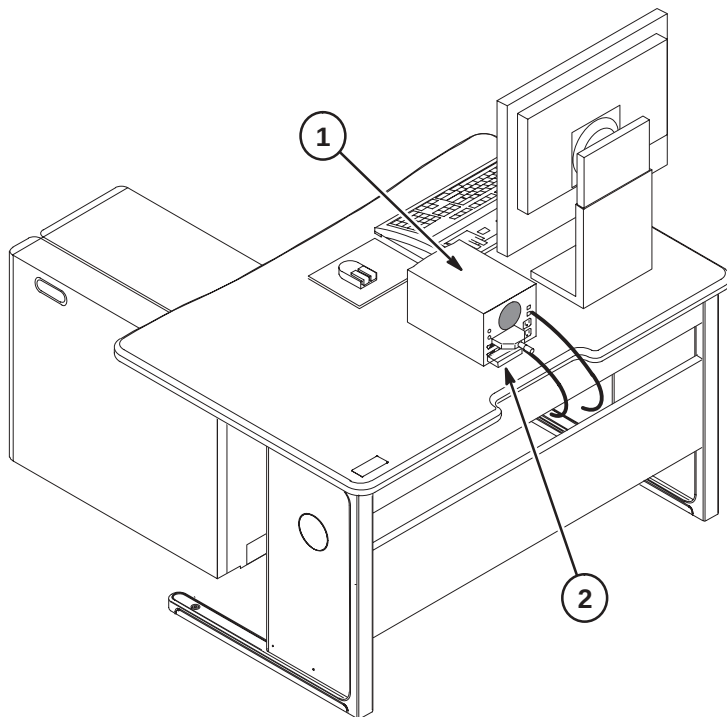


Note

The adhesive on the velcro takes 24 hours to completely adhere to the bottom of the keyboard. Allow the adhesive to dry completely before removing keyboard from SCIM base.

8-3-13 INSTALL SCSI TOWER

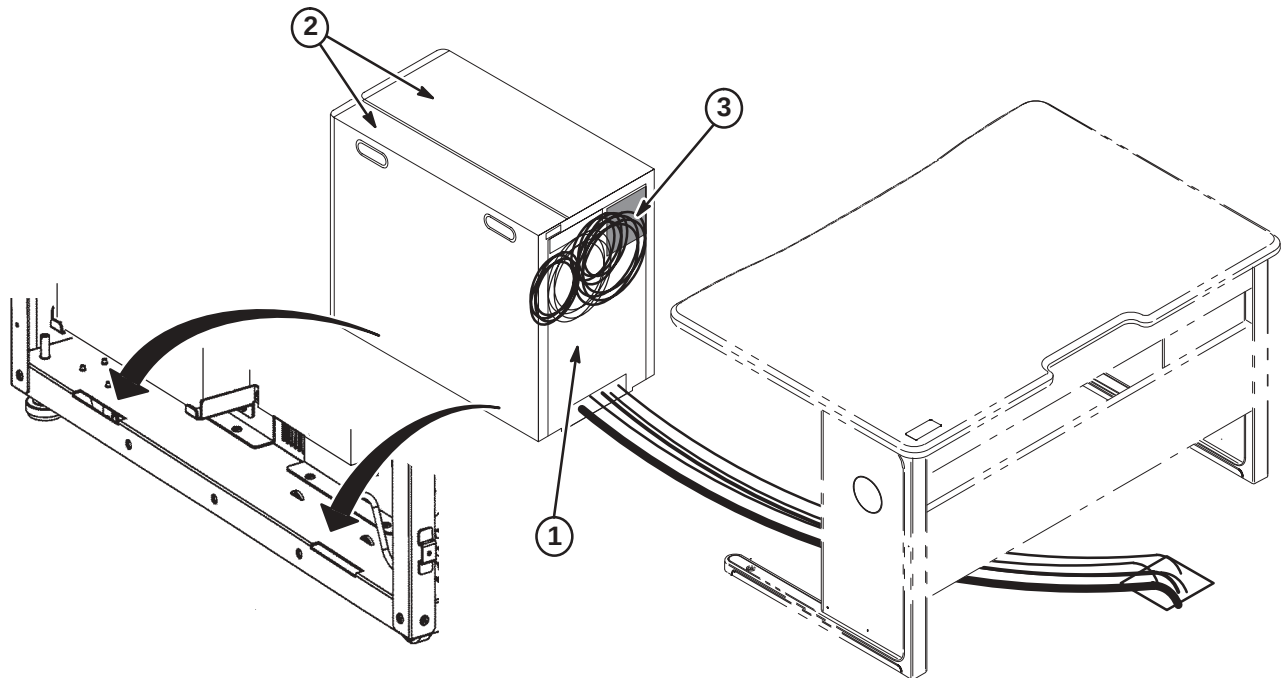
1. Place the SCSI Tower on the OW table and connect Runs 1064 and 1070 from GOC Cabinet. Run 1070 must be routed and secured thru ty-wrap located on the under side of table top.
2. Install terminator on back of SCSI Tower.



8-3-14 FINAL POSITIONING OF GOC CABINET

Complete positioning of GOC Cabinet.

1. Re-attach rear cover and tighten screws.
2. Install both side covers by inserting tabs at bottom of covers into slots (as indicated in illustration below) and slide forward. Install screws at rear of cover that were removed in earlier procedure.
3. Route remaining cables from top opening at rear of GOC Cabinet thru opening above Back Panel on OW table.
4. Place cabinet in final position as determined by customer.

**8-3-15 POWER ON INFORMATION****IMPORTANT**

When turning on the circuit breaker in the PDU, both the PC and Host circuit breakers must be energized.

Note

When powering up the GOC, there is a Main breaker located in the front of the GOC. There are also two switches for the WIM and Modem power.

SECTION 9 – COMPLETION CHECKLIST

TABLE OF CONTENTS

<u>SECTION</u>	<u>PAGE</u>
9-1 MECHANICAL INSTALLATION COMPLETION	9-2
9-1-1 RETURNING SHIPPING MATERIAL	9-2
9-1-2 FINAL MECHANICAL INSTALLATION STEPS	9-2
9-2 GE FIELD ENGINEER RESPONSIBILITIES	9-2
9-2-1 PHANTOMS AND SERVICE KITS	9-2
9-2-2 FINAL GE FIELD ENGINEER INSTALLATION STEPS	9-3
9-2-3 SUPPLIER PERFORMANCE REPORT	9-5

9-1 MECHANICAL INSTALLATION COMPLETION

The following steps need to be reviewed and completed by GE Field Engineer, especially if the system was installed by non-GE personnel. Check off boxes at the front of each item below when complete.

- ☐ Check mechanical flowchart to make sure all steps are complete
- ☐ Resolve shipment shortages
- ☐ Resolve omissions by mechanical contractors
- ☐ Have the on-site GE Healthcare Field Engineer (FE), Service Engineer (SE), or Project Manager of Installation (PMI) to confirm that the blower box has been anchored properly and installed outside the minimum service area.
- ☐ Make sure all rating plates are installed
- ☐ Recheck all wiring connections using cable map
- ☐ Correct for wiring errors if necessary
- ☐ Complete Section 9-1-1.
- ☐ Complete Section 9-1-2.

9-1-1 RETURNING SHIPPING MATERIAL

The shipping material must be returned **as soon as possible**. Prompt return is required so that production shipments can continue with these labor saving dollies and carts. These can be included with the "Return pile", Recycling Operation will route them to appropriate location.

9-1-2 FINAL MECHANICAL INSTALLATION STEPS

1. Complete assembly of PPG cable and probe head.
2. Make sure all Cabinet, Magnet Enclosure, and Rear Pedestal covers have been installed.
3. Make sure Patient Transport is in position at front of Magnet.

9-2 GE FIELD ENGINEER RESPONSIBILITIES

9-2-1 PHANTOMS AND SERVICE KITS

1. Place the MCR3 tool in service cabinet on site.
2. Locate Phantom Kit and SPT Phantom Cart Kit.
3. Unpack the phantoms carefully to reduce risk of damage, breakage, or chemical spill.
4. Locate the multilingual (provided in eight languages) phantom attention labels shipped with each of the phantoms.



Three types of multilingual phantom attention labels are provided and each type must be applied to the correct phantom. The directions at the top of each label sheet describe which phantoms that can receive that label.

9–2–1 PHANTOMS AND SERVICE KITS (Continued)

5. Read the directions provided at the top of each of the multilingual attention label sheets and apply the appropriate language attention label for the site to each of the phantoms that should receive that type of label.
 - a. Each phantom will receive the correct type of attention label written in the appropriate language for that site.
 - b. Note that the LVshim phantom assembly will receive 3 labels, one for each phantom component.
6. Place service phantoms into SPT Phantom Cart or service storage cabinet to be organized and ready to be used during applicable checks and calibrations. The DQA Phantom is to be stored in a customer designated location.
7. Locate TPS RF I/F Kit (if applicable) and Signa Spares Kit. Place kits into service storage cabinet for ready use.

9–2–2 FINAL GE FIELD ENGINEER INSTALLATION STEPS

1. Record and enter applicable data into applicable site configuration files and records.
2. Complete Product Locator information for all installed serialized components, new or updated, via either:
 - **(U.S. Only)** *FE Site Verification Web Site*
or
 - Process and return product locator installation cards for all serialized components to:
Product Locator File, P.O. Box 414, W–523, Milwaukee, WI 53201–0414

Refer to Section 1–10 for details on submitting Product Locator information.

Note

Failure to fill out and return Product Locator Cards may result in failure of your site to receive future FMI's.

3. Process the **Supplier Performance Report** located in Common Forms and Section 9–2–3 (reference copy).
4. Store the delivered site's set of service tools and spares kit in service cabinet at site.
5. Locate surface coil ("quick disconnect") adapters and any optional surface coils and positioning accessories and place on Patient Transport. Be sure only the correct polarity (normal or reverse **but not both**) head coil adapter is left at site.
6. Locate Box "To be opened by Applications Specialist" and set aside for later visit by Applications Specialist.
7. Leave the site's set of Manuals and CD–ROM at site. Install Service and Operator documentation on the host computer. Organize and set up reference cabinet for the Manuals.
8. Locate Material Safety Data Sheets (MSDS) packed with phantoms and gradient coil coolant. They must be retained on site. Customer is to be informed that material with MSDS was brought into site and customer should know/decide where MSDS should be retained at site.
9. Verify that coordination of application support for instruction of site users on the appropriate Software Release has been made before returning system to the users. Turn site over to applications who will instruct users.

This page left blank intentionally

9-2-3 SUPPLIER PERFORMANCE REPORT**GE Medical Systems Field Service**

Send completed forms to the 4 addresses at the end of this form.

Supplier Name: _____ Date: _____

Work Performed / Evaluated: _____

Commodity / Item Purchased: _____

GE Customer Name: _____

FDO Number: _____ Test Equipment Bar-code No; _

GEMS Evaluator Name: _____

Please rate suppliers as 1 to 5 on each of the 15 questions below:

1 = Unacceptable 2 = Poor 3 = Average 4 = Excellent 5 = Outstanding

There are 5 groups with three related questions in each group. Groups can be rated either as a group or 3 separate questions.

Any rating below 3 triggers a discrepancy review with the supplier.

Corrective actions will be faster and more effective when you provide examples and/or detail information.

Supplier schedules well and:					
1	starts on time?	1	2	3	4 5
2	finishes on time?	1	2	3	4 5
3	avoids disruptions?	1	2	3	4 5
Supplier is flexible and:					
4	responds to GEMS/customer requests?	1	2	3	4 5
5	cooperates with changes in schedule?	1	2	3	4 5
6	deals with unexpected problems?	1	2	3	4 5
Supplier tools and equipment are:					
7	appropriate for contracted work?	1	2	3	4 5
8	adequate for contracted work?	1	2	3	4 5
9	calibrated and in good working order?	1	2	3	4 5
Supplier meets GEMS quality standards with employees who:					
10	are trained, skilled and technically competent?	1	2	3	4 5
11	always keep job sites clean and free of debris?	1	2	3	4 5
12	ensure both function and appearance are as expected?	1	2	3	4 5
Supplier employees enhance GEMS image by their:					
13	appearance?	1	2	3	4 5
14	behavior?	1	2	3	4 5
15	attitude?	1	2	3	4 5

Examples / Details: _____

Any rating below 3 triggers a discrepancy review with the supplier.

SEND COMPLETED FORMS TO:

1. Senior Operations Specialist for your LCT
2. The contractor who performed the work
3. ISS (already in default distribution)
4. Sourcing Supplier Quality (already in default distribution)

SECTION 10 – APPENDIX

TABLE OF CONTENTS

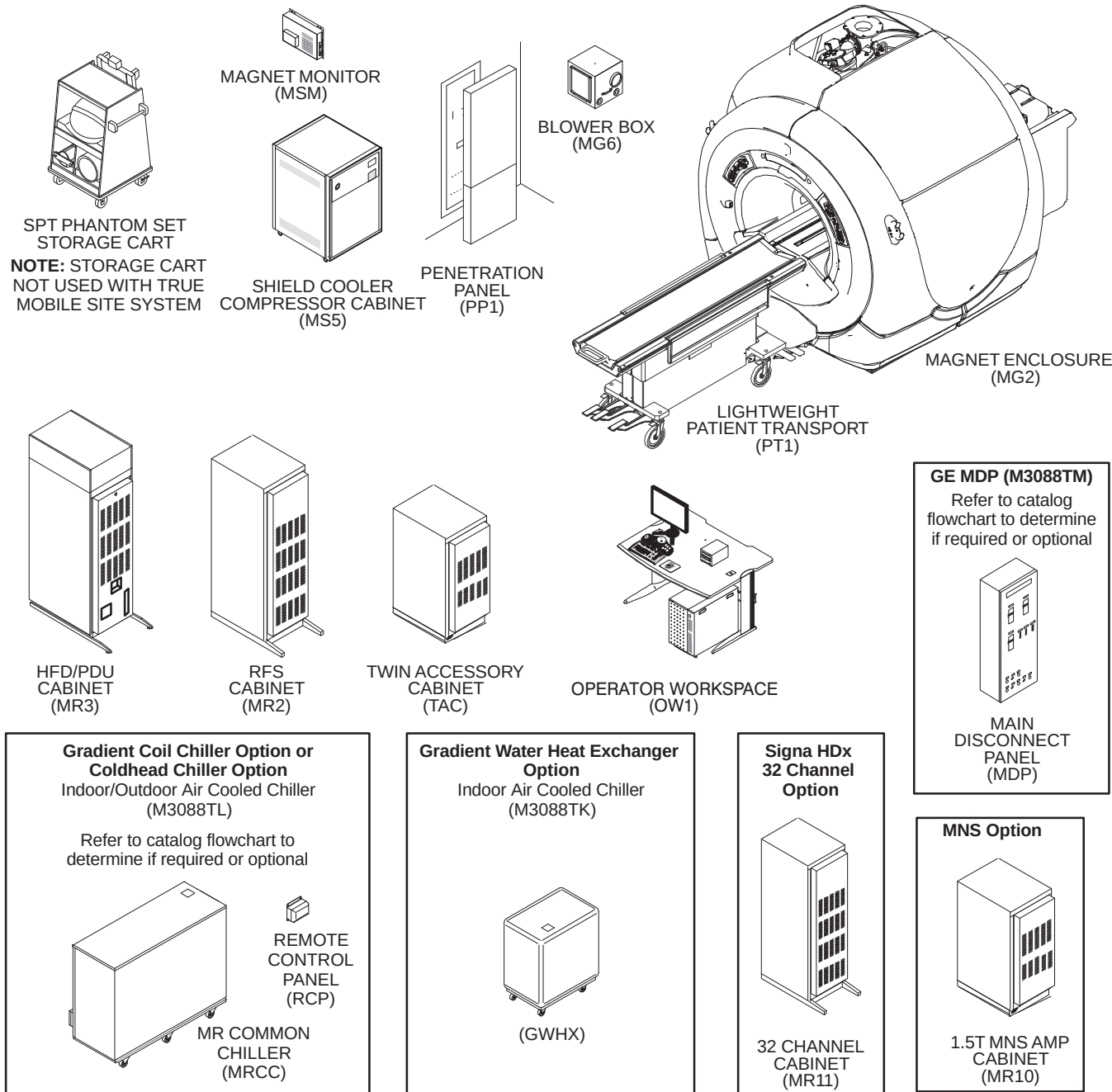
<u>SECTION</u>	<u>PAGE</u>
10-1 SYSTEM CONFIGURATION	10-2
10-1-1 BASIC SYSTEM	10-2
10-1-2 SYSTEM COOLING	10-5
10-2 CABLE IDENTIFICATION	10-7
10-2-1 PDU POWER AND GROUND CABLES	10-7
10-2-2 MAGNET ROOM CABLES NOT CUT TO LENGTH	10-8
10-2-3 MAGNET MONITORING CABLES	10-10
10-2-4 GE MAGNET CABLES	10-11
10-2-5 FIBER OPTIC CABLES	10-11
10-2-6 EQUIPMENT ROOM & MAGNET ROOM CABLES & HOSES CUT TO LENGTH ...	10-12
10-2-7 EQUIPMENT AND OPERATOR ROOM CABLES	10-13
10-2-8 CABLE SPARES (FRU'S)	10-14
10-2-9 EXCITE 4KW AND 8KW MNS CABLES (Optional)	10-15
10-3 FIBER OPTIC TERMINATION INSTRUCTIONS	10-16
10-3-1 FIBER OPTIC CABLE TERMINATION KIT	10-16
10-3-2 STRIPPING FIBER OPTIC OUTER JACKET	10-17
10-3-3 CRIMPING CONNECTOR TO FIBER OPTIC CABLE	10-17
10-3-4 TRIMMING EXCESS FIBER FROM CONNECTOR	10-18
10-3-5 POLISHING FIBER OPTIC END	10-18

10-1 SYSTEM CONFIGURATION

The tables in this section list the catalogs for the hardware equipment which comprise the Signa HDx 1.5T TwinSpeed system.

10-1-1 BASIC SYSTEM

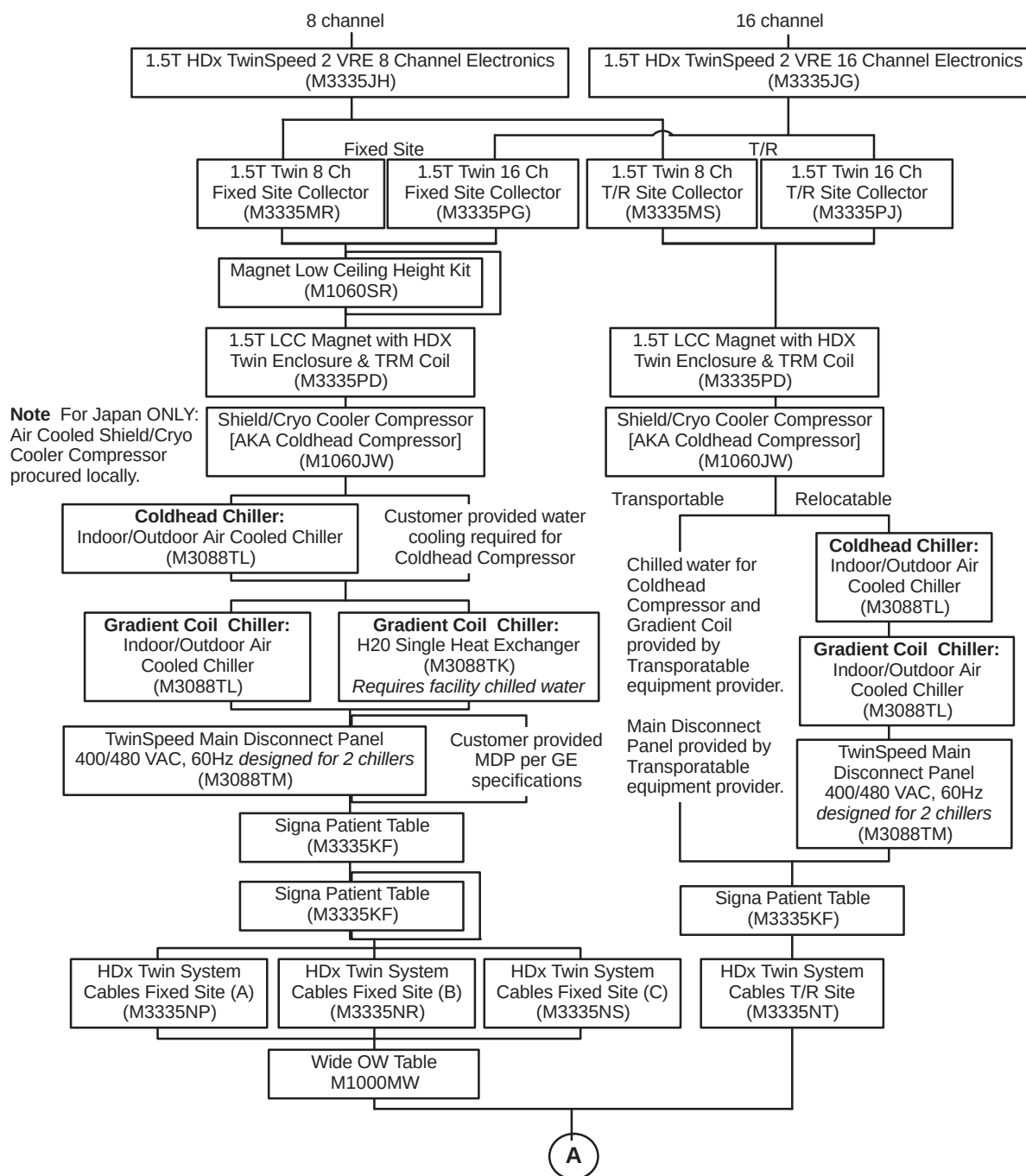
The basic system is configured from several catalogs. The following tables indicate what catalog(s) has been installed for this site. For those tables which offer more than one catalog, select the catalog which has been delivered to the site.



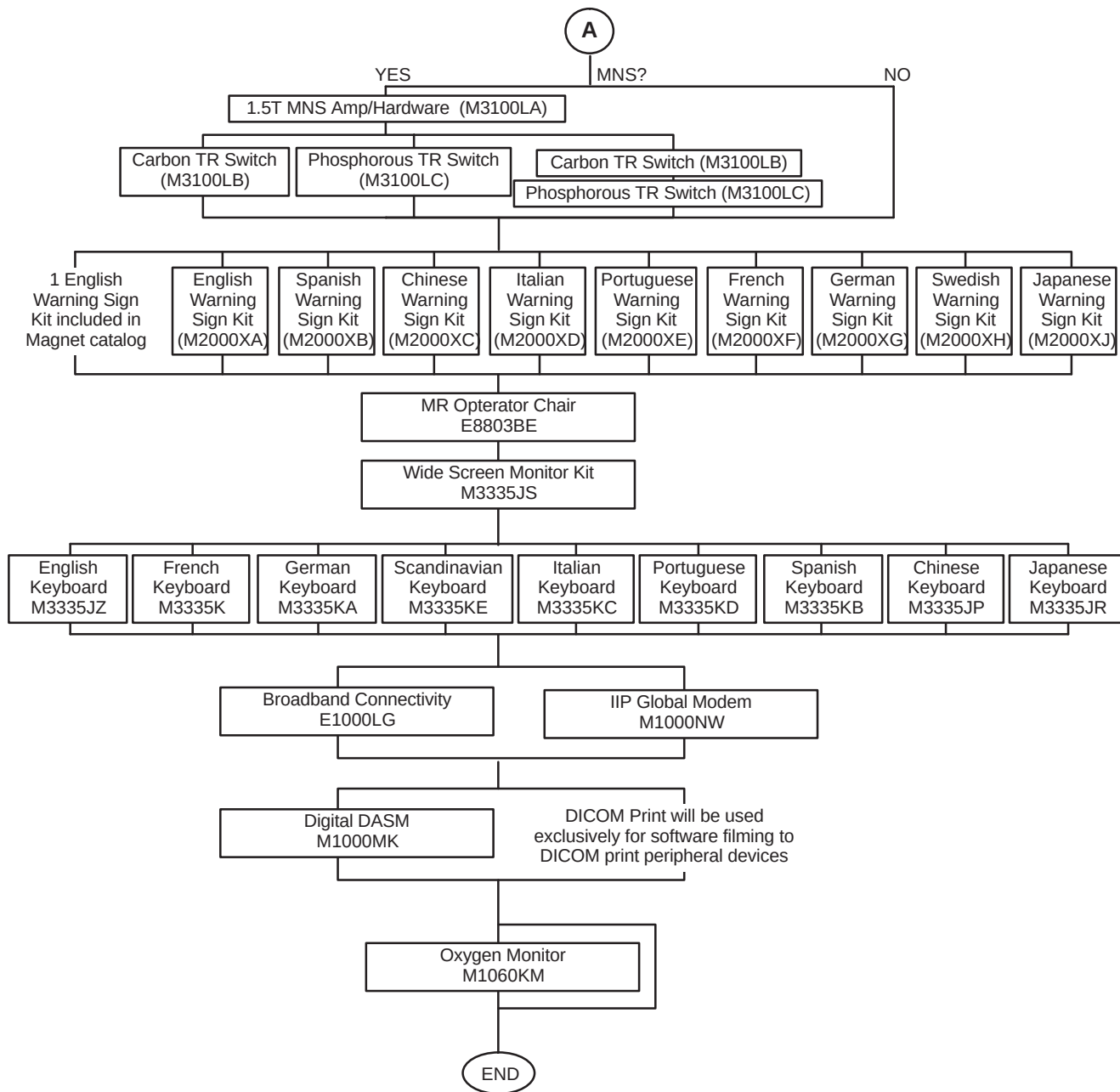
SIGNA HDx 1.5T TwinSpeed SYSTEM
ILLUSTRATION 10-1

10-1-1 BASIC SYSTEM (Continued)

TABLE 10-1
SIGNA HDx 1.5T TwinSpeed SYSTEM CATALOGS



10-1-1 BASIC SYSTEM (Continued)



10-1-2 SYSTEM COOLING

The Signa HDx 1.5T TwinSpeed system requires water cooling for the Magnet Shield/Cryo Cooler Compressor Cabinet and the TRM Gradient Coil located in the Magnet bore. The following system cooling equipment is available with the MR system:

Note

See Illustration 10-1 flowchart for system cooling equipment configurations options and relationship to other system equipment/options selections.

For water cooling specifications and requirements for Site Provided Water Cooling, refer to **Direction 5159902, Signa HDx 1.5T TwinSpeed Pre-Installation** manual.

- **MR Common Chiller (MRCC)**
The MRCC is a single-loop 10KW 50/60 Hz water chiller that can be used to cool either Shield/Cryo Cooler Compressor or the TwinSpeed Gradient Coil. The MRCC can be installed either indoors or outdoors. There is a Remote Panel to be installed in the MR system Equipment Room that can stop or restart the chiller and can display supply water temperature, chiller setpoints and alarms. The MRCC power is supplied by the MR system Main Disconnect Panel (MDP). This chiller will be serviced by a third party vendor.
- **Gradient Water Heat Exchanger (GWHX)**
The GWHX is a 10KW 50/60 Hz water-to-water heat exchanger that is used for cooling the TwinSpeed Gradient Coil. The GWHX is for indoor use only. The GWHX requires facility provided chilled water. The GWHX power is supplied by the MR system Power Distribution Unit (PDU). This heat exchanger will be serviced by GE Service. When the GWHX is used then the customer must supply water cooling for the Shield/Cryo Cooler Compressor.

Note

Gradient Coil water cooling must be supplied by cooling equipment (MRCC or GWHX) supplied with the MR system to prevent contamination/damage to the coil and for proper image quality.

There are several possible scenarios which a customer may utilize to meet the system water cooling requirements for the LCC Magnet Shield/Cryo Cooler Compressor Cabinet and the TRM Gradient Coil.

- Two MRCC installed outdoors where one unit provides Shield/Cryo Cooler Compressor cooling and the other provides Gradient Coil cooling. The 2 MRCC Remote Control Panels to be installed in the MR system Equipment Room.
- Two MRCC installed indoors where one unit provides Shield/Cryo Cooler Compressor cooling and the other provides Gradient Coil cooling. The 2 MRCC Remote Control Panels to be installed in the MR system Equipment Room. Customer supplied air cooling is required for the MRCC units.
- One GWHX installed indoors provides water cooling to the Gradient Coil (GWHX requires facility water cooling) and Customer provided facility water cooling for the LCC Magnet Shield/Cryo Cooler Compressor Cabinet.
- One MRCC installed outdoors provides water cooling to the Gradient Coil and Customer provided facility water cooling for the LCC Magnet Shield/Cryo Cooler Compressor Cabinet. The MRCC Remote Control Panel to be installed in the MR system Equipment Room.
- One MRCC installed indoors provides water cooling to the Gradient Coil and Customer provided facility water cooling for the LCC Magnet Shield/Cryo Cooler Compressor Cabinet. The MRCC Remote Control Panel to be installed in the MR system Equipment Room. Customer supplied air cooling is required for the MRCC unit.

10-1-2 SYSTEM COOLING (Continued)

TABLE 10-2
TWIN SYSTEM COOLING CABINET (SEE TABLE NOTE NO TAG)

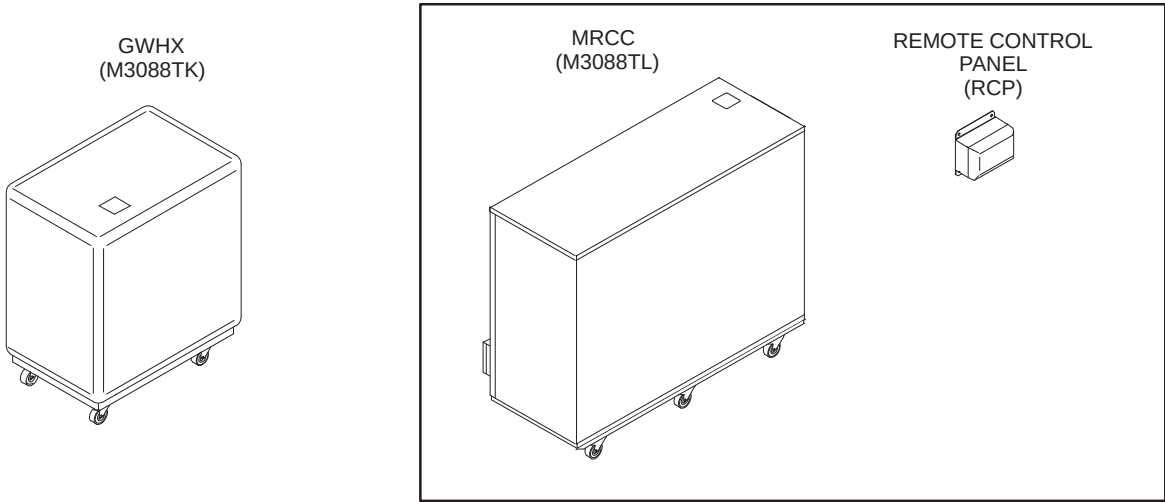
CATALOG	DESCRIPTION
M3088TK	Gradient Water Heat Exchanger (GWHX) See Note 1
M3088TL	MR Common Chiller (MRCC) See Note 2

Note 1 When GWHX is selected, customer supplied water cooling is required for the LCC Magnet Shield/Cryo Cooler Compressor. Refer to **Direction 5133302, Signa EXCITE HD 1.5T TwinSpeed Pre-Installation** for water cooling specifications.

2 Two M3088TL installed outdoors is standard cooling equipment configuration of Transportable/Relocatable Mobile systems.

Note

True Mobile system water cooling for the Gradient Coil and the Shield/Cryo Cooler Compressor is provided by the Mobile Van equipment.



SYSTEM COOLING EQUIPMENT SELECTIONS
ILLUSTRATION 10-2

10-2 CABLE IDENTIFICATION

Cables delivered to site for installation have specifications dependent on its usage. The following tables contain information on cable runs for this system as it was originally installed.

The table below provides definitions for the following cable tables.

COLUMN HEADING	DESCRIPTION	SPECIAL NOTES
RUN	The run number as seen on the delivered cable	The vendor cable may not have run numbers.
"FROM" & "TO"	Labels the destination of the cable	
DESCRIPTION	Could include information on length, number of conductors, type of cable, and end plug type	Total length of cable delivered to site, type of cable specifies power from data.
CABLE TYPE	Type designation for the cable	
VOLTAGE RATING	The highest voltage that may be continuously applied to a wire	Specific to the core cable manufacturer given.
ACTUAL VOLTAGE	Nominal for continuous; maximum for variable voltage	If voltage varies during use, the maximum is given.
TEMP RATING	Maximum temperature insulating material may be used without loss of basic properties	Specific to the core cable manufacturer given.
FLAME RATING	Measure of material's ability to support combustion	Specific to the core cable manufacturer given.
REMARKS	Describe the cable's use in the MR system	

10-2-1 PDU POWER AND GROUND CABLES

The cables listed in Table 10-3 are connected at one end to the PDU module in the ACGD/PDU Cabinet.

TABLE 10-3
PDU POWER AND GROUND CABLES

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
037	PD1 GND	MR2 A12 GND	36 ft Ground Wire (#10)	SO	600	0	105	VW-1	System Cabinet ground
040	MSI	RF COM GND	69 ft Ground Wire (#1/0)	SO	600	0	105	VW-1	Magnet Ground Cable
045	PP1 GND	MG6 GND	41 ft Ground Wire (#10)	SO	600	0	105	VW-1	Blower Cabinet Separate ground
968	MR1 A12 J1	MR3 PD1	35 ft Power (5-#8)	SO	600	208Y	75	FT-4	RFS Cabinet Power
988	PP1 A7 GND	TAC GND	60 ft. Ground (#8)	SO	600	208	80	VW-1	TAC to PP1 ground (3)
989	MR3 PD1	TAC J8	25 ft. Power (3-#10)	SO	600	120	90	FT-4	TAC Power
1081	OW1 A19 J1	MR3 PD1	65.5 ft Power (4-#10)	SO	600	120	75	FT-4	Operator Workspace Power
1182	PP1 PHPS PWR	MR3 PD1	61 ft. Power (3-#10)	AWM	600	120	90	FT-4	PHPS Module Power
1234	MR2 RF AMP	MR3 PD1	35.5 ft Power (4-#8)	SO	600	208	90	FT-4	RF Amplifier Power
1254	MR3 PD1	RF COM GND	69 ft Ground Wire (#1/0)	SO	600	0	105	VW-1	Magnet Ground Cable
N/A	PD1 MAG/SHIM OUTLET	MAG/SHIM POWER SUPPLY	50 ft Power (5-#8) With 30A Plug and Receptacle						Cable for PDU to GE Magnet Power Supply or SC Shim Supply Power

10-2-2 MAGNET ROOM CABLES NOT CUT TO LENGTH

The cables listed in Table 10-4 are routed and connected in the Magnet Room and are not cut to length.

TABLE 10-4
MAGNET ROOM RUNS (NOT CUT TO LENGTH)

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
282	PP1 J12	MG3 A2 J6	45 ft. 25 pin Sub-D	CI3	300	<30	80	FT-4	Patient Intercom
297	PP1 J18	EO1	90 ft. 9 pin Sub-D	CI3	300	<30	80	FT-4	Emergency off switch
300	PP1 J15	MG2 A15	60 ft. 25 pin Sub-D	CI3	300	<30	80	FT-4	Bore Light power
318	PP1 J11	MG3 A7	45 ft. 9 pin Sub-D	CI3	300	38.5 VDC	80	FT-4	Long Drive Motor power
385	PP1 J20	MG2 A30	67 ft. 37 pin Sub-D	CI3	300	<30	80	FT-4	Patient Alignment Light power
387	PP1 J47	MG2 A29 J1	60 ft. 25 pin Sub-D	CI3	300	38.5 VDC	80	FT-4	Table Elevation motor power
414	MG3 A35	MG3 A2 J9	30 ft. 9-pin Sub-D	CI3	300	<30	80	FT-4	Front Cover Microphone
421	MG2 A33 J1	MG3 A31 J1	15 ft. 15 pin Sub-D	CI3	300	<30	80	FT-4	Dock Limit
422	MG2 A33 J2	MG3 A1 J4	25 ft. 37 pin Sub-D	CI3	300	<30	80	FT-4	Long Drive Encoder/Limit Sw
458	PP1 J17	OM3	9 pin Sub-D	CI3	300	<30	80	FT-4	Optional Remote O2 Sensor
715	PP1 J14	MG2 A33 J1	65 ft. 37 pin Sub-D	CI3	300	<30	80	FT-4	SRI Power
716	PP1 J89	MG2 A11 A1 J2	65 ft. 15 pin Sub-D	CI3	300	<30	80	FT-4	PAC Power
743	MG2 A12 A1 J2	MG3 A3 A1 J3	10 ft Coax w/M&F "N"	STO	2500 VDC	950	85	VW-1	Body Coil Drive
744	MG2 A12 A1 J1	MG3 A3 A1 J4	10 ft Coax w/M&F "N"	STO	2500 VDC	950	85	VW-1	Body Coil Drive
778	PP1 A11 J72	MG2 A12 A2 J3	70 ft. MHV Coax	CI2	1900 VRMS	1000 VDC	80	VW-1	Dynamic Disable (body coil frnt)
779	PP1 A11 J73	MG2 A12 A2 J4	70 ft. MHV Coax	CI2	1900 VRMS	1000 VDC	80	VW-1	Dynamic Disable (body coil frnt)
780	PP1 A11 J74	MG3 A3 A3 J6	60 ft. MHV Coax	CI2	1900 VRMS	1000 VDC	80	VW-1	Inductive Drive
781	PP1 A11 J75	MG2 A12 A2 J5	60 ft. MHV Coax	CI2	1900 VRMS	1000 VDC	80	VW-1	Dynamic Disable (body coil rear)
782	PP1 A11 J76	MG2 A12 A2 J6	60 ft. MHV Coax	CI2	1900 VRMS	1000 VDC	80	VW-1	Dynamic Disable (body coil rear)
784	PP1 F5, F6	MG6 J1	40 ft. 2 cond power cable	SO	600	120	80	FT-4	Patient Cooling Blower power
848	MG2 A11 J4	MG2 A4 J5	16 ft. 25 pin Sub-D/rt. angl	CI3	300	<30	80	FT-4	Operator Display to PAC II
862	MG2 A4 J1	MG2 A5 J1	2 ft. 26 pin Rt. angle	CI2	300	<30	80	VW-1	Oper Display to Rt. Op Cntrl
863	MG2 A4 J2	MG2 A5 J2	2 ft. 26 pin Rt. angle	CI2	300	<30	80	VW-1	Oper Display to Lt. Op Cntrl
893	MG3 A3 J2	MG3 A3 A5 INPUT	6.6 Inch BNC		1900 VRMS	<30	80	VW-1	Splitter to Preamp Cable
897	MG2 A4 J3	MG2 A33 J4	20 ft. 26 pin Amplitude to 37 pin Sub-D	CI3	300	<30	80	FT-4	Display to SRI

10-2-2 MAGNET ROOM CABLES NOT CUT TO LENGTH (Continued)

TABLE 10-4 (CONT'D)
MAGNET ROOM RUNS (NOT CUT TO LENGTH)

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
1152	PP1 A5 J1	MG2 A12 A3 P1	29.5 ft. 37 pin CPC to 16 pin	PLTC	300		105	FT-4	Shim Filter Cable
1200	PP1 J200	MG3 16CHSW J1	8-Coax 8W8 Combo D	CATV R	1500	<30	85	FT-4	
1201	PP1 J201	MG3 16CHSW J2	8-Coax 8W8 Combo D	CATV R	1500	<30	85	FT-4	16 Channel only
1202	PP1 J202	MG3 A17 J3	2-Coax 8W8 Combo D	CATV R	1500	<30	85	FT-4	
1203	PP1 MC SW FLTR J203	MG3 16CHSW J11	49 ft. 37 pin Sub-D	CI3	300	<30	80	FT-4	
1206	MG3 IPED I/F J4	MG3 A36 J35 MG3 A3 A5 J11	1550mm/600mm 2 x 50 Ohm Coax Rd316-DS						
1215	MG3 16CHSW J20	MG2 SRI J20	37 pin Sub-D	CI3	300	<30	80	FT-4	
1216	PP1 A16 J77	MG3 A36 J5	80 ft. 15 Sub-D	CI3	300	<30	80	FT-4	Monitor to Compressor I/F
1217	PP1 A15 J78	MG3 A36 J10	80 ft. 15 Sub-D	CI3	300	<30	80	FT-4	SRI Power
1250	MG3 A3 J1	CONNECT TO RUN 257	Times Microwave LMR 600FR	CATV R	6000 VDC		85	FT-4	
1260	MG2 A33 J10	MG2 A36	35 ft 3 cond w/ 9-Sub D	CI3	300	<30	105	FT-4	Bore Temperature Sensor
1261	MG2 A4 J4	MG2 A33 J5	20 ft. Data Cable (80-#28)	CI2	300	<30	80	VW-1	SRI Splitter to Oper Display
1331	LPCA J2	16 CHSW #1 J23	3400mm						
1332	MC-9 TO MC-16	16 CHSW #1 J21	3100mm 8 x 50 Ohm Coax						
1333	MC J21	16 CHSW #1 J18	3500mm 37 pin Sub-D						
1334	IMAGE REJECT #1 J5	16 CHSW #1 J22	1000mm 8 x 50 Ohm Coax Rg316-DS						16 Channel only
1335	IMAGE REJECT #1 J6	16 CHSW #1 J17	1000mm 37 pin Sub-D						
1336	IMAGE REJECT #1 J7	16 CHSW #1 J24	1000mm 8 x 50 Ohm Coax Rg316-DS						16 Channel only
1337	IMAGE REJECT #1 J8	16 CHSW #1 J19	1200mm 25 pin Sub-D						
1338	IRF #1 J4	LPCA J3	3500mm						
1339	IRF #1 J1	LPCA J23	3400mm 8 x 50 Ohm Coax						16 Channel only
1340	IRF #1 J2	LPCA J22	3500mm 37 pin Sub-D						16 Channel only
1341	IRF #1 J3	LPCA J21	3400mm 8 x 50 Ohm Coax						16 Channel only
1370	MC j18	PED I/O J2	3250mm 9 pin Sub-D						
1378	MG3 A35	MG3 A2 J9	30 ft. 9-pin Sub-D	CI3	300	<30	80	FT-4	Front Cover Microphone

10-2-3 MAGNET MONITORING CABLES

The Magnet Monitoring cables listed in Table 10-5 are delivered with the Magnet.

TABLE 10-5
MAGNET MONITORING CABLES DELIVERED WITH MAGNET

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
823	MSM1 A1 J10	MR2 A11 J24	Data Cable (15-#22)	CI3	300	<30	80	FT-4	Monitor to System Cabinet I/F
824	PP1 J10	FJ1	Data Cable (9-#22)	CI3	300	<30	80	FT-4	Monitor Sensor I/F to Penetration Panel
825	PP1 J48	MSM1 A1 J8	Data Cable (25-#22)	CI3	300	<30	80	FT-4	Magnet Monitor connector to Penetration Panel
826	MSM1 A1 J9	FJ3	Data Cable (15-#22)	CI3	300	<30	80	FT-4	Monitor to Compressor I/F
827	FJ3	MS5 A5 A1 J1 / FJ4	Data Cable (8-#22) & Data Cable (4-#22)	CI3	300	<30	80	FT-4	Compressor I/F
828	PP1 J10	MA1 A3 A1 P403	Data Cable (9-#22)	CI3	300	<30	80	FT-4	Penetration Panel to Sensor
829	PP1 J48	MS1 FJ2	Data Cable (25-#22)	CI3	300	<30	80	FT-4	Penetration Panel to Magnet I/F
830	MS1 FJ2	MS1 A2 J2/ MS1 A5 A2 J1/ MG3 TB1	3cond/3cond/25cond	CI3	300	<30	80	FT-4	Magnet I/F
831	FJ1	MSM1 A1 J7	Data Cable (8-#24)	CI3	300	<30	80	FT-4	Monitor to Sensor I/F
832	MS1 A2 J1	MS1 SLEEVE/ MS1 COLDHEAD	Data Cable (8-#22)	CM	300	<30	80	FT-4	Buffer Amp to Cooling Sleeves
833	FJ4	MS5 A1 A6 JR	Data Cable (6-#24)	CM	300	<30	80	FT-4	Compressor to Compressor I/F
939	MDP UPS	MSM4	6 ft. Power (5-#15)	SO	600	120	85	VW-1	MDP to UPS
940	MSM3 RS232	MSM1 J1	6 ft. Data Cable (9-#22)	CI3	300	<30	80	FT-4	Monitor to Modem
1077	MSM1 J5	OW1 A2 A4 PRT7	80 ft. Cat 5 Ethernet	CI3	300	<30	80	FT-4	Monitor to OW Ethernet
	MR2 A11 J24	MR2 IPG J9/ MR2 PDU J5							System Cabinet I/F

10-2-4 GE MAGNET CABLES

The Magnet cables listed in Table 10-6 arrive on site with the magnet and are installed at the time of Magnet delivery.

TABLE 10-6
GE MAGNET CABLES

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
601	MS8	MS1	Power Cable (#4/0)	SO	600	208	80	VW-1	Ramp cables, Power Assembly Positive 4/0
602	MS8	MS1	Power Cable (#4/0)	SO	600	208	80	VW-1	Ramp cables, Power Assembly Negative 4/0
603	MS7	MS1 A3 A1	Power Cable (#4/0)	SO	600	208	80	VW-1	Ramp and Shim Cable Assembly
604	MS7 or MS8	MS1 A3 A1	Data Cable (12-#22)	CI3	150	60 VDC	80	FT-4	Switch Heaters
606	MS4	MS1 A3 A1	Data Cable (4-#22)	CI3	300	<30	105	FT-4	Emergency Run Down Unit
609	MS7 OR MS8	PDU	Power Cable (5-#16)	SO	300	208Y	105	FT-4	Feeder Cables
621	MS5 A1	MS1 A2	7/8" Gas Flex Line	STO	N/A	N/A	85	VW-1	Supply Flexible Gas Line (PP1 J60 or J61)
622	MS5 A1	MS1 A2	7/8" Gas Flex Line	STO	N/A	N/A	85	VW-1	Return Flexible Gas Line (PP1 J60 or J61)
623	MS5 A1	PP1 F1, F2, F3, F4	Power Cable (4-#18)	SO	600	200	105	VW-1	Compressor to Pen Panel Cable (equipment room)
624	MS1 A2	PP1 F1, F2, F3, F4	Power Cable (4-#18)	SO	600	200	105	VW-1	Cold Head Control Cable

10-2-5 FIBER OPTIC CABLES

The Fiber Optic cables listed in Table 10-7 are routed and connected in the Magnet Room and the Equipment Room.

TABLE 10-7
FIBER OPTIC CABLES

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
711/ 712	PP1-J91	MG2 A33 and MG2 A11 A1	100 ft. 3/4 in. fiber optic conduit	N/A	N/A	0	85	N/A	One duplex fiber optic cables and one simplex fiber optic cable and One duplex plastic fiber optic cable for PAC-II
1045	MR2 A11 J13	PP1 A17 A1	3/4 in. fiber optic conduit	N/A	N/A	0	85	N/A	Two duplex fiber optic and one simplex fiber optic cables
1179	MR2 A11 J17	MR3 A11 J2	20 ft. 3/4 in. fiber optic conduit	N/A	N/A	0	85	N/A	Four glass fiber optic cables

10-2-6 EQUIPMENT ROOM AND MAGNET ROOM CABLES AND HOSES CUT TO LENGTH

The Gradient cables and the RF Head and Body cables listed in Table 10-8 will be routed, cut, and terminated at designated cable ends according to instructions found in the CABLE INSTALLATION section of this manual.

TABLE 10-8
EQUIPMENT ROOM AND MAGNET ROOM CABLES & HOSES CUT TO LENGTH

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
235	CONNECT TO RUN 1259	PP1 J44	Times Microwave LMR-900FR	CATV R	5000 VDC		85	FT-4	1.5T Body Coil Run created from kit
257	PP1 J44	CONNECT TO RUN 1250	Times Microwave LMR-900FR	CATV R	5000 VDC		85	FT-4	1.5T Body Coil Transmit
536	WC1	MG2 A12 A1	Flexible Tubing	N/A	N/A	N/A	N/A	N/A	Gradient Coil Cooling Line
537	WC1	MG2 A12 A1	Flexible Tubing	N/A	N/A	N/A	N/A	N/A	Gradient Coil Cooling Line
976	MR3 CABI/F J3	TAC $\pm$ X	Conductor (3-#2)	STO	1000	600	90	FT-4	Twin X Gradient Power
977	MR3 CABI/F J4	TAC $\pm$ Y	Conductor (3-#2)	STO	1000	600	90	FT-4	Twin Y Gradient Power
978	MR3 CABI/F J5	TAC $\pm$ Z	Conductor (3-#2)	STO	1000	600	90	FT-4	Twin Z Gradient Power
979	TAC X+,WB-,ZM-	PP1X+,WB-,ZM-	Conductor (3-#2)	STO	1000	600	90	FT-4	Twin X Gradient Filter
980	TAC Y+,WB-,ZM-	PP1Y+,WB-,ZM-	Conductor (3-#2)	STO	1000	600	90	FT-4	Twin Y Gradient Filter
981	TAC Z+,WB-,ZM-	PP1Z+,WB-,ZM-	Conductor (3-#2)	STO	1000	600	90	FT-4	Twin Z Gradient Filter
982	PP1 ZZM+,ZM-	MG2 A12 A1 7/8	Conductor (3-#2)	STO	1000	600	90	FT-4	Twin Zzm Gradient Output
983	PP1 YZM+,ZM-	MG2 A12 A1 9/10	Conductor (3-#2)	STO	1000	600	90	FT-4	Twin Yzm Gradient Output
984	PP1 XZM+,ZM-	MG2 A12 A1 11/12	Conductor (3-#2)	STO	1000	600	90	FT-4	Twin Xzm Gradient Output
985	PP1 ZWB+,WB-	MG2 A12 A1 1/2	Conductor (3-#2)	STO	1000	600	90	FT-4	Twin ZWB Gradient Output
986	PP1 YWB+,WB-	MG2 A12 A1 3/4	Conductor (3-#2)	STO	1000	600	90	FT-4	Twin YWB Gradient Output
987	PP1 XWB+,WB-	MG2 A12 A1 5/6	Conductor (3-#2)	STO	1000	600	90	FT-4	Twin XWB Gradient Output
1238	PP1 J4	CONNECT TO RUN 1249	Times Microwave LMR-600FR	CATV R	6000 VDC		85	FT-4	RF Head Transmit
1328	PP1 J4	MG3 PEDI/F J2	Times Microwave LMR-600FR	CATV R	4000 VDC		85	FT-4	RF Head Transmit

10-2-7 EQUIPMENT AND OPERATOR ROOM CABLES

The cables listed in Table 10-9 are routed and connected between Penetration Panel, cabinets in equipment room and Operator Workspace.

**TABLE 10-9
EQUIPMENT & OPERATOR ROOM RUNS**

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
296	FACILITY DISCNCT	PP1 J18	45 ft 9 pin Sub-D	CI3	300	<30	80	FT-4	Emergency off switch
457	OM1	PP1 J17	9 pin Sub-D	CI3	300	<30	80	FT-4	Optional O2 Monitor cable
611	TAC J22	PP1 SHIM							Resistive Shim P.S. Option
701	MR2 A11 J15	RF Door switch	70 ft 9 pin Sub-D	CI3	300	<30	80	FT-4	RF Door switch
971	MR2 A11 J23	MR3 PD1 J2	40 ft 15 pin Sub-D	CI3	300	<30	80	FT-4	PDU Status (ACGD)
974	MR3 PD1 J5	TAC J12	49 ft. 9 pin Sub-D	CI3	300	<30	80	FT-4	Twin GIP/GP
1056	OW1 A17 PRT6	OW1-A12 ETHRNET	3.3 ft. Ethernet Cat 5	CI3	30	<30	80	FT-4	Laptop Access cable
1059	OW1 A17 PRT2	OW1-A15 ETHRNET	5.9 ft. Ethernet Cat 5	CI3	30	<30	80	FT-4	Lan Hub cable
1060	OW1 A19 J3	OW1 A21 J17	3.9 ft. 9 pin Sub-D	CI3	300	<30	80	FT-4	GOCAA Power cord
1061?	OW1 A15 KYBD/MS	OW1 A21 J8	2.9 ft. 25 pin Sub-D 2 PS/2s	CI3	300	<30	90	CL-2	Keyboard/Mouse cable
1063?	OW1 A15 Video 1	OW1 A6 HOST LCD	9.9 ft. 29 pin DVI, 15 pin VGA	CI3	300	<30	80	VW-1	Host LCD Display Video cable
1064	OW1 A15 SCSI 1	OW1 A16 SCSI TOWER	9.9 ft. SCSI cable 50	CI3	300	<30	80	VW-1	SCSI Tower Data cable
1065	OW1 A11 J2	OW1 A6 Pwr	9.9 ft. Power (3-#18)	SVT	300	120	80	VW-1	Host LCD Display Power cable
1066	OW1 A15	OW1 A13	3.9 ft. 25/9 pin Sub-D	CM	300	<30	80	FT-4	Modem I/O cable
1067	OW1 A15 SCSI 2	OW1 A5 DASM	5.9 ft. SCSI cable 50	CI3	30	<30	80	VW-1	DASM I/O cable
1070	OW1 A11 J11	OW1 A16 Pwr	9.9 ft. Power (3-#18)	SVT	300	120	80	VW-1	SCSI Tower Power cable
1071	OW1 A11 J1	OW1 A15 Pwr	9.9 ft. Power (3-#18)	SVT	300	120	80	VW-1	Host Computer Power cable
1074	OW1 A19 J2	OW1 A11	3.3 ft. Power (3-#14)	SVT	600	120	105	VW-1	Power Strip cord
1075	OW1 A19 J4	OW1 A11	3.3 ft. Power (3-#14)	SVT	600	120	105	VW-1	Power Strip cord
1085	OW1 A21 J7	PP1 J12	65 ft 25 pin Sub-D	CI3	300	<30	80	FT-4	Intercom
1180	MR3 J7	MR2 A11 J56	26 ft. 9 pin Sub-D	CI3	300	<30	80	FT-4	Can
1181	MR3 J8	PP1 PHPS J154	65 ft 9 pin Sub-D	CI3	300	<30	80	FT-4	Can
1184	MR2 A11 J125	PP1 PHPS J32	65 ft 25 pin Sub-D	CI3	300	<30	80	FT-4	
1185	MR2 A11 J200	PP1 J200	8-Coax 8W8 Combo D	CATV R	1500	<30	85	FT-4	
1186	MR2 A11 J201	PP1 J201	8-Coax 8W8 Combo D	CATV E	1500	<30	85	FT-4	16 Channel only
1187	MR2 A11 J202	PP1 J202	2-Coax 8W8 Combo D	CATV R	1500	<30	85	FT-4	
1188	MR2 DRV J3	PP1 A1 J77	65 ft 15 pin Sub-D	CI3	300	<30	80	FT-4	16 Channel SW Cntrl Power
1189	MR2 DRV J5	PP1 A17 J78	65 ft 37 pin Sub-D	CI3	300	<30	80	FT-4	MC Drive
1190	MR2 DRV J10	PP1 J203	65 ft 37 pin Sub-D	CI3	300	<30	80	FT-4	MC Drive
1191	MR2 DRV J21	PP1 A11 J92	65 ft. RG 58C/U Coax	CI2	1900 VRMS	1000 VDC	80	VW-1	Dynamic Disable

10-2-7 EQUIPMENT AND OPERATOR ROOM CABLES (Continued)

TABLE 10-9 (CONT'D)
EQUIPMENT & OPERATOR ROOM RUNS

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
1192	MR2 DRV J22	PP1 A11 J93	65 ft. RG 58C/U Coas	CI2	1900 VRMS	1000 VDC	80	VW-1	Dynamic Disable
1193	MR2 DRV J23	PP1 A11 J94	65 ft. RG 58C/U Coas	CI2	1900 VRMS	1000 VDC	80	VW-1	Dynamic Disable
1194	MR2 DRV J24	PP1 A11 J95	65 ft. RG 58C/U Coas	CI2	1900 VRMS	1000 VDC	80	VW-1	Dynamic Disable
1195	MR2 DRV J25	PP1 A11 J96	65 ft. RG 58C/U Coas	CI2	1900 VRMS	1000 VDC	80	VW-1	Dynamic Disable
1196	PP1 A17 J1	PP1 PHPS J35	2 ft 15 pin Sub-D	CI3	300	<30	80	FT-4	Repeater Board Power
1233	TAC J116 (J5)	PP1 PHPS J155	30 ft. 9 pin Sub-D	CI3	300	<30	80	FT-4	Can
1249	MR8 J3	CONNECT TO RUN 1238	5 ft. Times Microwave LMR	CATV R	6000 VDC		85	FT-4	RF Extension cable
1255	OW1 A21 J18	MR2 A11 J26	60 ft 9 pin Sub-D	CI3	300	<30	80	FT-4	E-stop
1259	MR2 RF AMP J4	CONNECT TO RUN 235	Times Microwave LMR 400FR	CATV R	2500 VCD		85	FT-4	
1320	OW1 A15 USB	HUB IN	2000mm RJ11						Temp/Humidity Sensor
1321	OW1 A2 USB	HUB OUT	2000mm RJ11						Temp/Humidity Sensor
1322	MR2 A11 Cabinet Sensor	HUB C2	100 ft. RJ11						Temp/Humidity Sensor
1329	MR2 DRV J10	PP1 A17 J78	65 ft 37 pin Sub-D	CI3	300	<30	80	FT-4	MC Drive
1330	MR2 DRV J5	PP1 J203	65 ft 37 pin Sub-D	CI3	300	<30	80	FT-4	MC Drive
1371	OW1 A15 ETH D	MR2 A11 J63	75 ft. Ethernet Cat 5	CI3	300	<30	80	FT-4	EXCITE Ethernet C'cable
2037	OW1 A15 Video 1	OW1 A6 MAIN LCD	9.9 ft. 29 pin DVI, 15 pin VGA	CI3	300	<30	80	VW-1	Host LCD Display Video cable
2038	OW1 A7 KYBD	OW1 A15 USB	1000mm USB or 2000mm USB						Keyboard extension cable
2039	MOUSE	HOST PC	1000mm USB						Mouse extension cable
2053	OW1 A20	OW1 A21 J6	13.2 ft. 50 pin SCSI, 9 pin Sub-D	CI3	300	<30	80	VW-1	SCIM cable

10-2-8 CABLE SPARES (FRU'S)

TABLE 10-10
EQUIPMENT & OPERATOR ROOM RUNS

CABLE PART NUMBER	CABLE RUN
46-244710G30	Run 1227 Run 1236
46-244712G30	Run 1188
46-244713G30	Run 1181
46-244714G30	Run 1180 Run 1181 Run 1233
46-258877G30	Run 1189 Run 1190

10-2-9 EXCITE 4KW AND 8KW MNS CABLES (Optional)

TABLE 10-11
EXCITE 4KW AND 8KW MNS RUNS (optional)

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
1129	CPC J8	MR3 PD1	35 ft. Power (3-#10)	STO	600	208	90	FT-4	4KW CPC Amp Power cable
1130	CPC J3	MR2 A11 J52	30 ft. RG 223/U Coax	CMX	1900V RMS	<30	80	VW-1	4KW MNS cable
1136	CPC J9	MR2 A11 J4	30 ft. RG 223/U Coax	CMX	1900V RMS	<30	80	VW-1	4KW MNS cable
1137	RUN 1158	PP1 J83	100 ft. Times Microwave LMR 600FR	CATV R	4000V DC		85	FT-4	4KW MNS cable
1138	MR6 J12	MR3 PD1	65 ft. Power (5-#8)	SO	600	208Y	75	FT-4	BB RF Amp Power Cable
1139	AMT J3	Mr2 A11 J52	30 ft. RG 223/U Coax	CMX	1900V RMS	<30	80	VW-1	8KW MNS cable
1140	MR1 J123	MR6 J1	30 ft. 25 pin Sub-D	CI3	300	<30	80	FT-4	8KW MNS cable
1141	MR1 J102	AMT J5	30 ft. RG 223/U Coax	CMX	1900V RMS	<30	80	VW-1	8KW MNS cable
1142	MR1 J103	AMT J6	30 ft. RG 223/U Coax	CMX	1900V RMS	<30	80	VW-1	8KW MNS cable
1143	MR1 J104	AMT J7	30 ft. RG 223/U Coax	CMX	1900V RMS	<30	80	VW-1	8KW MNS cable
1144	MR1 J105	AMT J8	30 ft. RG 223/U Coax	CMX	1900V RMS	<30	80	VW-1	8KW MNS cable
1145	MR2 A11 J4	AMT J11	30 ft. RG 223/U Coax	CMX	1900V RMS	<30	80	VW-1	8KW MNS cable
1146	RUN 1159	PP1 J83	100 ft. Times Microwave LMR 600FR	CATV R	4000V DC		85	FT-4	8KW MNS cable
1147	MG3 A17 J3	PP1 J83	100 ft. Times Microwave LMR 600FR	CATV R	4000V DC		85	FT-4	4KW & 8KW MNS cable
1158	RUN 1137	MR1 CPC J2	5 ft. Times Microwave LMR 400FR	CATV R	2500V DC		85	FT-4	4KW MNS Extension cable
1248	RUN 1146	MR9 J4	5 ft. Times Microwave LMR 400FR	CATV R	2500V DC		85	FT-4	8KW MNS Extension cable

10-3 FIBER OPTIC TERMINATION INSTRUCTIONS

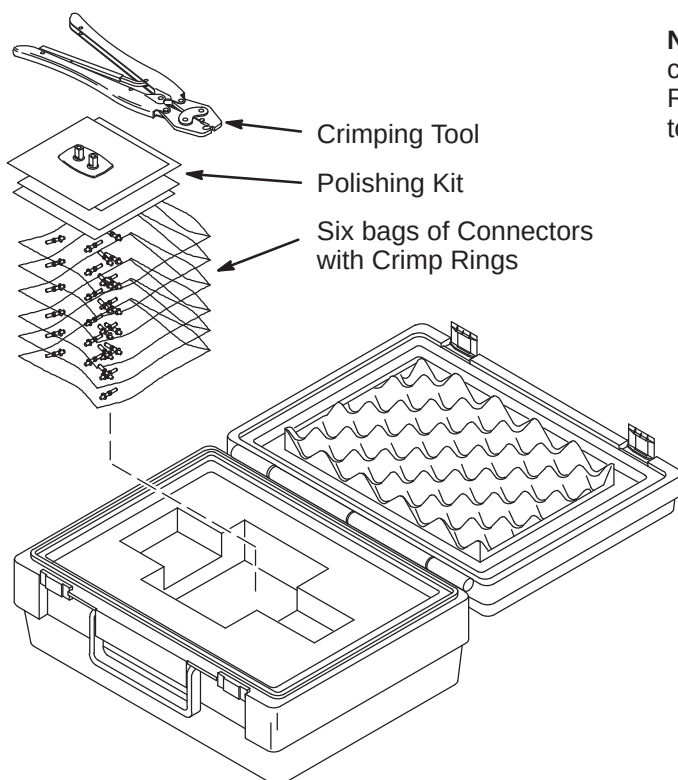
WARNING!

FERROUS MATERIAL HAZARD! CRIMP TOOL AND OTHER TOOLS REQUIRED FOR THIS PROCEDURE CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES. KEEP ALL FERROUS TOOLS AS FAR FROM THE MAGNET AS POSSIBLE.

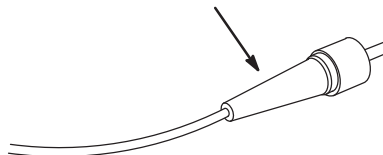
10-3-1 FIBER OPTIC CABLE TERMINATION KIT

The connectors on plastic fiber optic cables can be replaced if damaged or cut off to shorten.

1. The Fiber Optic Repair Kit, 46-301450G1 shown below, contains all the necessary materials to replace a damaged connector.
2. The "Spares Kit", delivered with the Shipping Collector, contains a 46-320119P1 Termination Kit for Fiber Optic Cables. This kit consists of consumables such as terminals and polishing sheets used to terminate cables during installation.
3. Remove the end of fiber optic cable to be re-terminated from magnet room if possible.
4. If it is not possible to remove the end to be repaired from the magnet room, the cable must be at least 10 feet away from the magnet before it can be worked on.
5. Cut cable to desired length.



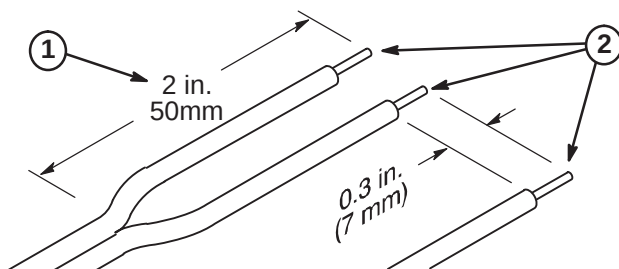
Note: The connectors on glass fiber optic cables **cannot** be replaced using the F/O Repair Kit. At this time, there is no kit available to repair glass fiber optic cable connectors.



10-3-2 STRIPPING FIBER OPTIC OUTER JACKET

The separated duplex cable must be stripped to equal lengths on each side of the cable. This allows proper seating of the cable into the connector.

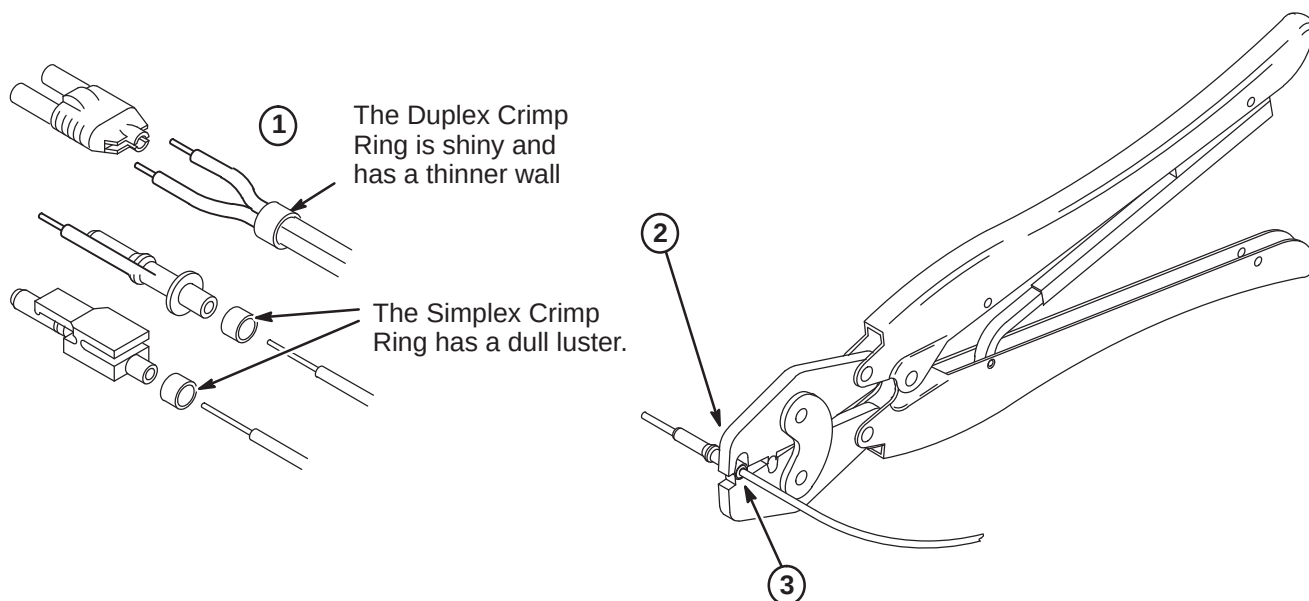
1. For Duplex Cables, separate the two fibers approximately 2 in. (50mm) back from the ends.
2. Strip off approximately 0.3 in. (7 mm) of the outer jacket with the 16 gauge wire strippers. Excess webbing on duplex cable may have to be trimmed to allow the connector to slide over the cable.



10-3-3 CRIMPING CONNECTOR TO FIBER OPTIC CABLE

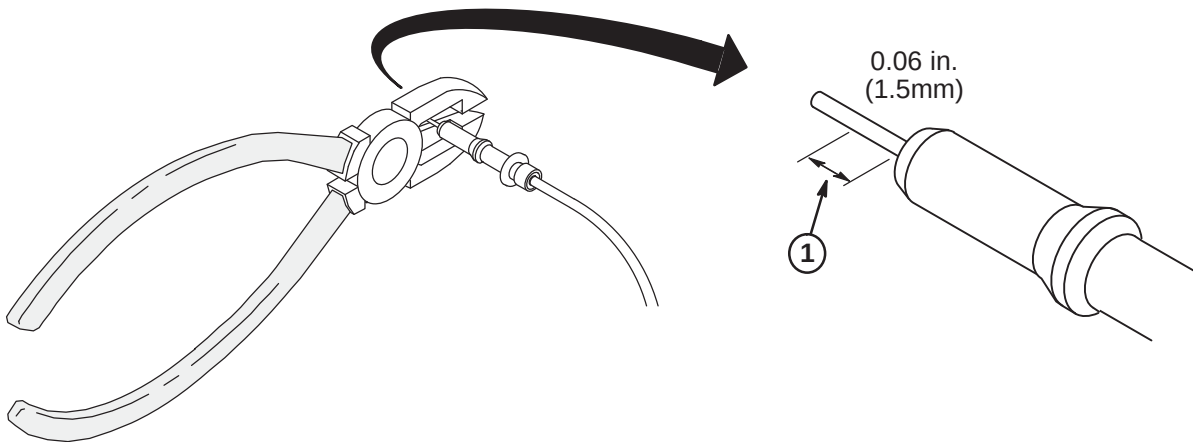
The Connector Crimp Rings cannot be interchanged between simplex and duplex connectors.

1. Place the Crimp Ring and Connector over the end of the cable.
2. The fiber should extend approximately 0.12 in. (3mm) through the end of the connector.
3. Carefully position the ring so that it is entirely on the connector and then crimp the ring in place with the crimping tool.



10-3-4 TRIMMING EXCESS FIBER FROM CONNECTOR

1. Cut off excess fiber protruding from the connector. The trimmed fiber should extend approximately 0.06 inch (1-1/2mm).



10-3-5 POLISHING FIBER OPTIC END

The four dots on the bottom of the polishing fixture are wear indicators. Replace the polishing fixture when any dot is no longer visible.

1. Insert the connector fully into the polishing fixture with the trimmed fiber end protruding from the bottom of the fixture.
2. Place the 600 grit abrasive paper on a flat smooth surface. Pressing down on the indicator, polish the fiber and the connector using a figure eight motion until the connector is flush with the end of the polishing fixture.
3. Wipe the connector and fixture with a clean cloth or tissue.
4. Place the flush connector and polishing fixture on the dull side of the 3 micron pink lapping film and continue to polish the fiber and connector for approximately 25 strokes.
5. The fiber end should be flat, smooth and clean. The cable end can now be connected.

